

SKG12.3 Reliability & Maintainability Engineering

Part 1 Level 2

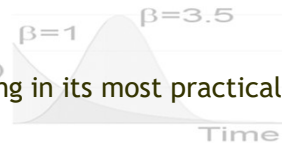
AP Ts Dr Masdi Muhammad CEng(UK), CMRP, CRE
 AP Ts Dr Ainul Akmar Mokhtar
 Ts Dr Hilmi Hussin

The Weibull Distribution with $\gamma=0$

$$f(t) = \frac{\beta}{\eta} \left(\frac{t}{\eta} \right)^{\beta-1} e^{-\left(\frac{t}{\eta} \right)^{\beta}} \rightarrow F(t) = 1 - e^{-\left(\frac{t}{\eta} \right)^{\beta}}$$

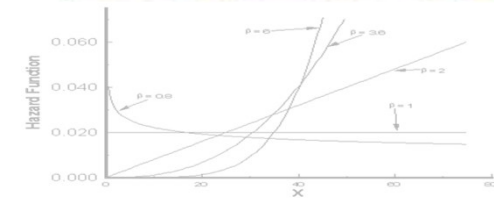


$$f(t) = \frac{\beta}{\eta} \left(\frac{t}{\eta} \right)^{\beta-1} e^{-\left(\frac{t}{\eta} \right)^{\beta}}$$



'Reliability is, after all, engineering in its most practical form.'

James R. Schlesinger
 Former US Secretary of State for Defence



Introduction

Name

Job function

Experience in reliability analysis

What do you expect from this course?



SKG12.3 TI & R

7-3-13-0003-QR1 v2.0

SKG 12 - Technology Inventory & Ruler
Discipline: 12.3 Reliability Engineering
Subdiscipline: -
Effective Date: 1st January 2020



Technology Inventory			Level 1	Level 2		Level 3			Level 4 Staff			Level 5 Principal			Level 6 Custodian	
				E1		E2			E3			E4			E5	
			20	21	22	E	23	24	E	25	26	E	27	28	E	29
Base	1	Reliability Integrity Management (RIM)	1	2	2	2	2	3	3	3	3	4	4	4	4	5
	2	Data Acquisition, Validation and Decision Analysis (DAVDA)*	1	1	2	2	3	3	3	4	4	4	4	4	4	4
	3	Asset Performance Analysis (APA)*	1	1	2	2	3	3	3	3	3	3	4	4	4	5
	4	Equipment Reliability Strategy (ERS)*	1	2	2	2	3	3	3	4	4	4	4	4	4	4
	5	Root Cause Analysis (RCA)*	1	2	2	2	3	3	3	4	4	4	4	4	4	5
	6	Life Cycle Costing (LCC)	1	1	2	2	2	3	3	3	4	4	4	4	4	4
	7	Benchmarking Process	1	1	1	2	2	2	3	3	3	3	4	4	4	4
Key	1	Risk Assessment (RA)	1	1	1	1	2	2	2	3	3	3	4	4	4	4
	2	System Reliability Modeling	1	1	2	2	2	2	2	3	3	3	4	4	4	5
	3	Advance Analytics	1	1	1	1	2	2	2	2	3	3	3	4	4	4
	4	Predictive Technology	1	1	1	1	2	2	2	3	3	3	3	4	4	4
Pacing	1	Multi State System Reliability Modeling	0	0	0	0	0	1	1	2	2	2	3	3	3	4
	2	Design for Reliability	0	0	0	0	0	0	0	1	1	2	2	3	3	4
Emerging	1	Digital Twins (DT) model in Asset Management	0	0	0	0	0	0	0	1	1	2	2	3	3	3

* indicates compulsory TI

Technology Competence Level

5 Expert

= Have the ultimate knowledge/Skill of Technology Inventory
= Network, originate - advise other's how to do.

4 Advanced

= Have advanced knowledge/Skill of Technology Inventory in question
= Mentor, integrate - teach others how to do

3 Skill

= Fully conversant with Technology Inventory in question
= Apply, demonstrate - do without help

2 Knowledge

= Have basic knowledge, able to perform according to instruction/method available.
= Contribute, participate - do without help

1 Awareness

= Aware of Technology Inventory in question, help needed.

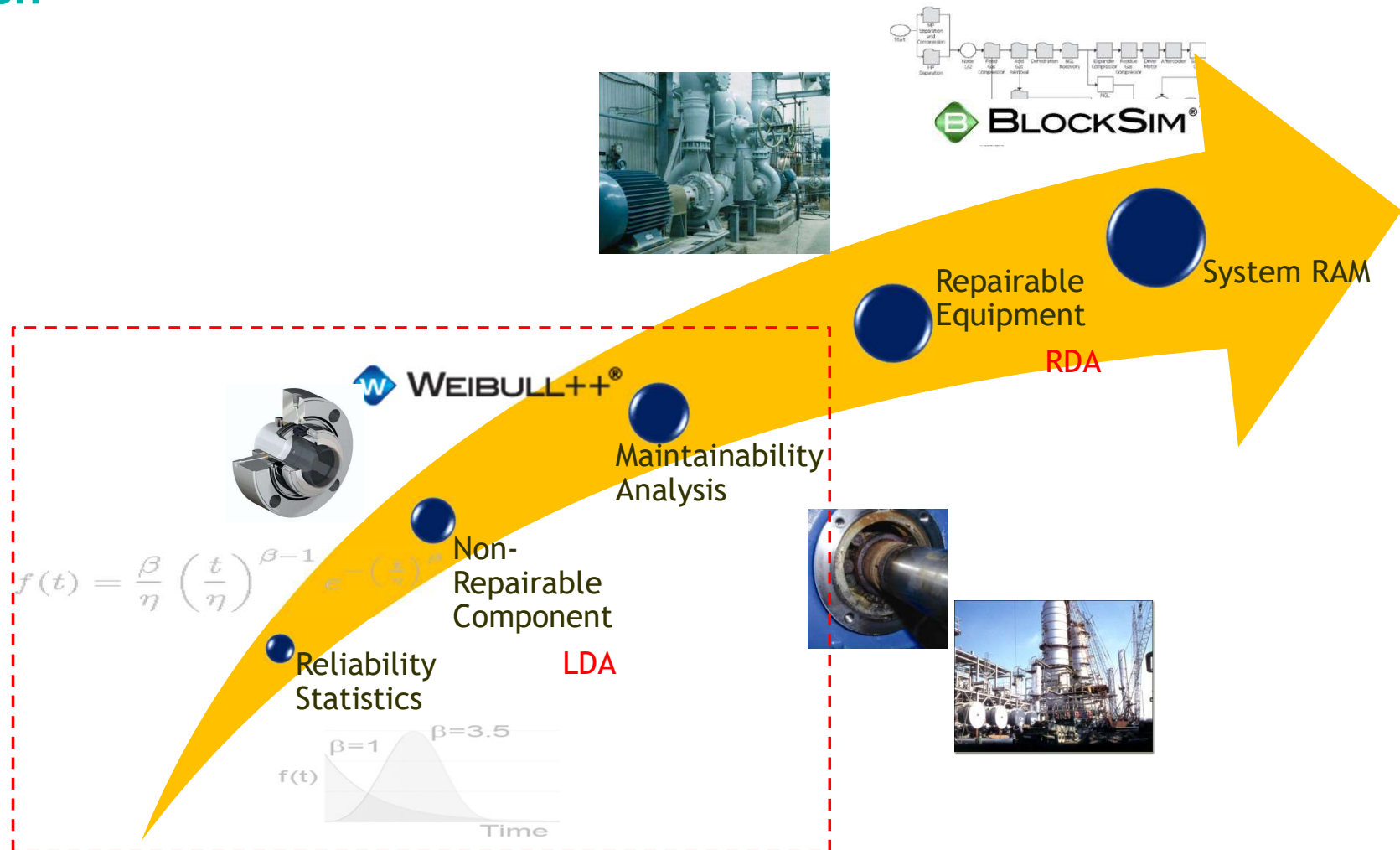
0 Not Applicable

= Recognise, identify - ask when help needed
= Technology Inventory in question is not applicable to Role Level in question.

Course Objectives

- By end of this course, participants will be able to:
 - Perform life data analysis (LDA) to estimate reliability measures such as failure rates, MTTF, B10 life
 - Apply LDA in assessing component life comparison, estimate optimum replacement and forecast number of spares required
 - Perform degradation analysis to estimate the remaining life based on condition monitoring data such as thickness data, crack length

Approach



Session schedule

Session	Content	Main Topics	Date	Time
0	Introduction	• SKG12.3 TI&R	Day 1	8:30am - 9:15am
1. 25/8/2021	Introduction to Reliability (Dr Masdi)	• Basic Terminologies/Acronyms • SKG12.3 TI&R • RIMS • Failure Definition • Reliability Measures • Bathtub Curve		9:15am - 10:00am Break (15 Min) 10:15am – 1:00pm Lunch/Prayer (1:00pm – 2:00pm)
2. 25/8/2021	Reliability Statistics (Dr Hilmi)	• Random variables • PDF,CDF and Reliability Function • Uniform Distribution • Exponential Distribution • Weibull Distribution • Normal Distribution • Lognormal Distribution	Day 1	2:00pm – 3:30pm Break (15 Min) 3:45pm – 5:00pm
3. 26/8/2021	Life Data Analysis (Dr Ainul)	• Parametric and Non-Parametric • Data Source • Data Types • Steps in conducting LDA	Day 2	8:30am - 10:00am Break (15 Min) Lunch/Prayer (1:00pm – 2:00pm)
4. 26/8/2021	Life Data Analysis- Cont (Dr Hilmi)	• Ranking of data for complete and censored data • Parameter Estimation • Goodness of Fit Test • LDA in Microsoft Excel • Case Study discussion	Day 2	10:15am – 1:00pm Lunch/Prayer (1:00pm – 2:00pm) 2:00pm – 3:30pm Break (15 Min) 3:45pm – 5:00pm

Session schedule

Session	Content	Main Topics	Date	Time
5. 1/9/2021	Application of Life data Analysis (Dr Ainul -AM/ Dr Hilmi-PM)	• Case Study Presentation (revision of session first session) • Confidence level and confidence Bound • Competing Failure Mode analysis • Life comparison • Preventive Replacement • Spare Part requirement	Day 3	8:30am - 10:00am Break (15 Min) 10:15am – 1:00pm Lunch/Prayer (1:00pm – 2:00pm) 2:00pm – 3:30pm Break (15 Min) 3:45pm – 5:00pm
6. 1/9/2021	Maintainability Engineering Analysis (Dr Ainul/Dr Masdi)	• Maintainability Analysis • Estimating maintainability function • Degradation Analysis • Weibull distribution in API • Weibull Analysis for PRD • Expert Elicitation	Day 4	8:30am - 10:00am Break (15 Min) 10:15am – 1:00pm Lunch/Prayer (1:00pm – 2:00pm) 2:00pm – 3:30pm Break (15 Min)
6.	Conclusion (Dr Masdi/Dr Ainul)	• Review • Post Test	Day 4	3:45pm – 5:00pm

Ground Rules

Be punctual

H/P in silent mode

Participate, ask questions, share experience



Introduction to RELIABILITY

Session Objectives

- To recognize the importance of reliability
- To understand the basic definitions of reliability and its measures
- To understand the concept of bathtub reliability curve
- To understand basic methodology in reliability analysis
- To differentiate system types

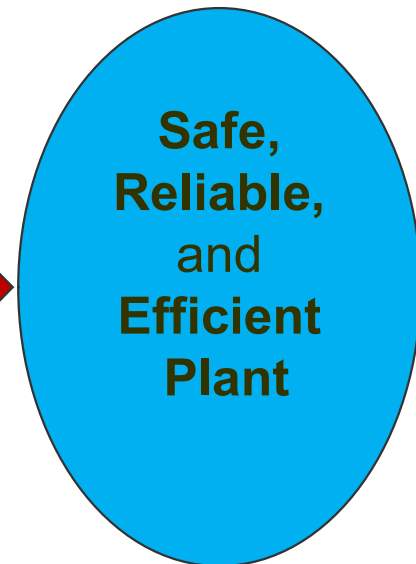
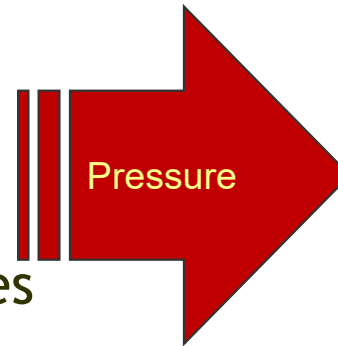
ACRONYMS

CCPS	Center for Chemical Process Safety
cdf	Cumulative distribution function
COF	Consequence of the failure
HPP	Homogeneous Poisson Process
LDA	Life Data Analysis
MTBF	Mean Time between failure
MTR	Mean time to restore
MTTF	Mean Time to Failure
MTTR	Mean time to repair
pdf	Probability distribution function
POF	Probability of Failure
RAM	Reliability Availability Maintainability
RBI	Risk Based Inspection
RDA	Recurrent Data Analysis
RIMS	Reliability Integrity Management System
TTF	Time to Failure
TTR	Time to repair

Highly competitive business environment

OPERATIONAL ISSUES

- ❖ Increased concern in safety and environment
- ❖ Tight profit margin
- ❖ Escalating operational cost
- ❖ Increased system complexity
- ❖ Depletion in oil and gas resources
- ❖ Increased in demand
- ❖ Changes in material, operating conditions, equipment ages



Why RELIABILITY?

PETROCHEMICAL BUSINESS DRIVERS

- ▶ Reduce operational cost
- ▶ Healthy, Safe and environmental friendly operation
- ▶ Maximize utilization
- ▶ Meeting operation target and customer demand
- ▶ Reduce wastes, failures and downtime
- ▶ High availability
- ▶ Continuously improve plant performance



**RELIABILITY
DIRECTLY IMPACTS
ALL THESE**

Reliability and Organization's profitability

Recent incident of oil spills in the Gulf of Mexico had caused an estimated of **USD 23 Billion** loss to BP



What causes it?

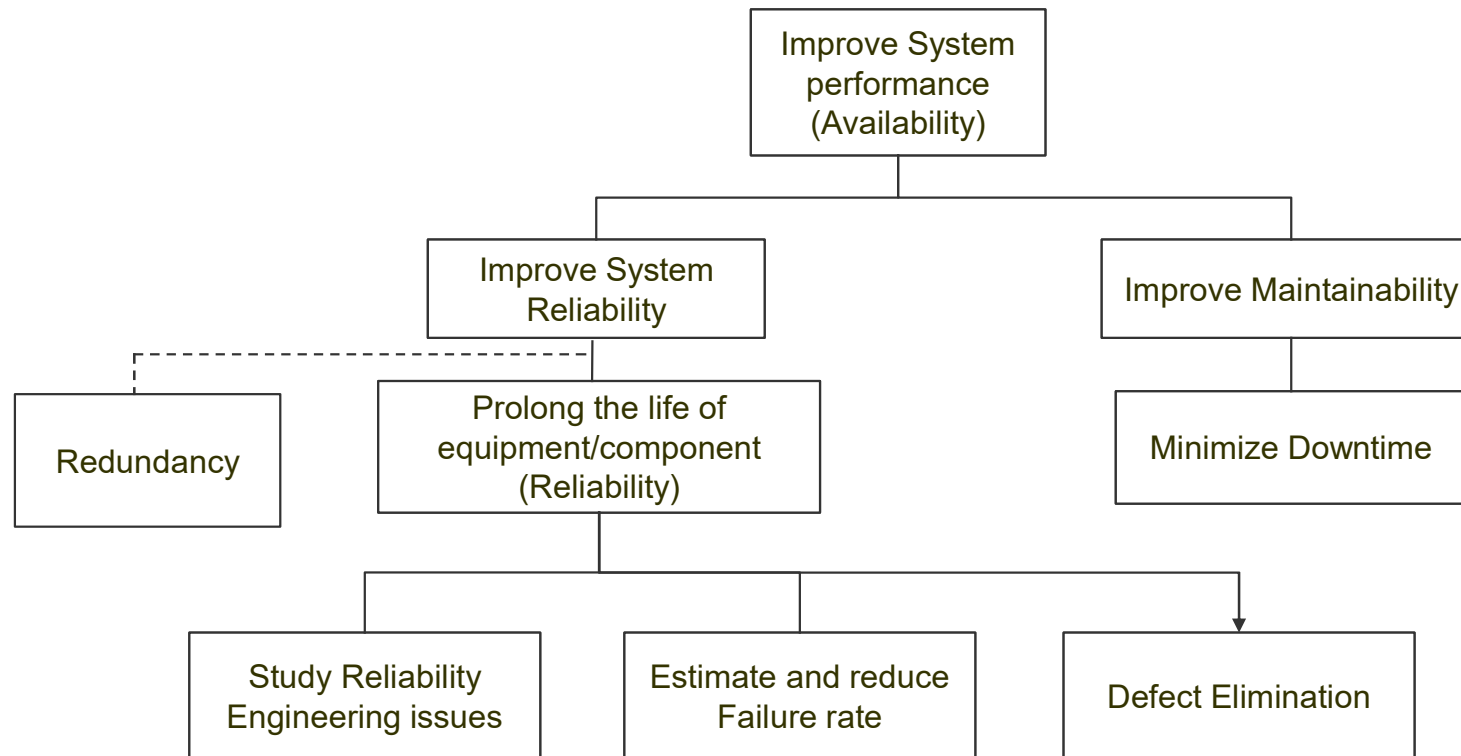
- Bad cement job
- **Failure of the shoe track barrier**
- The negative pressure test was accepted when it should not have been
- **Failure in well control procedures**
- **Failure in blow-out preventer failures**
- Rig's fire and gas system **failed** to prevent ignition

Source: BP report, www.bp.com

Objectives of Reliability Analysis

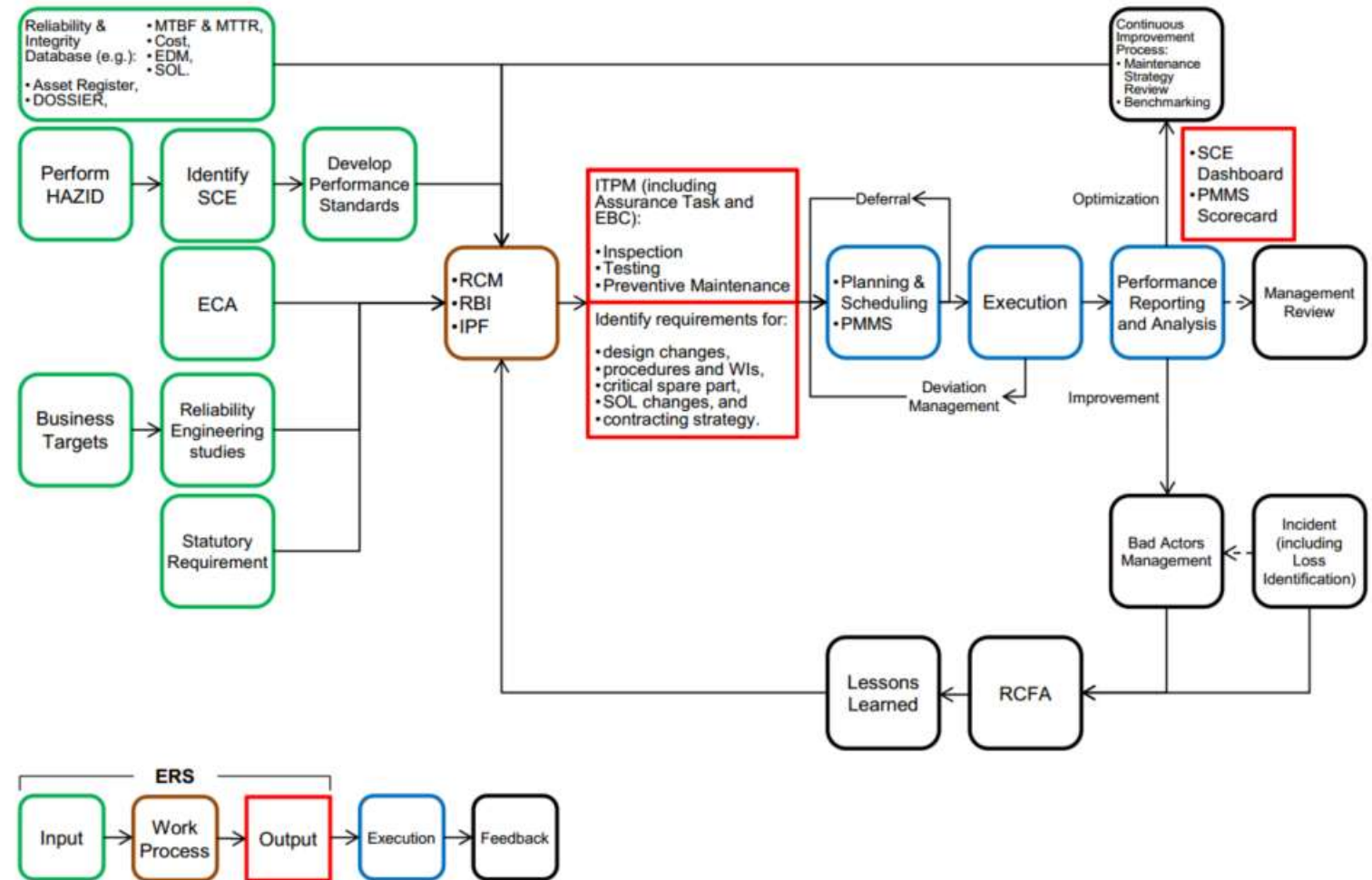
- Estimate probability of failure from which mitigation actions can be taken to prevent failure occurrence
- Determine time for replacement or overhaul through minimizing cost of ownership
- Compare reliability performance of component/system from different vendors for selection
- Implement the right maintenance strategy to improve system reliability and availability

System Performance Improvement



(Modarres, *et al* (1999))

Simplified Reliability and Integrity Management System



RIMS:OED-L1 -iPOCS-PAM-REL and INT, March 2015

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What is Failure?

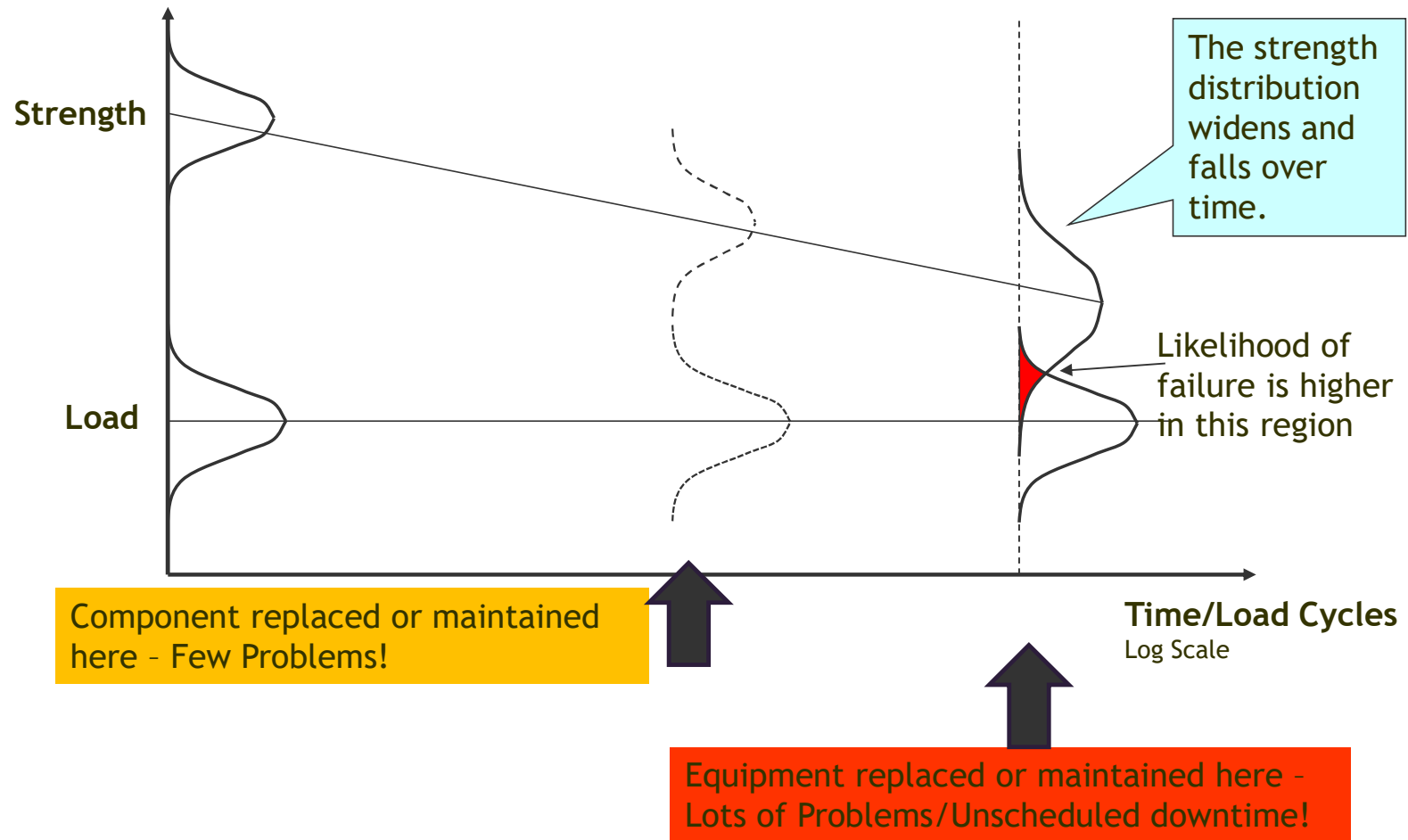
- Failed to perform the function that the equipment is designed for.
- Functional failures:
 - Unable to operate
 - Unable to operate at the required rate
 - Unable to meet the quality standard
 - Unable to meet the environment, health and safety standards

Category of Failure*

- Critical Failure
 - failure of component or system that causes an immediate cessation of the ability to perform a required function
- Degraded Failure
 - Failure does not cease the function but reduce. Delayed repair of such failure can develop into critical failure.
- Incipient Failure (crack, corrosion)
 - An imperfection in the state or condition of an item that is likely to result in a functional failure if not corrected.

*ISO 14224-2006

Why does component or system fail?



Potential causes of failure*

Causes	Descriptions
1. Poor design	Improper design, dimensions, tolerances, stress concentration, no interchangeability of parts
2. Improper Fabrication/installation	Improper foundation, excessive vibration, inadequate inputs (i.e. voltage etc.), wrong techniques/tools
3. Improper operation and maintenance	Under/over maintenance, wrong tools/technique, poor spare part management, insufficient skills and training
4. Poor operational instruction / SOP/ Management	Wrong instruction, lack of clarity, difficult to understand, poor language, planning, organization, quality assurance
5. Others/combined causes	Causes that do not fall into one of the categories listed above

*ISO 14224-2006

Brief History of RELIABILITY

❑ The word “reliability” was first coined by poet Samuel Taylor Coleridge in early 1800’s, but the concept of reliability can be traced back to early civilizations (Ancient Egypt, Babylon, China etc.)

❑ Early 1920’s, W.A. Shewhart introduced concept of statistical process control (SPC)

❑ Late 1930’, W. Weibull proposed a mathematical model for description, also known as Weibull distribution

❑ WWII - Reliability as a discipline played critical roles in the development and effectiveness of military equipment i.e. rockets, planes, electronics components

Since then Reliability field has been applied to various sectors: electronics, nuclear, automotive, oil and gas, etc.



What is RELIABILITY?

“the *probability* that the item will perform its *required function* under *given conditions* for the *time interval*”

- **Probability** - describe stochastic (random) behavior of occurrence of failure
- **Required function** - the designed function of the system
- **Given conditions** - the external condition in which the system usually operates
- **Time interval** - the design life period of the system

How to measure Reliability?

MEAN TIME TO FAILURE (MTTF)

The average time that elapses until a failure occurs. It is for **non-repairable** item

$$MTTF = \frac{1}{n} \sum_{i=1}^n t_i$$

Example 1:

Consider 6 similar type components have failure time of 23, 34, 32, 28, 19 and 27 days respectively

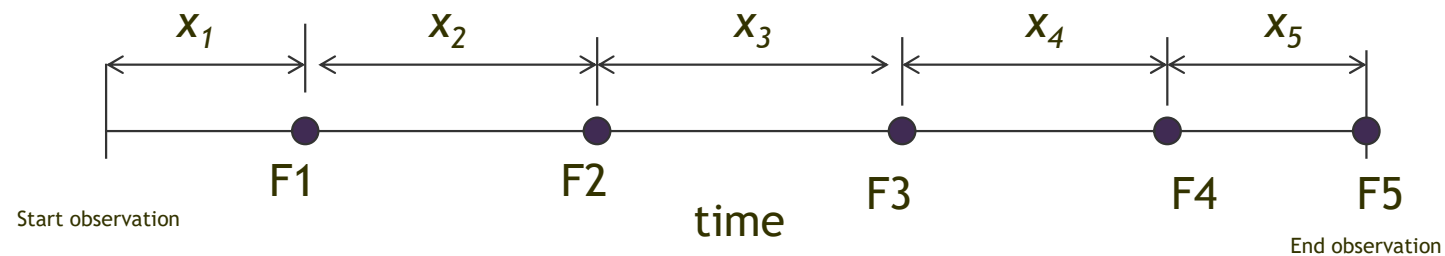
$$MTTF = (23+34+32+28+19+27) / 6 = 27.2 \text{ days}$$

How to measure Reliability?

MEAN TIME BETWEEN FAILURE (MTBF)

The average time between successive failures. It is used for **reparable systems** when **failure rate** is assumed to be **constant** (random failure).

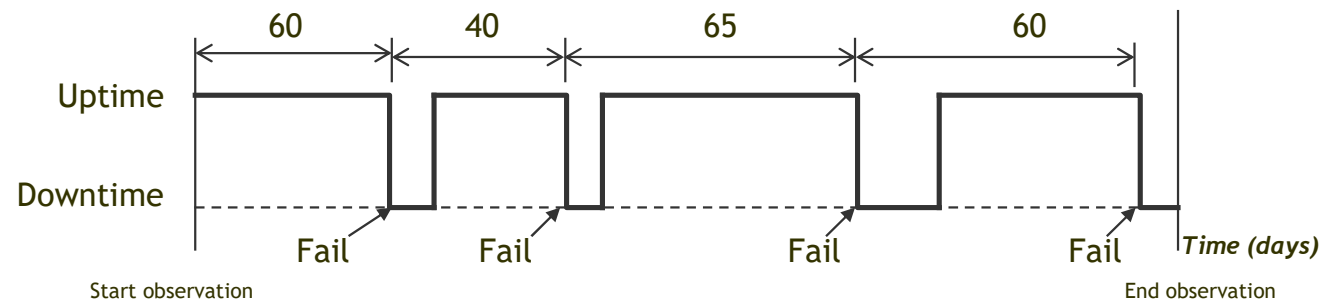
$$MTBF = \frac{1}{n} \sum_{i=1}^n x_i$$



$$MTBF = (x_1 + x_2 + x_3 + x_4 + x_5) / 5$$

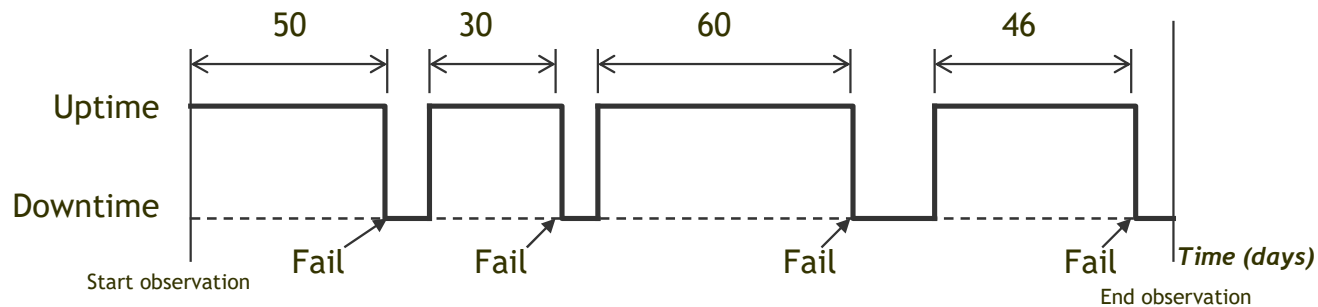
RELIABILITY MEASURES

Option 1: Ignore downtime



$$MTBF = (60+40+65+60) / 4 = 56.25 \text{ days}$$

Option 2: Consider downtime



$$MTBF = (50+30+60+46) / 4 = 46.5 \text{ days}$$

RELIABILITY MEASURES

FAILURE RATE (%)

Percentage of equipment failed during operation

Example 2:

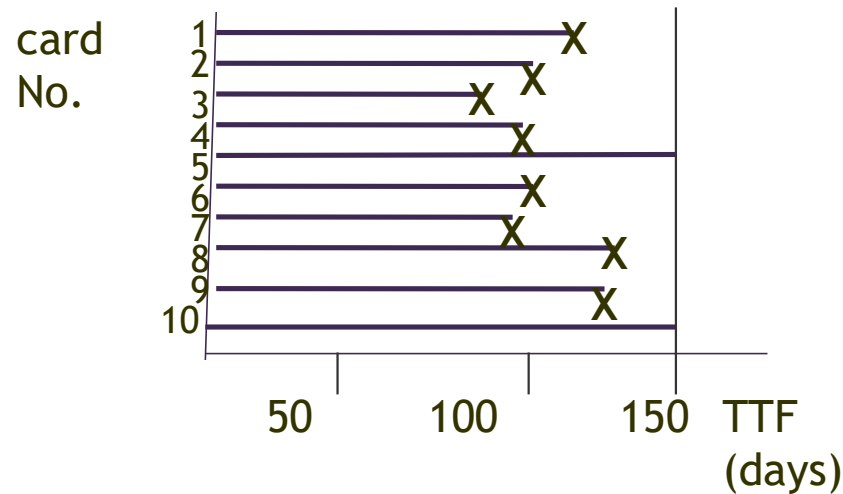
In 2013, 20 pumps were run. Out of 20, four pumps failed during operation. What is the FR?

$$FR = 4 / 20 = 20\%$$

RELIABILITY MEASURES

FAILURE RATE (FAILURE/UNIT TIME)

During 150 days of observation for 10 similar type communication cards, 8 cards failed during operation (data in table). Calculate what is the FR for the component?



$$FR = 8 / (110+100+85+98+150+100+95+120+118+150)$$

$$= 8 / 1126 = 0.0071 \text{ failure/day}$$

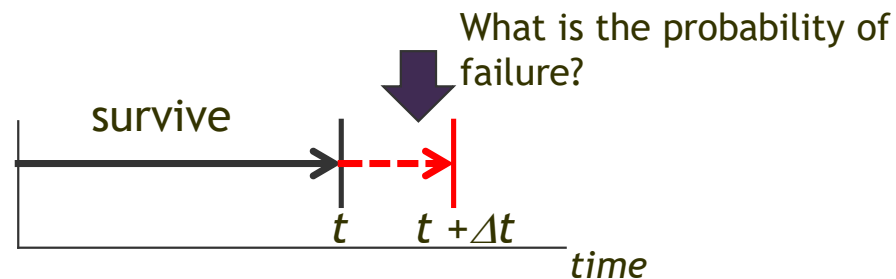
Card no	TTF (days)
1	110
2	100
3	85
4	98
5	Not fail
6	100
7	95
8	120
9	118
10	Not fail

Which situation is this calculation applicable?

RELIABILITY MEASURES

FAILURE RATE (HAZARD RATE)

Failure rate (hazard rate) is the conditional probability that a component fails in a small time interval given that it has survived from time zero until the beginning of the time interval.



Note : Failure rate term has been widely used to describe reliability of both non-repairable components and repairable system. The more appropriate term for **non-repairable** is **hazard rate**, and for **repairable** is **rate of occurrence of failure (ROCOF)**

RELIABILITY MEASURES

FAILURE RATE (HAZARD RATE) CT'D

Failure rate is an important function in Reliability study since it describes changes in the probability of failure over the lifetime of the item hence the item's reliability performance

Increasing rate = reliability deteriorates

Decreasing rate = reliability improves

Constant rate = reliability maintains

RELIABILITY MEASURES

RATE OF OCCURRENCE OF FAILURES (ROCOF)

ROCOF is measure of failure rate over time for repairable equipment / system.

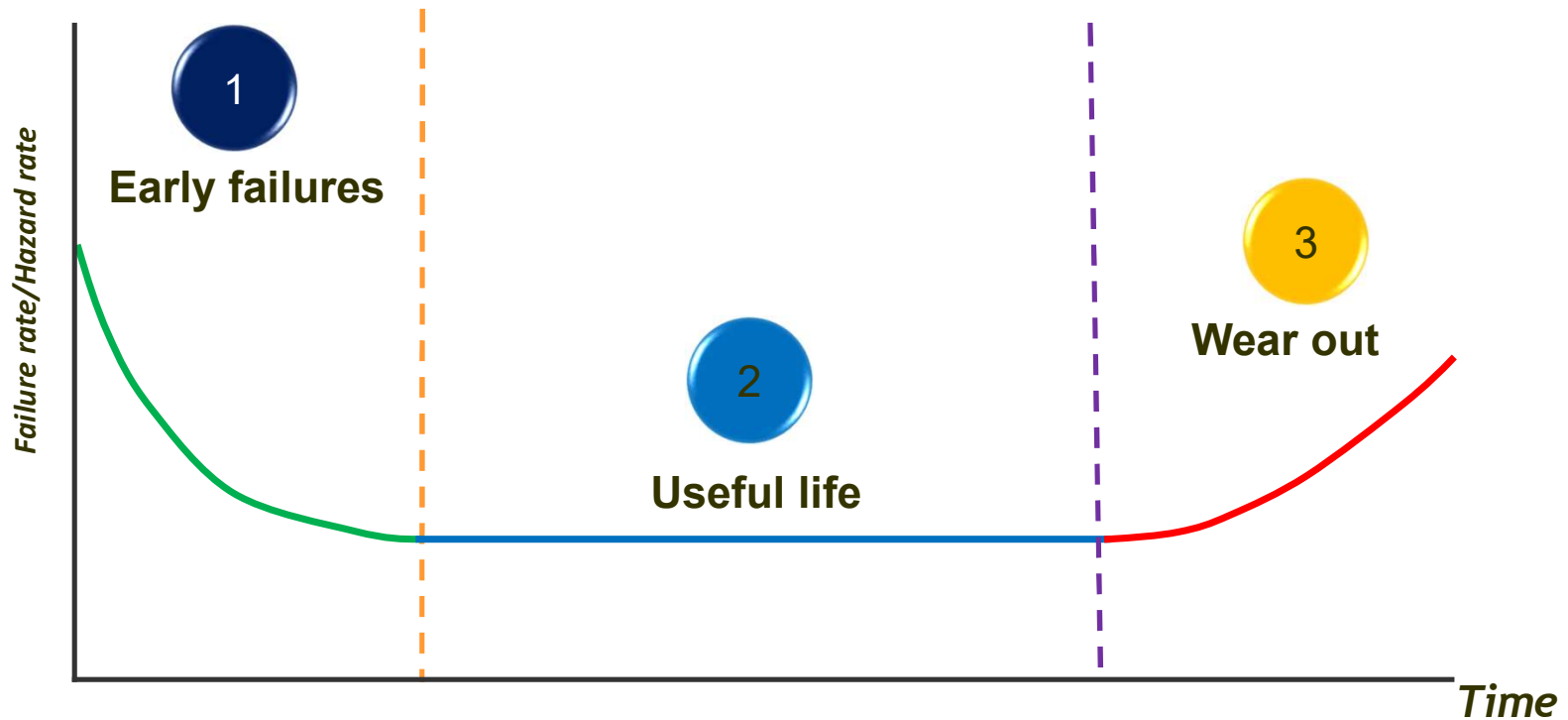
Example 3: Failures data for a gas turbine



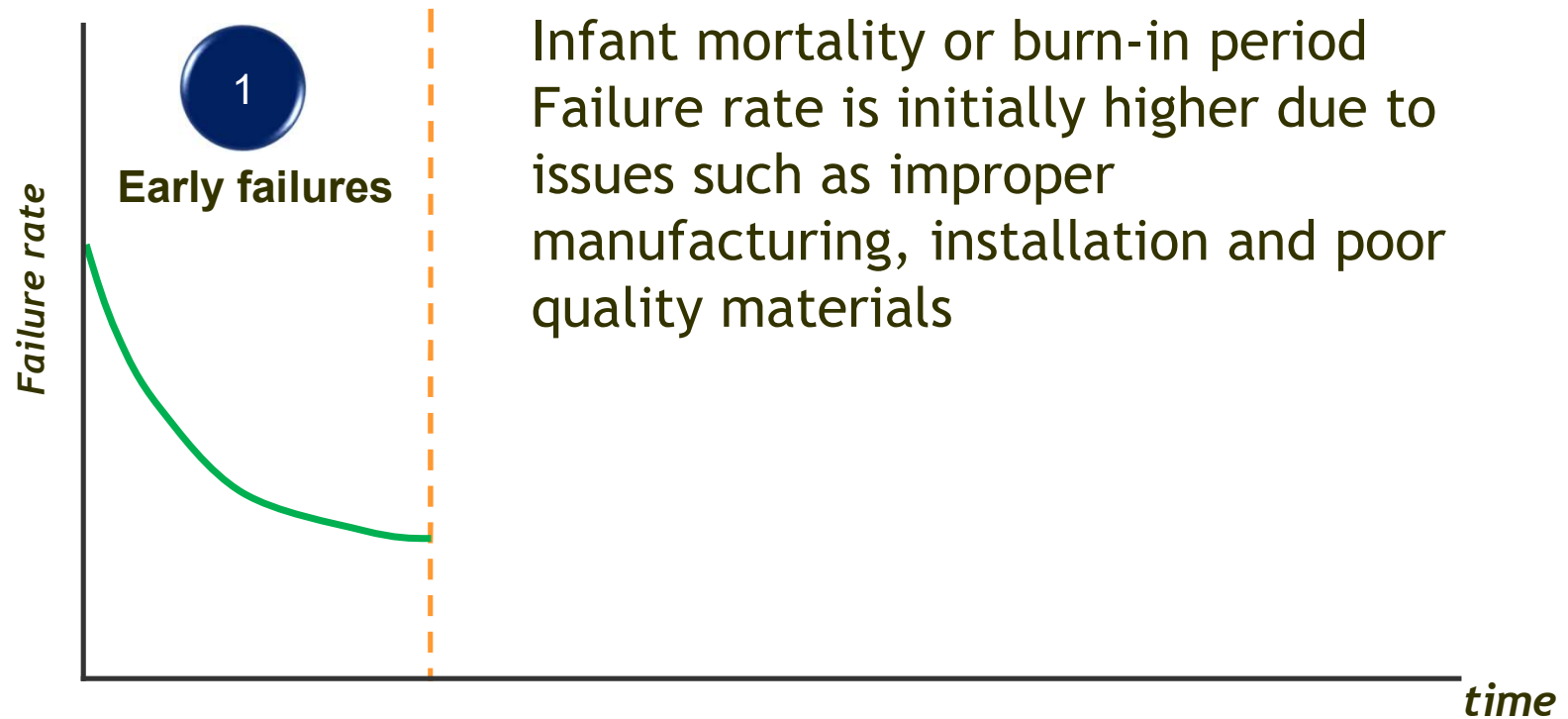
Year	2009	2010	2011	2012
ROCOF (failure per day)	$\frac{2}{365} = 0.0055$	$\frac{2}{365} = 0.0055$	$\frac{1}{365} = 0.0027$	$\frac{3}{365} = 0.0082$

Bathtub curve

Bathtub curve is a conceptual model of the reliability characteristics (failure rate) of a component or system over its lifetime. It is divided into three regions

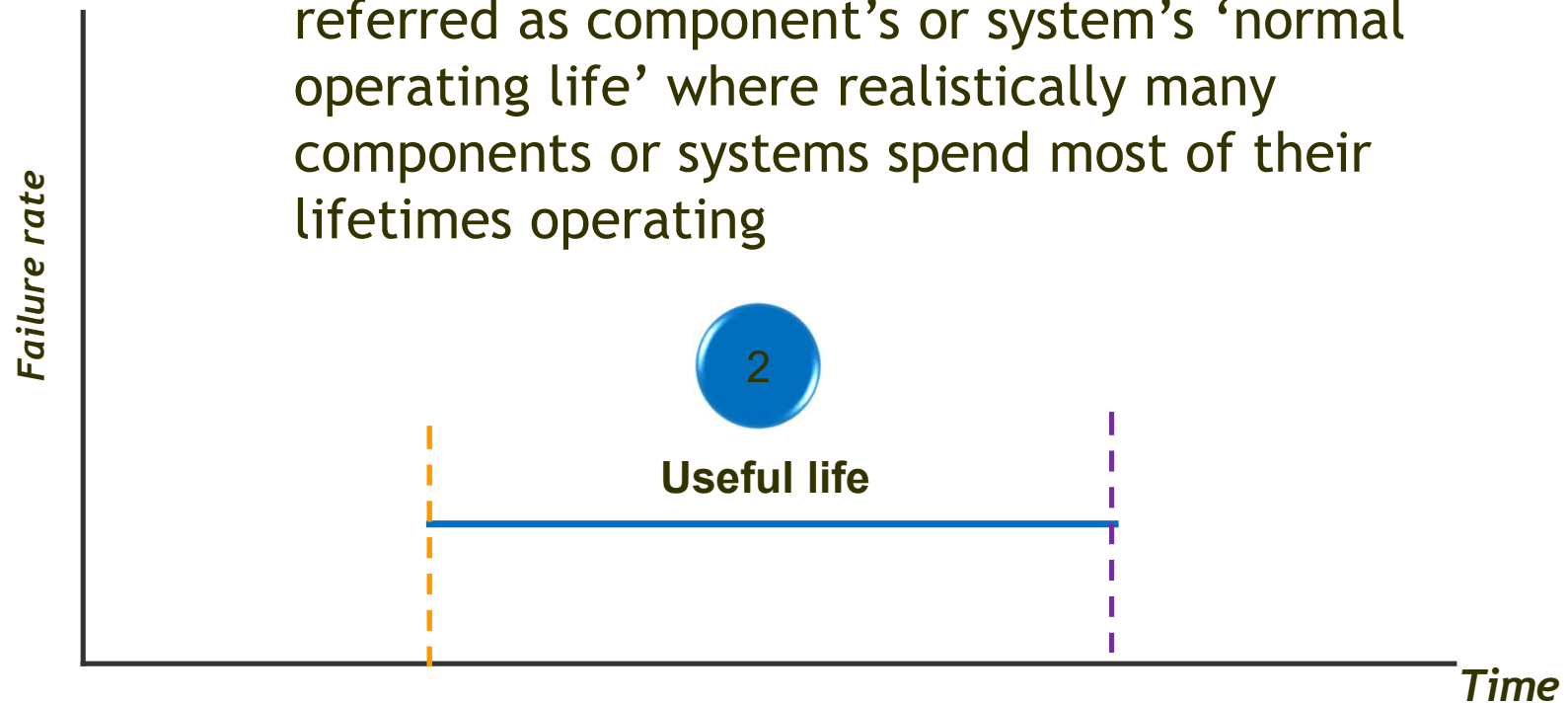


Bathtub curve

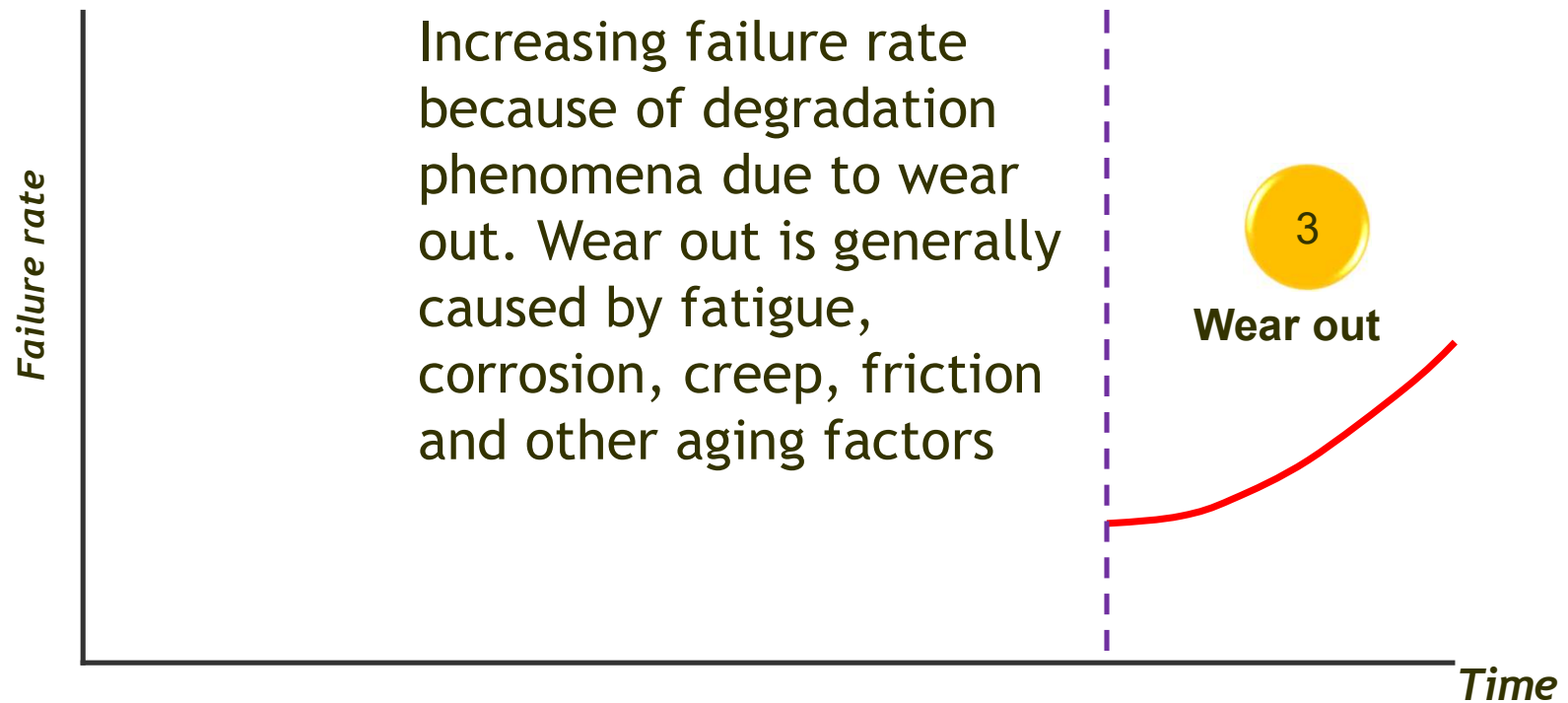


Bathtub curve

Failure rate is approximately constant as the failures, assumed mostly stress-related occur at random. This flat-portion of bathtub is also referred as component's or system's 'normal operating life' where realistically many components or systems spend most of their lifetimes operating



Bathtub curve



Various types of Failure rate curve

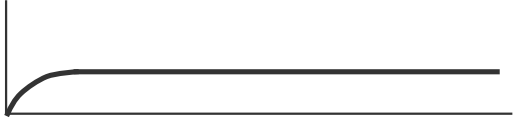
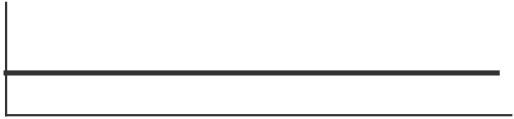
In late 1970's and early 1980's, research done on equipment used in airplanes and naval ships found that there were **SIX (6)** separate patterns that reflected how equipment failed. Similar investigations by the US Navy in the early 21st century confirmed the earlier findings.



Various types of Failure rate curve

1. Traditional view (random failure then wear out) 	Typical equipment : Belt, chains, impellers	Maintenance strategy: Preventive Maintenance
2. Bathtub curve 	Electro-mechanical components and motors	Condition monitoring
3. Slow aging (steady increase in failure rate) 	Turbine, engines, compressors, piping	Condition monitoring

Various types of Failure rate curve

4. Best New (sharp increase in failure rate, then level off) 	Typical equipment: Hydraulic and pneumatic equipment	Maintenance strategy: Condition based maintenance
5. Random failure (failure rate is constant, no age related failure pattern) 	Ball and roller bearing	Condition based maintenance
6. Worst New (high infant mortality, then random failure) 	Electronics equipment /components	Condition based maintenance

Failure rate (Hazard rate), $h(t)$

The conditional probability that the item will fail in the time interval $(t, t + \Delta t)$ given that it has survived up to the beginning of that time interval

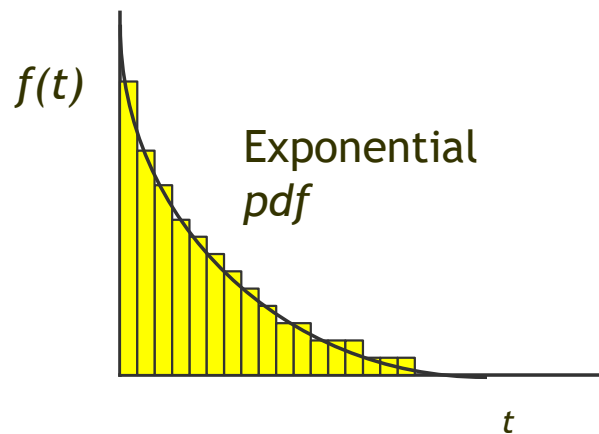
$$h(t) = \frac{f(t)}{R(t)} \quad \text{For } t > 0$$

Failure rate (Hazard rate), $h(t)$

For negative exponential *pdf*:

$$f(t) = \lambda e^{-\lambda t}, \quad F(t) = \int f(t) = 1 - e^{-\lambda t}, \quad R(t) = e^{-\lambda t}$$

Hence,
$$h(t) = \frac{f(t)}{R(t)} = \frac{\lambda e^{-\lambda t}}{e^{-\lambda t}} = \lambda$$



Failure Rate is constant - resembles the useful life region of the bath tub curve

Failure rate and MTTF

If non-repairable component has an **exponential distribution**
(constant failure rate)

$$\text{Failure rate} = \lambda, \quad R(t) = e^{-\lambda t}$$

MTTF is the average time that elapses until component fails and is given by

$$MTTF = \int_0^{\infty} R(t) dt = \int_0^{\infty} e^{-\lambda t} dt = \frac{1}{\lambda}$$

Thus, MTTF of a component is the reciprocal of the failure rate, and vice-versa

$$MTTF = \frac{1}{\lambda} \quad \text{or} \quad \lambda = \frac{1}{MTTF}$$

If “ λ ” is expressed in the number of failures per hour, the MTTF is expressed in hours

Failure rate and MTBF

In the useful life region of the bath tub curve (random failure), where the *pdf* is **exponential**

$$\text{Failure rate} = \lambda, \quad R(t) = e^{-\lambda t}$$

MTBF is the average time a **system** will run between failures and is given by

$$MTBF = \int_0^\infty R(t) dt = \int_0^\infty e^{-\lambda t} dt = \frac{1}{\lambda}$$

Thus,

$$MTBF = \frac{1}{\lambda} \quad \text{or} \quad \lambda = \frac{1}{MTBF}$$

If “ λ ” is expressed in the number of failures per hour, the MTBF is expressed in hours

Failure rate, MTTF and MTBF

Only when the Failure data has **exponential distribution** (failure time is random), where the **failure rate is constant**, this relationship is valid

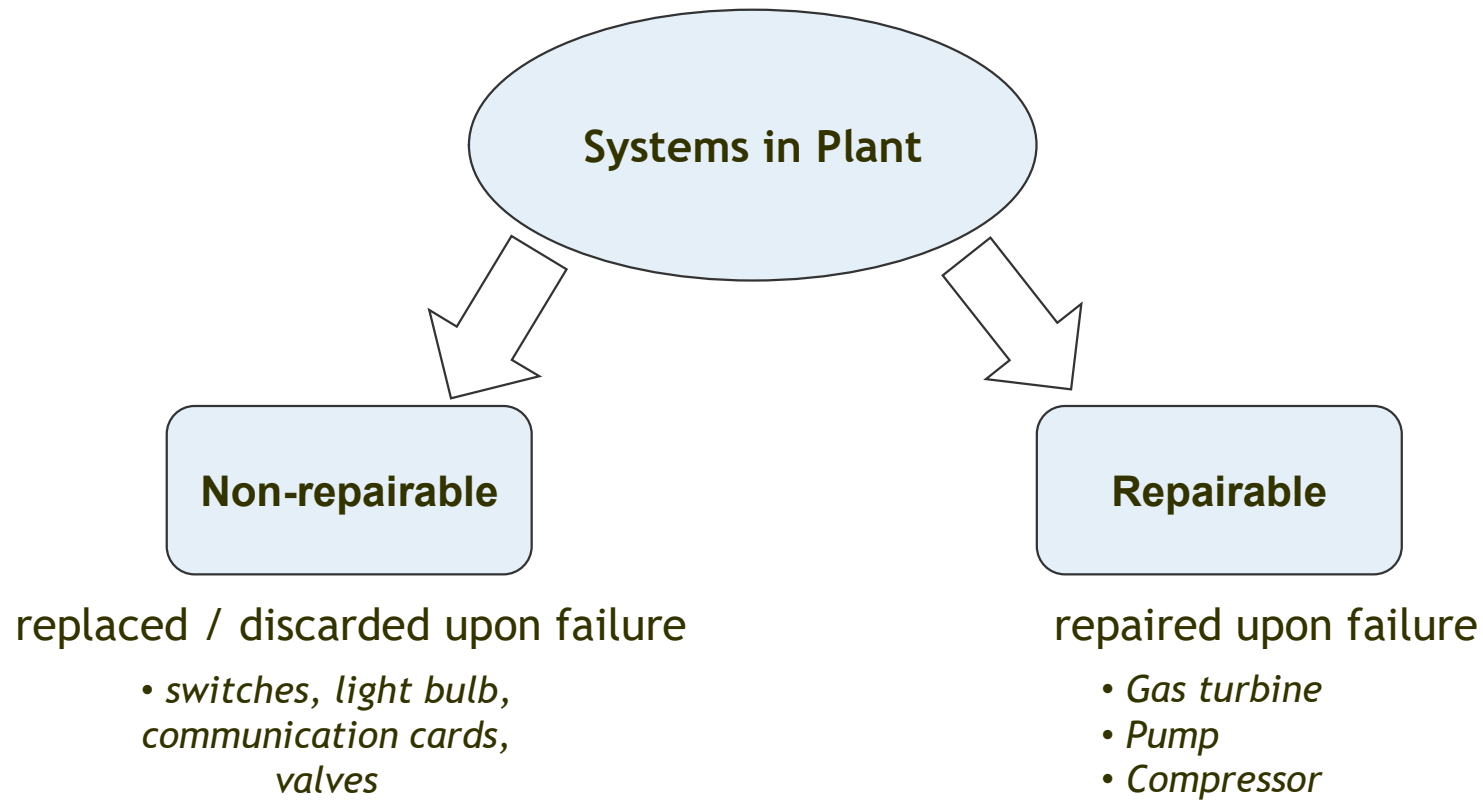
Non-repairable

$$MTTF = \frac{1}{\lambda} \quad \text{or} \quad \lambda = \frac{1}{MTTF}$$

Repairable

$$MTBF = \frac{1}{\lambda} \quad \text{or} \quad \lambda = \frac{1}{MTBF}$$

System Types



Different approach is used to model and analyze the reliability of each equipment type !

Non-Repairable System

When a component fails, it is replaced with identical component and not repaired

- Analysis on non-repairable data is also known as **lifetime data analysis (LDA)**
- The time to failure (lifetime) is a random variable, T described by a **single time to failure**
- For a group of identical components, the lifetimes are assumed independent and come from a single population
- The lifetimes data are considered : *independent* and *identically distributed (IID)*

Repairable Systems

A repairable item is an item which upon failure, is restored to operation by a repair action instead of replacing it

The lifetime of the system is the age, T of the system, that is, the total hours of operation

Many equipment and systems in the plant are repairable



Non-Repairable vs. Repairable

	Non-repairable	Repairable
Definition	Discarded / replaced upon failure	Restored to operating condition without replacement
Measures	Reliability	Availability (Reliability, Maintainability)
Reliability Measures	MTTF	MTBF
Reliability Measures (Failure rate)	Hazard Rate	Rate of occurrence of failures (ROCOF)
Lifetime	Random variable described by single time to failure	Age of the system or total operation hours
Equipment / system	One or more (having similar characteristics /conditions)	Single equipment/system
Analysis approach	Life data analysis (LDA)	Point Process, RDA
Constant Failure rate Assumption	Exponential	Homogeneous Poisson Process (HPP)

Reliability Statistics

Session Objectives

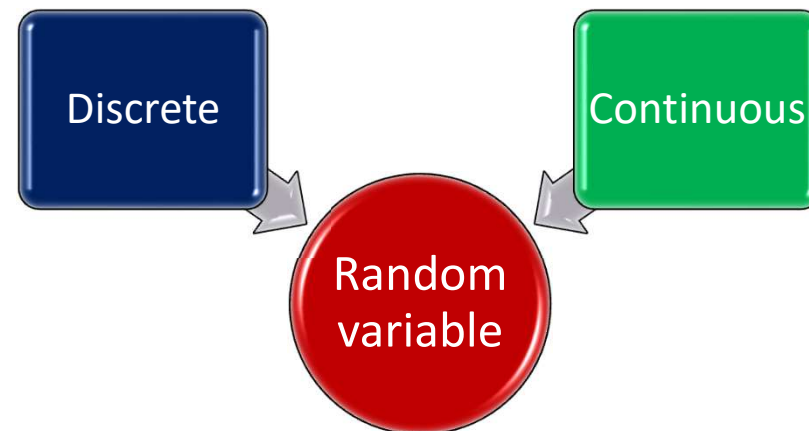
- To understand basic statistics related to reliability
- To understand the concept of reliability function
- To recognize common distributions related to reliability
- To understand reliability measures

Random Variables

a variable whose value results from a measurement on some types of random process

a numerical description of the outcome of an experiment (e.g., the possible results of rolling two dice: (1, 3), (2, 4), etc.)

2 types of random variables:



Discrete Random Variables

a random variable that may assume either a finite number of values or an infinite sequence of values)

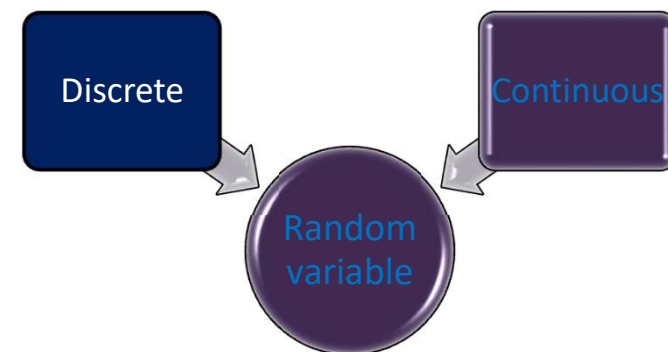
Example: The possible outcomes for one coin toss can be described by the state space $\Omega = \{\text{heads, tails}\}$. We can introduce a real-valued random variable Y as follows:

$$Y(w) = \begin{cases} 1 & \text{if } w = \text{tail} \\ 0 & \text{if } w = \text{head} \end{cases}$$



If the coin is equally likely to land on either side then it has a probability mass function given by:

$$P_Y(y) = \begin{cases} 1/2 & \text{if } y = 1 \\ 1/2 & \text{if } y = 0 \end{cases}$$



Continuous Random Variables

A random variable where the data can take infinitely many values.

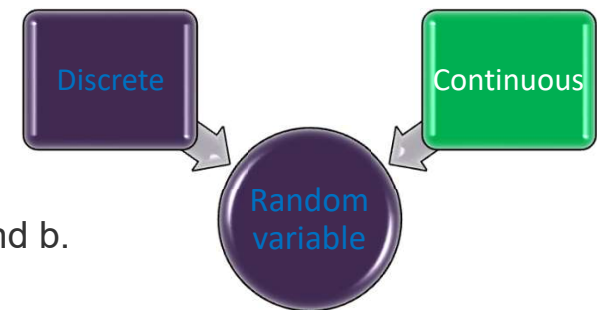
For example, a random variable measuring the time taken for something to be done is continuous since there are an infinite number of possible times that can be taken.

For a continuous random variable X , a probability density function is a function such that

1. $f(x) \geq 0$

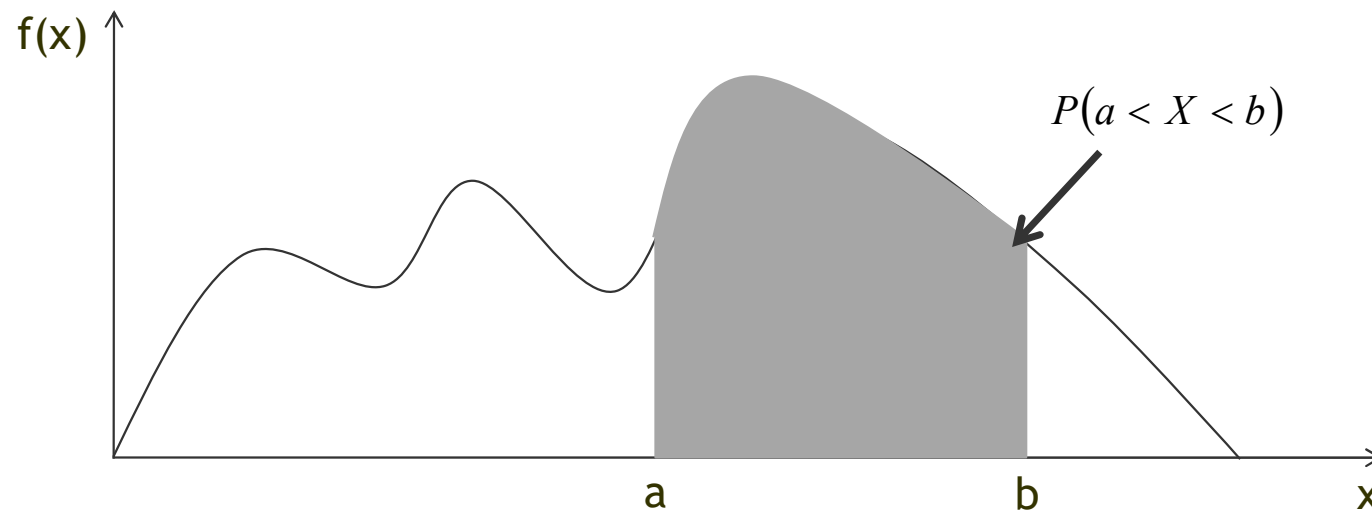
2. $\int_{-\infty}^{\infty} f(x) dx = 1$

3. $P(a \leq X \leq b) = \int_a^b f(x) dx$ = area under $f(x)$ from a to b for any a and b .



Continuous Random Variables

$$P(a \leq X \leq b) = \int_a^b f(x) dx = \text{area under } f(x) \text{ from } a \text{ to } b \text{ for any } a \text{ and } b.$$



$$\int_{-\infty}^{\infty} f(x) dx = 1$$

Reliability function, $R(t)$

Reliability function $R(t)$ is the probability that an item or system will be successfully operating without failure in the interval from time 0 to time t

$$R(t) = \Pr [T \geq t], \quad t \geq 0$$

where

$R(t)$ = a non increasing reliability function, where $R(t) \geq 0$ and $R(0) = 1$

T = a continuous random variable of the time of occurrence of a failure, where $T \geq 0$

t = time period

Thus, for a given value of t , $R(t)$ is the probability of the time to failure, T , is greater or equal to t .

Probability of failure, $F(t)$

The unreliability or probability of a failure will occur before time t can be denoted as $F(t)$, and defined as

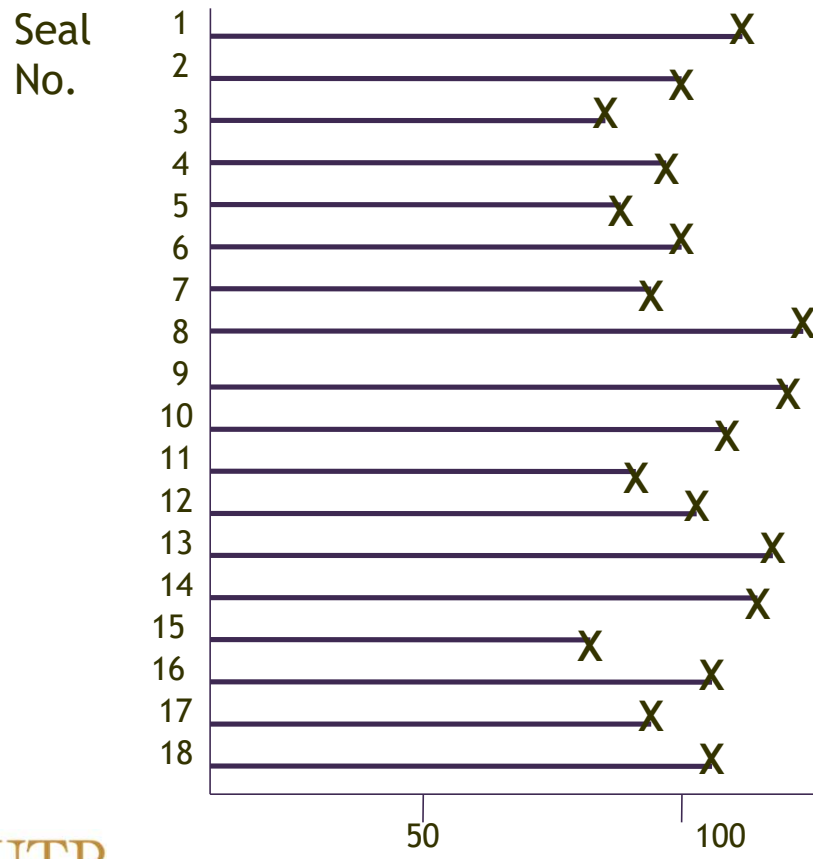
$$F(t) = 1 - R(t) = \Pr [T < t], \quad F(0) = 0 \text{ and } \lim_{t \rightarrow \infty} F(t) = 1$$

$F(t)$ is the cumulative distribution function (CDF) of the time to failure distribution. If the time to failure, T , has a probability density function (PDF) of $f(t)$, then

$$F(t) = \int_0^t f(x)dx$$

Example 1

Consider **time to failure (TTF)** for mechanical seals (similar type)

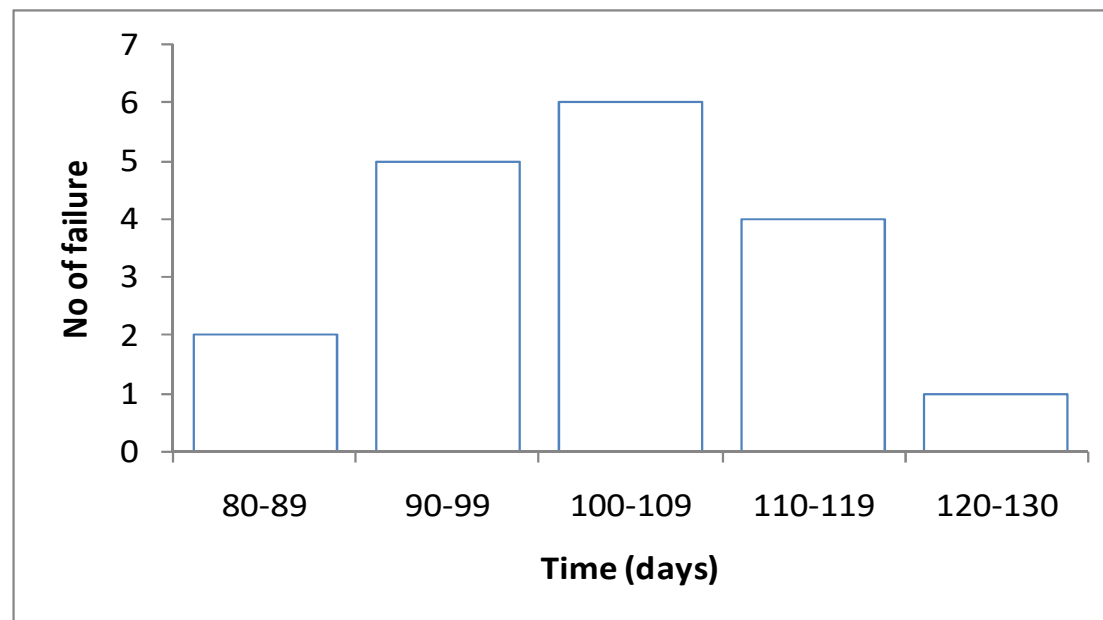


Pump no	TTF (days)	Pump no	TTF (days)
1	110	10	108
2	100	11	92
3	85	12	102
4	98	13	115
5	90	14	114
6	100	15	82
7	95	16	105
8	120	17	96
9	118	18	107

Histogram

Construct a
histogram

Range (days)	Qty Failure
80-89	2
90-99	5
100-109	6
110-119	4
120-130	1



Histogram in Excel

The screenshot illustrates the process of creating a PivotTable and PivotChart in Excel. The 'INSERT' tab is selected, and the 'PivotTable' button is highlighted. The 'Create PivotTable' dialog box is open, showing the data range 'Sheet1!\$C\$2:\$C\$19' and the location 'Sheet1!\$F\$5'. The 'PivotTable Fields' task pane is on the right, showing the 'Data' field. The 'Value Field Settings' dialog box is also open, showing the 'Sum' calculation for the 'Count of 110' field.

Excel Data (Sheet1):

	A	B	C	D	E	F	G	H
1								
2			110					
3			100					
4			85					
5			98					
6			90					
7			100					
8			95					
9			120					
10			118					
11			108					
12			92					
13			102					
14			115					
15			114					
16			82					
17			105					
18			96					
19			107					

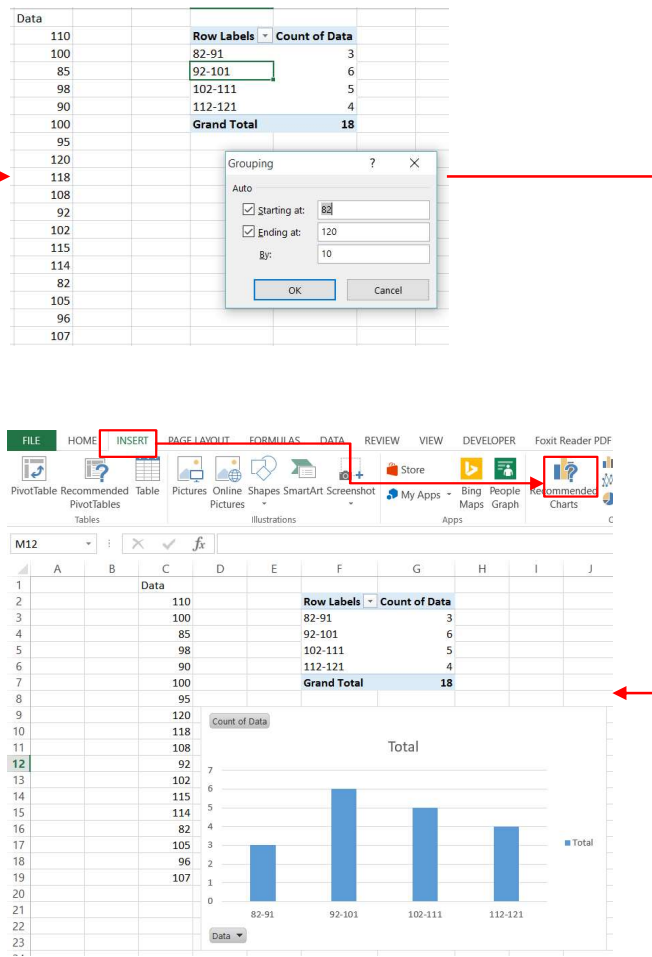
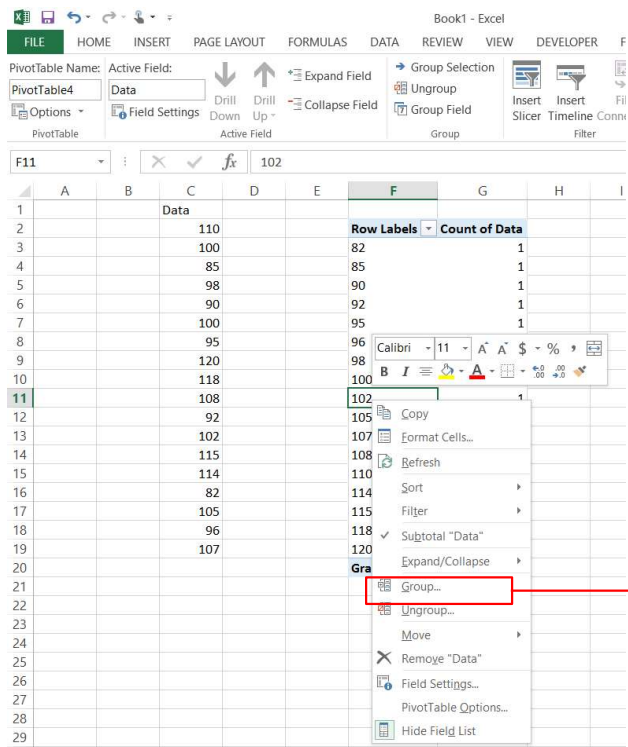
PivotTable Fields Task Pane:

- Choose fields to add to report:
 - ☒ Data
- MORE TABLES...

Value Field Settings Dialog Box:

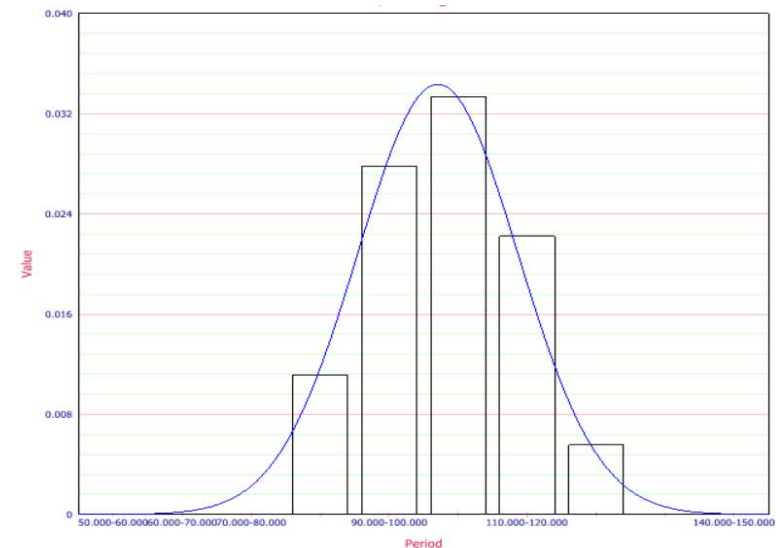
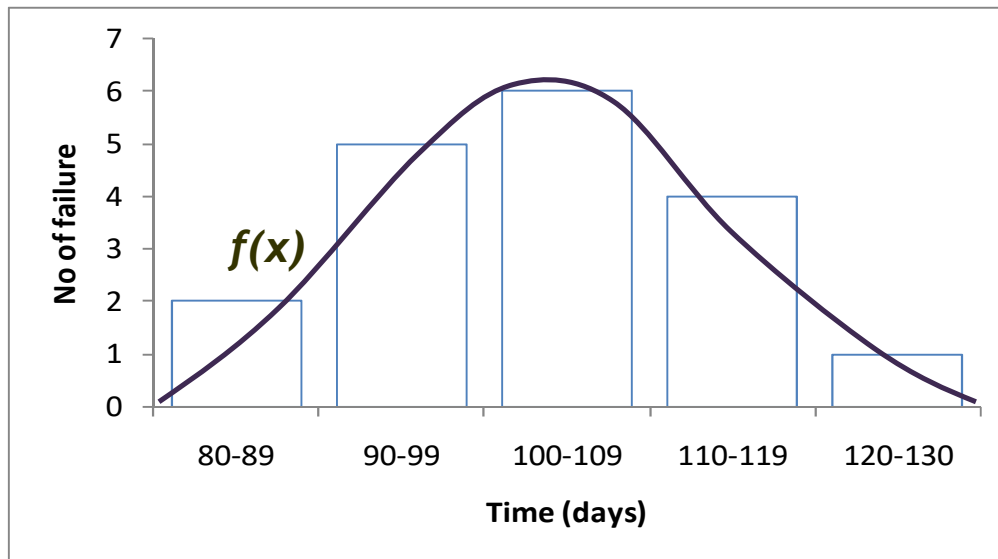
- Source Name: 110
- Custom Name: Count of 110
- Summarize Values By: Show Values As
- Summarize value field by: Choose the type of calculation that you want to use to summarize data from the selected field.
 - Sum
 - Average
 - Max
 - Min
 - Product
- Number Format

Histogram in Excel



Time to failure distribution, $f(x)$

This histogram can be modeled by a continuous function, $f(x)$



This $f(x)$ is called **Time to Failure (TTF) Distribution function**

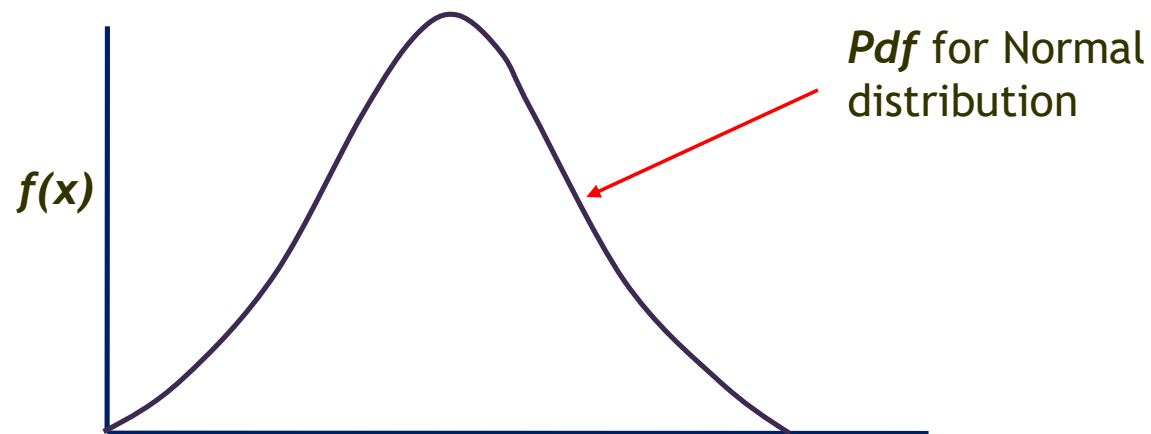
It is basically a **probability density function (pdf)**

Probability distribution function (*pdf*)

Recall,

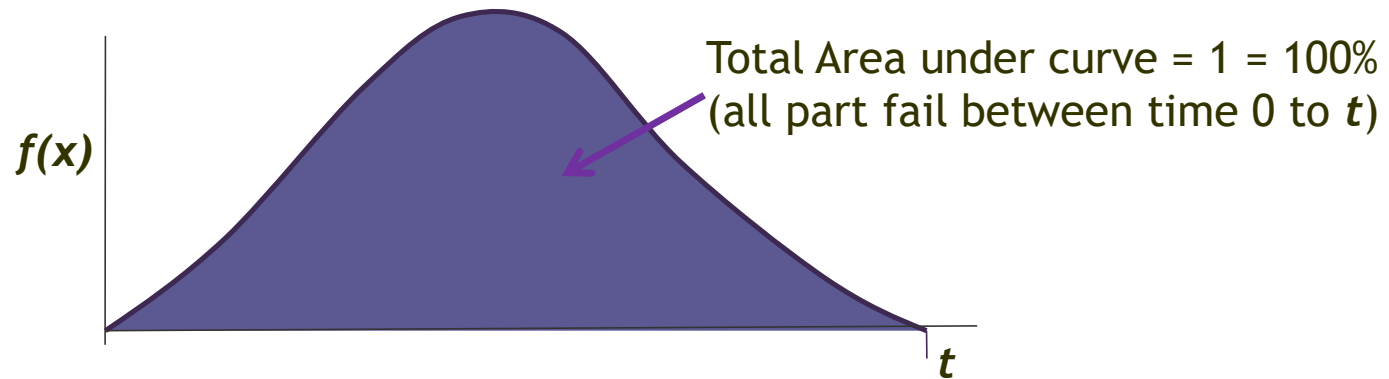
For example, $f(x)$ for normal distribution is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$



Probability distribution function (*pdf*)

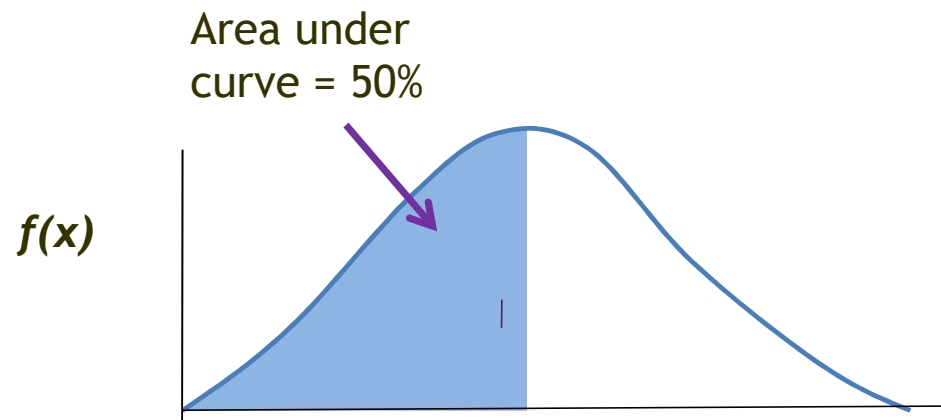
Pdf is an important function, since the area under curve represents the probability of failure (unreliability) of an item at the corresponding time interval



Probability distribution function (pdf)

Example:

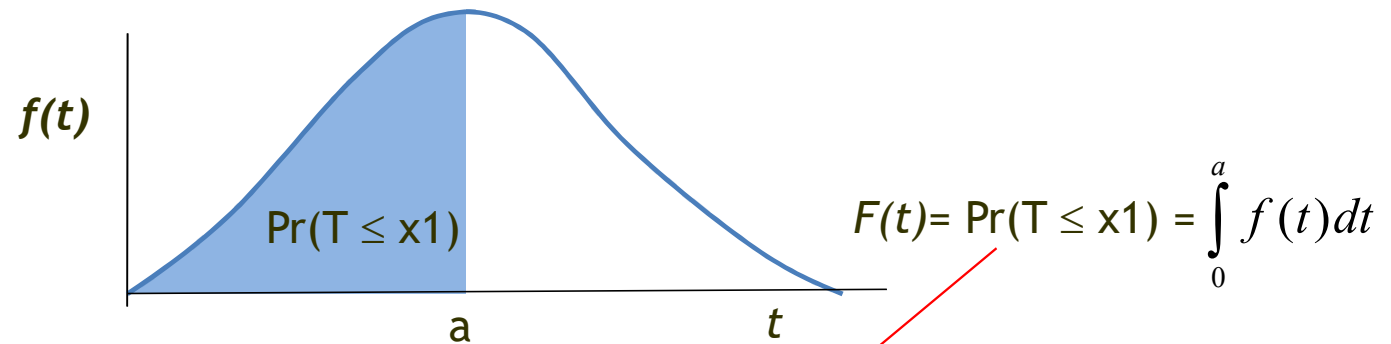
When area under curve is 50% , probability of failure up to the time a , is 0.5



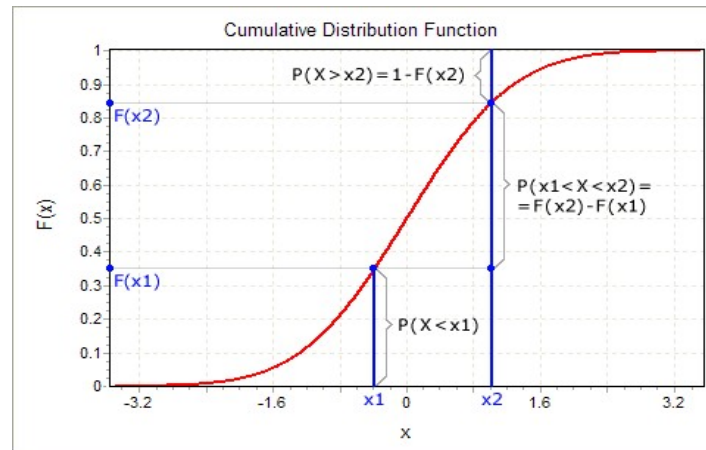
Cumulative distribution function (cdf)

The area under the curve (probability of failure) can also be computed by integration of *pdf*

$$\text{Probability of failure} = \int pdf = F(t)$$



$$F(a) = \Pr(T \leq x_1) \rightarrow$$



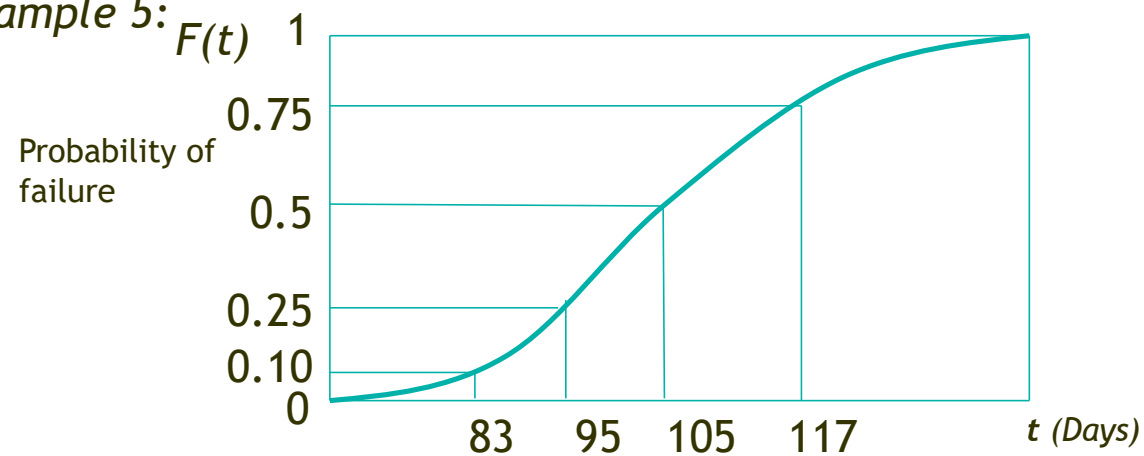
$F(t) = \text{cumulative distribution function (cdf)}$

Cumulative distribution function (*cdf*)

From a *cdf* plot we can determine

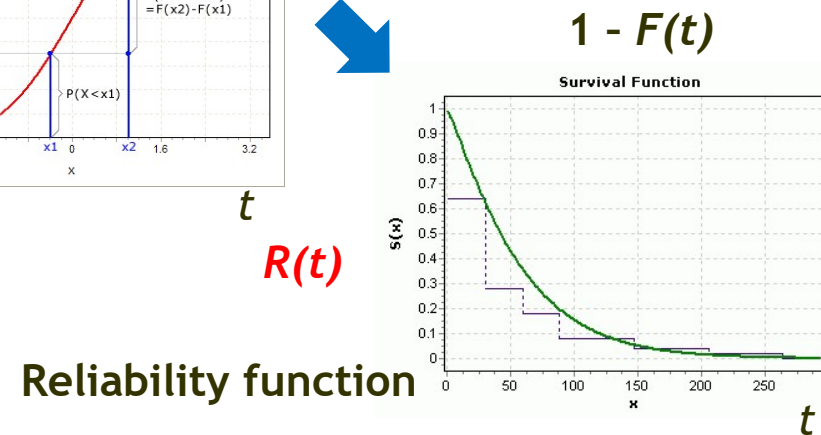
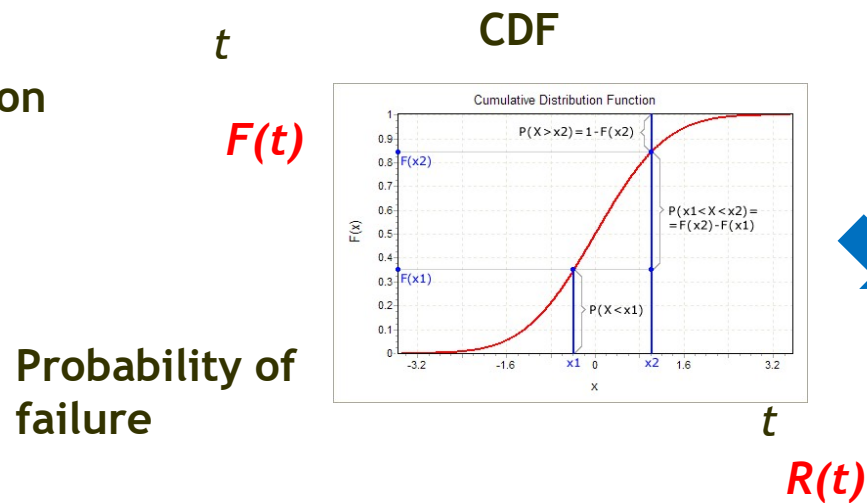
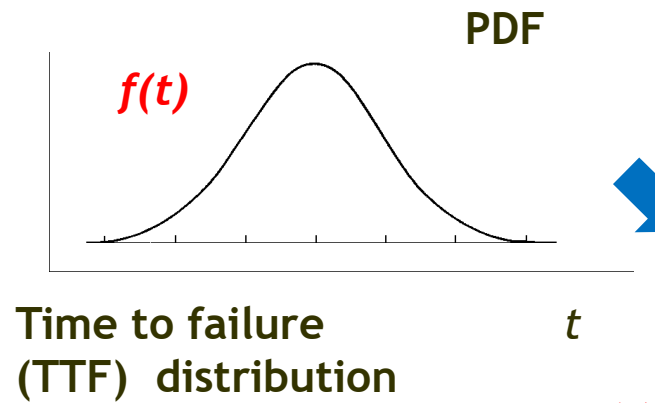
- The probability of failure of a component at any given time
- The time when probability is at certain level

Example 5:

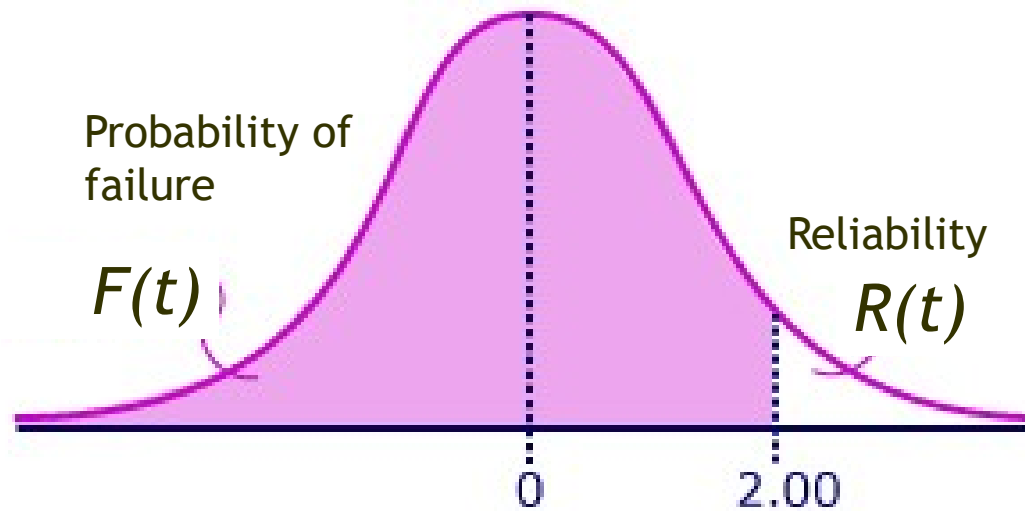


1. When time, $t = 83$ days, the probability of failure is 0.10 (10%)
2. When time, $t = 105$ days, the probability of failure is 0.5 (50%)
3. The probability of failure is 25%, when time, $t = 95$ days
4. The probability of failure is 75%, when time, $t = 117$ days

Relationship between TTF, F(t) and R(t)



Relationship between $F(t)$ and $R(t)$



$$F(t) = 1 - R(t)$$

Common Failure Distributions

Exponential Distribution

- **Very common used in reliability engineering due to its simplicity, even in cases where it does not apply.**
- **Used to describe component/equipment/system having a constant failure rate.**
- **Can be thought of as a special case of the Weibull distribution with $\beta = 1$.**

Exponential Distribution

- pdf and cdf of 1-parameter exponential distribution

pdf $f(t) = \lambda e^{-\lambda t} = \frac{1}{m} e^{-\frac{1}{m}t}$

cdf $F(t) = 1 - e^{-\lambda t} = 1 - e^{-\frac{1}{m}t}$

$$MTBF = m = \frac{1}{\lambda}$$

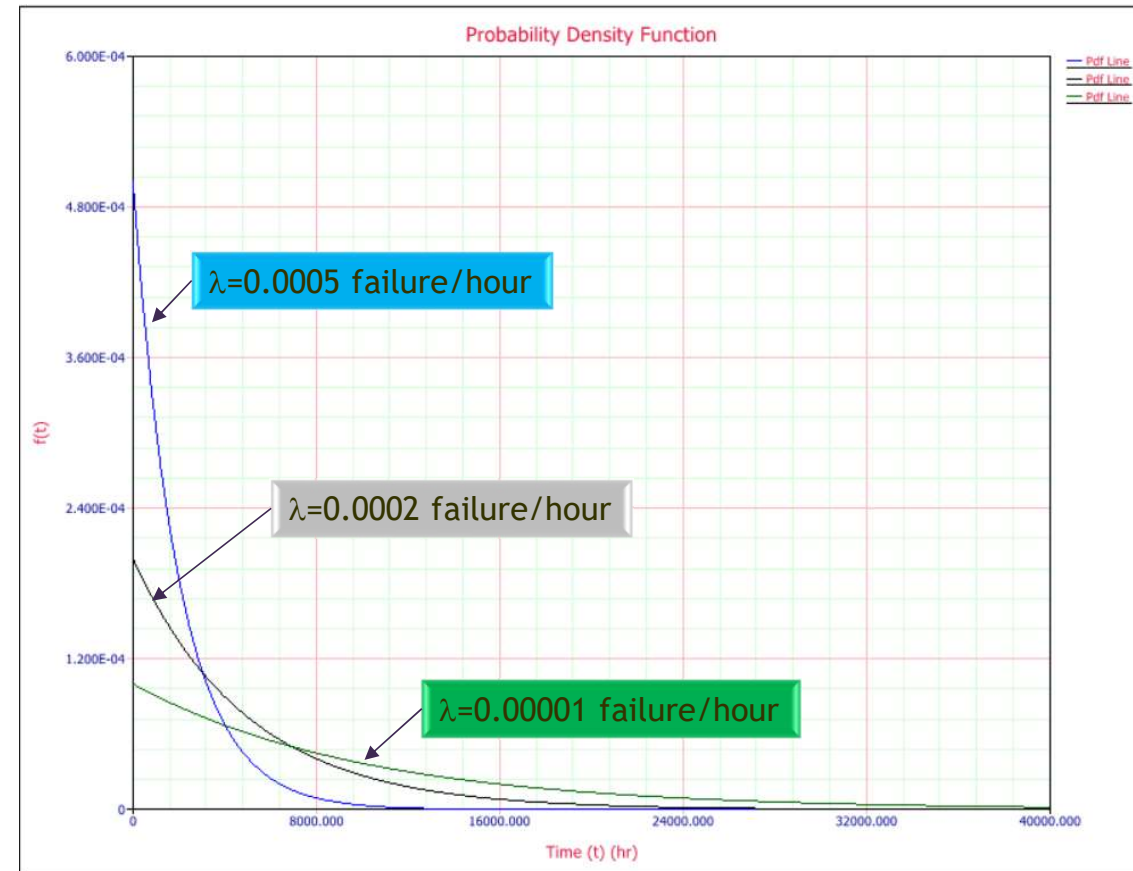
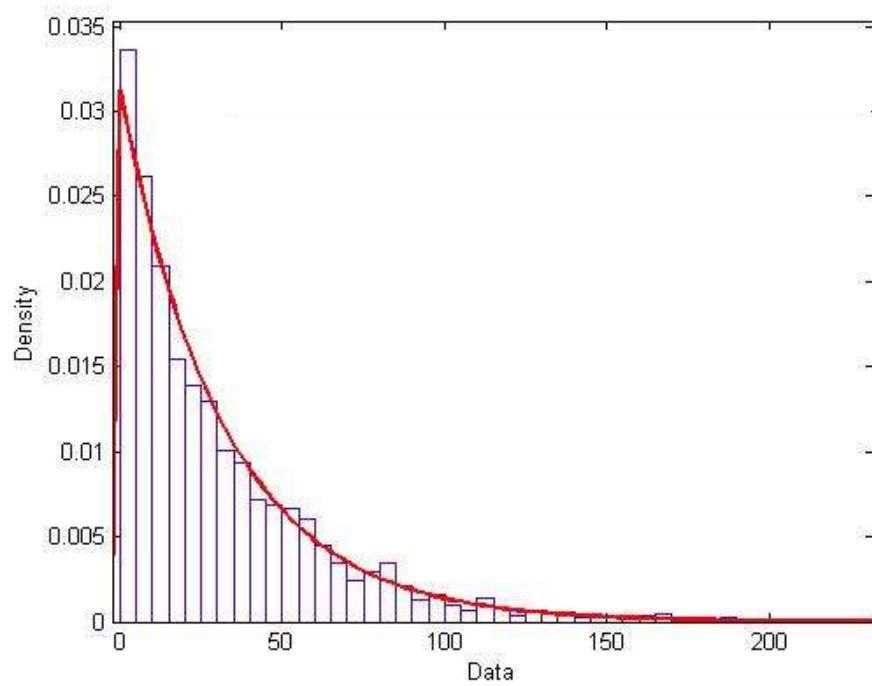
where

λ = constant failure rate, failures per unit of measurement (e.g. failures per hour, per cycle, etc)

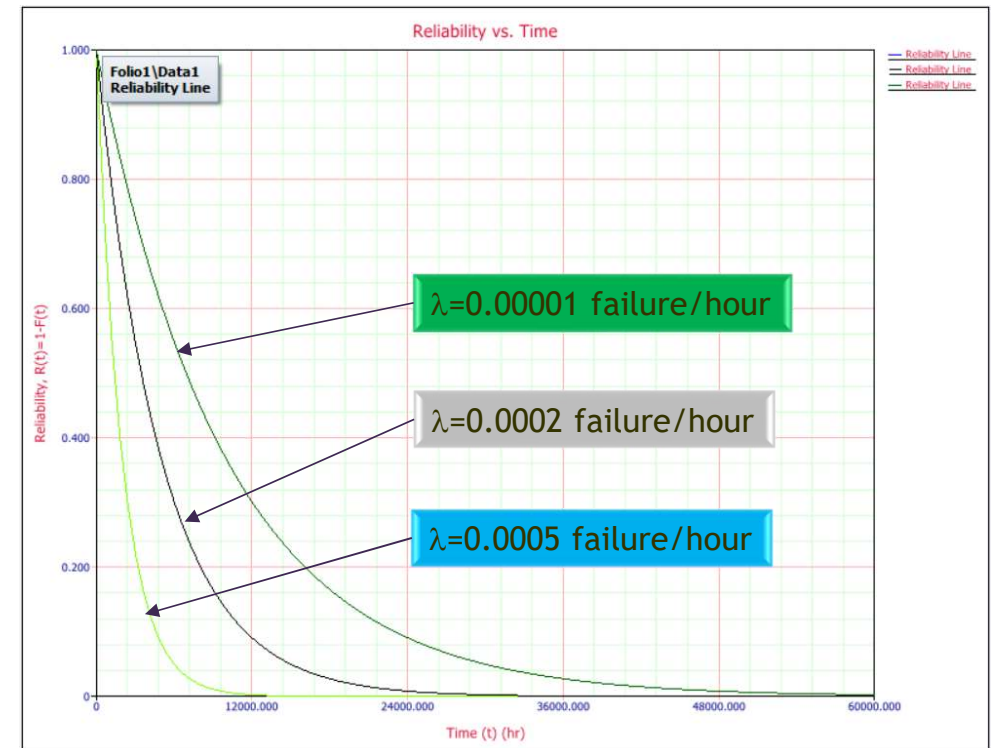
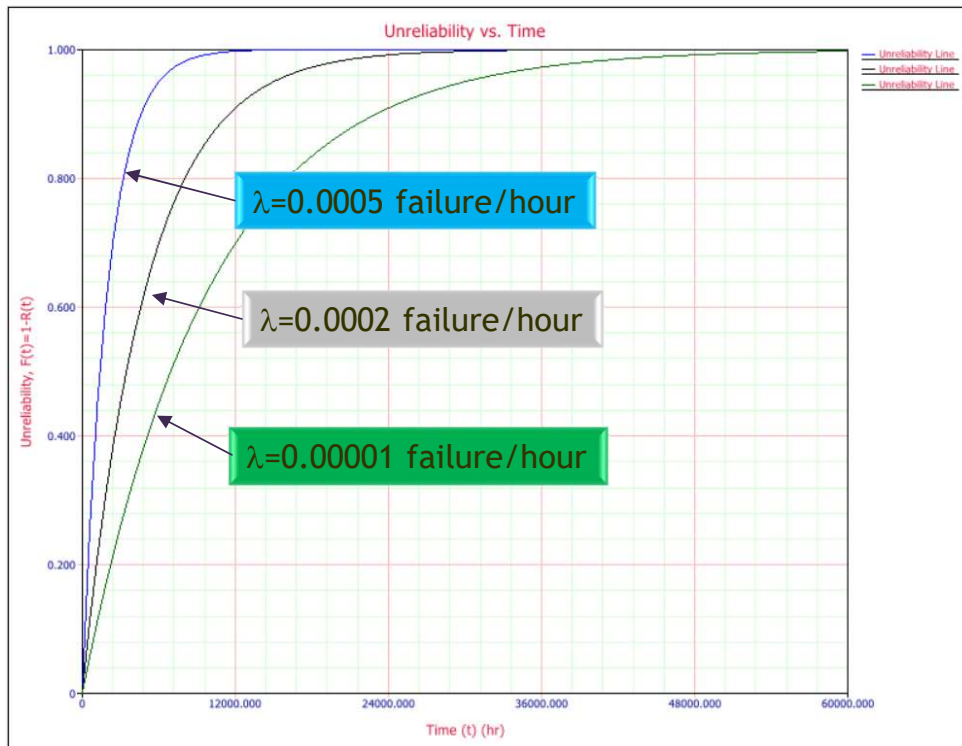
m = mean time between failures, or to a failure

t = operating time, life or age, in hours, cycles, etc

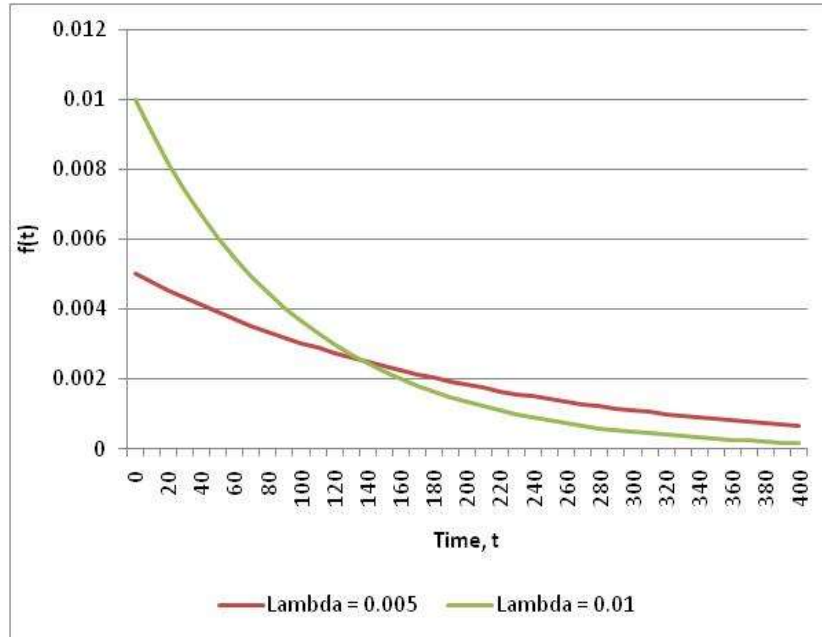
Exponential Distribution-PDF



Exponential Distribution-CDF and Reliability

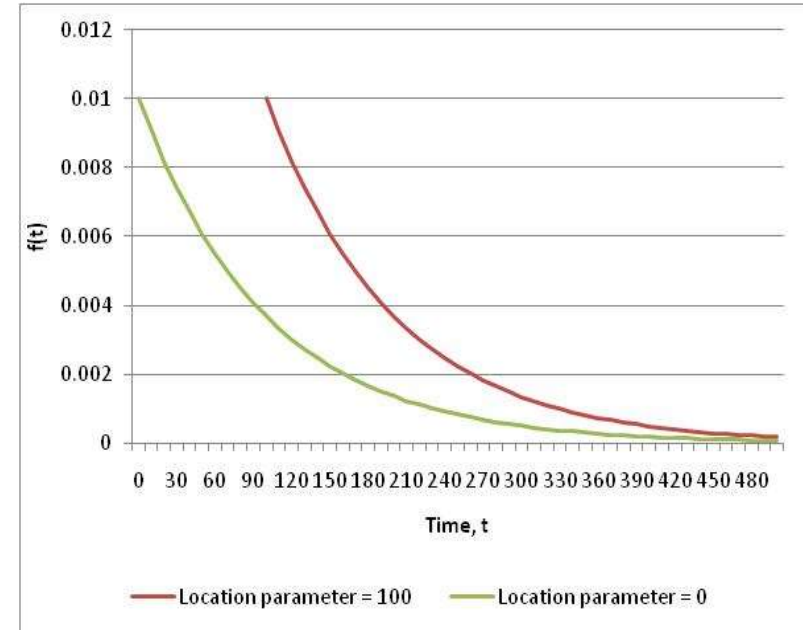


Exponential Distribution



Effects of λ on the exponential distribution

- $\lambda = 0.005 \rightarrow m = 200$
- $\lambda = 0.01 \rightarrow m = 100$



1-parameter vs. 2-parameter exponential distribution with location parameter = 100

Exponential Distribution - Excel Function

= EXPON.DIST(x , lambda, cumulative)

The screenshot shows an Excel spreadsheet with the following data:

	A	B	C	D	E	F	G	H	I	
1	Exponential Distribution									
2	A railway train is fitted with three engine/transmission units that can be assumed to exhibit a constant failure rate of 0.005 failure/hour. Assess the probability that the transmitter will last until 150 hours of operations.									
3										
4										
5										
6										
7	Time to failure, t	150								
8	Lambda	0.005								
9	pdf, f(t)	0.00236								
10	cdf, F(t)	0.528								
11	R(t)	0.472								
12										

The formula bar shows: `=EXPON.DIST(B7,B8,FALSE)`

Exponential Distribution - Excel Function

= EXPON.DIST(x , lambda, cumulative)

The screenshot shows an Excel spreadsheet titled "Types of distributions.xlsx - Excel". The "DATA" tab is active. The formula bar displays the formula `=EXPON.DIST(B7,B8,TRUE)`. The spreadsheet contains the following data:

	A	B	C	D	E	F	G	H	I	
1	Exponential Distribution									
2	A railway train is fitted with three engine/transmission units that can be assumed to exhibit a constant failure rate of 0.005 failure/hour. Assess the probability that the transmitter will last until 150 hours of operations.									
3										
4										
5										
6										
7	Time to failure, t	150								
8	Lambda	0.005								
9	pdf, f(t)	0.00236								
10	cdf, F(t)	0.528								
11	R(t)	0.472								
12										

Normal distribution

- Most widely known distribution. However, it is somewhat less useful in life data analysis due to its symmetry.
- pdf and cdf of 2 parameter distribution:

pdf
$$f(t) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{t-\mu}{\sigma}\right)^2}$$

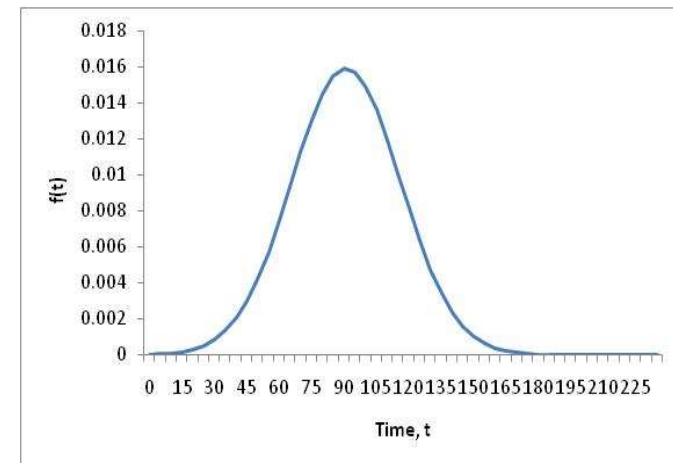
cdf
$$F(t) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^t e^{-\frac{1}{2}\left(\frac{t-\mu}{\sigma}\right)^2} dy$$

where

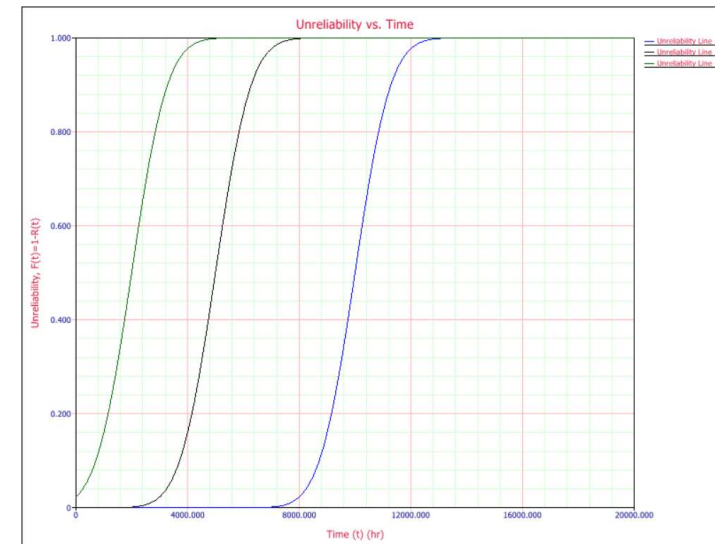
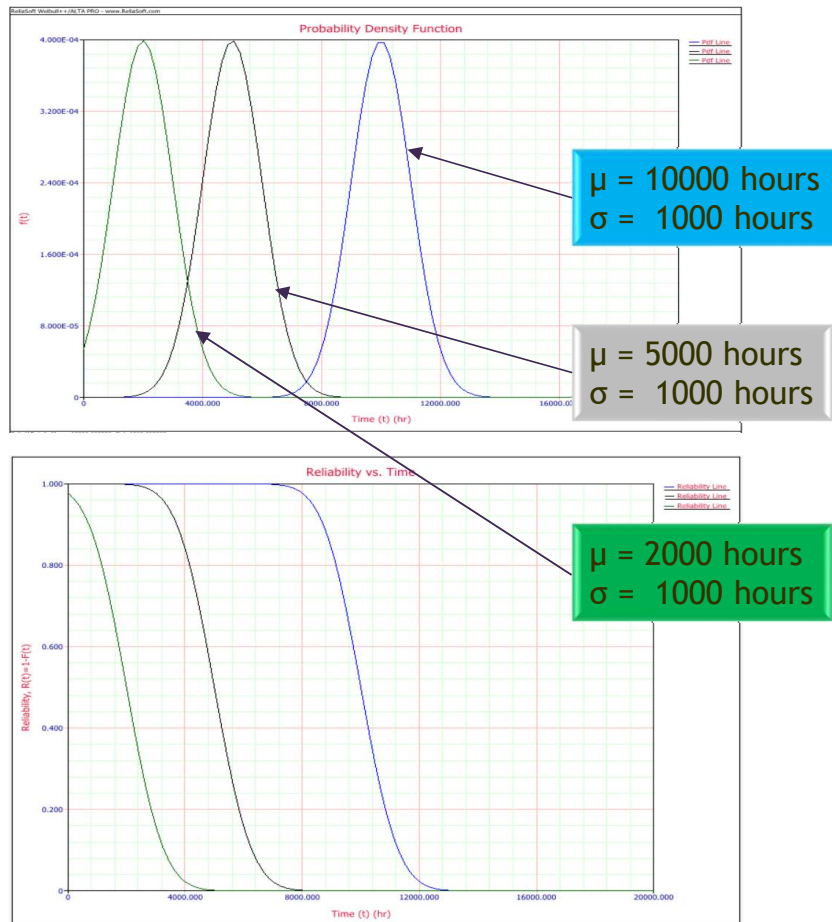
μ = mean or average

σ = standard deviation

t = operating time, life or age, in hours, cycles, etc



Normal Distribution: PDF, CDF and Reliability

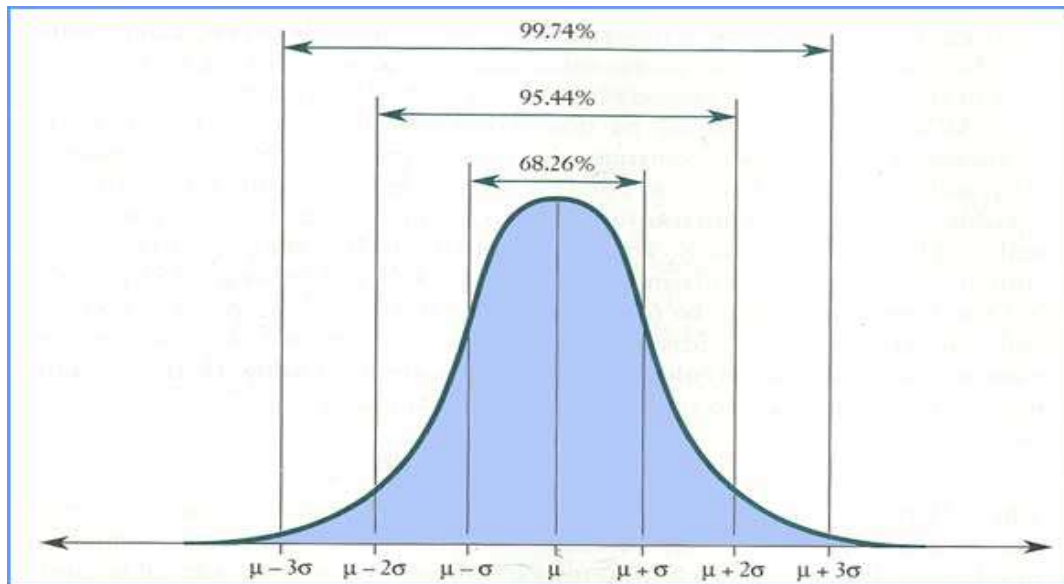


Normal distribution

- Mean = mode = median life= MTTF
- No shape parameter → only one shape and this shape does not change

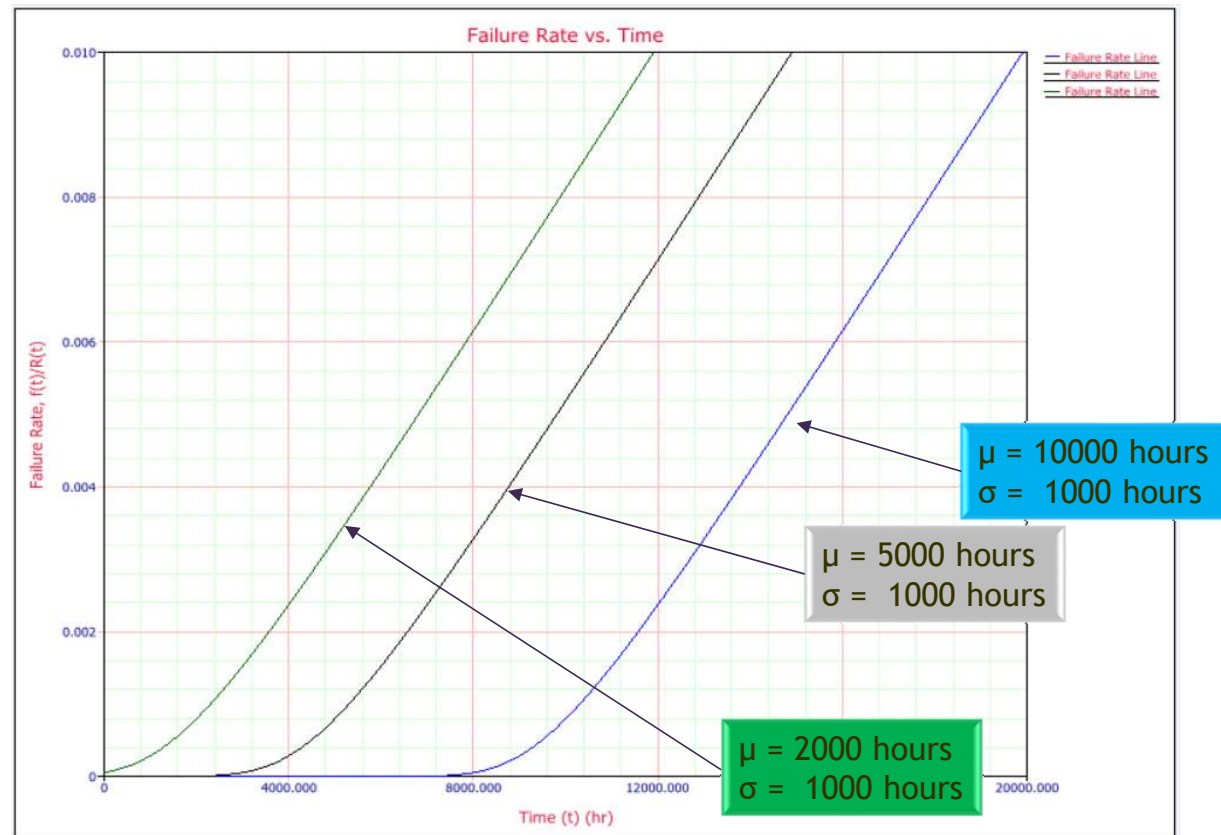


What is area under curve for 6 sigma?



Normal distribution

- The failure rate for normal distribution has an **increasing failure rate**.



Example 2: Normal Distribution

- The life of a bearing is normally distributed with mean at 1200 hrs and standard deviation of 200 hrs. Determine:
 - Reliability at 800 hrs
 - Probability that the bearing will fail after running for 1200 hours

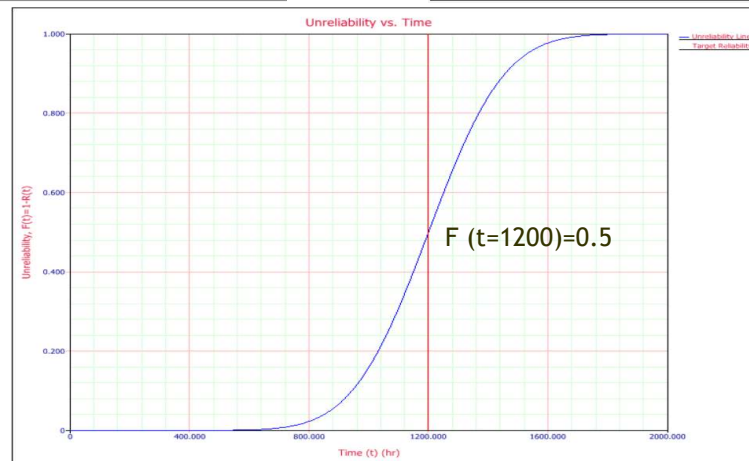
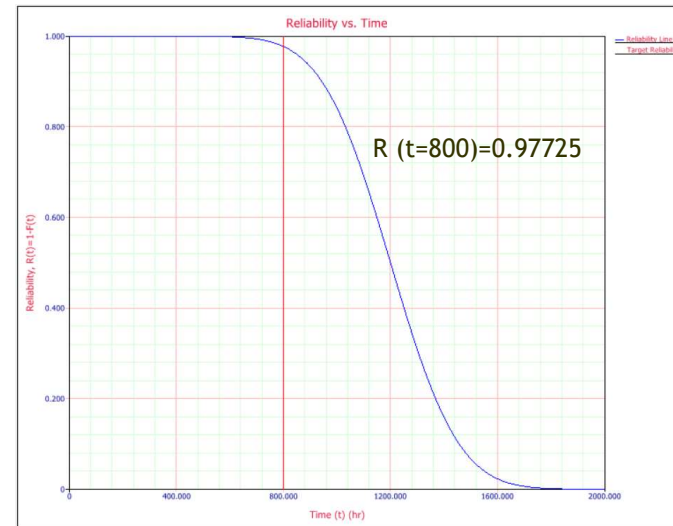
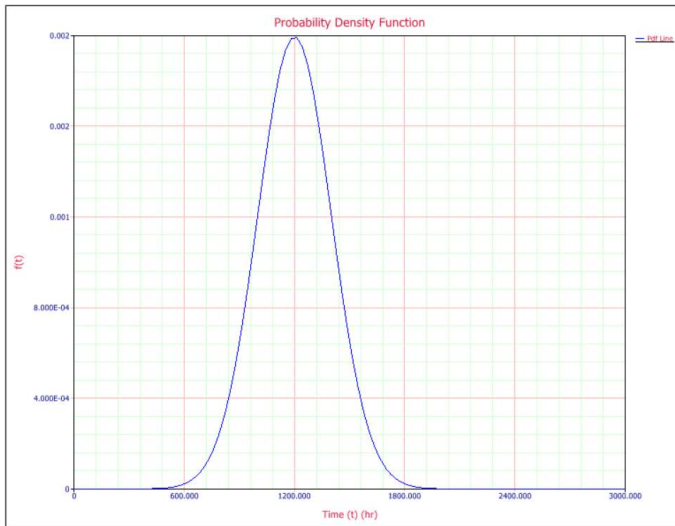
Use Excel to calculate

Standard Normal Probability Table

	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

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Example 2: Solution



Normal Distribution - Excel Function

= NORM.DIST(x , mean, standard_dev, cumulative)

The screenshot shows an Excel spreadsheet with the following data:

	A	B	C	D	E	F	G	H	I	J	K	
1	Normal Distribution											
2	A study reveals that the transmitter's time to failure can be described by a normal distribution with mean of 1800 hours and standard deviation of 120 hours. Assess the probability that the transmitter will last until 2000 hours of operations.											
7	Time to failure, t	2000										
8	Mean	1800										
9	Std dev	120										
10	pdf, f(t)	0.00083										
11	cdf, F(t)	0.952										
12	R(t)	0.048										

The formula bar shows the formula: `=NORM.DIST(B7,B8,B9,FALSE)`

Normal Distribution - Excel Function

= NORM.DIST(x , mean, standard_dev, cumulative)

The screenshot shows an Excel spreadsheet with the following data:

	A	B	C	D	E	F	G	H	
1	Normal Distribution								
2	A study reveals that the transmitter's time to failure can be described by a normal distribution with mean of 1800 hours and standard deviation of 120 hours. Assess the probability that the transmitter will last until 2000 hours of operations.								
7	Time to failure, t	2000							
8	Mean	1800							
9	Std dev	120							
10	pdf, f(t)	0.00083							
11	cdf, F(t)	0.952							
12	R(t)	0.048							
13									

The formula bar shows the formula: `=NORM.DIST(B7,B8,B9,TRUE)`. A red box highlights the formula bar and the cell B11, which contains the result 0.952. A red arrow points from the formula bar to cell B11.

Lognormal Distribution

A random variable is lognormally distributed if the logarithm of the random variable is normally distributed.

pdf and cdf of lognormal distribution

$$\begin{aligned} \text{pdf} \quad f(t') &= \frac{1}{\sigma' \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{t' - \mu'}{\sigma'} \right)^2} \\ \text{cdf} \quad F(t) &= \frac{1}{\sigma' \sqrt{2\pi}} \int_{-\infty}^t e^{-\frac{1}{2} \left(\frac{t' - \mu'}{\sigma'} \right)^2} dy \end{aligned}$$

where

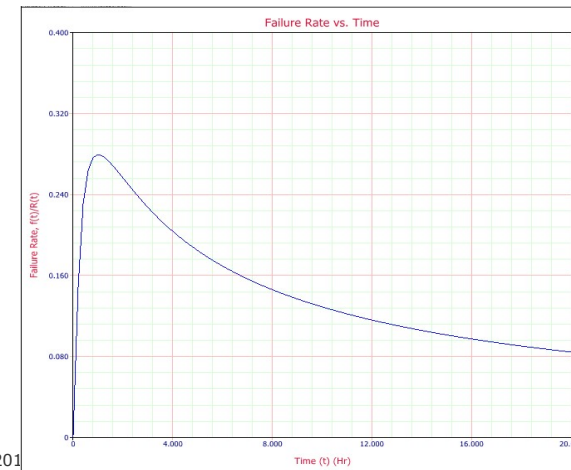
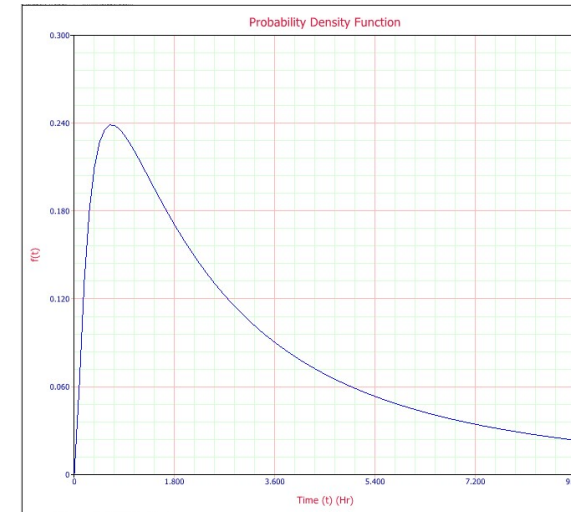
$t' = \ln(t)$ and t is the time to failure

μ' = the mean of the natural logarithms of the times to failure

σ' = the standard deviation of the natural logarithms of the times to failure

Lognormal Distribution

- Unlike the normal distribution which is symmetrical, the lognormal is non-negative and skewed positively, which makes it more suitable for modeling life data.
- The failure rate for lognormal distribution initially increases and then decreases with time



Lognormal Distribution - Excel Function

= LOGNORM.DIST(x , mean, standard_dev, cumulative)

Types of distributions.xlsx - Excel

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Proofing Language Comments

B10 =LOGNORM.DIST(B7,B8,B9,FALSE)

	A	B	C	D	E	F	G	H	I	J	K
1	Lognormal Distribution										
2	The life of an electronic controller is distributed lognormally, with the parameters $\mu = 20$ and $\sigma = 10$. Assess the probability that the controller transmitter will last until 200 hours of operations.										
3											
4											
5											
6											
7	Time to failure, t	200									
8	Mean	20									
9	Std dev	10									
10	pdf, f(t)	0.00007									
11	cdf, F(t)	0.07076									
12	R(t)	0.929									
13											

Lognormal Distribution - Excel Function

= LOGNORM.DIST(x , mean, standard_dev, cumulative)

The screenshot shows an Excel spreadsheet titled "Types of distributions.xlsx - Excel". The formula bar displays the formula `=LOGNORM.DIST(B7,B8,B9,TRUE)`. A text box in the spreadsheet provides the context: "The life of an electronic controller is distributed lognormally, with the parameters $\mu = 20$ and $\sigma = 10$. Assess the probability that the controller transmitter will last until 200 hours of operations."

	A	B	C	D	E	F	G	H	I	J	K	
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Lognormal Distribution Example

- The data below is the time to fail for 4 light bulbs, and is known to have a lognormal distribution. What is the reliability at 100 hours?

Time to failure (hrs)	Ln (time to failure)
115	4.7449
155	5.0434
183	5.2095
217	5.3799

mean

$$\mu = \frac{4.7449 + 5.0434 + 5.2095 + 5.3799}{4} = 5.0944$$

Sample standard deviation

$$s_x = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}$$

n = The number of data points

$\bar{x}$ = The mean of the x_i

x_i = Each of the values of the data

The reliability is then: $R(100) = 1 - \Phi\left(\frac{\ln(100) - 5.0944}{0.2708}\right) = 1 - \Phi(-1.807)$

$$= 1 - (0.0354) \\ = 0.9646$$

Use standardized
normal table

Weibull Distribution

Widely used in reliability engineering and life data analysis due to its versatility.

Describe by three parameters (3-parameter Weibull):

- shape parameter, β (also known as slope)
- scale parameter, η (characteristic life)
- location parameter, γ (location of distribution along the abscissa)

pdf and cdf of 3-parameter Weibull distribution

$$\text{pdf} \quad f(t) = \frac{\beta}{\eta} \left(\frac{t - \gamma}{\eta} \right)^{\beta-1} e^{-\left(\frac{t - \gamma}{\eta} \right)^{\beta}}$$

$$\text{cdf} \quad F(t) = 1 - e^{-\left(\frac{t - \gamma}{\eta} \right)^{\beta}}$$

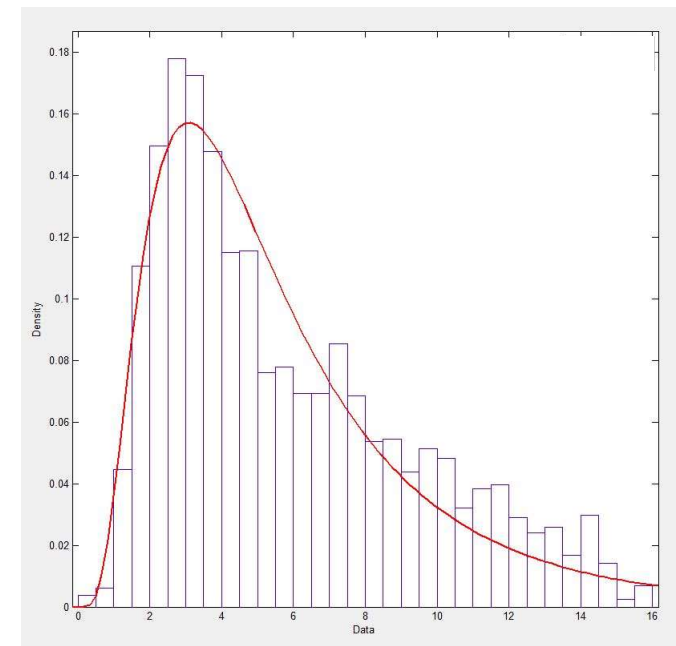
Weibull Distribution

If $\gamma = 0$, then it becomes 2-parameter Weibull

pdf and cdf of 2-parameter Weibull distribution

pdf
$$f(t) = \frac{\beta}{\eta} \left(\frac{t}{\eta} \right)^{\beta-1} e^{-\left(\frac{t}{\eta} \right)^{\beta}}$$

cdf
$$F(t) = 1 - e^{-\left(\frac{t}{\eta} \right)^{\beta}}$$



Shape Parameter, β

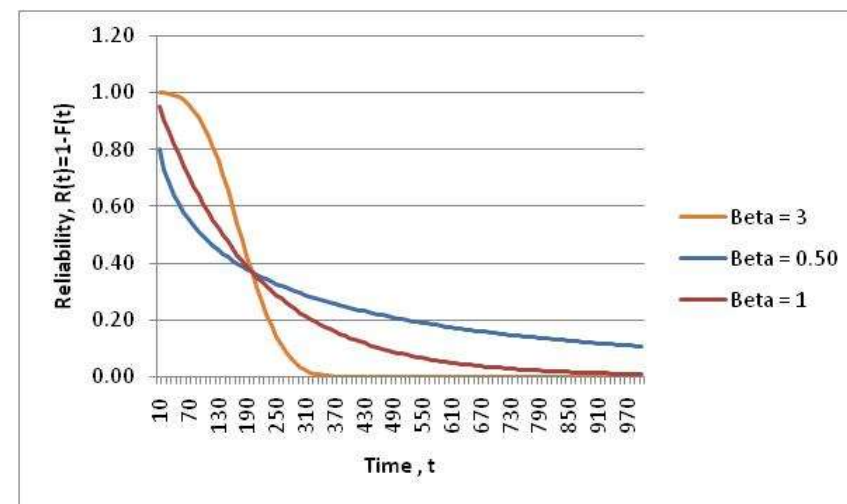
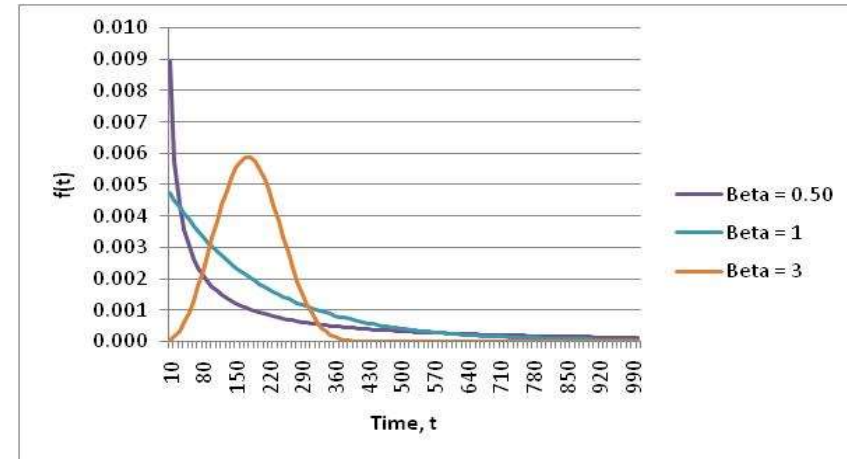
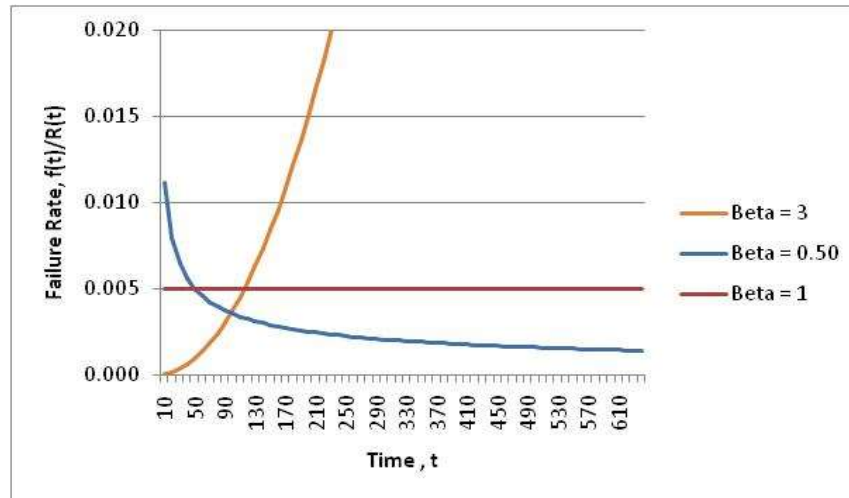
Slope of Weibull distribution (on Weibull probability plotting)

Dimensionless

$\beta < 1$: decreasing failure rate

$\beta = 1$: constant failure rate

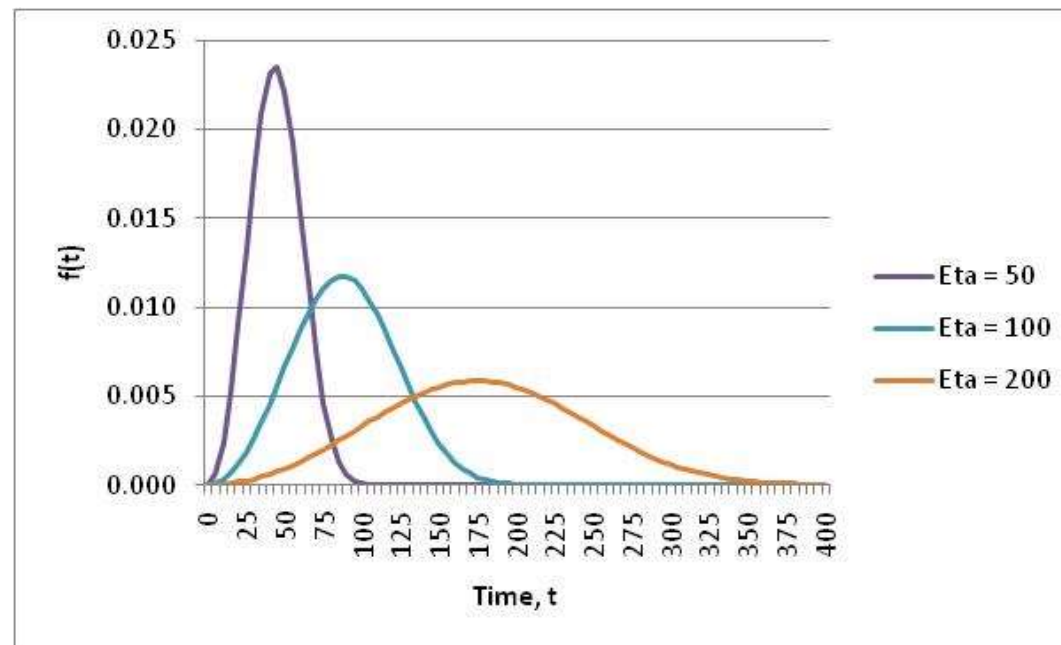
$\beta > 1$: increasing failure rate



Scale Parameter, η

affect such distribution characteristics as the shape of the pdf curve, the reliability and the failure rate.

Has same unit as t , such as hours, miles, cycles, etc.



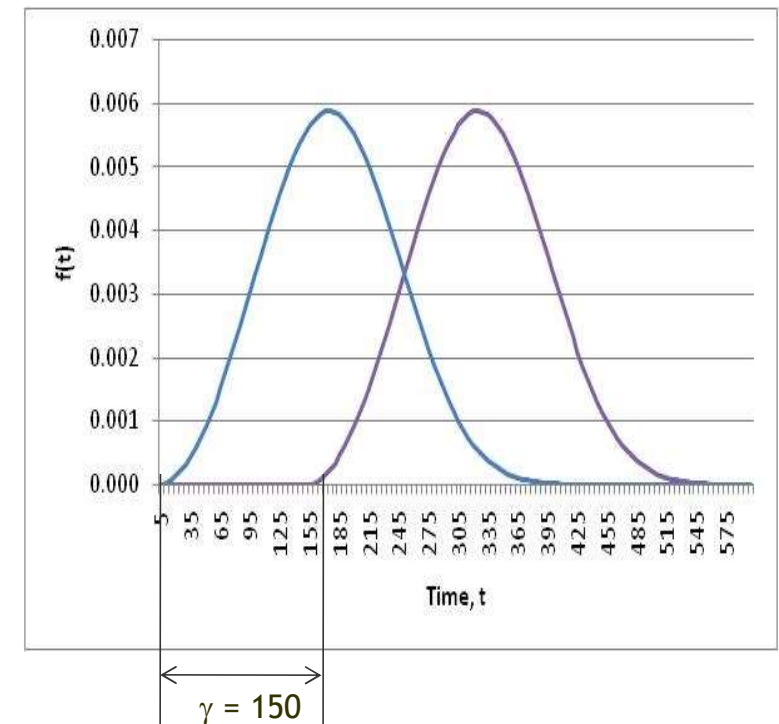
Location Parameter, γ

Locates the distribution along the abscissa

Changing the value of γ has the effect of "sliding" the distribution and its associated function either to the right (if $\gamma > 0$) or to the left (if $\gamma < 0$)

$\gamma > 0$: no failure will occur before the time equivalent to γ i.e. reliability = 1 up to time γ

$\gamma < 0$: system starts out with reliability < 1 . This may indicate quiescent failure modes, damage due to shipping, manufacturing issues etc.



Weibull Distribution - Excel Function

= WEIBULL.DIST(x , alpha, beta, cumulative)

The life of product B is distributed by Weibull distribution with $\beta = 0.5$ and $\eta = 1000$ hours. Assess the probability that the controller transmitter will last until 200 hours of operations.

7	Time to failure, t	200
8	Beta	0.5
9	Eta	1000
10	pdf, f(t)	0.00071
11	cdf, F(t)	0.36059
12	R(t)	0.639

Weibull Distribution - Excel Function

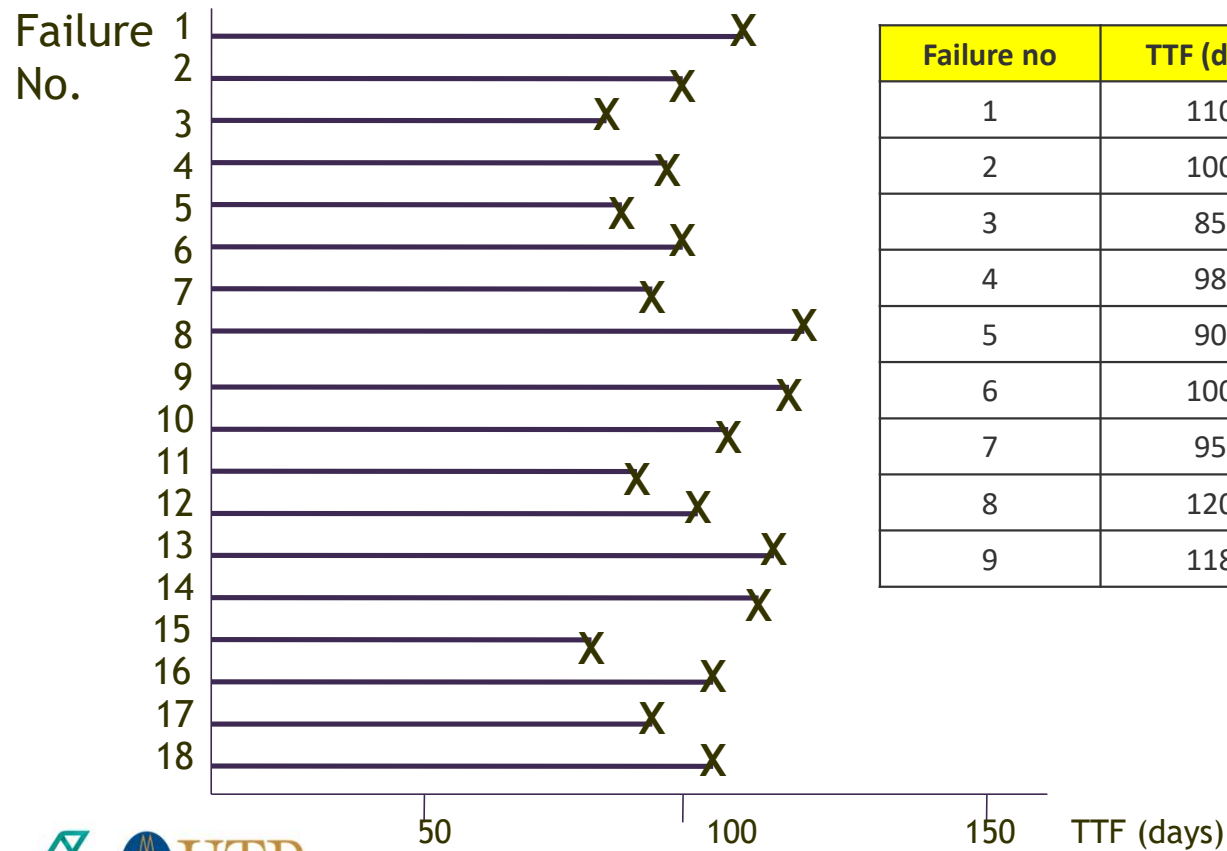
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Example 3

Consider time to failure (TTF) for mechanical seal



Failure no	TTF (days)	Failure no	TTF (days)
1	110	10	108
2	100	11	92
3	85	12	102
4	98	13	115
5	90	14	114
6	100	15	82
7	95	16	105
8	120	17	96
9	118	18	107

Example 3 (ct'd)

Calculate the MTTF and λ . Assume the data are exponentially distributed.

$$MTTF = \frac{1}{n} \sum_{i=1}^n t_i = \frac{(110+100+85 + \dots + 107)}{18} = 102.1 \text{ days}$$

$$\lambda = \frac{1}{MTTF} = \frac{1}{102.1} = 0.0098 \text{ failure/day}$$

Example 4

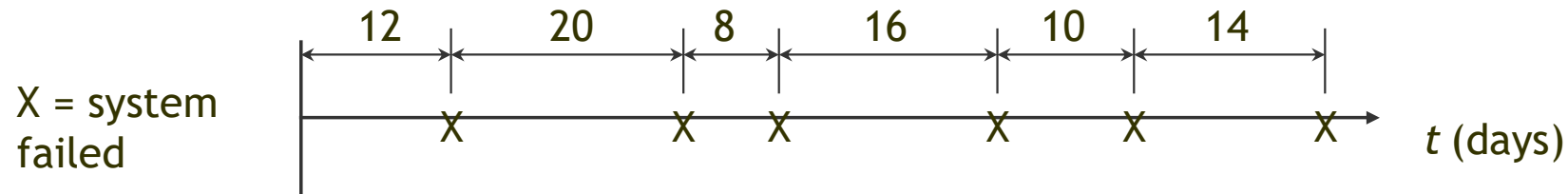


Figure above describes the failure time (time between failure) of a repairable system. **Assume that the failure rate is constant**, calculate MTBF and λ

$$MTBF = \frac{(12 + 20 + 8 + 16 + 10 + 14)}{6} = \frac{80}{6} = 13.33 \text{ days}$$

$$\lambda = \frac{1}{MTBF} = \frac{1}{13.33} = 0.075 \text{ failure per day}$$

or

$$\lambda = \frac{\text{Number of failures}}{\text{Total time}} = \frac{6}{(12 + 20 + 8 + 16 + 10 + 14)} = 0.075 \text{ failure per day}$$

Example 5

A centrifugal pump has a MTBF of 3,500 hours. What is the probability of operating for a period of 2,000 hours without a failure?

$$\lambda = 1/3,500 = 0.000286 \text{ failure per hour}$$
$$t = 2,000 \text{ hours}$$

$$R(t) = e^{-\lambda t} = e^{-(0.000286)(2000)}$$
$$= e^{-0.572} = 0.5644$$

Hence, the probability of the pump operating 2,000 hours without a failure is 56.44%

Probability of the pump to fail before operating 2000 hours is 43.56%

Exercise

Based on the mechanical seal data in Example 1, determine the following:

1. What is the probability of failure at 100, 150 and 250 hours ?
2. What are the operating time when reliability of the pump at 25%, 50% and 75%?

Life Data Analysis (LDA)

Session Objectives

- By end of the session, the students should be able to:
 - Understand the general approach in LDA and probability plotting
 - Perform parametric LDA for complete and censored data
 - Evaluate goodness of distribution fit test for selected distribution

Life Data Analysis (LDA)

- Making predictions about the life of all products/component in the population by **fitting a statistical distribution to life data from a representative sample of units**. The parameterized distribution for the data set can then be used to estimate important life characteristics of the product such as reliability or probability of failure at a specific time, the mean life and the failure rate.
 - Gather life data for the product.
 - Select a lifetime distribution that will fit the data and model the life of the product.
 - Estimate the parameters that will fit the distribution to the data.
 - Generate plots and results that estimate the life characteristics of the product, such as the reliability or mean life.

Non-Parametric Estimation

- Let n be the initial number of units (sample size) and d_i denote the number of units that dies or failed in the i^{th} interval (t_{i-1}, t_i) . The non-parametric estimator of $F(t)$, based on simple binomial distribution is

$$F(t) = \frac{\text{No of failure up to time } t_i}{n} = \frac{\sum_{j=1}^i d_j}{n}$$

Non-Parametric Estimation

- Example 9:
 - Test on 100 heat exchanges show the following data: Year 1- 1 failure, year 2- 2 failures and year 3- 2 failure. Estimate the CDF.
- Solution
 - $F(1) = 1/100$
 - $F(2) = 3/100$
 - $F(3) = 5/100$

Non-Parametric Estimation

- To derive directly from plant failure data:
 - the failure distribution
 - reliability function
 - failure rate (hazard rate)
- Uses histogram and survival function such as Kaplan Meier , to estimate the distribution and reliability function
- Usually applied when there is **no probability distribution that can adequately fit the data**
- Note: Nonparametric is also called distribution free statistics because they do not require that the data fit certain type of distribution, usually normal distribution

Parametric Estimation

- More preferable method compared to non-parametric since non-parametric model does not provide information beyond the range of the sample data
- the interest is on establishing the probabilistic nature of the underlying failure process, not the sample data
- often failure process is a result of some physical phenomena that can be associated with a particular distribution
- small sample sizes provide limited information concerning the failure process

Source of Data

- **Historical Data** – test and field data on the same components /equipment
- **Vendor data** – Data from manufacturer / vendor / consultant
- **Test data** – experimental data of the parts
- **Operational data** – Field data collected under actual operating conditions
- **Handbook data** – theoretical data from standard engineering handbook, Reliability database i.e. OREDA, MIL-HDBK 217F
- **Judgmental data** – information based on expert opinion inputs

Source of Data and Accuracy

For a prediction using site-/company-specific data (illustrated in Figure 4.1)

<i>One can be this confident</i>	<i>That the eventual field failure rate will be BETTER than:</i>
95%	$3\frac{1}{2}$ times the predicted
90%	$2\frac{1}{2}$ times the predicted
60%	$1\frac{1}{2}$ times the predicted
<i>One can be this confident</i>	<i>That the eventual field failure rate will be in the range:</i>
90%	$3\frac{1}{2}:1$ to $2/7:1$

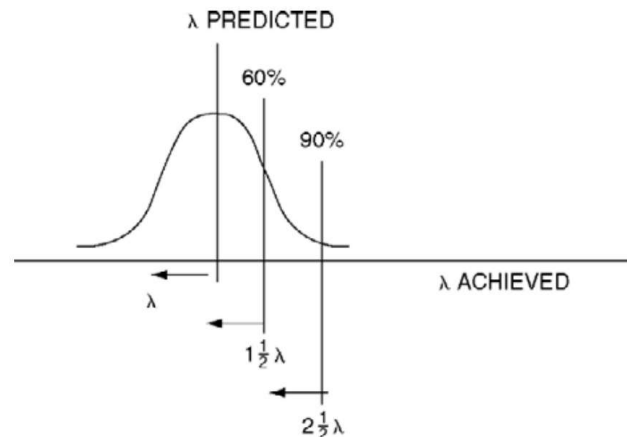


Figure 4.1: Using site/company data

David J Smith, Reliability, Maintainability and Risk, *Practical methods for engineer*, 2011, Elsevier, Oxford, UK

Source of Data and Accuracy

For a prediction using industry-specific data (illustrated in Figure 4.2)

<i>One can be this confident</i>	<i>That the eventual field failure rate will be BETTER than:</i>
95%	5 times the predicted
90%	4 times the predicted
60%	$2\frac{1}{2}$ times the predicted
<i>One can be this confident</i>	<i>That the eventual field failure rate will be in the range:</i>
90%	5:1 to 1/5:1

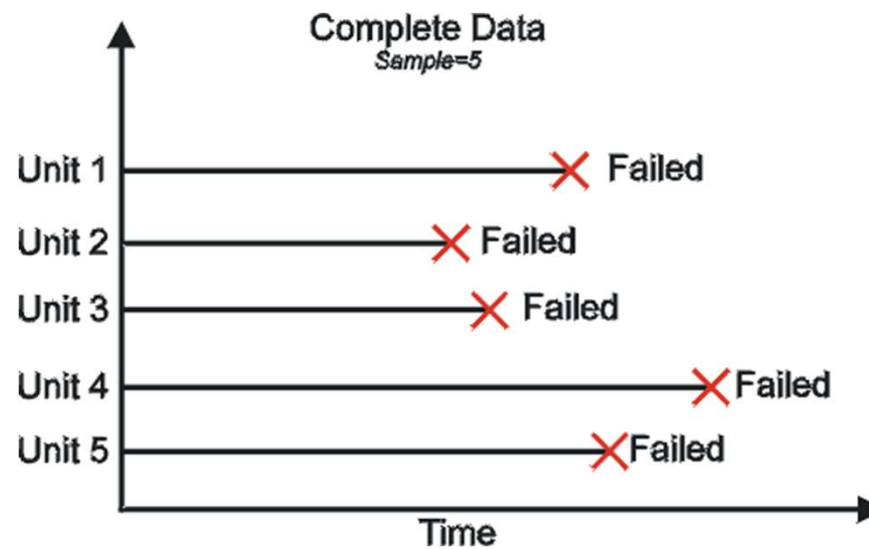
For a prediction using generic data (illustrated in Figure 4.3)

<i>One can be this confident</i>	<i>That the eventual field failure rate will be BETTER than:</i>
95%	8 times the predicted
90%	6 times the predicted
60%	3 times the predicted
<i>One can be this confident</i>	<i>That the eventual field failure rate will be in the range:</i>
90%	8:1 to 1/8:1

David J Smith, Reliability, Maintainability and Risk, *Practical methods for engineer*, 2011, Elsevier, Oxford, UK

Data Types : Complete Data

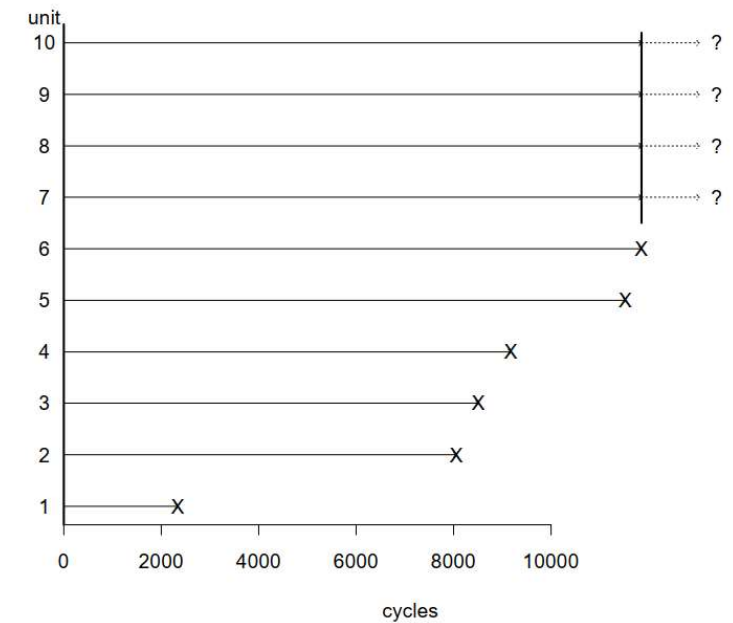
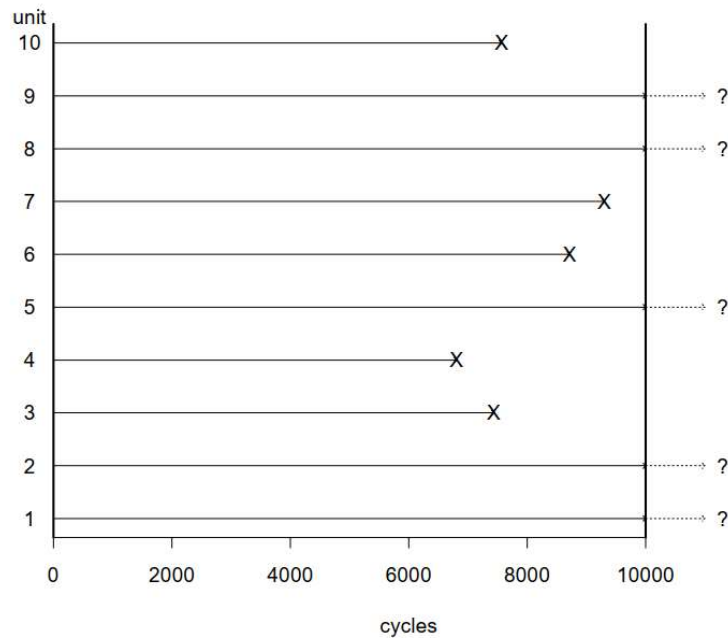
- The value of each sample unit is observed or known
- For LDA, the TTF for all the units in the sample are known.



Data Type: Censored Data

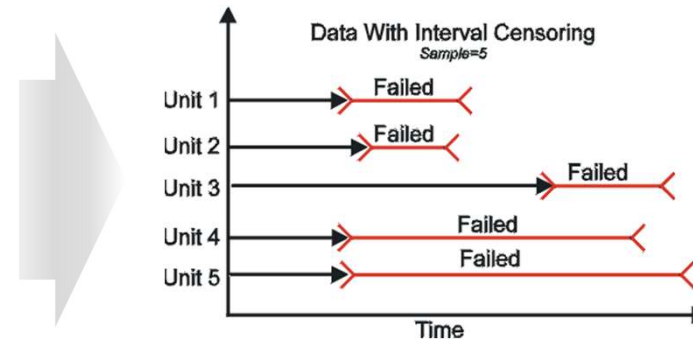
- All the units may not have failed during the observation period.
- The exact TTF of all the units are not known

Right Censored : The time to failure for the non-failed units occur after the observation is completed

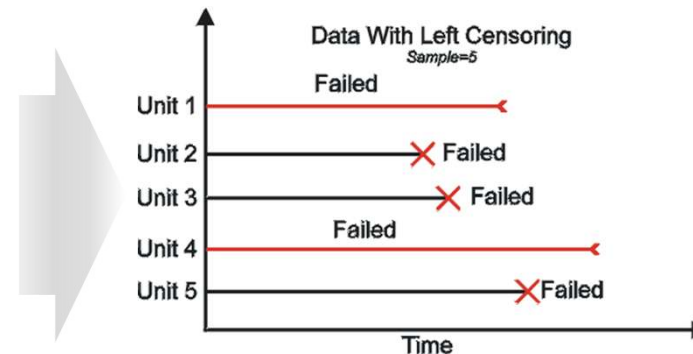


Data Type: Censored Data

- Interval censored: Uncertainty of the exact time of failure within an interval. This is often referred to as inspection data.



- Left Censored: The failure time is only known to happen before the data point. This is similar to interval censored data with starting time of the interval is zero.



Case Study

- 34 fire sensors are installed in an offshore installation and are inspected in 6 weeks interval. The state of sensors after the inspection is as shown below. The sensors are replaced upon failure. Estimate the life distribution.

34 sensors inspected in batches over 6-week cycle																																	
Seria No:	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30			
FS1	S						F						S						S						F								
FS2	S						F						S						S						S								
FS3	S						S						S						S						F								
FS4	S						S						F						S						S								
FS5	S						S						S						F						S								
FS6		S						F						S						S						F							
FS7		S						S						S						S						F							
FS8		S						S						S						S						S							
FS9		S						S						S						S						F							
FS10		F						S						S						S						F							

Collecting field data

Planning plant	Order	Description	Notifictn type	Notificat ion	Equipment	Description	Functional Loc.	Description	Code group	Prob. grp. text	Damage Code	Prob. code text	Cause code	Cause text	Cause code text	Priori ty	ABC indic.	Catalog profile	Caus e	Cause grp. text	MaintPI ant	Code group	ObjectPart Code	Planner group	Plant section	User status	Notif.date
M099	982034	To troubleshoot 72MOV4020	PM	2369319	11-72MOV-9 4020	T-7246 IN MOV	72MOV-4020	T-7246 IN MOV	0E-MOV	MOTOR OPERATED VALVE	SHCU	SHORT CIRCUIT TRIP	YILD		Overloading / yield	2	2	0E-MOV 000	000	Inspection-Mechanical Failure	M099	0E-MOV CTRL	AP6	A6	SCRN	02/02/2018	
M099	982034	72MOV4021 FALSE INDICATION.	PM	2369324	11-72MOV-0 4021	T-7246 OUT MOV	72MOV-4021	T-7246 OUT MOV	0E-MOV	MOTOR OPERATED VALVE	FTST	FAIL TO START	PWSG		POWER SURGE	2	2	0E-MOV 000	000	INSTRUMENT FAILURE	M099	0E-MOV CTRL	AP6	A6	SCRN	02/02/2018	
M102	981905	K23202A motor purging malfunction	PM	2368654	5 20-KM-23202A	MOTOR NET GAS COMPRESSOR	M102-KM-23202A	MOTOR NET GAS COMPRESSOR	0E-	MOTOR ELECTRIC MOTOR	FTST	FAIL TO START	OPCU		OPEN CIRCUIT (DISCONNECTION)	2	2	0E-MOTOR 000	000	ELECTRICAL FAILURE	M102	MOTOR CTRL	AP2	A2	SCRN	01/19/2018	
M102	981905	P22202A suddent stop	PM	2368381	6 20-PM-22202A	MOTOR WASH WATER INJECTION PUMP	M102-PM-22202A	MOTORWASH WATER INJECTION PUMP	0E-	MOTOR ELECTRIC MOTOR	FTST	FAIL TO START	OPCU		OPEN CIRCUIT (DISCONNECTION)	3	2	0E-MOTOR 000	000	ELECTRICAL FAILURE	M102	MOTOR CTRL	AP2	A2	SCRN	01/15/2018	
M102	981431	MM2 Damage A-51221-PM3A	PM	2367409	6 20-A-51221-P3AE-CELL PUMPS		A-51221-P3A	E-CELL PUMPS	0E-	MOTOR ELECTRIC MOTOR	WORN	WORN	WEAR		ABRASIVE AND ADHESIVE WEAR	3	3	0M-PMPCFG 000	000	MATERIAL FAILURE	M102	MOTOR CTRL	AP5	A5	SCRN	12/19/2017	
M102	981740	stop button not functioning	PM	2367359	1 20-P-25201A	PUMP CHARGE	M102-P-25201A	PUMP CHARGE	0E-	MOTOR ELECTRIC MOTOR	FTST	FAIL TO START	OPCU		OPEN CIRCUIT (DISCONNECTION)	3	3	0M-PMPCFG 000	000	ELECTRICAL FAILURE	M102	MOTOR CTRL	AP2	A2	SCRN	12/18/2017	
M102	981072	EDI stack orange #5 require plug	PM	2366433	7 20-A-51221-B2	E-CELL RECTIFIER SKID B	A-51221-B2	E-CELL RECTIFIER SKID B	0E-	MOTOR ELECTRIC MOTOR	SHCU	SHORT CIRCUIT TRIP	SHCU		SHORT CIRCUIT	3	1	0E-MOTOR 000	000	ELECTRICAL FAILURE	M102	MOTOR CTRL	AP5	A5	SCRN	11/24/2017	
M102	981026	59-KSV6517 Motor tork trip	RM	2366037	6		1101-F59601-CM	CONTROL & MONITORING,HRSG BOILER	0E-MOV	MOTOR OPERATED VALVE	OVLD	OVERLOAD PROTECTION TRIP	DC	Replace Main card	Defective component (cards, electronic c	3	3	000000001 000	000	FAULTY signal / indication / alarm	M102	0E-MOV CTRL	AP5	A5	SCRN	11/15/2017	
M102	981137	motor manager faulty	PM	2365821	2 20-PM-26202A	MOTOR MDEA RECYCLE PUMP	M102-PM-26202A	MOTOR MDEA RECYCLE PUMP	0E-	MOTOR ELECTRIC MOTOR	FTST	FAIL TO START	EROS		EROSIVE WEAR	3	3	0E-MOTOR 000	000	MATERIAL FAILURE	M102	MOTOR CTRL	AP2	A2	SCRN	11/09/2017	
M102	981137	K23202A purging not complete	PM	2365774	7		M102-K/KM23202A	COMPRESSOR NET GAS	0E-	MOTOR ELECTRIC MOTOR	FTST	FAIL TO START	OPCU		OPEN CIRCUIT (DISCONNECTION)	2		000000001 000	000	ELECTRICAL FAILURE	M102	MOTOR CTRL	AP2	A2	SCRN	11/08/2017	
M102	981137	K23202B cannot start	PM	2365774	6		M102-K/KM23202B	COMPRESSOR NET GAS	0E-	MOTOR ELECTRIC MOTOR	FTST	FAIL TO START	OPCU		OPEN CIRCUIT (DISCONNECTION)	2		000000001 000	000	ELECTRICAL FAILURE	M102	MOTOR CTRL	AP2	A2	SCRN	11/08/2017	
M102	980794	EDI orange stack #5 fuse blown	PM	2365483	0 20-A-51221-B2	E-CELL RECTIFIER SKID B	A-51221-B2	E-CELL RECTIFIER SKID B	0E-	MOTOR ELECTRIC MOTOR	SHCU	SHORT CIRCUIT TRIP	SHCU		SHORT CIRCUIT	2	1	0E-MOTOR 000	000	ELECTRICAL FAILURE	M102	MOTOR CTRL	AP5	A5	SCRN	11/02/2017	
M102	980704	MOV547 - cannot operate Remote and Local	PM	2365078	7 20-32MOV-547	MOVV-32202 QUENCH WATER INLET	M102-32MOV-547	MOV V-32202 QUENCH WATER INLET	0E-MOV	MOTOR OPERATED VALVE	FTST	FAIL TO START	SFFL		FAULTY DUE TO SOFTWARE FAILURE	2	1	0E-MOV 000	000	INSTRUMENT FAILURE	M102	0E-MOV CTRL	AP3	A3	SCRN	10/24/2017	
M102	980819	REM CANNOT MOVING	PM	2365116	3 20-65REM-2019 No.12	REMOTE ELEVATED MONITOR - S/S	65REM-2019	REMOTE ELEVATED MONITOR - S/S	0E-	MOTOR ELECTRIC MOTOR	FTST	FAIL TO START	OPCU		OPEN CIRCUIT (DISCONNECTION)	3	1	05-FFEQPT 000	000	ELECTRICAL FAILURE	M102	MOTOR CTRL	AP2	A3	SCRN	10/24/2017	
M102	980819	motor manager faulty	PM	2365037	4 20-EM-24215C	MOTOR DEISOHEXANISER OVERHEAD CONDENSER	M102-EM-24215C	MOTOR DEISOHEXANISER OVERHEAD CONDENSER	0E-	MOTOR ELECTRIC MOTOR	FTST	FAIL TO START	EROS		EROSIVE WEAR	3	3	0E-MOTOR 000	000	MATERIAL FAILURE	M102	MOTOR CTRL	AP2	A2	SCRN	10/23/2017	
M102	980634	MOV-541 - Actuator loose	PM	2364960	6 20-32MOV-541	MOVV-32201 DRAIN TO COKE PIT	M102-32MOV-541	MOV V-32201 DRAIN TO COKE PIT	0E-MOV	MOTOR OPERATED VALVE	NOIS	NOISY / HARMING	OPCU		OPEN CIRCUIT (DISCONNECTION)	3	1	0E-MOV 000	000	ELECTRICAL FAILURE	M102	0E-MOV CTRL	AP3	A3	SCRN	10/20/2017	
M102	980856	P22202A suddenly trip	PM	2364192	6		M102-P/PM22202A	PUMP WASH WATER INJECTION	0E-	MOTOR ELECTRIC MOTOR	FTST	FAIL TO START	OPCU		OPEN CIRCUIT (DISCONNECTION)	3		000000001 000	000	ELECTRICAL FAILURE	M102	MOTOR CTRL	AP2	A2	SCRN	10/01/2017	
M102	980288	72mov2005(outlet T72206) malfunction	PM	2368311	6 2005	MOVVT-72206 OUTLET	M102-72MOV-2005	MOV T-72206 OUTLET	0E-MOV	MOTOR OPERATED VALVE	FTST	FAIL TO START	OPCU		OPEN CIRCUIT (DISCONNECTION)	3	2	0E-MOV 000	000	ELECTRICAL FAILURE	M102	0E-MOV CTRL	AP4	A4	SCRN	09/21/2017	
M102	980227	MOV's for Loopl problem	PM	2363605	11MOV-68-2 2015	MOVVT-68204 Outlet y	M102-68MOV-2015	MOV T-68204 Outlet y	0E-MOV	MOTOR OPERATED VALVE	FTST	FAIL TO START	OPCU		OPEN CIRCUIT (DISCONNECTION)	2	3	0E-MOV 000	000	ELECTRICAL FAILURE	M102	0E-MOV CTRL	AP4	A4	SCRN	09/16/2017	
M102	980658	EM-29216C breaker relay fault	RM	2363599	1 20-EM-29216C	MOTOR COLD HP SEPARATOR FEED COOLER	M102-EM-29216C	MOTOR COLD HP SEPARATOR FEED COOLER	0E-	MOTOR ELECTRIC MOTOR	FTST	FAIL TO START	EROS		EROSIVE WEAR	3	3	0E-MOTOR 000	000	MATERIAL FAILURE	M102	MOTOR CTRL	AP2	A2	SCRN	09/16/2017	
M102	980228	73MOV2047 local selector switch loose	PM	2363415	11-73MOV-7 2047	MOVVT-73211 Outlet	73MOV-2047	MOV T-73211 Outlet y	0E-MOV	MOTOR OPERATED VALVE	FTST	FAIL TO START	OPCU		OPEN CIRCUIT (DISCONNECTION)	3	3	0E-MOV 000	000	ELECTRICAL FAILURE	M102	0E-MOV CTRL	AP4	A4	SCRN	09/12/2017	
M102	980228	68MOV204 wrong indication - remote fail	PM	2363425	11MOV-68-1 2004	MOVVT-68202 Inlet y	M102-68MOV-2004	MOV T-68202 Inlet y	0E-MOV	MOTOR OPERATED VALVE	FTST	FAIL TO START	OPCU		OPEN CIRCUIT (DISCONNECTION)	3	3	0E-MOV 000	000	ELECTRICAL FAILURE	M102	0E-MOV CTRL	AP4	A4	SCRN	09/12/2017	
M102	980227	68MOV2012 wrong indication - remote fail	PM	2363415	11MOV-68-8 2012	MOVVT-68203 Inlet y	M102-68MOV-2012	MOV T-68203 Inlet y	0E-MOV	MOTOR OPERATED VALVE	FTST	FAIL TO START	OPCU		OPEN CIRCUIT (DISCONNECTION)	3	3	0E-MOV 000	000	ELECTRICAL FAILURE	M102	0E-MOV CTRL	AP4	A4	SCRN	09/12/2017	
M102	980227	68MOV2013 wrong indication- remote fail	PM	2363415	11MOV-68-9 2013	MOVVT-68204 Inlet y	M102-68MOV-2013	MOV T-68204 Inlet y	0E-MOV	MOTOR OPERATED VALVE	FTST	FAIL TO START	OPCU		OPEN CIRCUIT (DISCONNECTION)	3	3	0E-MOV 1 000	000	ELECTRICAL FAILURE	M102	0E-MOV CTRL	AP4	A4	SCRN	09/12/2017	

Which info is required for the analysis?

SAP Data field

“Notif Long Text” Files (32 parameters)

- | | |
|----------------|----------------|
| 1. NOTIFNO | 18. PRIORITY |
| 2. NOTIFTYP | 19. BREAKIND |
| 3. SYSSTATUS | 20. DOWNTIME |
| 4. USRSTATUS | 21. OBJPART |
| 5. NOTIFDESC | 22. OBJPARTCOD |
| 6. FUNCLOC | 23. DAMAGE |
| 7. EQPTNO | 24. DAMAGECOD |
| 8. ASSEMBLY | 25. DAMAGETXT |
| 9. MPG | 26. CAUSE |
| 10. MAINWKCEN | 27. CAUSECOD |
| 11. MAINTPLT | 28. CAUSETXT |
| 12. REPORTBY | 29. ORDNO |
| 13. NOTIFCRED | 30. DESCLTXT |
| 14. MALFUNSTD | 31. DAMAGELTXT |
| 15. MALFUNENDD | 32. CAUSELTXT |
| 16. NOTIFCOMP | |
| 17. NOTIFREFD | |

“WO Header Long Text” files (8 parameters)

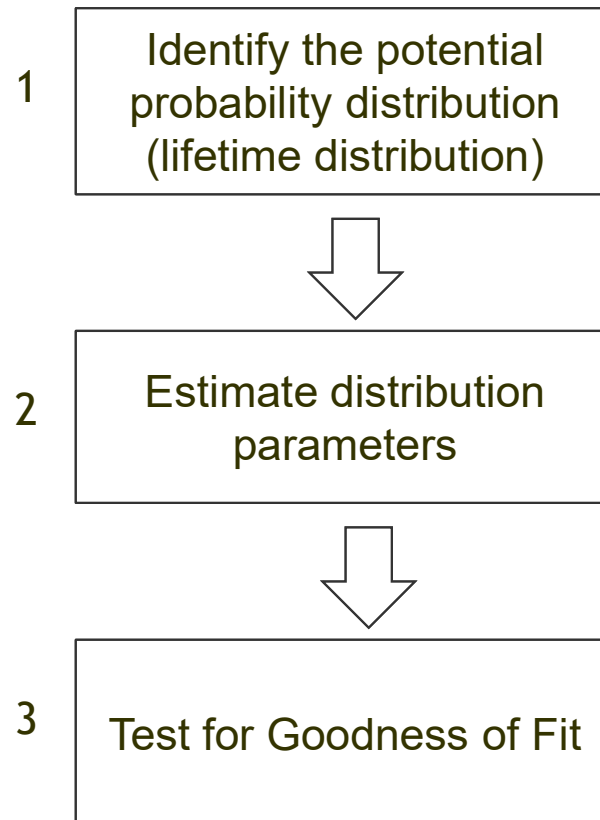
1. MAINTPLT
2. ORDTYPE
3. ORDDESC
4. SYSTATUS
5. USTATUS
6. ORDERNO
7. FORMAT
8. LONGTEXT

“WO Service & Material” files (67 parameters)

- | | | |
|---------------|---------------|---------------|
| 1. ORDERNO | 21.SUBOPN | 41.RESVITMENO |
| 2. ORDERTYP | 22.WRKCTR | 42.RESVNO |
| 3. ORDERSTTXT | 23.CRTLKEY | 43.COMTDQTY |
| 4. SYSSTATUS | 24.OPNDESC | 44.ISSUEQTY |
| 5. USRSTATUS | 25.EARSCHDATE | 45.STORAGELOC |
| 6. MPG | 26.EARFINDT | 46.RECVQTY |
| 7. PLANPLANT | 27.COMPITMENO | 47.MATGRP |
| 8. MAINWRKCTR | 28.MATNO | 48.PRNO |
| 9. MAINACTTYP | 29.MATDESC | 49.PRITEMNO |
| 10.BSCSTARTDT | 30.REQQTY | 50.PRICE |
| 11.FINDATE | 31.UOM | 51.CSTELEMENT |
| 12.PRIORITY | 32.ITEMCAT | 52.PURCHGRP |
| 13.FUNCLOC | 33.PURCHGRP | 53.PURCHORG |
| 14.EQPTNO | 34.VALPRICE | 54.REQUESTOR |
| 15.ASSEMBLY | 35.GL | 55.PONO |
| 16.ORDCRTNDT | 36.UNLOADPNT | 56.VENDORNO |
| 17.LOCATION | 37.MATPRNO | 57.VENDORNAME |
| 18.ORDRELDAT | 38.MATPRITME | 58.POITEM |
| 19.OPNCNFDAT | 39.RECIPIENT | 59.POQTY |
| 20.OPNACT | 40.POINDC | 60.POITMPRICE |
| | | 61.POVALUE |
| | | 62.CURRENCY |
| | | 63.EXCHGRATE |
| | | 64.OANO |
| | | 65.OAITMNO |
| | | 66.POPOSTDATE |
| | | 67.MOVETYP |

Probability distribution analysis steps

Consists of **THREE** main steps:



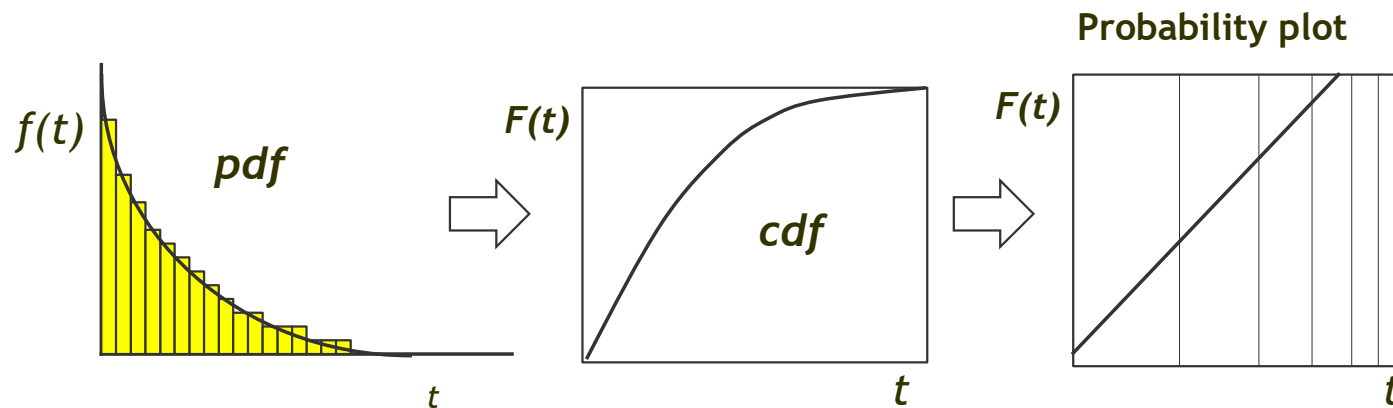
Step 1: Identify the potential distribution

- Identification of the right candidate for distribution is both an art and a science
- Understanding and knowledge on these will help the process:
 - failure process (chemical, electrical, physical, mechanical, structural, or thermal processes)
 - maintenance practice and improvement actions
 - theoretical distribution characteristics
 - statistical analysis of data
- Possible approaches to identify the distribution:
 - construct histogram
 - compute descriptive statistics
 - analyze empirical failure rate
 - probability plotting
- The use of statistical software will greatly assist the process, however, at the end it is up to the analysts to select the best distribution representing the failure process

Step 1: Identify the potential distribution (cont'd)

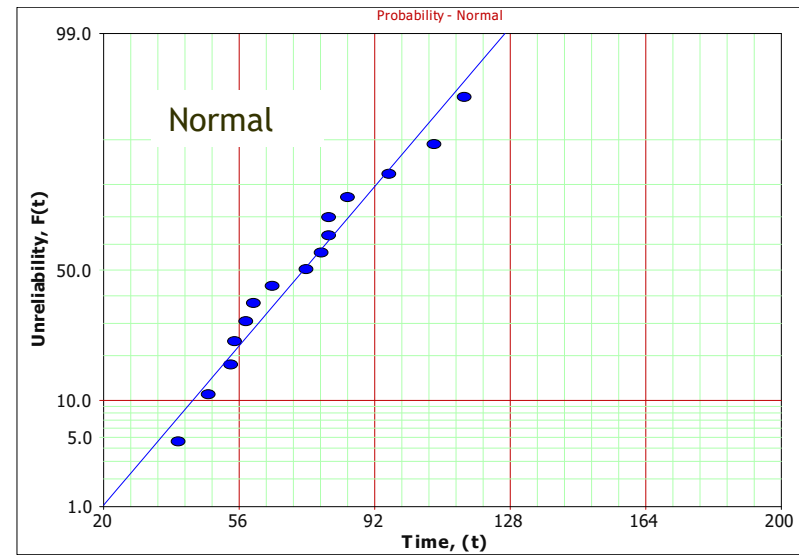
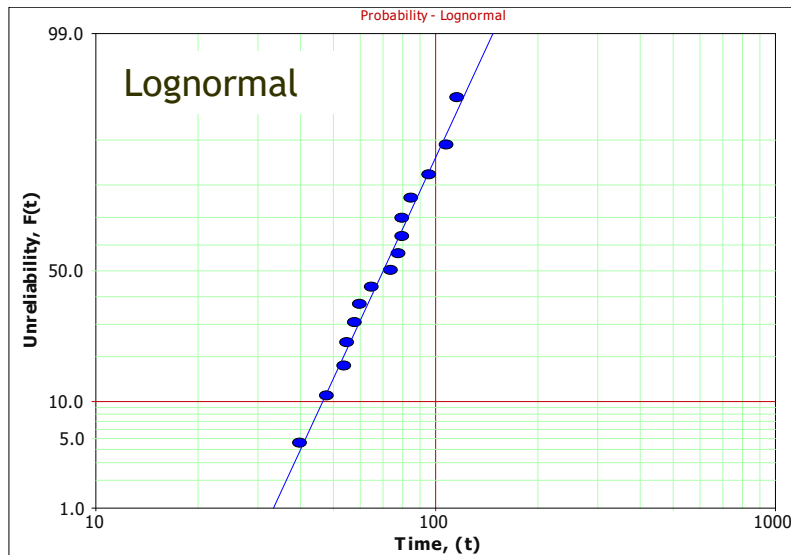
Probability plotting

probability plotting takes the **cdf** of the distribution and attempts to **linearize** it by employing a specially constructed paper.



Different probability papers exist for each type of distribution

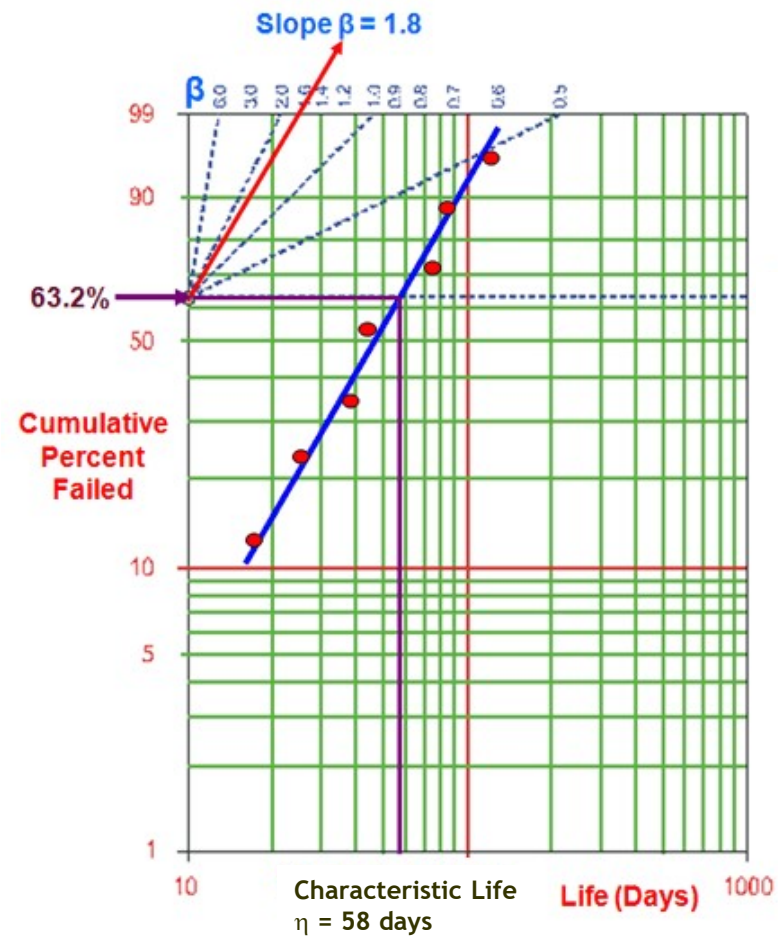
Example: Probability plots



Which one is better?

Step 1: Identify the potential distribution (cont'd)

Example:
Weibull Probability plot



Linearizing CDF

- Exponential Distribution

$$F(t) = 1 - e^{-\lambda t}$$

$$\ln \left[\frac{1}{1 - F(t)} \right] = \lambda t$$

$$y = Bx + C \quad \left\} \quad y : \ln \left[\frac{1}{1 - F(t)} \right] \right.$$

$$x : t$$

$$B : \text{Slope} = \lambda$$

Linearizing CDF

- Weibull Distribution

$$F(t) = 1 - e^{-\left(\frac{t}{\eta}\right)^\beta}$$

$$\ln(1 - F(t)) = \ln\left(e^{-\left(\frac{t}{\eta}\right)^\beta}\right)$$

$$\ln(-\ln(1 - F(t))) = \beta \ln\left(\frac{t}{\eta}\right)$$

$$\ln\left(\ln\left(\frac{1}{1 - F(t)}\right)\right) = \beta \ln(t) - \ln(\eta)$$

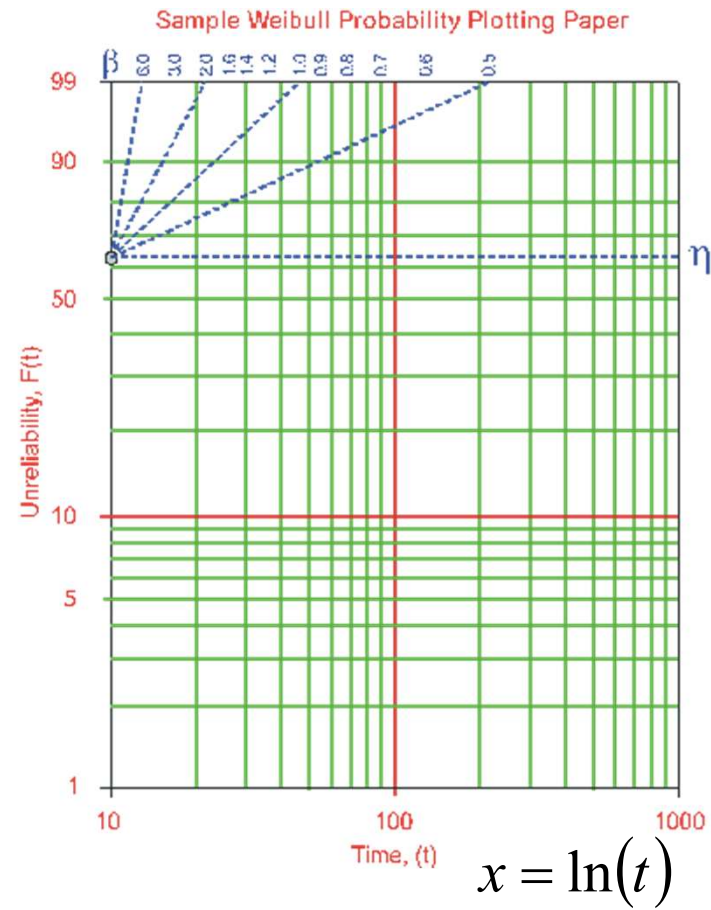
$$y = \ln\left(\ln\left(\frac{1}{1 - F(t)}\right)\right)$$

$$x = \ln(t)$$

$$y = \beta x - \beta \ln(\eta)$$

Linearizing CDF - Weibull

$$y = \ln \left(\ln \left(\frac{1}{1 - F(t)} \right) \right)$$



Ranking Of Data

- Probability plotting involves charting the variable of interest against CDF.
- Thus, data need to be ordered and the cumulative probability of each data point is calculated.
- Ranking of data provides an estimate of what percentage is represented by the particular sample.

Ranking Methods

- Mean Rank
 - Used mostly for symmetrical statistical distribution such as normal distribution

$$\text{mean rank} = \frac{i}{N+1}$$

- Median Rank
 - Used mostly for non-normal distributions (exponential, Weibull, etc).
 - Cumulative Binomial Method

$$P = \sum_{k=j}^N \binom{N}{k} Z^k (1-Z)^{N-k} \quad \xrightarrow{\text{For median rank}} \quad 0.50 = \sum_{k=j}^N \binom{N}{k} Z^k (1-Z)^{N-k}$$

- Algebraic Approximation

$$\xrightarrow{\text{Bernard's Approximation}} \quad \text{Median Rank } r_j = \frac{j-0.3}{N+0.4}$$

Ranking Example

- For complete data

$$\text{Median Rank } r_j = \frac{j-0.3}{N+0.4}$$

Time to failure
140
37
69
73
108
63



Rank Number, j	Time to failure
1	37
2	63
3	69
4	73
5	108
6	140

↑
x



Median Ranking
$\frac{1-0.3}{6+0.4} = 0.109$ or 10.9%
$\frac{2-0.3}{6+0.4} = 0.266$ or 26.6%
$\frac{3-0.3}{6+0.4} = 0.422$ or 42.2%
$\frac{4-0.3}{6+0.4} = 0.578$ or 57.8%
$\frac{5-0.3}{6+0.4} = 0.734$ or 73.4%
$\frac{6-0.3}{6+0.4} = 0.891$ or 89.1%

↑
y

Assuming the TTF follows a Weibull distribution estimate the parameters using Weibull Probability paper.

Ranking Censored Data

- Censored data are not plotted but their existence will affect the rank of other data
- Thus, ranking need to be adjusted to reflect the uncertainty of the unknown failure times.
- The steps for adjusted ranking are as follows:
 - List order number (i) of failed items (i=1,2,...) (Column 1)
 - List increasing ordered sequence of life values (ti) of failed items (Column 2)
 - List reverse order number (Column 3)
 - Calculate adjusted ranking using :

$$i = \text{Adjusted Rank} = \frac{\text{Reverse Rank} \times \text{Previous adjusted Rank} + (N + 1)}{1 + (\text{Reverse Rank})}$$

- Calculate the median rank for each failed item using the following:

$$r_{t_i} = \frac{i_{t_i} - 0.3}{n + 0.4} \%$$

Ranking Censored data-Example

Item #	Time	Reverse Ranking	Adjusted Ranking	Median Rank $r_{t_i} = \frac{i_{t_i} - 0.3}{n + 0.4} \%$
1	10 (S)	8	Suspended	
2	30 (F)	7	$(7 \times 0) + (8+1)/(7+1) = 1.125$	$(1.125 - 0.3)/(8 + 0.4) = 9.8\%$
3	45 (S)	6	Suspended	
4	49 (F)	5	$(5 \times 1.125) + (8+1)/(5+1) = 2.438$	25.5%
5	82 (F)	4	$(4 \times 2.438) + (8+1)/(4+1) = 3.750$	41.1%
6	90 (F)	3	$(3 \times 3.750) + (8+1)/(3+1) = 5.063$	56.7%
7	96 (F)	2	$(2 \times 5.063) + (8+1)/(2+1) = 6.375$	72.3%
8	100 (S)	1	suspended	

Once the ranking is calculated, proceed with plotting the data

$$i = \text{Adjusted Rank} = \frac{\text{Reverse Rank} \times \text{Previous adjusted Rank} + (N + 1)}{1 + (\text{Reverse Rank})}$$

Step 2: Estimate distribution parameters

Estimating parameters using manual probability plotting is inaccurate, time consuming and does not provide repeatability.

Two other common approaches in estimating the parameters are:

- **Rank Regression**
- **Maximum Likelihood Estimation (MLE)**

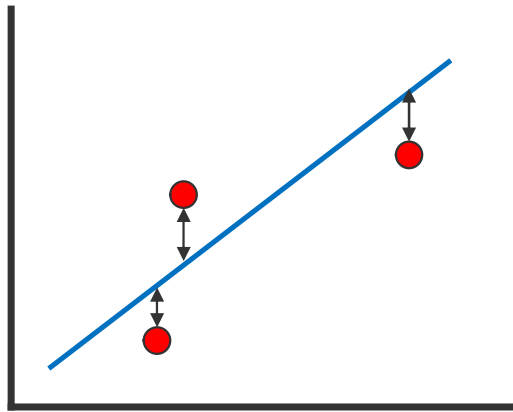
Step 2: Estimate distribution parameters

Rank Regression method

- Use the idea of probability plotting to mathematically estimate the parameters that result in the straight line best fitting of data
- Also known as “**Least Squares method**”
- Use for **complete data** and **small sample size**

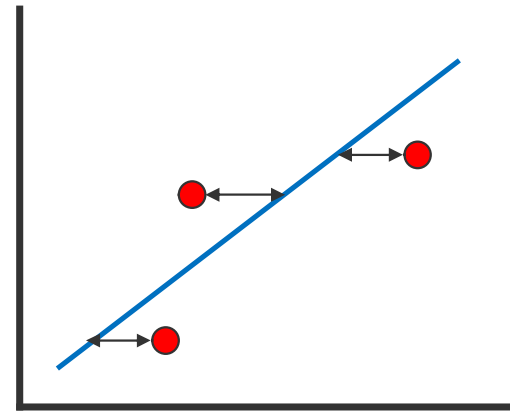
Step 2: Estimate distribution parameters

Rank Regression method



Rank Regression Y

Minimizing the sum of the squares of the distance of the points to the fitted line in **Y-axis** direction



Rank Regression X

Minimizing the sum of the squares of the distance of the points to the fitted line in **X-axis** direction

Rank Regression

The fitting of straight line will take the form $y = bx + a$

And we want to calculate the values of a and b that minimize the square of the distance from the point to the line. Mathematically for regression on y :

$$\sum_{i=1}^N (\hat{a} + \hat{b}x_i - y_i)^2 = \min(a, b) \sum_{i=1}^N (a + bx_i - y_i)^2$$
$$\hat{a} = \frac{\sum_{i=1}^N y_i}{N} - \hat{b} \frac{\sum_{i=1}^N x_i}{N} = \bar{y} - \hat{b}\bar{x}$$
$$\hat{b} = \frac{\sum_{i=1}^N x_i y_i - \frac{\sum_{i=1}^N x_i \sum_{i=1}^N y_i}{N}}{\sum_{i=1}^N x_i^2 - \frac{\left(\sum_{i=1}^N x_i\right)^2}{N}}$$

Rank Regression

- The measurement of how good the fit is known as correlation coefficient which is given by

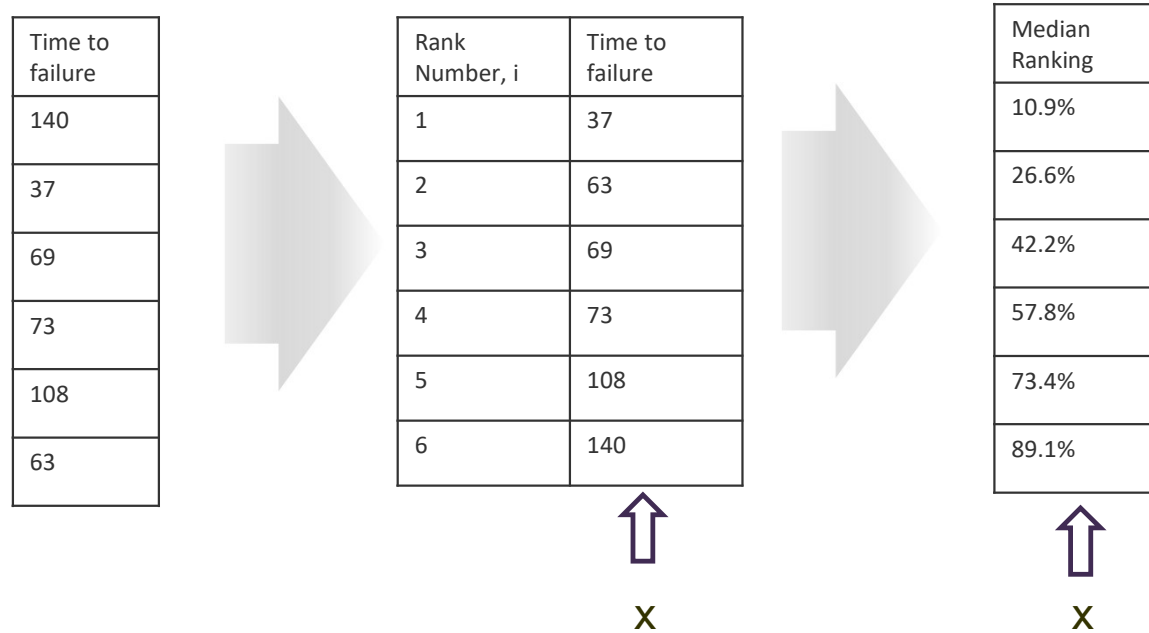
$$\rho = \frac{\sigma_{xy}}{\sigma_x \sigma_y}$$

And can be estimated with the following equation.

$$\hat{\rho} = \frac{\sum_{i=1}^N x_i y_i - \frac{\sum_{i=1}^N x_i \sum_{i=1}^N y_i}{N}}{\sqrt{\left(\sum_{i=1}^N x_i^2 - \frac{\left(\sum_{i=1}^N x_i \right)^2}{N} \right) \left(\sum_{i=1}^N y_i^2 - \frac{\left(\sum_{i=1}^N y_i \right)^2}{N} \right)}}$$

ρ can range from +1 and -1 for perfect fit with positive slope negative slope respectively. Zero value indicates not correlation.

Rank Regression Example



Rank Regression Example-Weibull Distribution

N =6

i	Time to failure	Median ranking (F(t))	$x=\ln(t)$	$y=\ln(\ln(1/(1-F(t))))$	$x_i y_i$	x_i^2
1	37	0.109	3.611	-2.156	-7.784	13.039
2	63	0.266	4.143	-1.175	-4.869	17.166
3	69	0.422	4.234	-0.602	-2.547	17.928
4	73	0.578	4.290	-0.147	-0.632	18.408
5	108	0.734	4.682	0.282	1.320	21.922
6	140	0.891	4.942	0.794	3.925	24.420

$$\hat{b} = \frac{\sum_{i=1}^N x_i y_i - \frac{\sum_{i=1}^N x_i \sum_{i=1}^N y_i}{N}}{\sum_{i=1}^N x_i^2 - \frac{\left(\sum_{i=1}^N x_i\right)^2}{N}}$$

Sum

25.902	-3.003	-10.587	112.882
--------	--------	---------	---------

$$\hat{b} = \hat{\beta} = 2.245$$

$$\hat{a} = -\hat{\beta} \ln(\eta) = -10.2$$

$$\therefore \eta = e^{\frac{10.2}{2.245}} = 93.7 \text{ hours}$$

$$\hat{a} = \frac{\sum_{i=1}^N y_i}{N} - \hat{b} \frac{\sum_{i=1}^N x_i}{N} = \bar{y} - \hat{b} \bar{x}$$

Rank Regression Example-Exponential Distribution

For one parameter exponential distribution

$$\hat{a} = 0$$

$$\hat{b} = \frac{\sum_{i=1}^N x_i y_i}{\sum_{i=1}^N y_i^2}$$

$$b = \frac{1}{\hat{b}} = -\lambda \Rightarrow \lambda = -\frac{1}{\hat{b}}$$

Goodness of fit test

$$\hat{\rho} = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^N (x_i - \bar{x})^2 \cdot \sum_{i=1}^N (y_i - \bar{y})^2}}$$

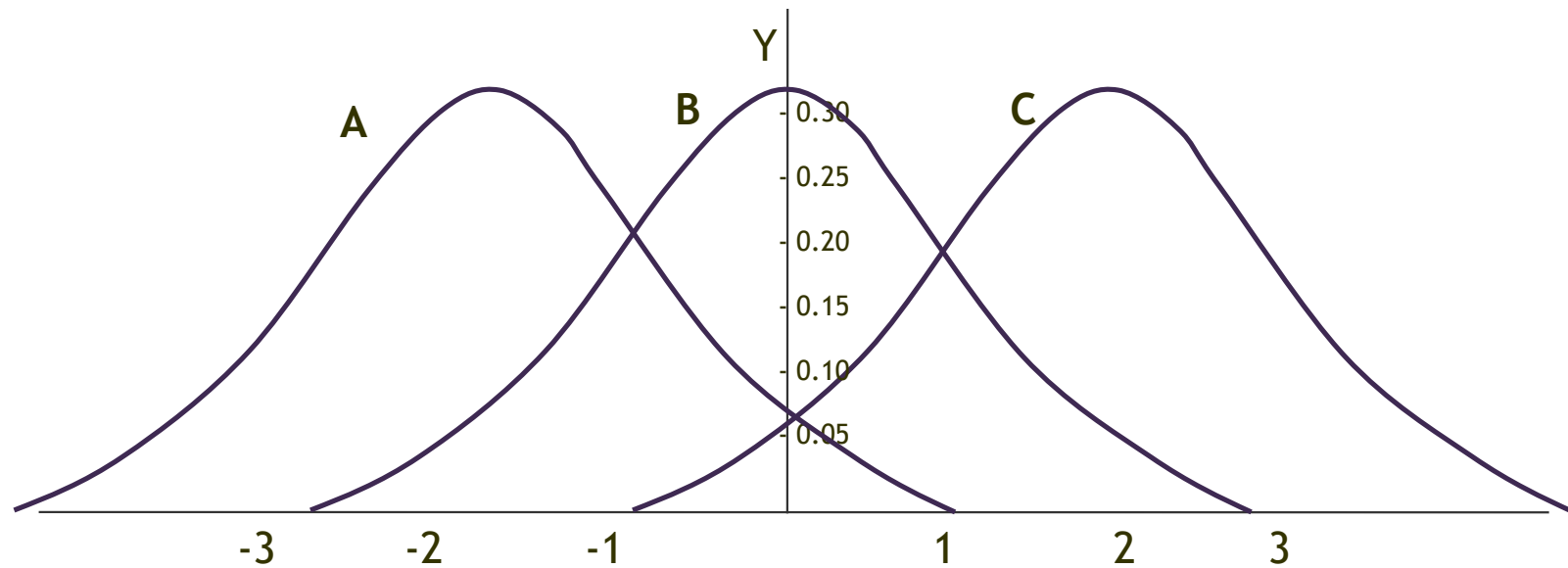
Step 2: Estimate distribution parameters

Maximum Likelihood Estimation (MLE)

- Is statistical (non-graphical) method of estimating the parameters of statistical model
- For a given set of data and underlying statistical model, the method selects values of the model parameters that produce a distribution that gives the observed data the greatest probability (i.e., parameters that **maximize the likelihood function**)
- Can be used for Suspension (censored) and mixed data
- Larger sample size (>30 data)

Maximum Likelihood Estimation (MLE)

If two values are observed: -2, 2
Which is the most likely model?



Maximum Likelihood

- Consider a generic pdf function, where x is the TTF and θ_k is the parameters to be estimated.

$$f(x; \theta_1, \theta_2, \dots, \theta_k)$$

- The likelihood function for a complete data is given as

$$L = \prod_{i=1}^n f(x_i; \theta_1, \theta_2, \dots, \theta_k)$$

- The function is converted to log for easy manipulation

$$\Lambda = \ln L = \sum_{i=1}^n \ln f(x_i; \theta_1, \theta_2, \dots, \theta_k)$$

- Then the maximum value is calculated by finding the partial derivatives for each parameter and setting it equal to zero:

$$\frac{\partial \Lambda}{\partial \theta_j} = 0, \quad j = 1, 2, \dots, k$$

Maximum Likelihood-Example

Let $X_1, X_2, \dots, X_n$ be a random sample from the exponential distribution with p.d.f.

$$f(x; \theta) = \frac{1}{\theta} e^{-x/\theta}, 0 < x < \infty, \theta \in \Omega = \{\theta; 0 < \theta < \infty\}.$$

The likelihood function is given by

$$L(\theta) = L(\theta; x_1, x_2, \dots, x_n) = \left(\frac{1}{\theta} e^{-x_1/\theta}\right) \left(\frac{1}{\theta} e^{-x_2/\theta}\right) \cdots \left(\frac{1}{\theta} e^{-x_n/\theta}\right) = \frac{1}{\theta^n} \exp\left(-\frac{\sum_{i=1}^n x_i}{\theta}\right), 0 < \theta < \infty.$$

The natural logarithm of $L(\theta)$ is

$$\ln L(\theta) = -(n) \ln(\theta) - \frac{1}{\theta} \sum_{i=1}^n x_i, 0 < \theta < \infty.$$

Thus,

$$\frac{d[\ln L(\theta)]}{d\theta} = \frac{-n}{\theta} + \frac{\sum_{i=1}^n x_i}{\theta^2} = 0.$$

The solution of this equation for θ is

$$\theta = \frac{1}{n} \sum_{i=1}^n x_i = \bar{x}.$$

Step 3: Test for Goodness of Fit

To determine whether the specified distribution fit the data

- It is a statistical test in which the validity of one hypothesis is tested without specification of an alternative hypothesis
- Hypothesis test :
 - H_o (*null*) = the failure times come from the specified distribution
 - H_a (*alternative*) = the failure times did not come from the specified distribution
- In the test, a statistic based on the data is computed and then compared with a critical value from a table of such values. Typically, if the test statistic is less than a critical value, the null hypothesis is accepted, otherwise alternative hypothesis is accepted

Step 3: Test for Goodness of Fit (ct'd)

Common Goodness of Fit Tests

Types	Description	Distribution
Chi-Square test	General test for discrete and continuous distribution, requires large sample size and binned data (histogram)	Exponential, Weibull, Normal, Lognormal, Binomial
Kolmogorov-Smirnov (K-S) test	General test for continuous distribution, more sensitive near the center than the tails. For small sample size	Normal, Lognormal, Exponential, Weibull, Logistics
Bartlett's test	Specific test	Exponential
Mann's test	Specific test	Weibull
Anderson-Darling test	Modification of K-S test, gives more weight to the tails, limited tables of critical values	Normal, Lognormal, Exponential, Weibull, Logistics

Weibull++ GOF Test

Distribution Wizard Setup

Weights for Rank Regression		Weights for Maximum Likelihood (MLE)	
Goodness of Fit	50 %	Goodness of Fit	40 %
Plot Fit	20 %	Plot Fit	10 %
Likelihood Ratio	30 %	Likelihood Ratio	50 %
Total: 100 %		Total: 100 %	

Other Options

☐ Discard distribution if location parameter is less than zero

☒ Discard 3-parameter Weibull if location parameter is less than: 1 % of min. time

☒ Discard Weibull if shape parameter is greater than: 15

Default OK Cancel



Distribution Wizard

	Distribution	AVGOF	AVPLOT	LKV
1	1P-Exponential	59.58086022	7.74887329	-130.2769526
2	Normal	92.27754045	11.08006669	-137.09092
3	Lognormal	40.22708547	7.06333327	-131.4650075
4	2P-Weibull	48.38718316	6.295370144	-130.2031534
5				
6				
7				
8				
9				
10				
11				

Analyze Implement Close

Print Setup

Initial Intermediate Final Report



Distribution Wizard

Main Analysis Details

Distributions and Rankings

☒ 1P-Exponential	3
☐ 2P-Exponential	
☒ Normal	4
☒ Lognormal	2
☒ 2P-Weibull	1
☐ 3P-Weibull	
☐ Gamma	
☐ G-Gamma	
☐ Logistic	
☐ Loglogistic	
☐ Gumbel	

Select All Clear All

Analysis is complete.

The panel on the left shows the rankings for the distributions that were considered. Rank = 1 is the best fit for the data.

Click the 'Analysis Details' tab for specific calculation results.

Click 'Implement' to return to the data sheet and apply the highest ranking distribution to analyze the data.

Analyze Implement Close

Print Setup

Analysis method selected in the data sheet: RRX

Kolmogorov-Smirnov Test

- Steps in performing K-S test
 - Tabulate ranked failure data (from smallest to largest)
 - Calculate the value of $|x_i - E_i|$ where x_i is the i th cumulative rank and E_i is the expected cumulative rank for the assumed distribution
 - Calculate the maximum value
 - Compare the value to the K-S value table.

K-S Test Example

Event	TTF (H)	$X_i = (i-0.3)/(N+0.4)$	E_i (based on selected distribution)	$Abs(X_i - E_i)$
1	12.2	0.056	0.027	0.030
2	13.1	0.137	0.105	0.032
3	14	0.218	0.282	0.064
4	14.1	0.298	0.308	0.009
5	14.6	0.379	0.450	0.071
6	14.7	0.460	0.480	0.020
7	14.7	0.540	0.480	0.060
8	15.1	0.621	0.599	0.022
9	15.7	0.702	0.759	0.057
10	15.8	0.782	0.782	0.000
11	16.3	0.863	0.876	0.013
12	16.9	0.944	0.946	0.002

Bx Life

- The term BX is derived from the terminology used by bearing manufacturers (in the early days of reliability engineering), specifically the B10 life
- The time at which X% of the units in a population will have failed. For example, if an item has a B10 life of 100 hours, that means that 10% of the population will have failed by 100 hours of operation.
- B10 life refers to the time by which 10% of the bearings would fail (or 90% of population survives)

Example 6

The time to failures (days) of 20 switches are shown below:

50.1	20.9	31.1	96.5	36.3
42.6	84.9	6.2	32.0	30.4
14.2	4.6	2.5	1.8	11.5
88.6	10.7	99.1	87.7	84.6

Determine the best fit distribution.

Example of LDA using Microsoft Excel

Example – Microsoft Excel

1.Key in Data in the Weibull Excel template

Key-in
data in
column B
in
ascending
order

Distribution Fitting Template_Optimal.xlsx - Excel

FILE HOME INSERT PAGE LAYOUT FORMULAS DATA REVIEW VIEW FOXIT PDF

From Access From Web From Text From Other Sources Existing Connections Refresh All Connections Sort Filter Clear Reapply Advanced Text to Columns Flash Fill Remove Duplicates Data Validation Consolidate

F24 : X ✓ fx =LN(LN(E24))

Sample	Time-to-Failure	Rank	Median Rank	1/1-Median Rank	Input Y Range ln(ln(1/(1-Median Rank)))	Input X Range ln(Time-to-Failure)
1	1.80	1	0.034	1.036	-3.355	0.588
2	2.50	2	0.083	1.091	-2.442	0.916
3	4.60	3	0.132	1.153	-1.952	1.526
4	6.20	4	0.181	1.222	-1.609	1.825
5	10.70	5	0.230	1.299	-1.340	2.370
6	11.50	6	0.279	1.388	-1.116	2.442
7	14.20	7	0.328	1.489	-0.921	2.653
8	20.90	8	0.377	1.606	-0.747	3.040
9	30.40	9	0.426	1.744	-0.587	3.414
10	31.10	10	0.475	1.907	-0.438	3.437
11	32.00	11	0.525	2.103	-0.297	3.466
12	36.30	12	0.574	2.345	-0.160	3.592
13	42.60	13	0.623	2.649	-0.026	3.752
14	50.10	14	0.672	3.045	0.107	3.914
15	84.60	15	0.721	3.579	0.243	4.438
16	84.90	16	0.770	4.340	0.384	4.441
17	87.70	17	0.819	5.514	0.535	4.474
18	88.60	18	0.868	7.556	0.704	4.484
19	96.50	19	0.917	12.000	0.910	4.570
20	99.10	20	0.966	29.143	1.216	4.596

Note: Generated result is comparable with result from Weibull ++ with setting RRY (rank regression Y) and SRM (standard regression method), median ranking
Enter TTF failure data with suspension data removed. Copy ranking order based on rank adjustment
Use excel addin for regression (under data-->data analysis) and SMS2 as the target output cell

Example – Microsoft Excel

2. Use Excel Add-in for regression (under data-->data analysis) and \$M\$2 as the target output cell

Choose Data

Choose Data Analysis

The screenshot shows the Microsoft Excel interface with the 'DATA' tab selected in the ribbon. The 'Data Analysis' button is highlighted in the 'Data Tools' group. The spreadsheet contains a Weibull Distribution data set with columns for Sample, Time-to-Failure, Rank, Median Rank, Wt-Median Rank, Input Y Range, and Input X Range. The 'Data Analysis' dialog box is open, showing the 'Data Analysis' tab with various statistical tools.

Sample	Time-to-Failure	Rank	Median Rank	Wt-Median Rank	Input Y Range	Input X Range
1	1.80	1	0.034	1.036	-3.355	0.588
2	2.50	2	0.083	1.091	-2.442	0.916
3	4.60	3	0.132	1.153	-1.952	1.526
4	6.20	4	0.181	1.222	-1.609	1.825
5	10.70	5	0.230	1.299	-1.340	2.370
6	11.50	6	0.279	1.388	-1.116	2.442
7	14.20	7	0.328	1.489	-0.921	2.853
8	20.90	8	0.377	1.606	-0.747	3.040
9	30.40	9	0.426	1.744	-0.587	3.414
10	31.10	10	0.475	1.907	-0.438	3.437
11	32.00	11	0.525	2.103	-0.297	3.466
12	36.30	12	0.574	2.345	-0.160	3.592
13	42.60	13	0.623	2.649	-0.026	3.752
14	50.10	14	0.672	3.045	0.107	3.914
15	84.60	15	0.721	3.579	0.243	4.438
16	84.90	16	0.770	4.340	0.384	4.441
17	87.70	17	0.819	5.514	0.535	4.474
18	88.60	18	0.868	7.556	0.704	4.484
19	96.50	19	0.917	12.000	0.910	4.570
20	99.10	20	0.966	29.143	1.216	4.596

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.9842
R Square	0.9687
Adjusted R Square	0.9669
Standard Error	0.2131
Observations	20

ANOVA

	df	SS	MS	F
Regression	1	25.272	25.272	556.2
Residual	18	0.8178	0.0454	
Total	19	26.09		

Coefficient and Standard Error

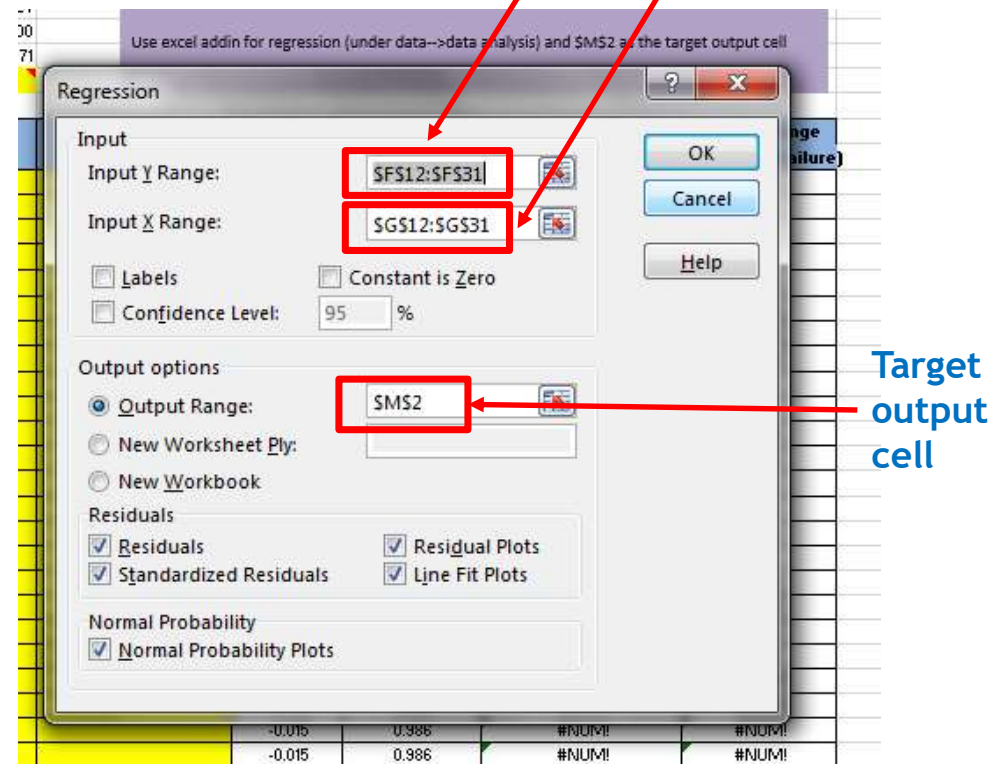
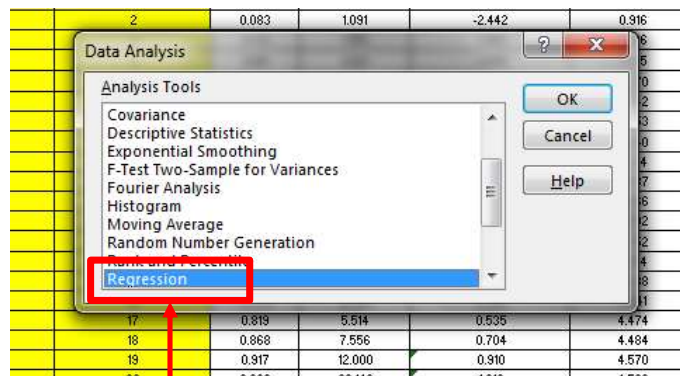
	Coefficient	Standard Error	t Stat	P-value
Intercept	-3.437	0.1333	-26.11	9E-1
X Variable 1	0.9235	0.0392	23.585	5E-1

RESIDUAL OUTPUT

Observation	Predicted	Residual	Standard Residual
1	-2.954	-0.401	-1.932
2	-2.651	0.2089	1.0072
3	-2.088	0.1354	0.6526
4	-1.812	0.2031	0.9788
5	-1.308	-0.032	-0.154
6	-1.241	0.1256	0.6055

Example – Microsoft Excel

2. Use Excel Add-in for regression (under data-->data analysis) and \$M\$2 as the target output cell



Example – Microsoft Excel

3. Generate results

Calculated
Weibull
parameters
 $\beta = 0.924$
 $\eta = 44.1$
MTTF = 45.771

Distribution Fitting Template_Optimal.xlsx - Excel

Ainul Akmar Bt Mokht

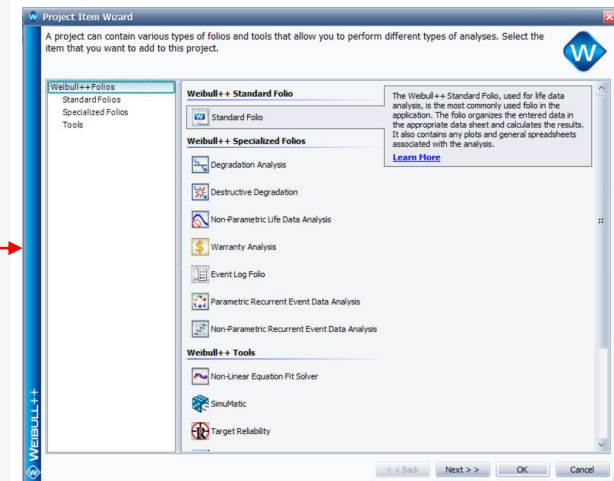
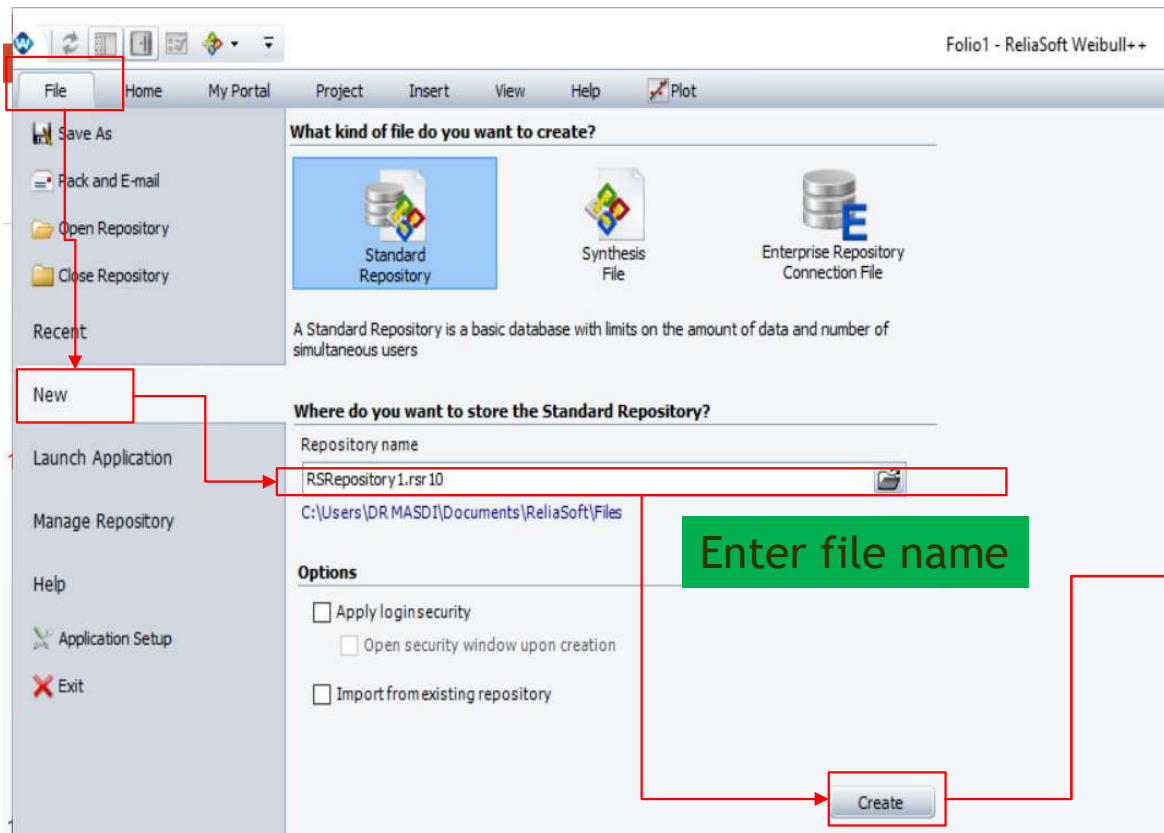
	A	B	C	D	E	F
1	Weibull Distribution					
2						
3	Objective: To estimate parameter for Weibull distribution.					
4						
5	Beta (Shape parameter)	0.924				
6	Eta (characteristic life)	44.100				
7	MTTF	45.771				
8	n	20				
9						
10						
11	Sample	Time-to-Failure	Rank	Median Rank	1/1-Median Rank	Input Y Rank
12	1	1.80	1	0.034	1.036	-3.355
13	2	2.50	2	0.083	1.091	-2.442
14	3	4.60	3	0.132	1.153	-1.952
15	4	6.20	4	0.181	1.222	-1.609
16	5	10.70	5	0.230	1.299	-1.340
17	6	11.50	6	0.279	1.388	-1.116
18	7	14.20	7	0.328	1.489	-0.921

Note: Generated result is comparable with result from Weibull ++ with setting RRY (rank regression Y) and SRM (standard regression method). Enter TTF failure data with suspension data removed. Copy ranking on adjustment. Use excel addin for regression (under data-->data analysis) and \$M\$2.

LDA using Weibull++

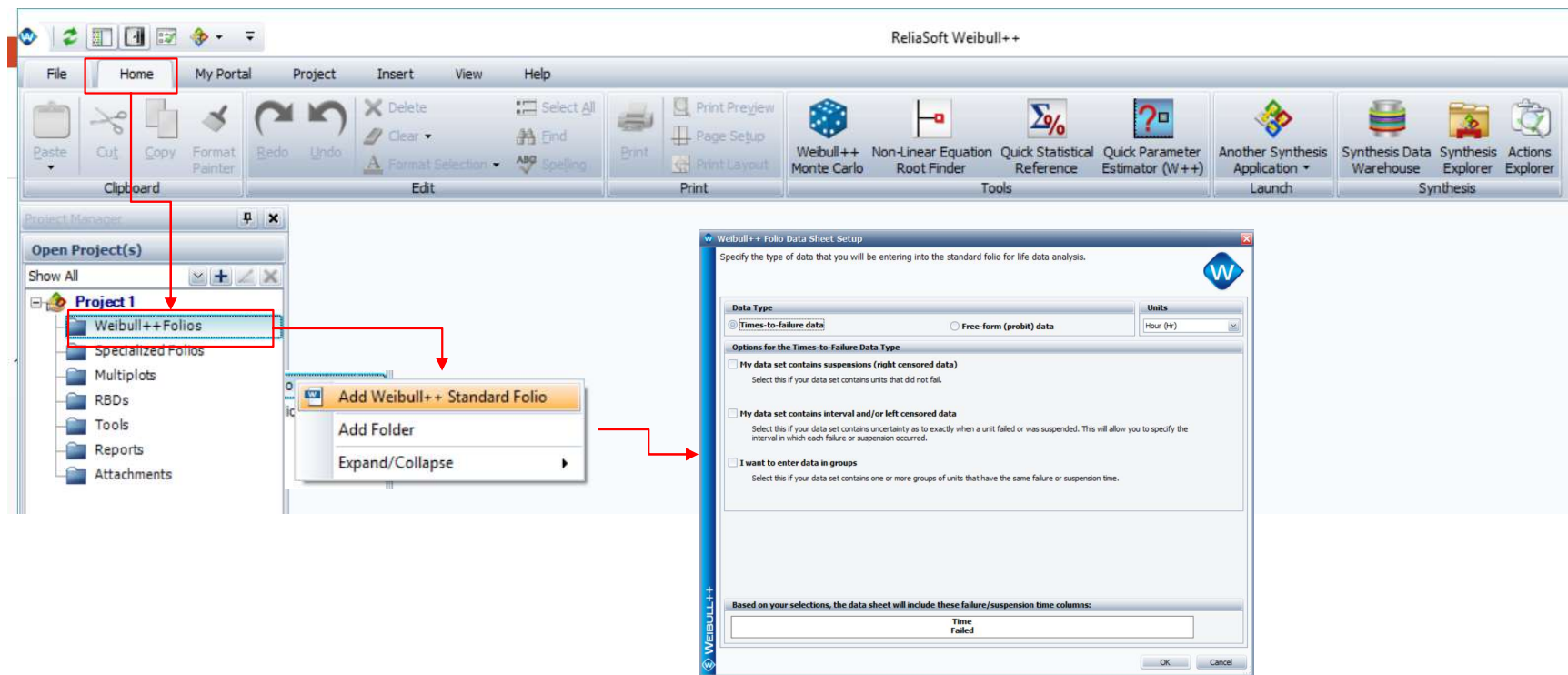
Introduction to Weibull ++

- Start Menu:



Introduction to Weibull ++

- Opening analysis folio



Introduction to Weibull ++

- Key in Data

The screenshot displays the ReliaSoft Weibull++ software interface. The main window shows a data sheet for 'Folio1' with columns for 'Time Failed (Day)' and 'Subset ID 1'. The data is as follows:

Time Failed (Day)	Subset ID 1
50.1	1
42.6	1
14.2	1
88.6	1
20.9	1
84.9	1
4.6	1
10.7	1
31.1	1
6.2	1
2.5	1
99.1	1
96.5	1
32	1
1.8	1
87.7	1
36.3	1
30.4	1
11.5	1
84.6	1

The 'Distribution Wizard' dialog box is open, showing the 'Analysis Details' tab. The 'Distributions and Rankings' list shows the following results:

Distribution	Ranking
1P-Exponential	2
2P-Exponential	3
Normal	5
Lognormal	4
2P-Weibull	1
3P-Weibull	
Gamma	
G-Gamma	
Logistic	
Loglogistic	
Gumbel	

A red box highlights the '2P-Weibull' distribution with a ranking of 1. A red arrow points from the 'Analysis Settings' tab in the main window to the '2P-Weibull' distribution in the wizard. A green box with the text 'Weibull 2P is the best fit' is overlaid on the wizard. The 'Analysis Details' tab also shows the 'Analysis Settings' section with 'RRX' and 'SRM' selected, and 'FM' and 'MEV' unselected. The 'Not Analyzed' section is empty.

Introduction to Weibull ++

- Key in data and estimate parameters

The screenshot displays the ReliaSoft Weibull++ software interface. The main window is titled 'Folio1 - ReliaSoft Weibull++'. The 'File' menu is open, and the 'Calculate' option is highlighted. The 'Project Manager' pane on the left shows a tree view with 'Project 1' expanded, containing 'Weibull++ Folios', 'Specialized Folios', 'Multiplots', 'RBDs', 'Tools', 'Reports', and 'Attachments'. The 'Folio1' folder is selected. The central pane shows a data table with columns 'Time Failed (Day)' and 'Subset ID 1'. The data is as follows:

Time Failed (Day)	Subset ID 1
1.8	1
2.5	1
4.6	1
6.2	1
10.7	1
11.5	1
14.2	1
20.9	1
30.4	1
31.1	1
32	1
36.3	1
42.6	1
50.1	1
84.6	1
84.9	1
87.7	1
88.6	1
96.5	1
99.1	1

The right pane shows the 'STANDARD FOLIO' window. The 'Distribution' tab is selected, showing '2P-Weibull'. The 'Analysis Settings' section shows 'RRX' and 'SRM' selected. The 'Analysis Summary' section shows the following parameters:

Parameters	Value
Beta	0.956305
Eta (Day)	43.265430

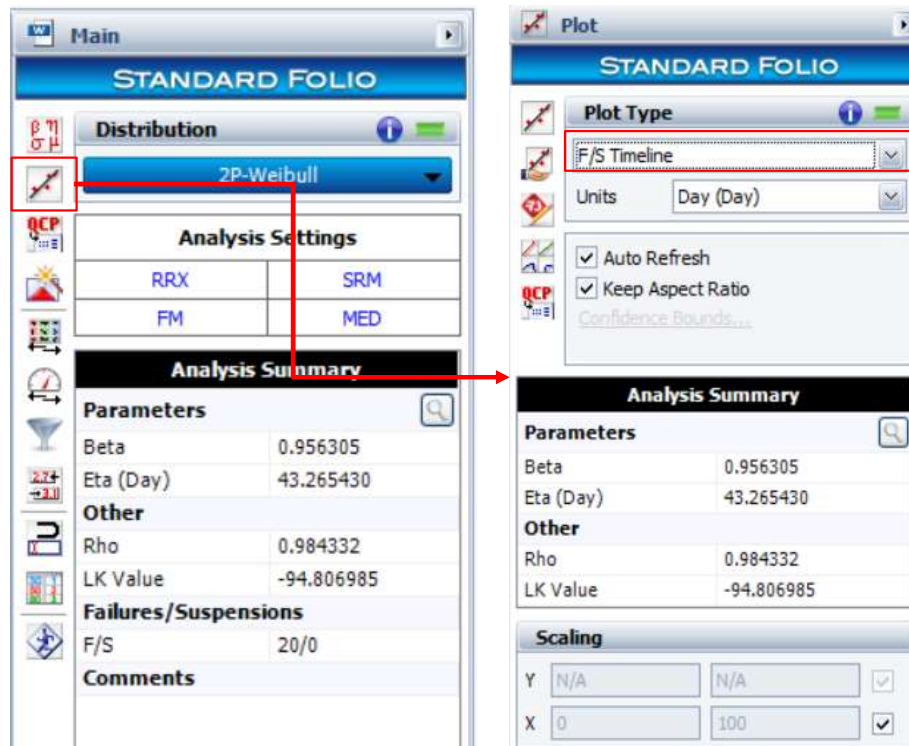
The 'Other' section shows the following values:

Other	Value
Rho	0.984332
LK Value	-94.806985

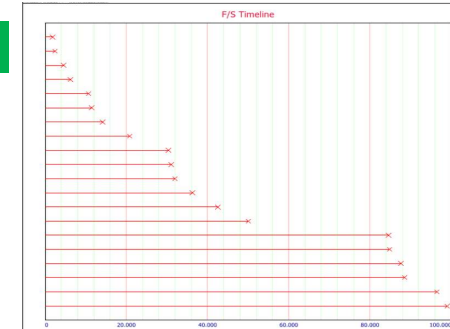
The 'Failures/Suspensions' section shows 'F/S' as '20/0'. The 'Comments' section is empty. Red arrows point from the 'Results' and 'Goodness of fit' labels to the 'Analysis Summary' and 'Other' sections respectively.

Introduction to Weibull ++

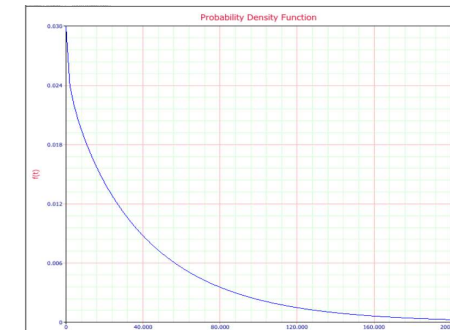
- Plotting results



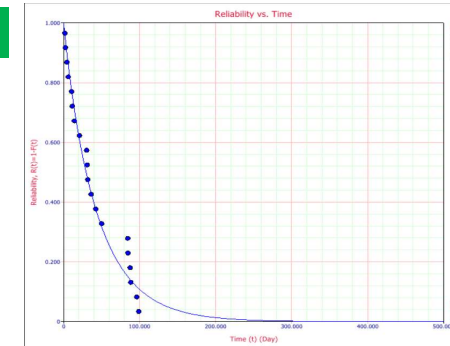
Timeline plot



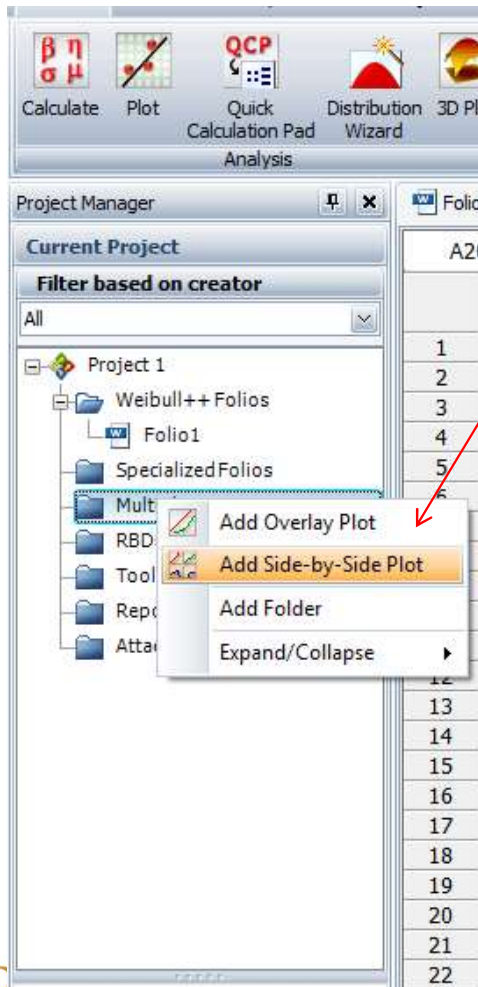
PDF Plot



Reliability Plot

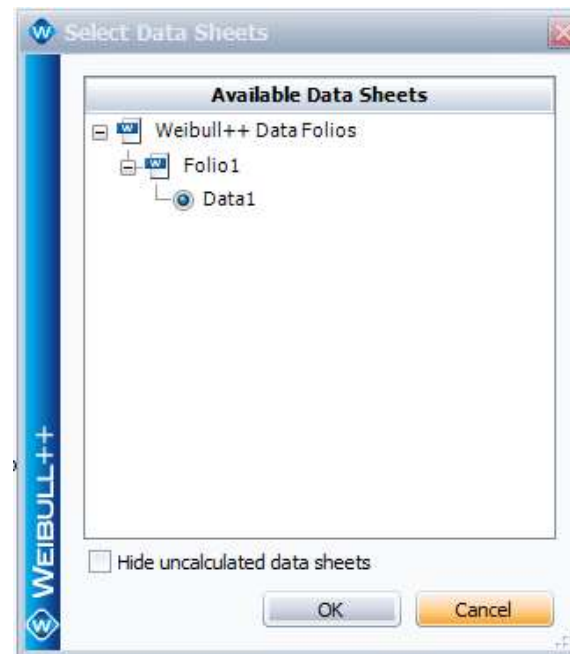


Plot graphs : Multiplot - side by side



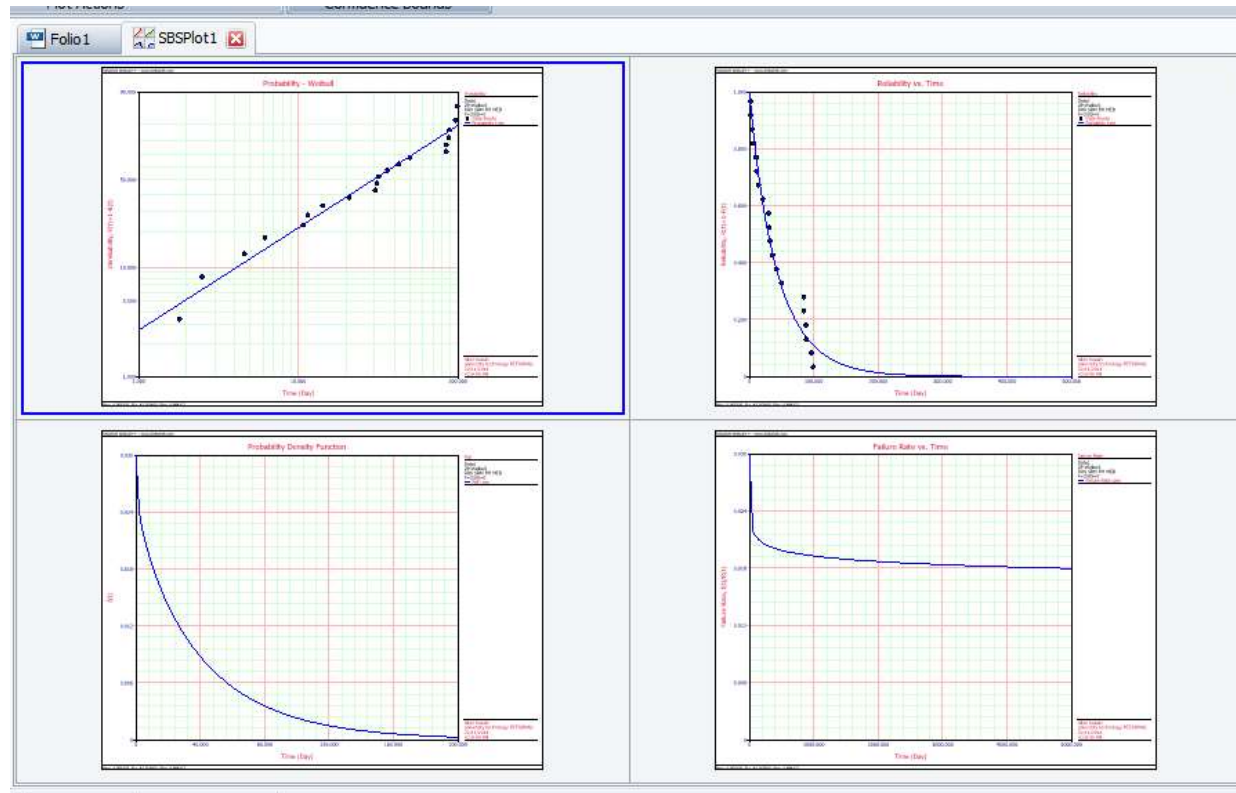
Right click “Multiplot” folder

Select “Add Side-by-Side plot”



To view multiplot of single data set

Plot graphs : Multiplot - side by side



Plot

SIDE-BY-SIDE PLOT

☒ Keep Aspect Ratio

[Confidence Bounds...](#)

Scaling

X ☒

Vary

☒ Plot Types ☐ Distributions

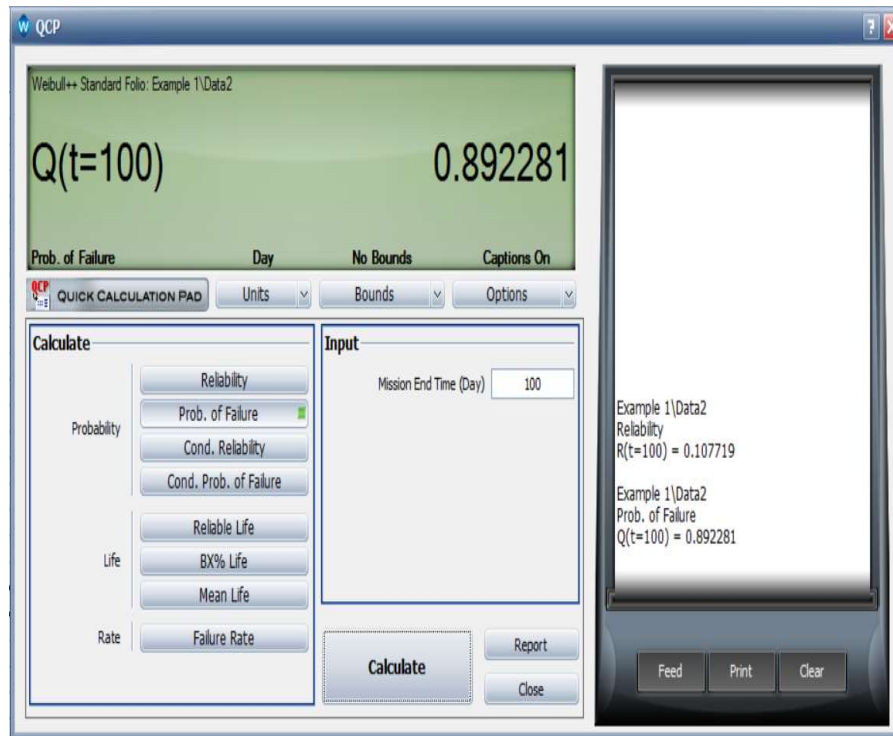
Plot Types

- ☒ Probability
- ☒ Reliability
- ☐ Unreliability
- ☒ Pdf
- ☒ Failure Rate
- ☐ Contour
- ☐ F/S Histogram
- ☐ F/S Pie
- ☐ F/S Timeline

[Folio1\Data1](#)

Introduction to Weibull ++

- Using calculation pad to calculate reliability measures



Probability	-Reliability -Prob of failure -Cond reliability -Cond Prob of failure	-R(t) -F(t) -Reliability for additional period of time - Unreliability for additional period of time
Life	-Reliable life -BX% Life -Mean life	-Expected life for given R -Expected life for given unreliability -Average time to failure
Rate	-Failure rate	Instantaneous FR for specified time

Introduction to Weibull ++

- Calculation with right censored data (use data from example 2)

The image shows the Weibull++ Folio Data Sheet Setup dialog box on the left and the main Weibull++ software interface on the right. The dialog box is titled "Weibull++ Folio Data Sheet Setup" and contains the following options:

- Data Type:** Times-to-failure data (selected), Free-form (probit) data
- Units:** Hour (h)
- Options for the Times-to-Failure Data Type:**
 - ☒ My data set contains suspensions (right censored data). Select this if your data set contains units that did not fail.
 - ☐ My data set contains interval and/or left censored data. Select this if your data set contains uncertainty as to exactly when a unit failed or was suspended. This will allow you to specify the interval in which each failure or suspension occurred.
 - ☐ I want to enter data in groups. Select this if your data set contains one or more groups of units that have the same failure or suspension time.

Based on your selections, the data sheet will include these failure/suspension time columns:

State F or S	Time to F or S
-----------------	-------------------

The main software interface shows the "Example 2" data sheet. The data is as follows:

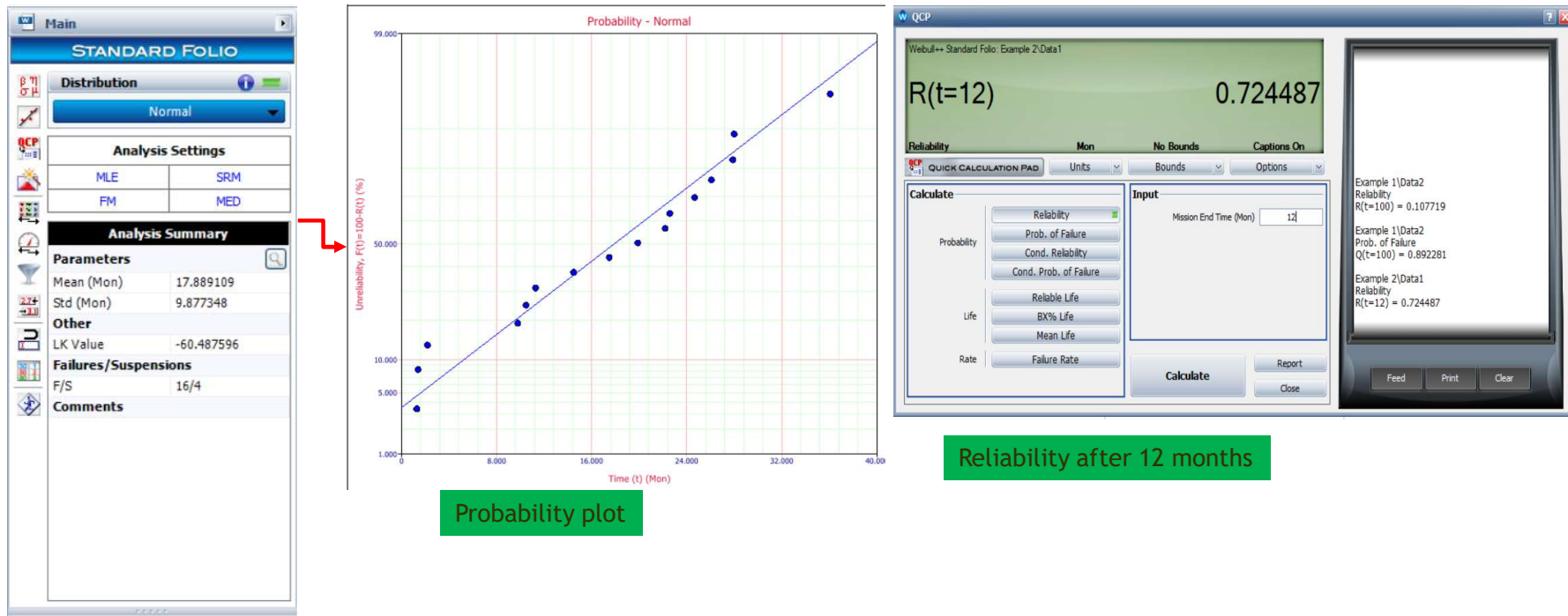
State F or S	Time to F or S (Mon)	Subset ID 1
1	F	1.3
2	F	1.4
3	F	2.2
4	S	2.9
5	F	9.8
6	F	10.5
7	F	11.3
8	S	10.5
9	F	14.5
10	F	17.5
11	F	19.9
12	F	22.2
13	F	22.6
14	F	24.7
15	S	4.4
16	F	26.1
17	F	27.9
18	F	28
19	S	9.3
20	F	36.1

The "Analysis Settings" panel on the right shows the following options:

- Distribution:** 2P-Weibull
- Analysis Settings:**
 - ☒ MLE
 - ☐ SRM
 - ☐ FM
 - ☐ MED
- Not Analyzed**
- Comments**

Introduction to Weibull ++

- Calculation with right censored data (use data from example 2)-results



Introduction to Weibull ++

- Calculation with right and interval/left censored data (use data from example 3)

Weibull++ Folio Data Sheet Setup

Specify the type of data that you will be entering into the standard folio for life data analysis.

Data Type
☒ Times-to-failure data
☐ Free-form (probit) data

Units
 Hour (H+)

Options for the Times-to-Failure Data Type

☒ My data set contains suspensions (right censored data)
 Select this if your data set contains units that did not fail.

☒ My data set contains interval and/or left censored data
 Select this if your data set contains uncertainty as to exactly when a unit failed or was suspended. This will allow you to specify the interval in which each failure or suspension occurred.

☐ I want to enter data in groups
 Select this if your data set contains one or more groups of units that have the same failure or suspension time.

Based on your selections, the data sheet will include these failure/suspension time columns:

Last Inspected	State F or S	Time to For S
----------------	--------------	---------------

OK Cancel

Project Manager

Open Project(s)

Show All

- Project 1
 - Weibull++ Folios
 - Example 1
 - Example 2
 - Folio 1
 - Folio 2
 - Example 3
 - Specialized Folios
 - Multiplots
 - RBDs
 - Tools
 - Reports
 - Attachments

Open Project(s)

Project List

Example 1

	Last Inspected	State F or S	Time to For S (Mon)	Subset ID 1
1	0	F	6	
2	5	F	5	
3	6	F	9	
4	0	F	10	
5	4	F	11	
6	5	S	5	
7	4	S	4	
8	9	F	13	
9	13	F	13	
10	0	F	19	
11	7	F	20	
12	8	F	19	
13	14	S	14	
14	1	F	25	
15	9	F	33	
16	18	F	18	
17	4	F	21	
18	0	F	34	
19	3	F	38	
20	10	F	33	
21	13	S	13	
22	16	F	34	
23	12	S	12	
24	5	F	42	
25	23	F	38	
26	25	F	25	

Data 1

EXERCISES

Exercise # 1

You have collected the time to failures (MONTHS) of 20 fuses as shown in Table below. The fuse once it failed, will not be repaired (not economic), instead it will be replaced with the new one.

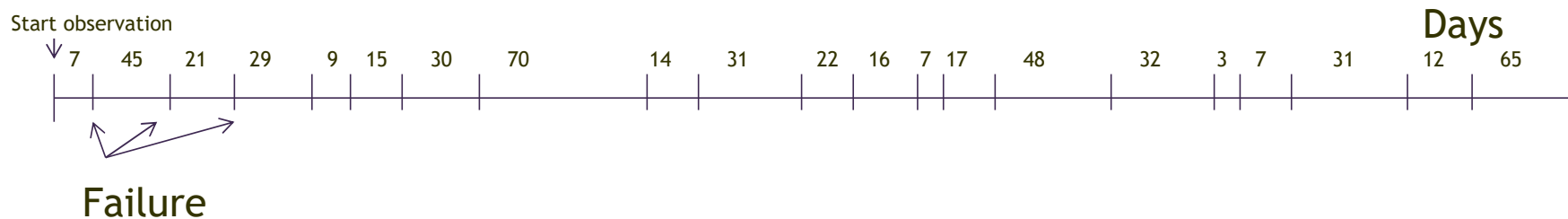
3	1.25	2	6	2.5
2.75	5.5	0.5	2	2
1	0.25	0.15	0.1	0.75
5.5	0.75	7	6	5.75

Questions:

1. What is the best distribution to fit these failure data?
2. What can you say about its failure rate?
3. Your superior want you to determine the time when 50% and 75% of switches will fail.
4. How many percent of the switches will fail after being in operation for 10, 25 and 50 hours?
5. How many percent of switches will survive after being in operation for 50 and 70 hours?

Exercise # 2

You have been monitoring the Corrective maintenance data for a centrifugal compressor. You have calculated the time between failures (days) and the data are shown in the Figure below. There were 21 failures recorded. The equipment undergone some kind of repair when it failed.



Questions:

Assume the data is IID (HPP), hence exponentially distributed.

1. Using Weibull ++ , what is the estimated reliability after it is being in operation for 500 hours?
2. What is the expected operation time when reliability is at 50% and 75%?

Application of Life Data Analysis

Confidence Interval

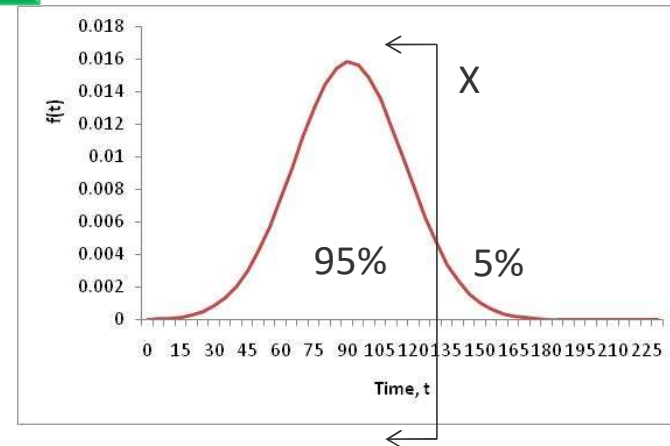
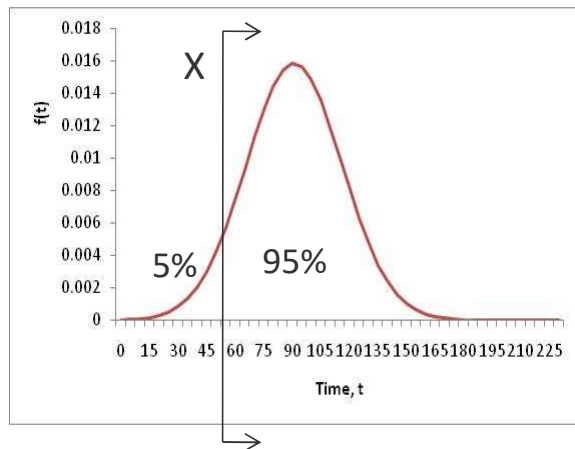
Confidence Interval

- An interval estimate for a population parameter.
- We cannot be certain that the interval contains the true, unknown population parameter – we only use a sample from the full population to compute the point estimate and the interval.
- CI is constructed so that we have high confidence that it does contain the unknown population parameter.
- The selection of a confidence level determines the probability that the confidence interval calculated will contain the true parameter value.
- Common choices of confidence level is 90%, 95% or 99% which corresponds to the percentages of the area under PDF curve.
- 2 types:
 1. One-sided intervals
 2. Two-sided intervals

One-sided Intervals

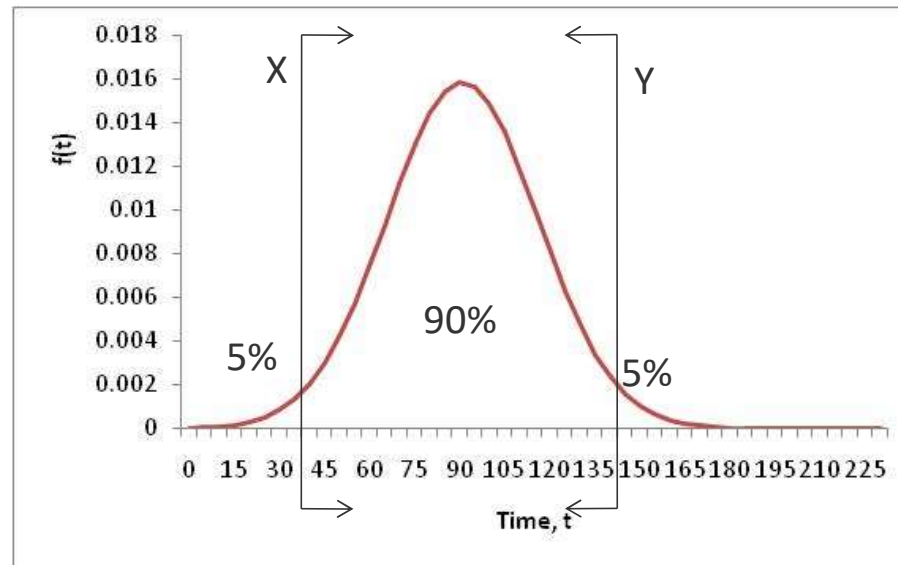
- When using one-sided intervals, we are looking at the percentage that is greater than (upper) or less than (lower) a certain point X .
- For example, 95% one-sided bounds imply that 95% of the population is greater than X if lower bounds, or that 95% is less than X if upper bounds.
- When estimating reliability, we are usually not worried if the actual reliability is higher than the estimate – only if it is lower. So we tend to use one-sided lower limits.

When would this be used?



Two-sided Intervals

- Use two-sided confidence intervals to see where most of the values are likely to lie.
- Example: when we use 90% two-sided bounds, we are saying that 90% of the population lies between X and Y with 5% less than X and 5% more than Y.



Confidence Interval for Exponential Distribution

$$\text{MTBF}_L = \frac{2 \sum_{i=1}^n x_i}{\chi_{\alpha/2, k}^2} \qquad \text{MTBF}_U = \frac{2 \sum_{i=1}^n x_i}{\chi_{1-\alpha/2, k}^2}$$

where

x_i = Time to failure

$1 - \alpha$ = selected confidence level

r = number of failures

$$k = \begin{cases} 2(r+1) & \text{for Type I data} \\ 2r & \text{for Type II or complete data} \end{cases}$$

Exponential Distribution Example

A bearing manufacturer claims that the useful lives (in hours) for the component follow an **exponential distribution**. If the sum of the lives of the 10 sample tested is equal to 1740 hours, what is the 95% confidence for the population MTBF.

❖ **Solution:**

❖ **Based on Chi-Squared table**

$$\begin{aligned}\chi^2_{0.025,20} &= 34.2 & \chi^2_{0.975,20} &= 9.59 \\ &= \text{CHISQ.INV.RT}(0.025,20) & &= \text{CHISQ.INV.RT}(0.975,20)\end{aligned}$$

❖ **Then we can conclude that with 95% confidence that the MTBF shall lie between**

$$\begin{aligned}MTBF &\in \left(\frac{3480}{34.2}, \frac{3480}{9.59} \right) \\ MTBF &\in (101.75, 362.88)\end{aligned}$$

Confidence Interval for Weibull Distribution

$$\hat{\beta} \exp\left(\frac{-0.78z_{\alpha/2}}{\sqrt{n}}\right) \leq \beta \leq \hat{\beta} \exp\left(\frac{0.78z_{\alpha/2}}{\sqrt{n}}\right)$$

$$\hat{\eta} \exp\left(\frac{-1.05z_{\alpha/2}}{\hat{\beta}\sqrt{n}}\right) \leq \eta \leq \hat{\eta} \exp\left(\frac{1.05z_{\alpha/2}}{\hat{\beta}\sqrt{n}}\right)$$

Where

z = standardized normal deviate

$1 - \alpha$ = selected confidence level

n = sample size

$\hat{\beta}$ and $\hat{\eta}$ are the estimated parameters for Weibull distribution

Confidence Interval in Weibull ++

Confidence interval calculation setting

The image shows two windows from the STANDARD FOLIO software. The left window, titled 'Main', displays the 'Analysis Settings' tab. Under 'Analysis Settings', the 'FM' option is selected and highlighted with a red box. The right window, titled 'Analysis', displays the 'Confidence Bounds Method' dropdown menu, which is also set to 'Fisher Matrix' and highlighted with a red box. A red arrow points from the 'FM' selection in the left window to the 'Fisher Matrix' option in the right window.

Analysis Settings

MLE	SRM
FM	MED

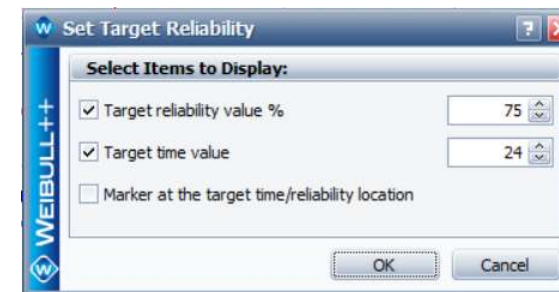
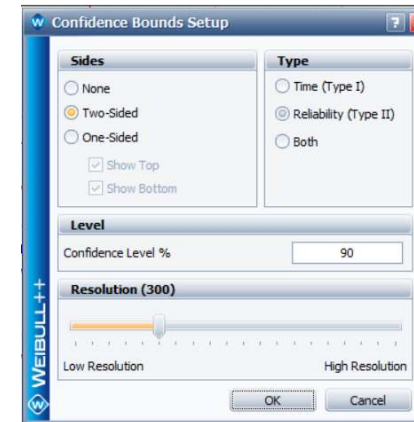
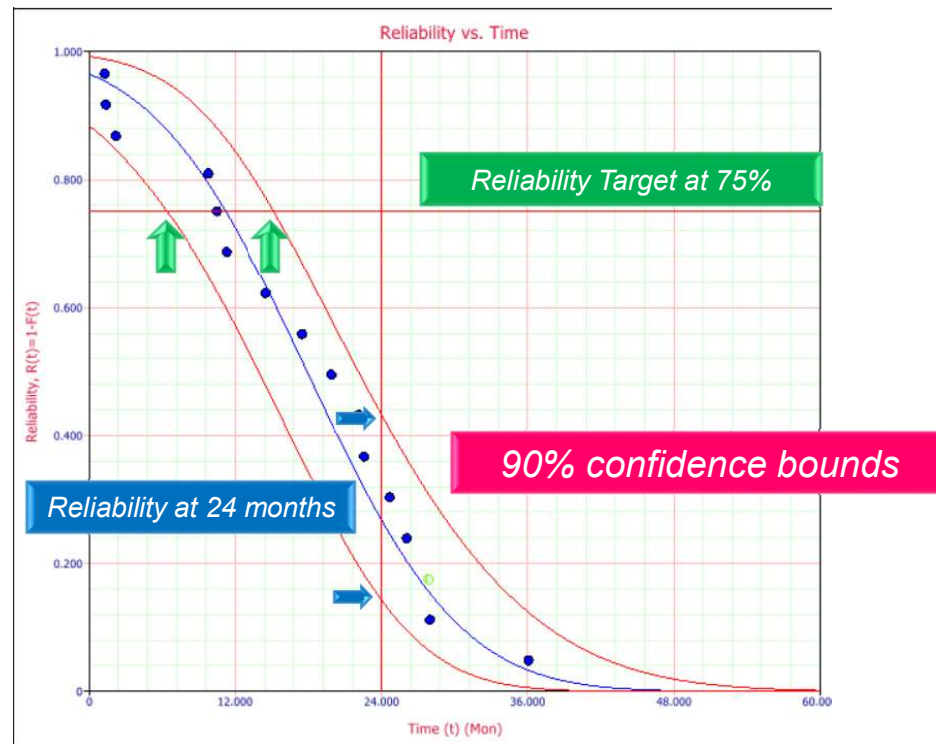
Confidence Bounds Method

- Fisher Matrix
- Fisher Matrix
- Likelihood Ratio
- Bayesian

Use default setting: Fisher-Matrix (FM)

Confidence Interval in Weibull ++

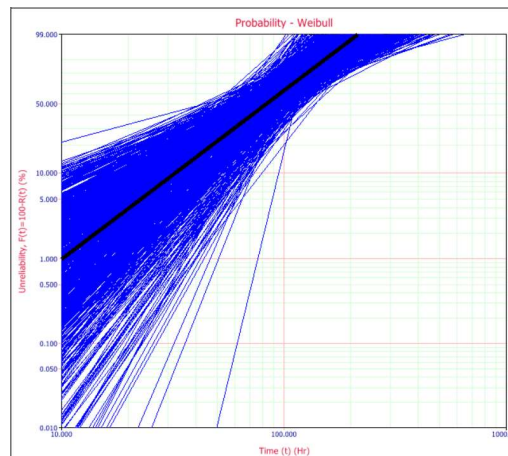
Refer to example 2



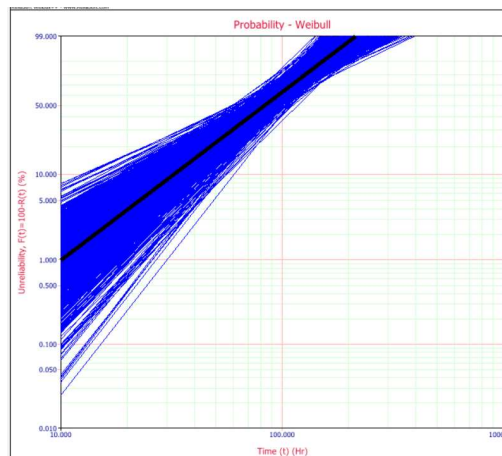
Effect of sample size to CI

Analysis Summary	
General	
Beta	2
Eta (Hr)	100

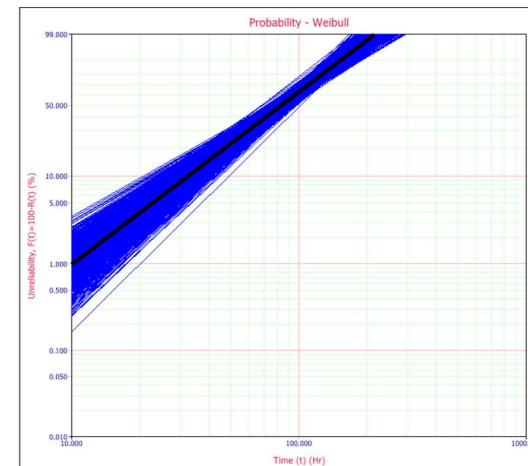
90% confidence level



Sample size = 10



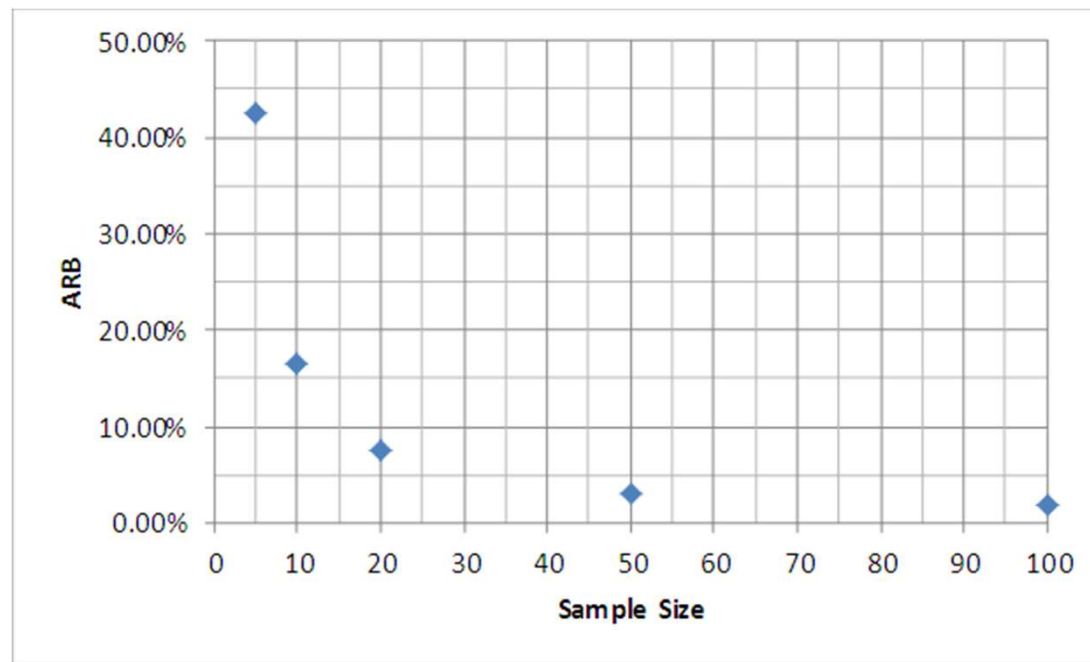
Sample size = 30



Sample size = 100

The larger the sample size, the narrower the CI

Sample Size Vs Accuracy



$$RB = \frac{(\hat{\beta} - \beta)}{\beta} \times 100\%$$

Competing Failure Modes

Competing Failure Modes Analysis

Component can fail due to one of possible failure modes.

- Eg bearing can fail due to crack, pitting corrosion, overheating, fatigue.

Separate analysis of each failure mode is needed, with other failure modes not analysed are assumed to be in suspension

Competing Failure Modes Analysis

Data Collected on bearing failures due to different failure modes

Estimate the probability that the bearing will fail after running 100,000 cycles.

Bearing Failure Modes			
Number in State	State (F or S)	State end Cycle ('000)	FM
1	F	80	Crack
1	F	28	Crack
1	F	147	Pitting
1	F	245	Pitting
1	F	10	Crack
1	F	143	Crack
1	F	88	Crack
1	F	30	Crack
1	F	247	Crack
1	F	275	Pitting
1	F	13	Crack
1	F	293	Pitting
1	F	181	Pitting
1	F	2	Crack
1	F	173	Crack
1	F	266	Pitting
1	F	212	Pitting
1	F	261	Crack
1	F	65	Crack
1	F	106	Crack
2	F	23	Crack
8	S	300	

Competing Failure Modes Analysis

D15	Pitting			
	Number in State	State F or S	State End Time	Subset ID 1
1	1	F	2	Crack
2	1	F	10	Crack
3	1	F	13	Crack
4	2	F	23	Crack
5	1	F	28	Crack
6	1	F	30	Crack
7	1	F	65	Crack
8	1	F	80	Crack
9	1	F	88	Crack
10	1	F	106	Crack
11	1	F	143	Crack
12	1	F	147	Pitting
13	1	F	173	Crack
14	1	F	181	Pitting
15	1	F	212	Pitting
16	1	F	245	Pitting
17	1	F	247	Crack
18	1	F	261	Crack
19	1	F	266	Pitting
20	1	F	275	Pitting
21	1	F	293	Pitting
22	8	S	300	
23				
24				
25				
26				

Main

STANDARD FOLIO

Distribution

CFM-Weibull

Analysis Settings

MLE	SRM
FM	MED

Analysis Summary

Parameters

Beta[1]	0.671072
Eta[1] (Cyc)	449.427230
Beta[2]	4.337278
Eta[2] (Cyc)	340.384242

Other

LK Value [1]	-101.364207
LK Value [2]	-47.162210

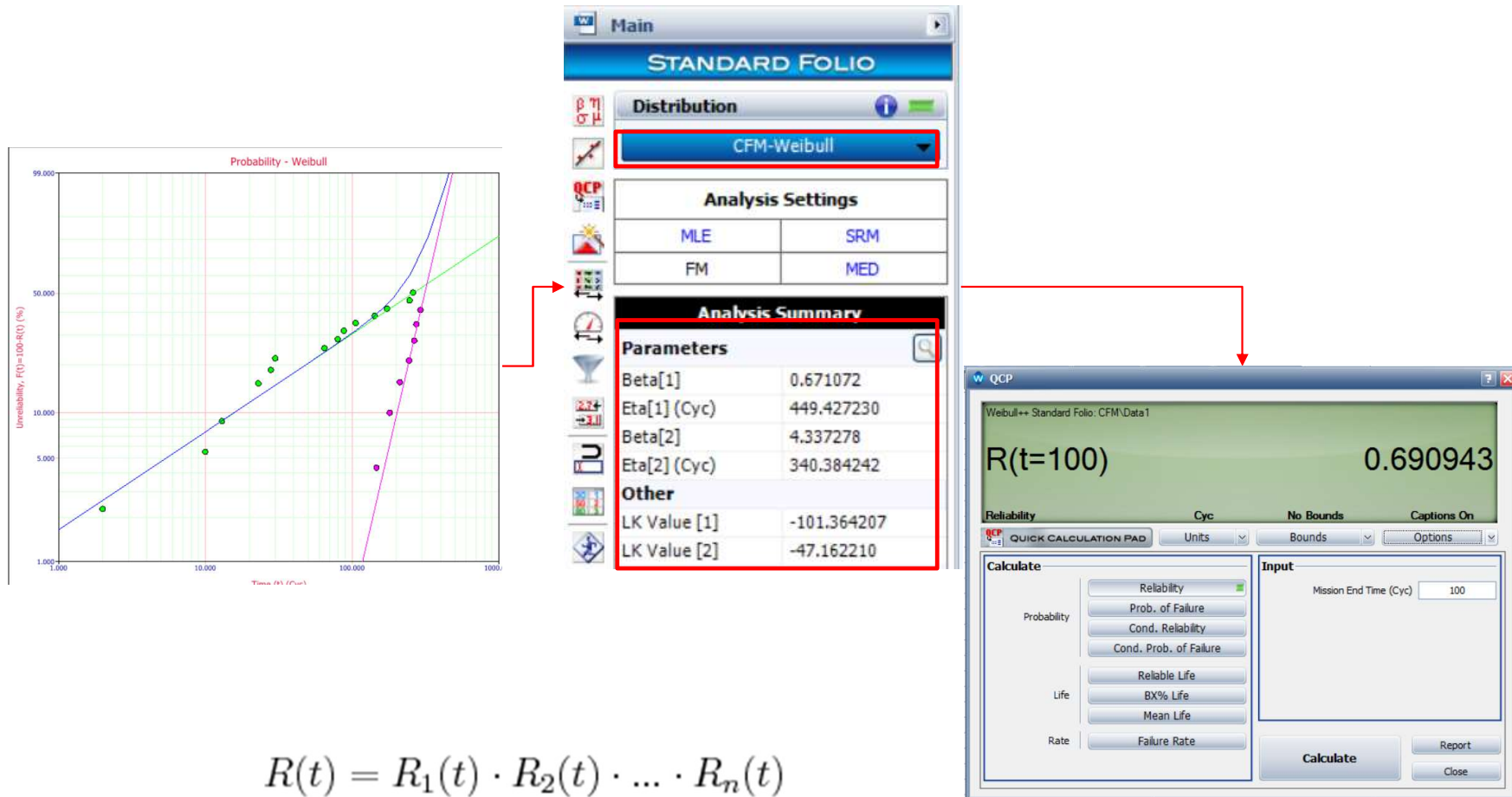
Competing Failure Modes - Select Subsets

Move subset IDs from available subsets to the appropriate mode.

Available Subset IDs	Modes
	Mode 1 Crack Mode 2 Pitting Mode 3 [no subsets] Mode 4 [no subsets]

OK Cancel

Competing Failure Modes Analysis

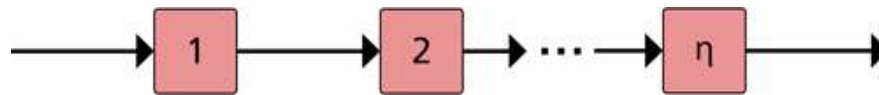


$$R(t) = R_1(t) \cdot R_2(t) \cdot \dots \cdot R_n(t)$$

Competing Failure Modes Analysis

Generally, each failure modes are arranged in series. Any one of them occurs, the component will fail.

CFM analysis, analyse each FM to evaluate reliability $[R_1(t), R_2(t) \dots]$ and calculate the overall reliability, $R(t)$



$$R(t) = R_1(t) \cdot R_2(t) \cdot \dots \cdot R_n(t)$$

Exercise

- Failure data for Mechanical seals were collected as shown in the Table.
 - Analyze the data
 - What are the conclusion and recommendation?

Failure No	TTF (Days)	Failure No	TTF (Days)
1	7	11	140
2	12	12	150
3	15	13	160
4	20	14	500
5	23	15	520
6	100	16	530
7	105	17	550
8	110	18	570
9	120	19	590
10	130	20	610

Component Life Comparison

Objectives

- Life data comparison will help to answer the following questions:
 - Does the new component perform better than old component?
 - Does the change in design really improve the component reliability?
 - Are components from one supplier is better than the other?

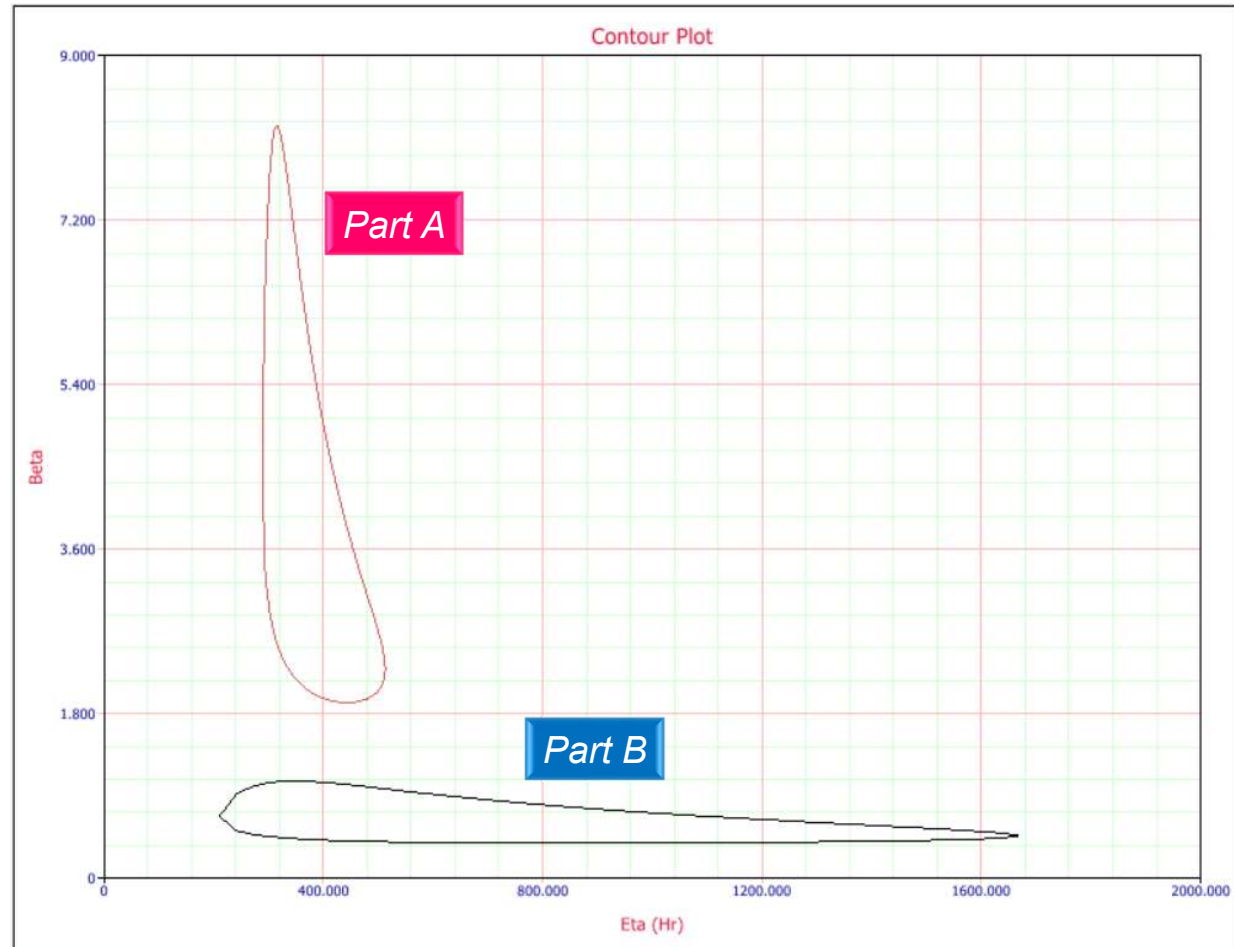
Life Comparison in Weibull ++

- Two different methods available in Weibull ++
 - Life comparison using contour plot. Limitation: both component must have similar 2-parameter distributions such as Weibull, Lognormal or normal
 - Life comparison using life comparison tool
 - This method compares the data sets over the entire range of the random variable (from 0 to ∞).
 - The two data sets must be of similar sample size.
 - This method estimates the probability of whether the times-to-failure of one population are better or worse than the times-to-failure of the second.

$$P[t_2 \geq t_1] = \int_0^{\infty} f_1(t) \cdot R_2(t) \cdot dt$$

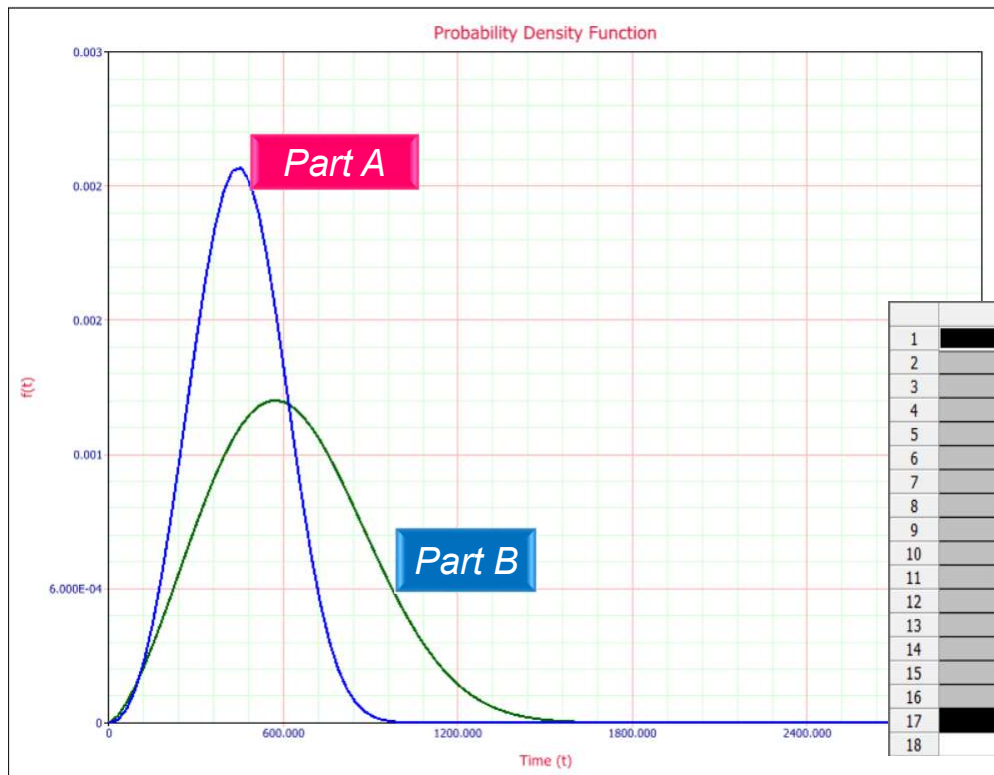
- $f_1(t)$ is the *pdf* of the first distribution.
- $R_2(t)$ is the reliability function of the second distribution.

Life Comparison-Contour Plot



Life Comparison Tool

- Plot



Results

	A	B	C
1	Quick Results Report		
2	Report Type	Life Comparison Results	
3	User Info		
4	User	Masdi Muhammad	
5	Company	Universiti Teknologi PETRONAS	
6	Date	22/11/2014	
7	Life Comparison		
8	Comparing:	Folio2!Data1	
9	To	Folio3!Data1	
10	Comments During Comparison		
11	The probability that items from:	Folio2!Data1	
12	will last longer than items from:	Folio3!Data1	
13	is:	29.97409148%	
14	Comparison Results		
15	Population:	Folio3!Data1 will last longer with probability of 70%	
16	Convergence Delta at Solution:	0.00000599%	
17	End of Quick Results Report		
18			

Case Study

- New mechanical seal design has been proposed to reduce the failure rate of a pump due to leakage. The historical failure data for old design is as shown in following table. The supplier, on the other hand has provided the following leak data for the new design. Is the new design significantly better?

Mechanical Seal Failure Data (Leaks)		
Old Design		New design
Time to failure (Week)		Time to failure (Week)
2	23	15
2	28	32
3	33	61
4	34	67
6	34	75
9	37	116
11	38	148
17	40	178
19	45	181
21	55	183
57	56	
67	115	
76	126	
90	197	

Step 1: Key in data

The screenshot displays the ReliaSoft Weibull++/ALTA Standard software interface. The main window shows a data table for 'Old Design' with columns for 'Time Failed (Wk)' and 'Subset ID 1'. The data is entered for 15 rows. A secondary window, 'Old Design', is also visible, showing a similar data table for 'New Design' with 13 rows. The software interface includes a menu bar, a toolbar, and a project manager pane on the left.

	Time Failed (Wk)	Subset ID 1
1	2	1
2	2	1
3	3	1
4	4	1
5	6	1
6	9	1
7	11	1
8	17	1
9	19	1
10	21	1
11	23	1
12	28	1
13	33	1
14	34	1
15	34	1
16	37	1
17	38	1
18	40	1
19	45	1
20	55	1
21	56	1
22	57	1
23	67	1
24	76	1
25	90	1
26	115	1
27	126	1
28	197	1

	Time Failed (Wk)	Subset ID 1
1	15	1
2	32	1
3	61	1
4	67	1
5	75	1
6	116	1
7	148	1
8	178	1
9	181	1
10	183	1
11		
12		
13		

Step 2: Calculate distribution Parameters

Main

STANDARD FOLIO

Distribution

2P-Weibull

Analysis Settings

RRX	SRM
FM	MED
F=28/S=0	

Analysis Summary

Parameters

Beta	0.987335
Eta (Wk)	44.849552

Other

Rho	0.985848
LK Value	-134.270961

Main

STANDARD FOLIO

Distribution

2P-Weibull

Analysis Settings

RRX	SRM
FM	MED
F=10/S=0	

Analysis Summary

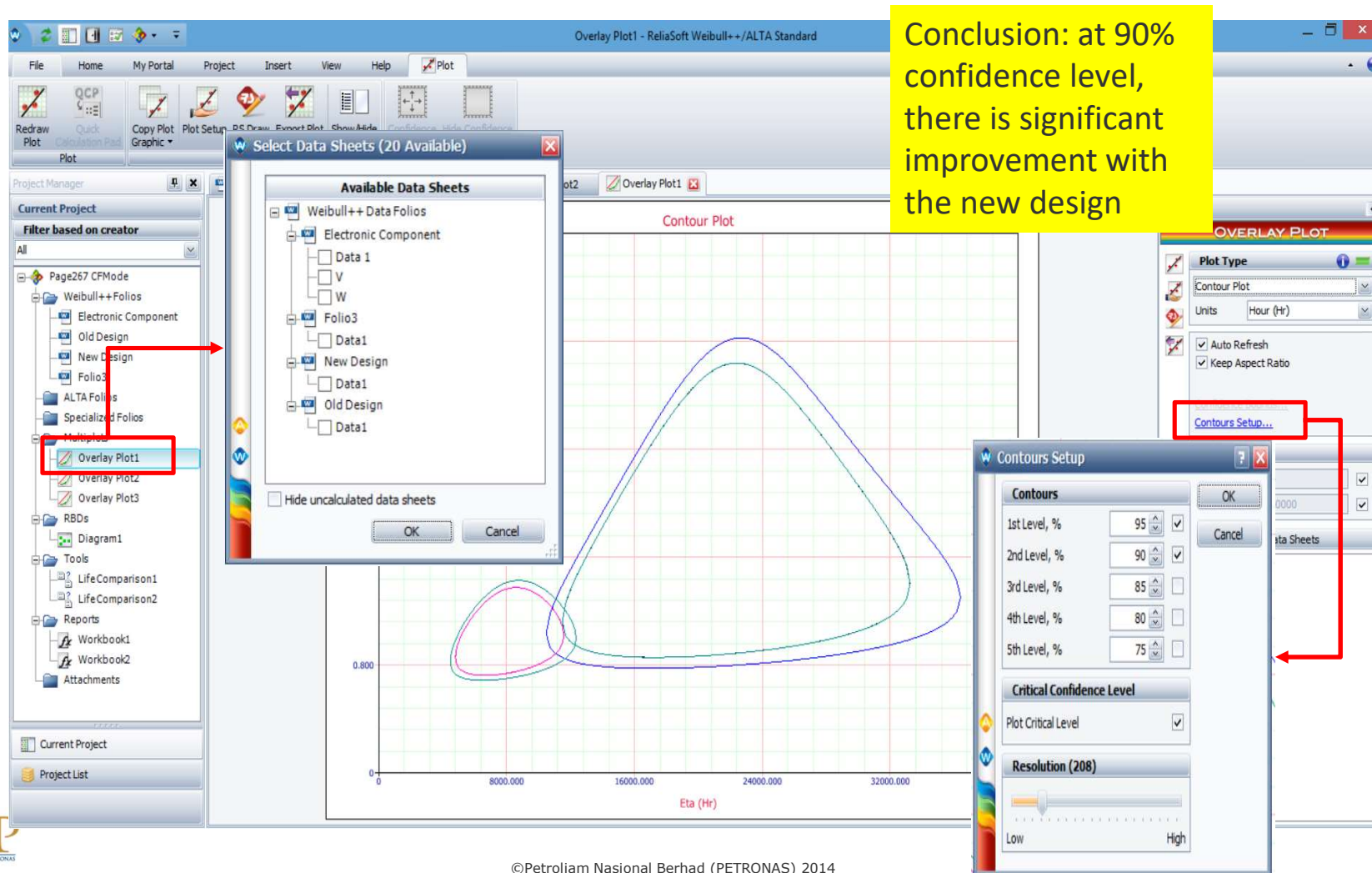
Parameters

Beta	1.369869
Eta (Wk)	121.185102

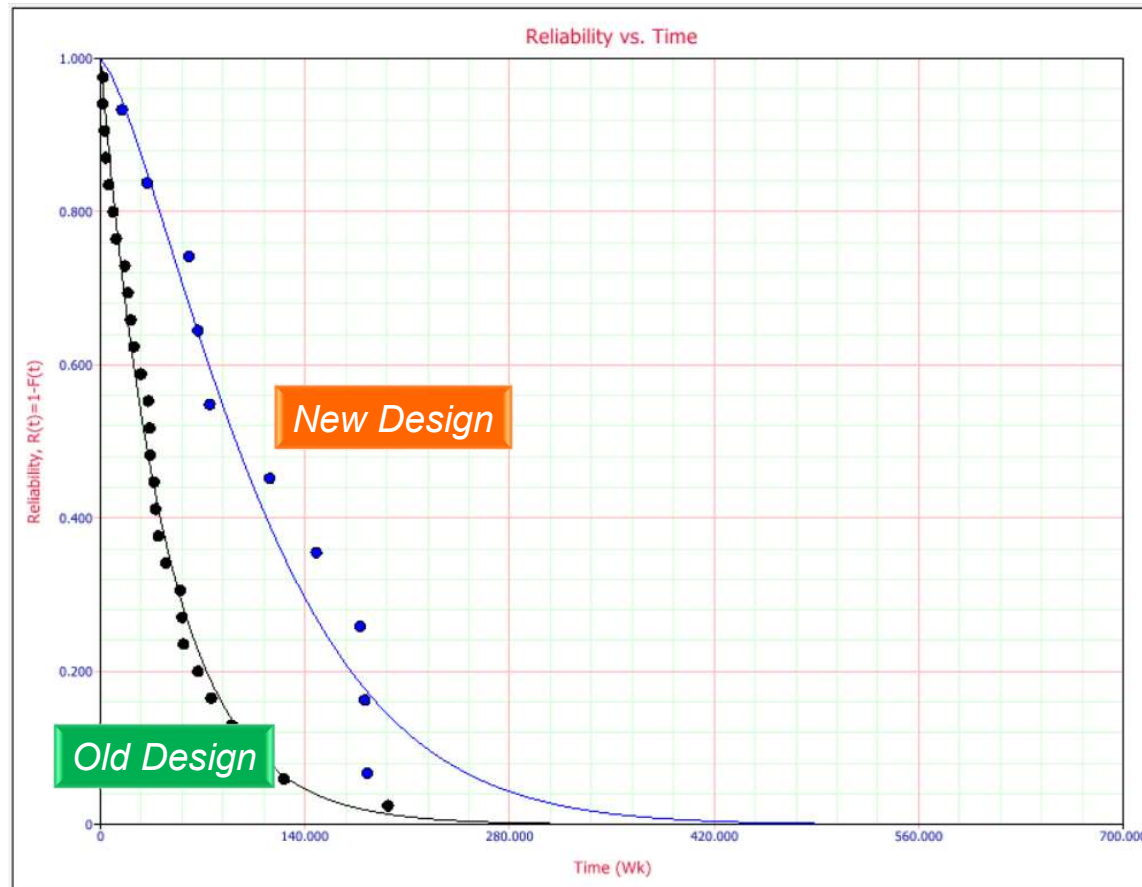
Other

Rho	0.980076
LK Value	-55.253767

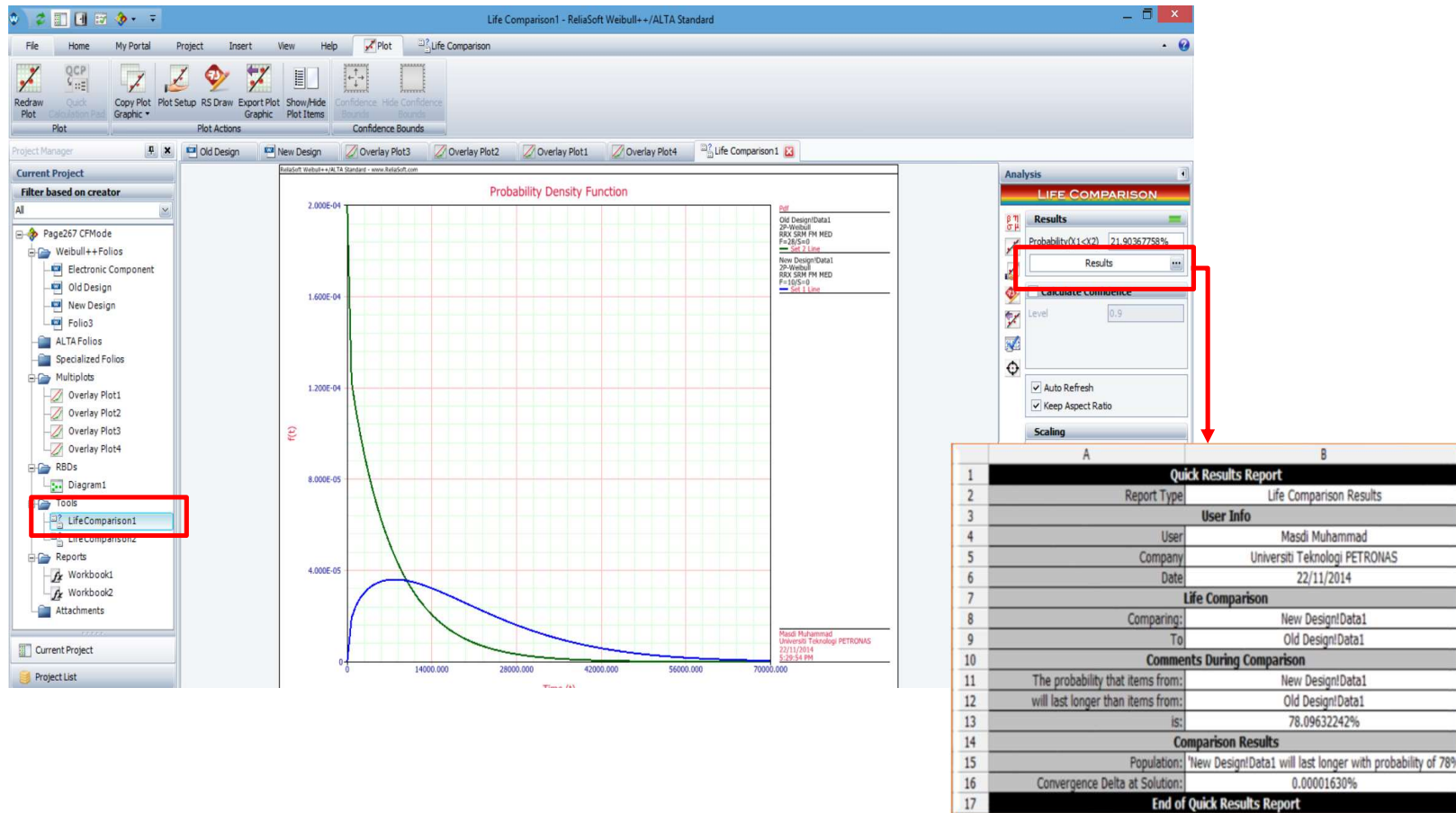
Step 3: Create Contour Plot



Step 3a: Create Reliability Plot



Step 4: Life Comparison



Exercise

- A benchmarking process is initiated to compare the performance of similar MOV communication card operating at different area. One is close to cooling tower and the other is situated new heat exchanger. Failure data are collected as below. How would you conclude the result?

Cooling Tower (Days)	Heat Exchanger (Days)
601	823
614	1284
952	1394
1039	1451
1414	2358

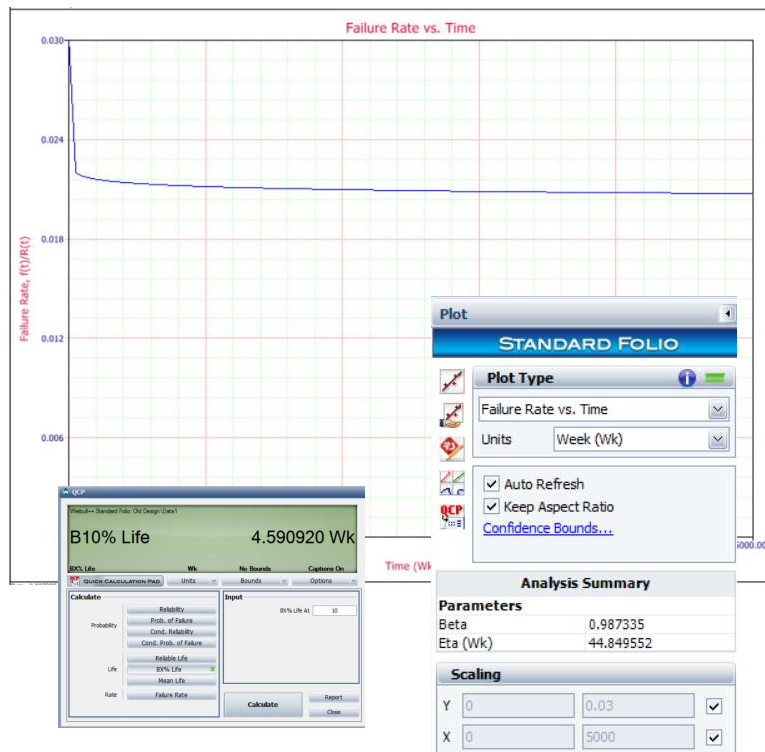
Optimum Replacement

The right time to Replace

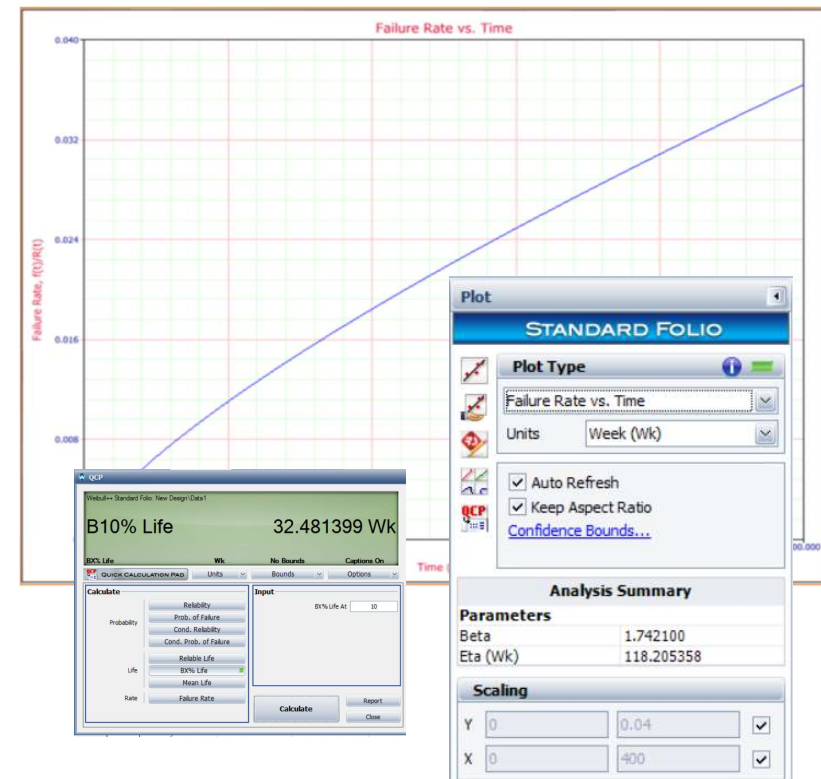
- Based on component's failure distribution, cost of planned replacement and cost of unplanned replacement, the optimum time to replace can be calculated.
- The right time to replace is based on the lowest cost.
- Consider the comparison between the proposed new mechanical seal design versus old design.

New versus old design

Old Design



New Design



Optimum Replacement Workbook

The screenshot shows the 'Optimum Replacement Time Computation' workbook in the background and the 'Analysis Workbook Wizard' dialog box in the foreground. The wizard is used to create a new report based on a template.

Background Workbook:

Optimum Replacement Time Computation									
Chart Start Time:	1			Cost for Planned Replacement:		\$10			
Increment by:	1			Cost for Unplanned Replacement:		\$5,000			
Time (Week)	Cost / Unit Time								

Analysis Workbook Wizard:

Click 'Select' to choose the first default data source for the new report. Then select the check box if you want to create the new report based on a previously saved template.

Default Data Source:

Weibull!New Design!Data1 Select...

☒ **Based on Existing Template**

A list of standard templates shipped with the software appears below. To use one of your own saved templates, click the 'User' tab.

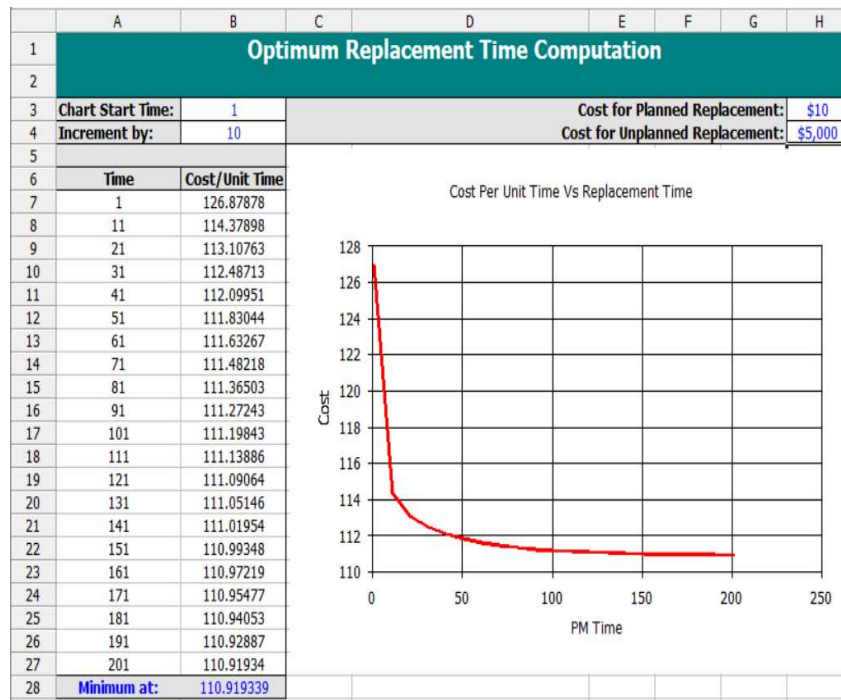
- ALTA Single Constant Stress Summary Template
- ALTA Two Stress Summary Template
- ALTA Warranty Forecasts Template
- Weibull Compare BX Life Template
- Weibull Compare Mean Template
- Weibull Intangible Asset Valuation Template
- Weibull Optimum PM Replacement Template**
- Weibull Optimum PM Replacement Template

Standard User

OK Cancel

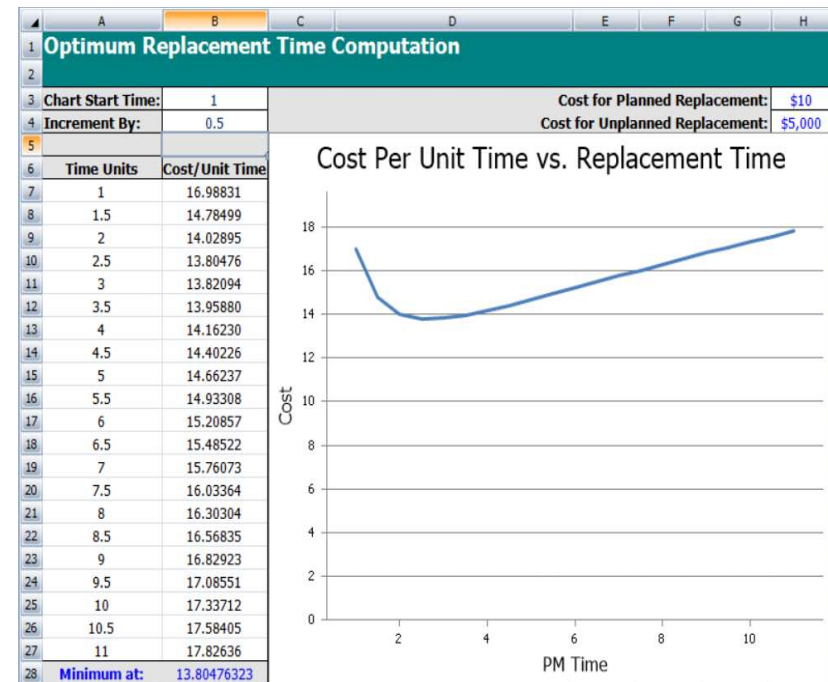
Optimum replacement

Old Design



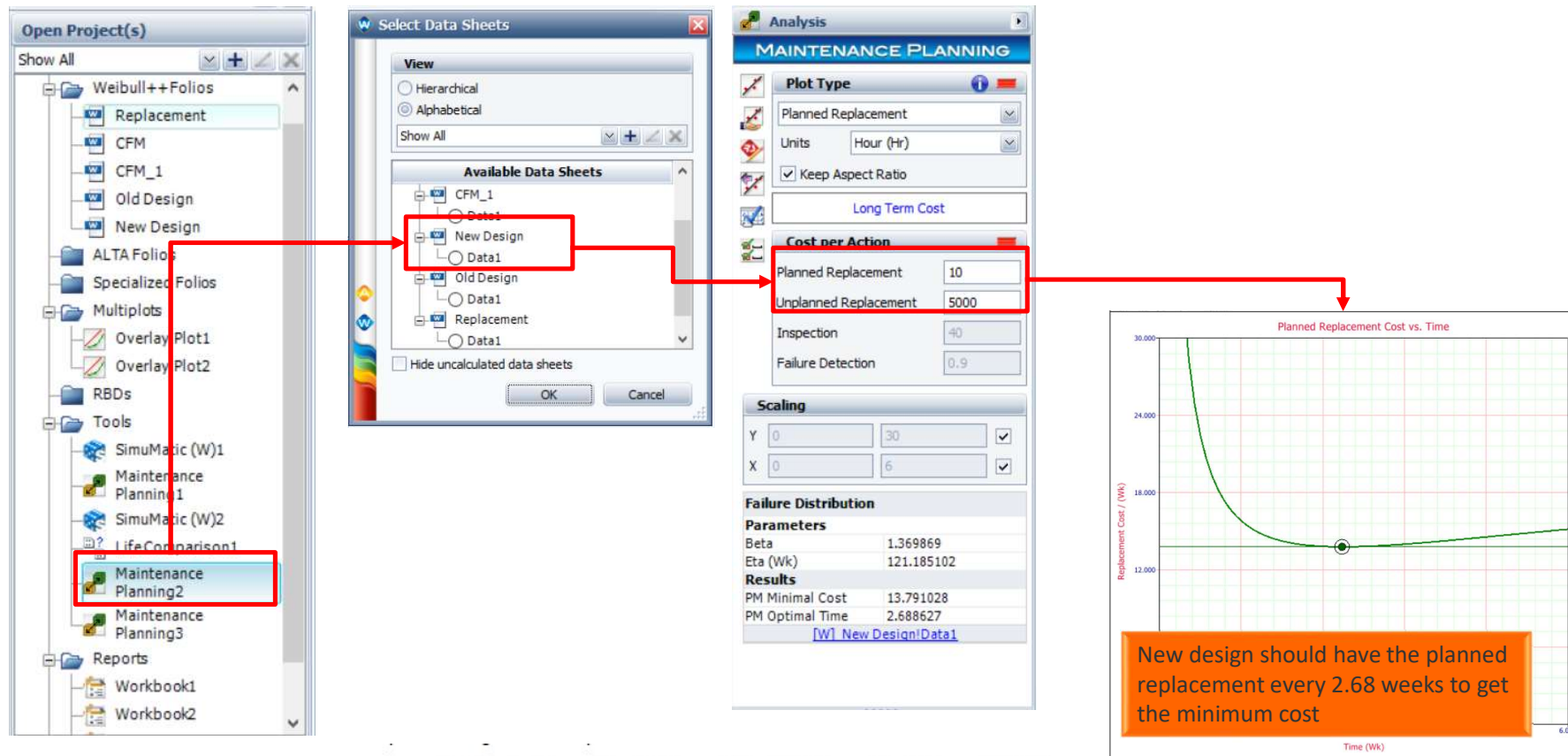
Since the old design follow a constant failure rate, run to fail is the right strategy

New Design



New design should have the planned replacement every 2.5 weeks to get the minimum cost

Maintenance Planning



Equation:
$$CPUT_L(x) = \frac{E(\text{Cost Per Cycle})}{E(\text{Cycle Length})} = \frac{C_P R(x) + C_U (1 - R(x))}{\int_0^x R(t) dt}$$

- $f(x)$ is the probability density function of the failure time.
- $R(x)$ is the reliability function of the failure time.
- C_P is the average cost due to a planned maintenance.
- C_U is the average cost due to an unplanned maintenance.

Example-Optimum Replacement

Failure data for lean solution filter were collected as shown in table below. Assume that the average cost for a planned replacement of the filter is \$200 and for unplanned replacement, the cost is \$5000. The objectives are:

- Find the optimal scheduled replacement interval for minimizing the cost in the long term.

Failure No	Time to Failure (Days)
1	94
2	129
3	112
4	57
5	104
6	52
7	124
8	98
9	47
10	149

Spare part and Renewal Process

- When to use?
 - Renewal process, **AND**
 - Distribution of time-between-failures is exponential (CFR)
- The **probability distribution** for the **number of failures** by time t , is given by:

$$\Pr\{N(t) = j\} = \frac{(\lambda t)^j e^{-\lambda t}}{j!}$$

j = number of failures
 λ = failure rate

- Expected number of failures by time t , is given by:

$$E[N(0, t)] = \lambda t$$

- The cumulative probability that **k number of failures or fewer** is going to take place by time t , is given by:

$$\Pr_{\text{cum}}\{N(t) = k\} = \sum_{j=0}^k \Pr\{N(t) = j\}$$

Homogeneous Poisson process

Lambda=	0.000050	per hr
MTTF=	20000	hr
No. of Brgs=	20	
Ops run, t=	216000	hr
(Lambda)x(t)=	11	

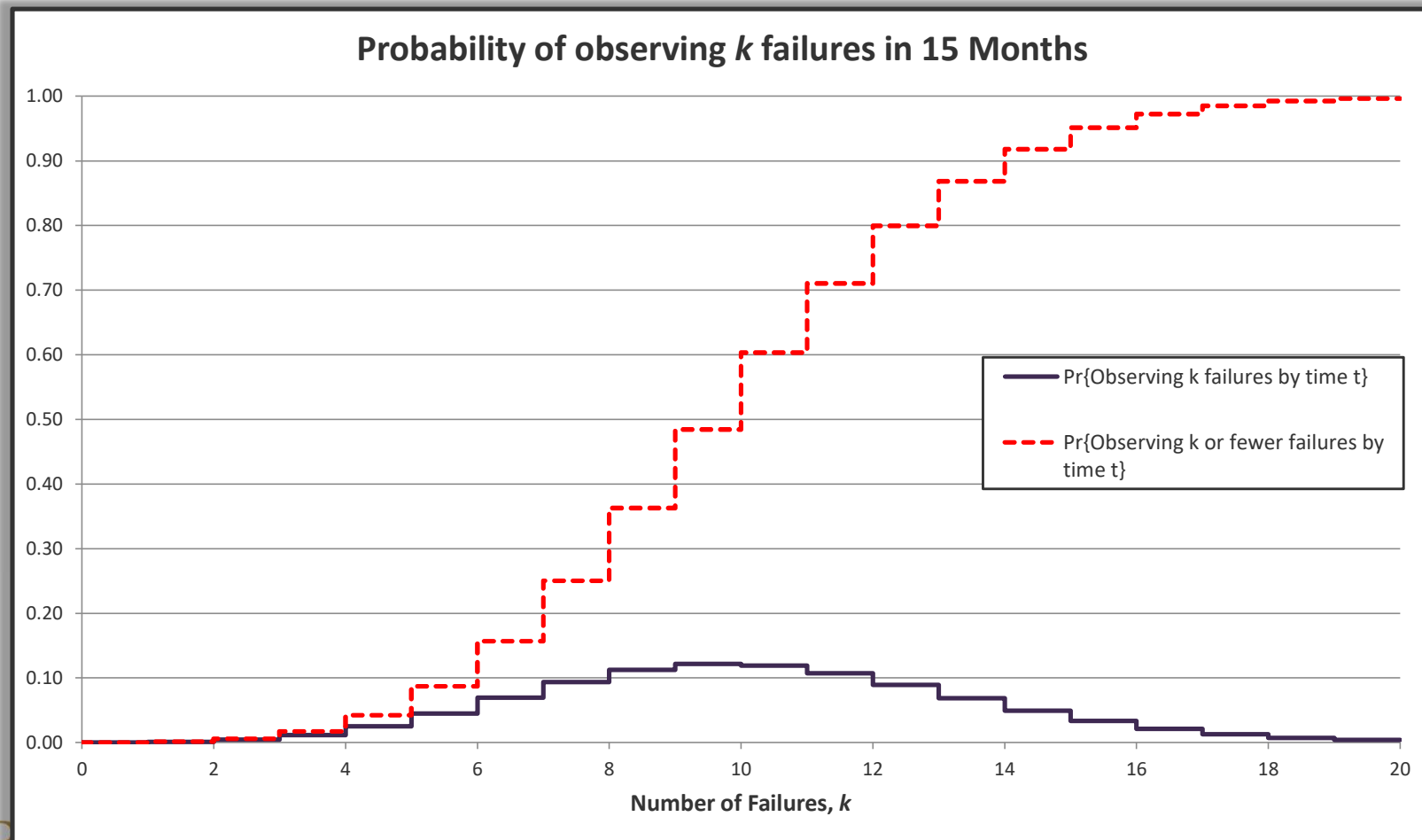
- Failure records for the past three years for a certain bearing (both DE and NDE) on a group of motors indicate random critical failures.
- There are 20 such bearings with CFR of 0.00005 per operating hour.
- The maintenance team plans to stock **10 units of spare bearings.**
- If the period until the next shutdown for the process unit is 15 months from now, what is the probability that the number of spare bearings is enough?
- Establish the probability distribution and the cumulative probability of failures for the coming 15 months.
- How many bearings would you recommend that maintenance keep?

$$\Pr\{N(t) = j\} = \frac{(\lambda t)^j e^{-\lambda t}}{j!}$$

No. of Failures k	Pr{Observing k failures by time t }	Pr{Observing k or fewer failures by time t }
No. of Spare Bearings k		Pr{k or fewer bearings is enough by time t }
1	0.00	0.00
2	0.00	0.00
3	0.00	0.01
4	0.01	0.02
5	0.02	0.04
6	0.04	0.09
7	0.07	0.16
8	0.09	0.25
9	0.11	0.36
10	0.12	0.48
11	0.12	0.60
12	0.11	0.71
13	0.09	0.80
14	0.07	0.87
15	0.05	0.92
16	0.03	0.95
17	0.02	0.97
18	0.01	0.99
19	0.01	0.99
20	0.00	1.00

$$\Pr_{\text{cum}}\{N(t) = k\} = \sum_{j=0}^k \Pr\{N(t) = j\}$$

Homogeneous Poisson process



Maintainability Analysis

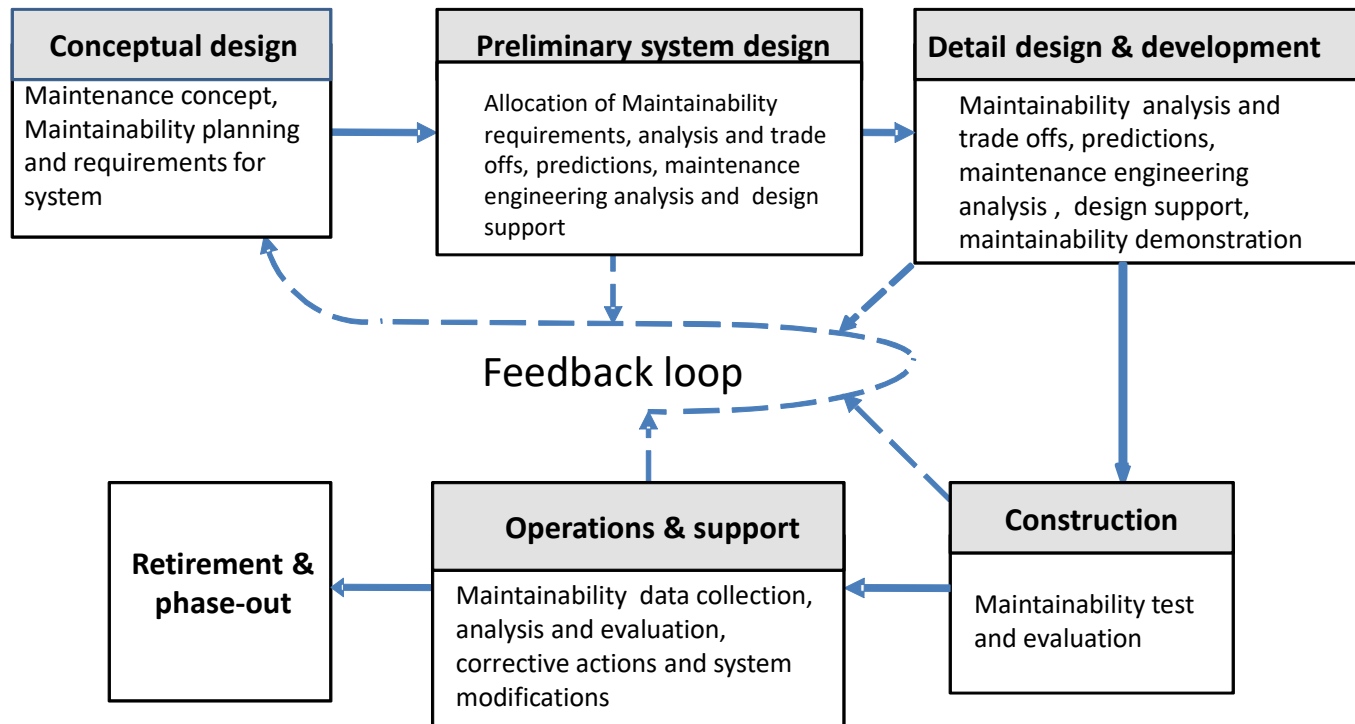
Session objectives

- To understand the concept and methodology of maintainability analysis
- To recognize the applications of maintainability analysis as a strategic tool for improving system performance and operating cost

Why Maintainability?

- Generally, the costs associated with system operation and support of many large systems constitute from 60% - 75% of the total life cycle cost for that system
- The main portion comes from the cost of maintaining the equipment and system
- As such, maintenance characteristics and performance should be considered in every phase of system life cycle
- Besides reliability (frequency of failures), maintainability (duration of downtime) plays equally critical roles in achieving high operational availability

Maintainability Requirements in System Life cycle



(Blanchard et al., 1995)

Maintainability Analysis at Operational phase

At the **operation and support** phase, the on-going maintainability analysis provides :

- ✓ quantifiable assessment of the performance and effectiveness of the maintenance and support system
- ✓ identification of critical equipment system and process high cost and downtime drivers
- ✓ evaluation of maintainability measures and prediction

The analysis results are then can be used as valuable information for **operation , maintenance** and **design** personnel to :

- plan logistic support requirement (i.e workers, tools and materials)
- carry out improvement actions to reduce the maintenance and support costs
- make the maintenance system more effective
- achieve current system operation performance targets

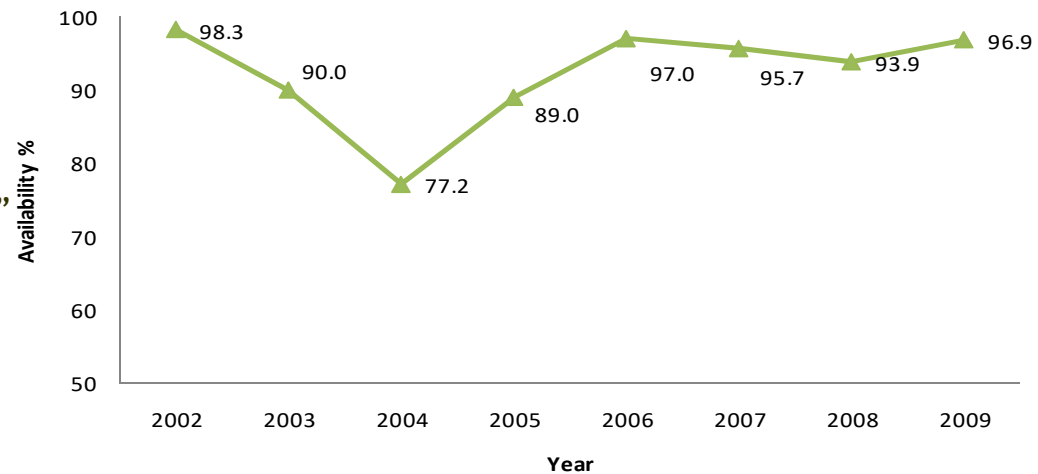
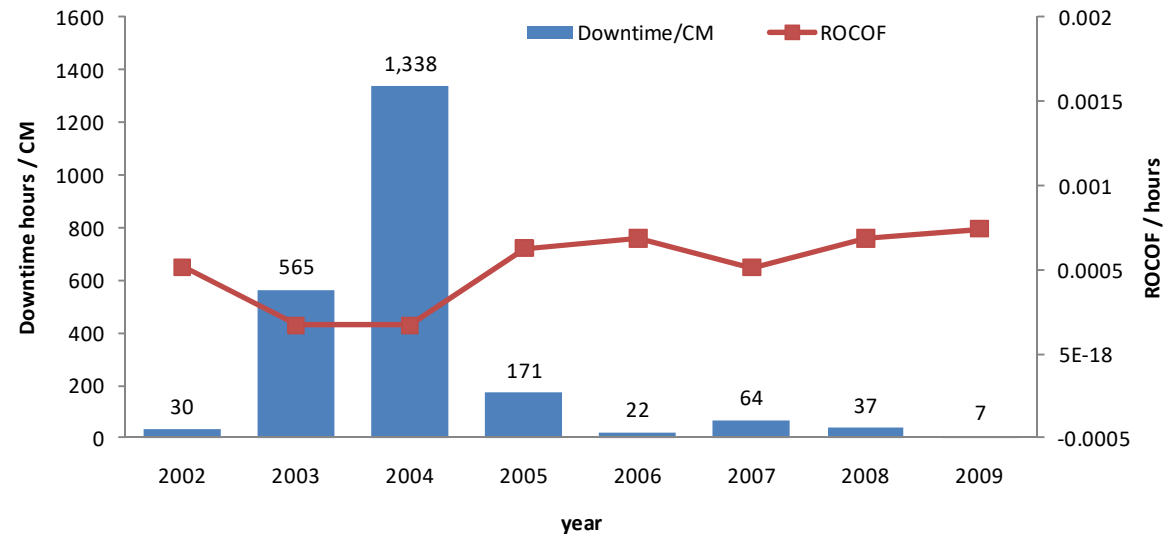
Importance of Maintainability - Example

Gas Compression
Train system
(Offshore)

Low availability in
2003-2005 is mainly
due to maintainability
issues, not reliability

Improvement actions done include:

- Spare part management - “Pit crew”
- Upgrading tech & operators skills
(Hasnan et. al (2004))
- Long term supplier contract



What is MAINTAINABILITY?

- Maintainability (of an item):

“the measure of the ability of an item to be retained in or restored to specified condition when maintenance is performed by personnel having specified skill levels, using prescribed procedures and resources, at each prescribed level of maintenance and repair ” (MIL-STD-721C, 1981)

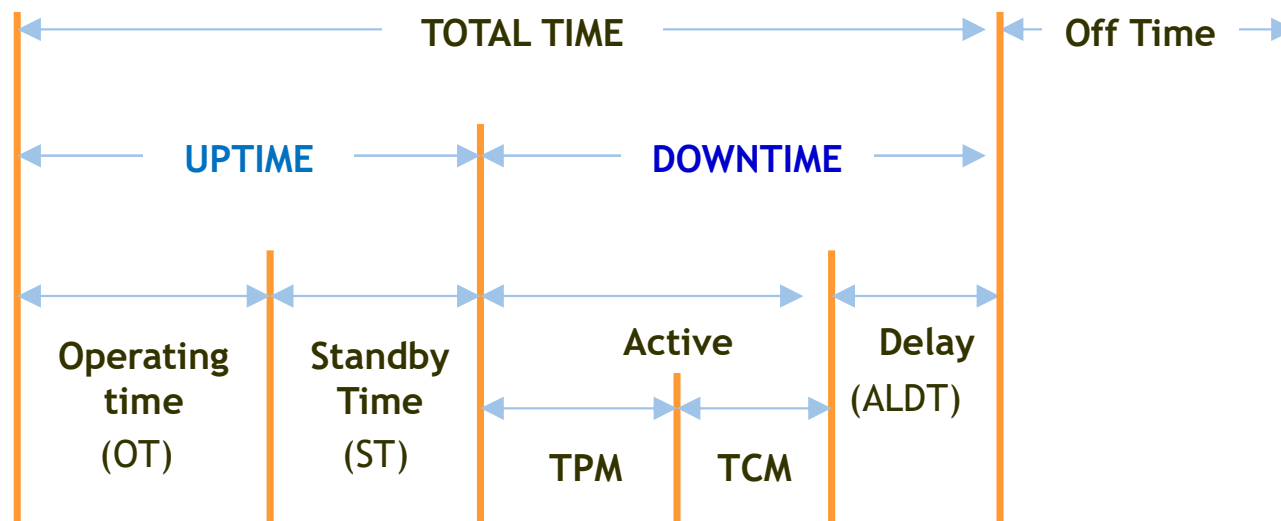
- Maintainability (as a measure) :

- a ***probability*** that a given maintenance action, performed under stated conditions and using stated procedures and resources, can be carried out within a stated time interval (ISO 60050-191)
- a ***measure of the time*** required to repair a certain percentage of all system failures (MIL-HDBK-338B, 1998)

What is MAINTAINABILITY? (ct'd)

- At the highest level, maintainability can be seen as a product of overall support programme in the system where high maintainability reflects the effectiveness of in the design approach, manpower allocations, training delivery and supply chain management (Knezevic, 1997).

Breakdown of total system or equipment time



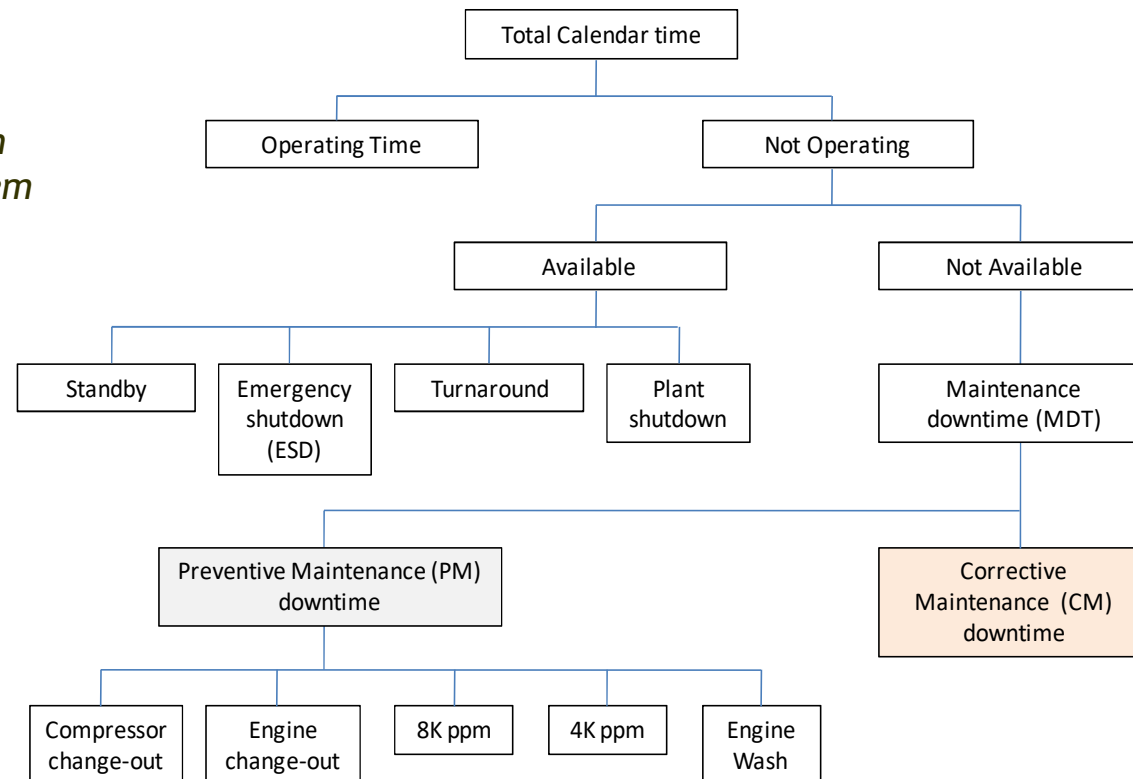
TPM = total preventive maintenance
TCM = total corrective maintenance

ALDT = Administrative and Logistic delay time

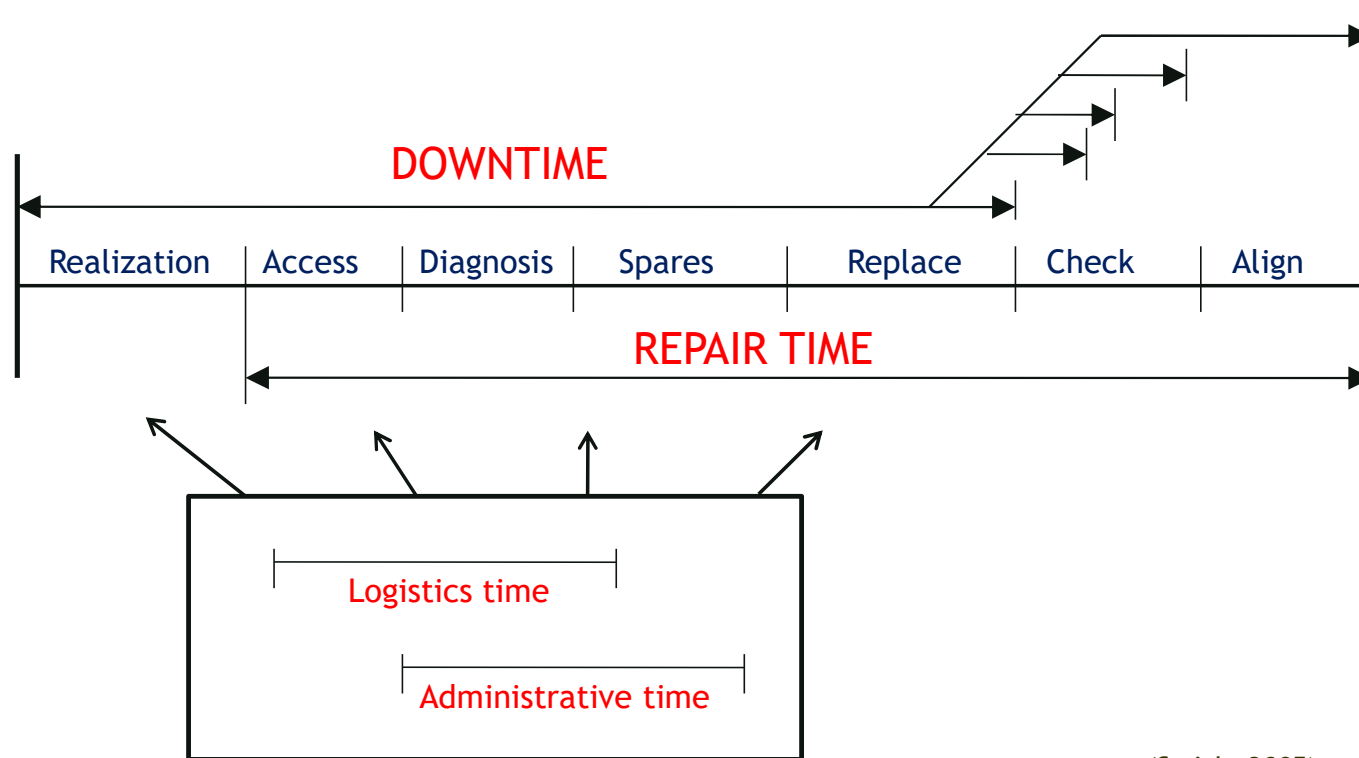
System time breakdown

Exact downtime period should be identified from plant data before it can be analyzed

*Example:
Breakdown of time on
gas compression system*



Breakdown of Downtime



(Smith, 2005)

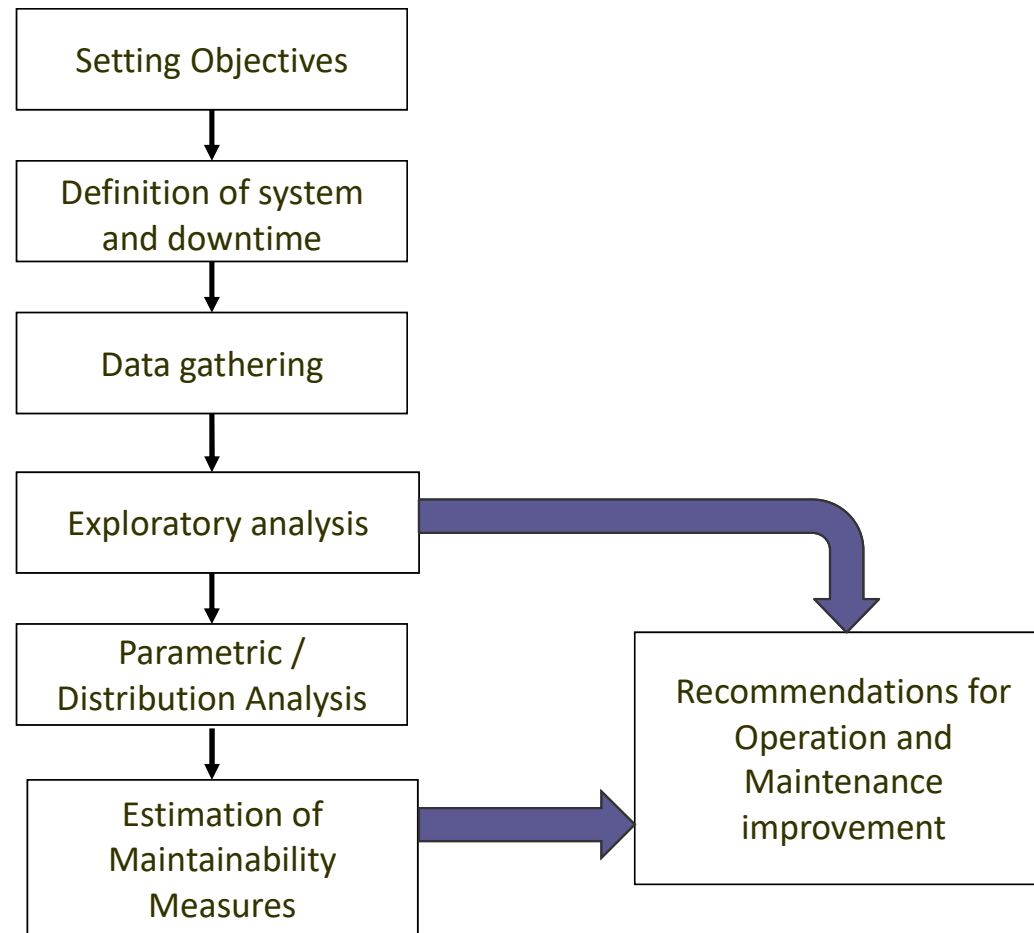
Related TERMS

- **DOWNTIME** - the time interval for which the system is unable to perform as required due to fault or maintenance activity
- **REPAIR TIME** - that part of the maintenance time during which a maintenance action is performed on an item, either automatically or manually, excluding logistic and administrative delays
- **LOGISTIC DELAY TIME**- the accumulated time during which maintenance cannot be carried out due to necessity to acquire maintenance resources, excluding any administrative delay
- **ADMINISTRATIVE DELAY TIME** – the accumulated time during which maintenance cannot be carried out due to administrative reasons
- **UPTIME** – the time interval for which the system can perform a required function, assuming that the external resources, if required, are provided
- **OPERATING TIME** – the time interval during which the system is performing the required function
- **STANDBY TIME** – the time interval during which the system is not performing the required function, but assumed operable

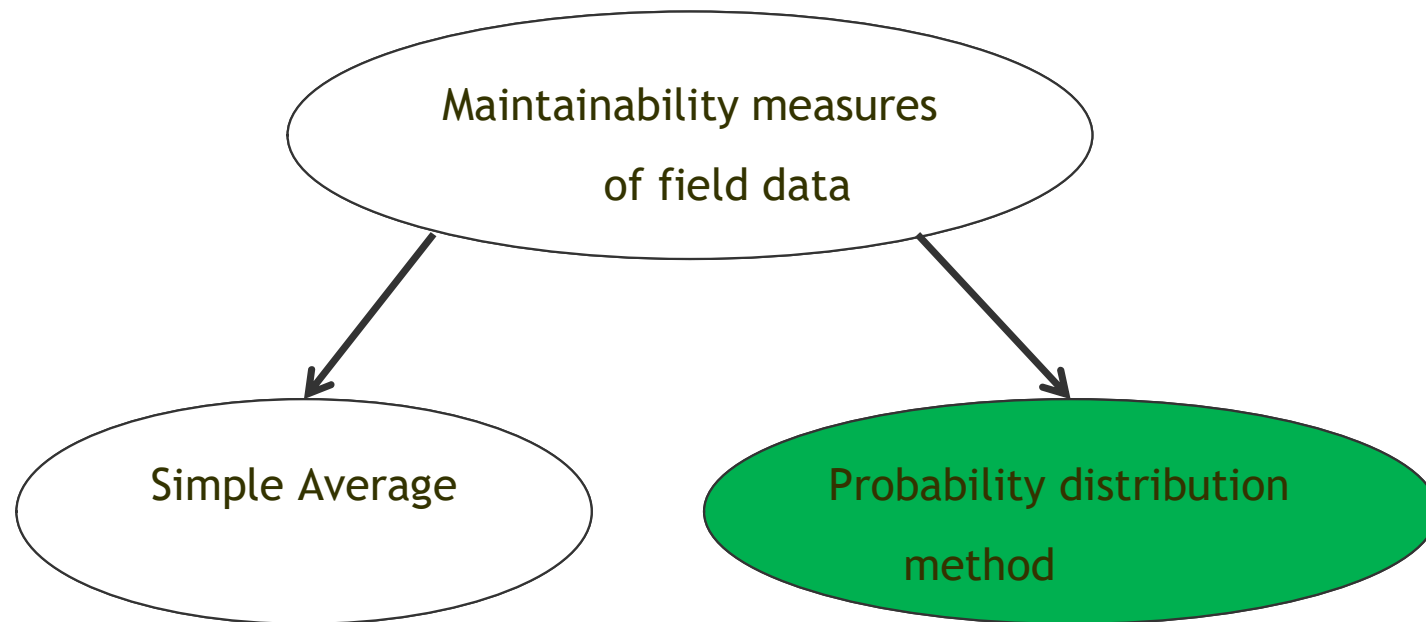
Maintainability Measures

- Mean time to repair (MTTR)
- Mean preventive maintenance time
- Mean corrective maintenance time
- Mean maintenance downtime
- Measures based on maintainability function
 - probability of task completion
 - success of task completion
 - percentual duration of restoration or downtime
 - mean duration of maintenance task or downtime

General Maintainability Analysis Methodology



Common Approach to Maintainability Measures



(Knezevic ,2009)

Probability Distribution Approach

- ❑ Downtime is expressed in term of probability distributions, where the downtime is treated as random variable since every failure event will always result in different downtime duration due to different failure modes, components failure and skill level of maintenance people.
- ❑ Maintainability function, derived from the probability distribution, is used for determining various maintainability measures
- ❑ Distribution approach offers more information ,thus is the preferred method in evaluating maintainability measures
- ❑ Common distributions : **exponential, normal, and lognormal**

Maintainability Function

Maintainability functions can provide the probability that a repair, starting at time $t=0$, will be completed in a time t .

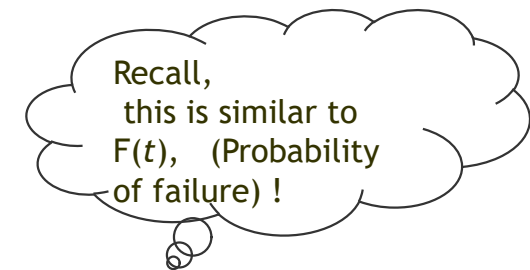
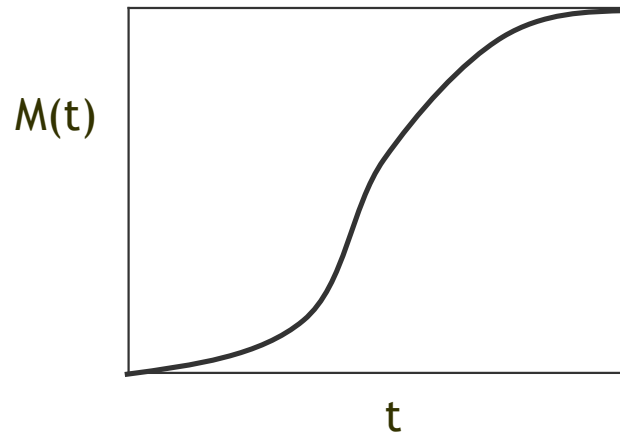
The maintainability function for any distribution is defined by:

$$M(t) = \int fr(t) dt$$

Where:

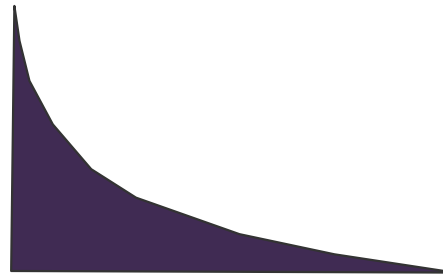
t = time

$fr(t)$ = probability density function of the repair time



Common Maintainability Distribution

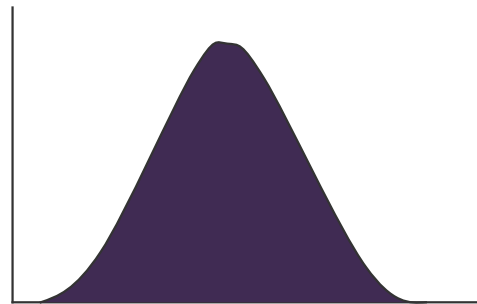
Exponential distribution



It is usually applied to electronics parts with build-in test capability and have fast remove and replace maintenance scheme. It is however not realistic for many downtimes situations, except in the case where most part of the downtimes are attributed to failures searching actions

Common Maintainability Distribution

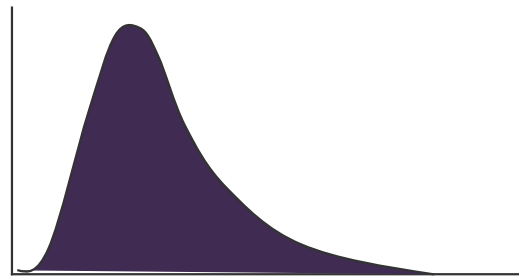
Normal distribution



The normal distribution is sometimes assumed for equipment having relatively simple removal and replacement tasks that usually can be completed within a fixed amount of time with little variations .

Common Maintainability Distribution

Lognormal distribution



The lognormal distribution is the most common model for repair time or downtime distribution for both electronics and mechanical equipment. The shape of the lognormal is skewed to the right meaning that most of the downtimes are distributed about the center and few will be at the right-tail of the distribution. This characteristic seems logical in many downtime events since some downtime is very long due to the unavailability of spare parts at site or difficulty of maintenance crew to get access to or repair the failure

Example : Distribution approach

Consider the PM downtime data below, determine :

1. The appropriate distribution and its parameters
2. Mean downtime (MDT)
3. Downtime at 10%, 50% and 90% completion rate

no	Downtime (hrs)	no	Downtime (hrs)	no	Downtime (hrs)
1	40	6	48	11	54
2	55	7	58	12	60
3	65	8	74	13	78
4	116	9	108	14	96
5	80	10	80	15	85

Example : Distribution approach

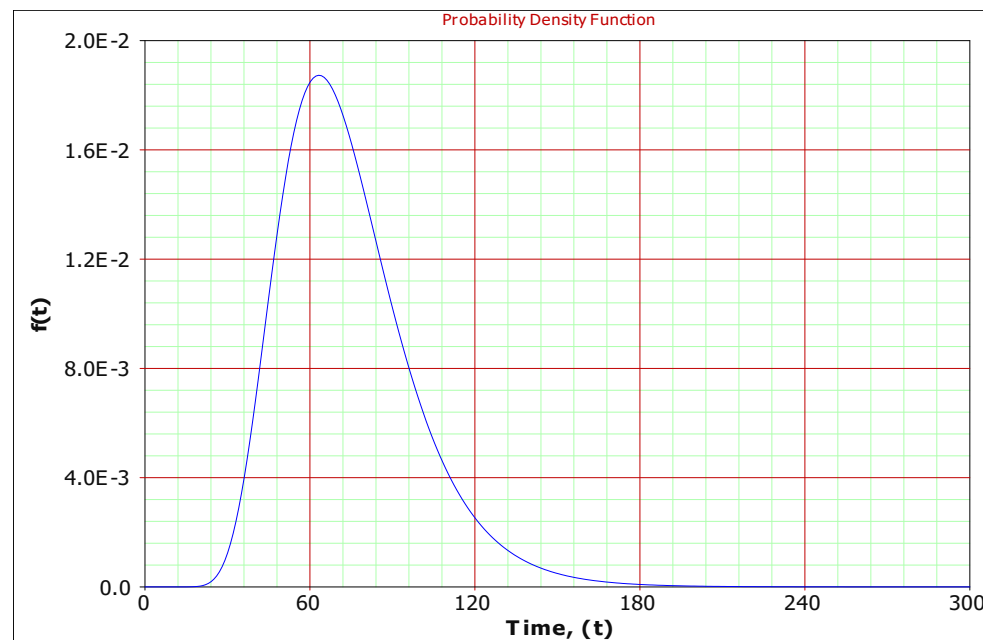
Solution:

Using Weibull ++ software

1. The most appropriate distribution is lognormal,
parameters: mean = 4.2502, std = 0.3197

2. Mean Downtime (MDT)
= 73.7954 hrs

95% confidence =
(62.77 - 86.75 hrs)

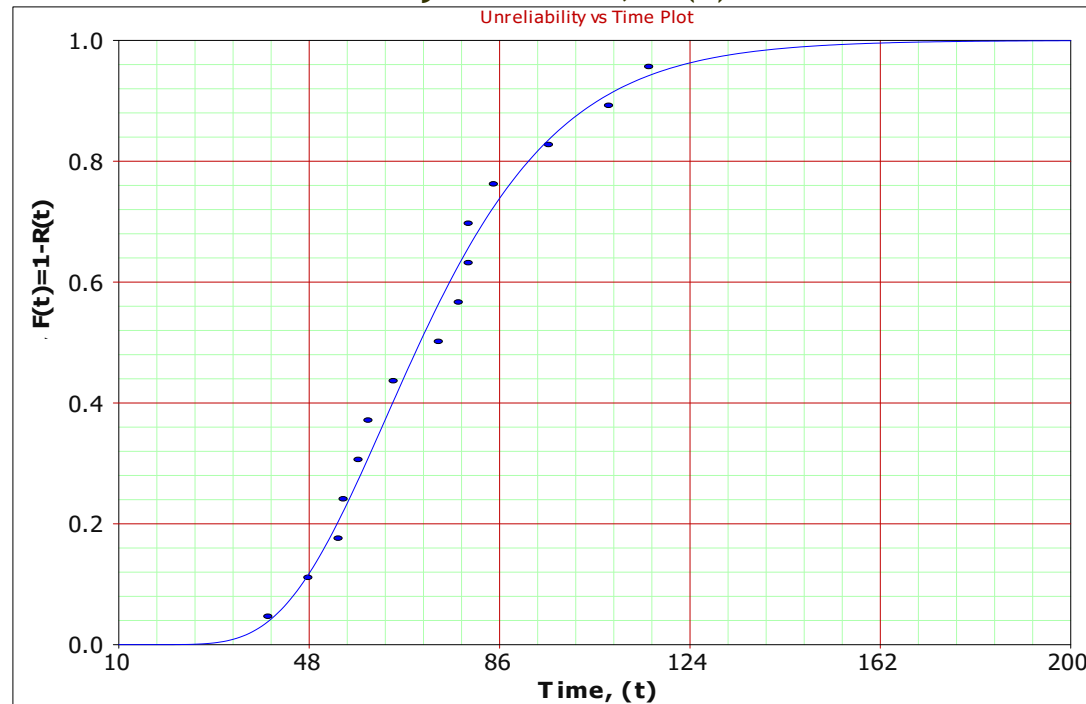


Example : Distribution approach

Solution (ct'd):

3. Downtime (Maintainability) at 10%, 50% and 90% completion rate

Maintainability function, $M(t)$



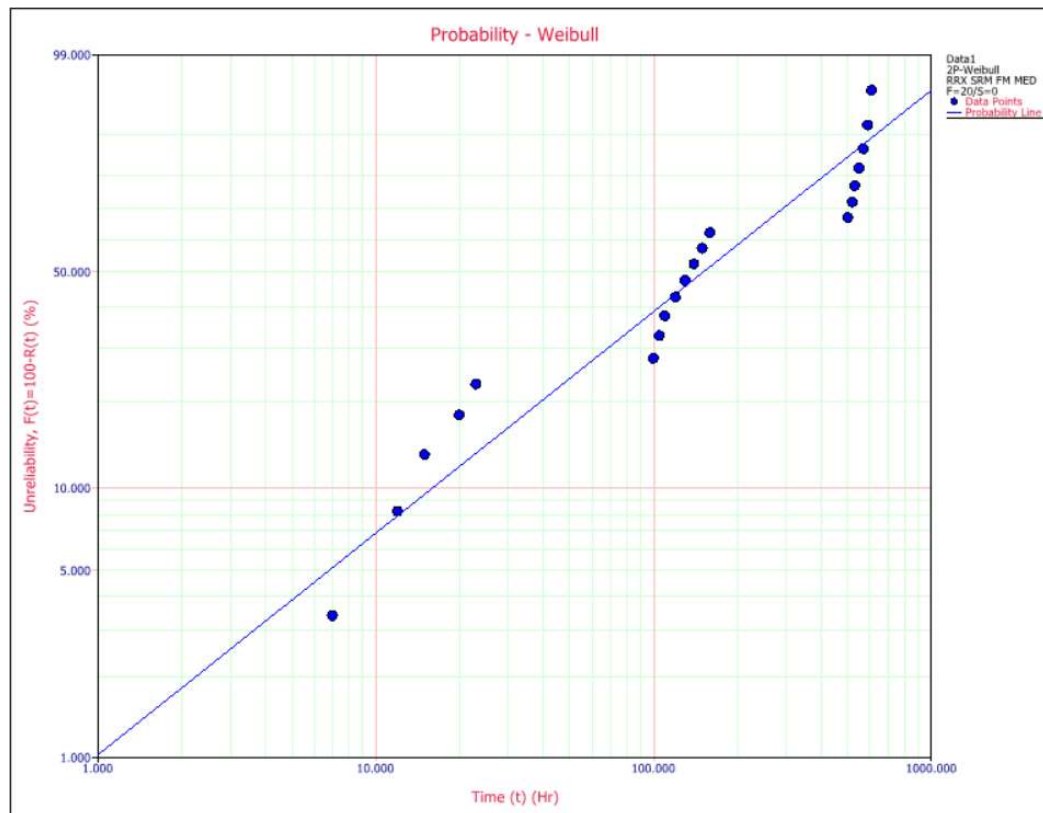
$$DT_{10} = 46.55 \text{ hrs}$$

$$DT_{50} = 70.12 \text{ hrs}$$

$$DT_{90} = 105.6 \text{ hrs}$$

Meaning that up to 105.6 hrs ,
approximately 90% of PM tasks
are completed

Multi Failure modes



EXERCISES

CASE STUDY

GCT (Gas compression system) at an Offshore platform had the following PM downtime data for year 2002-2008

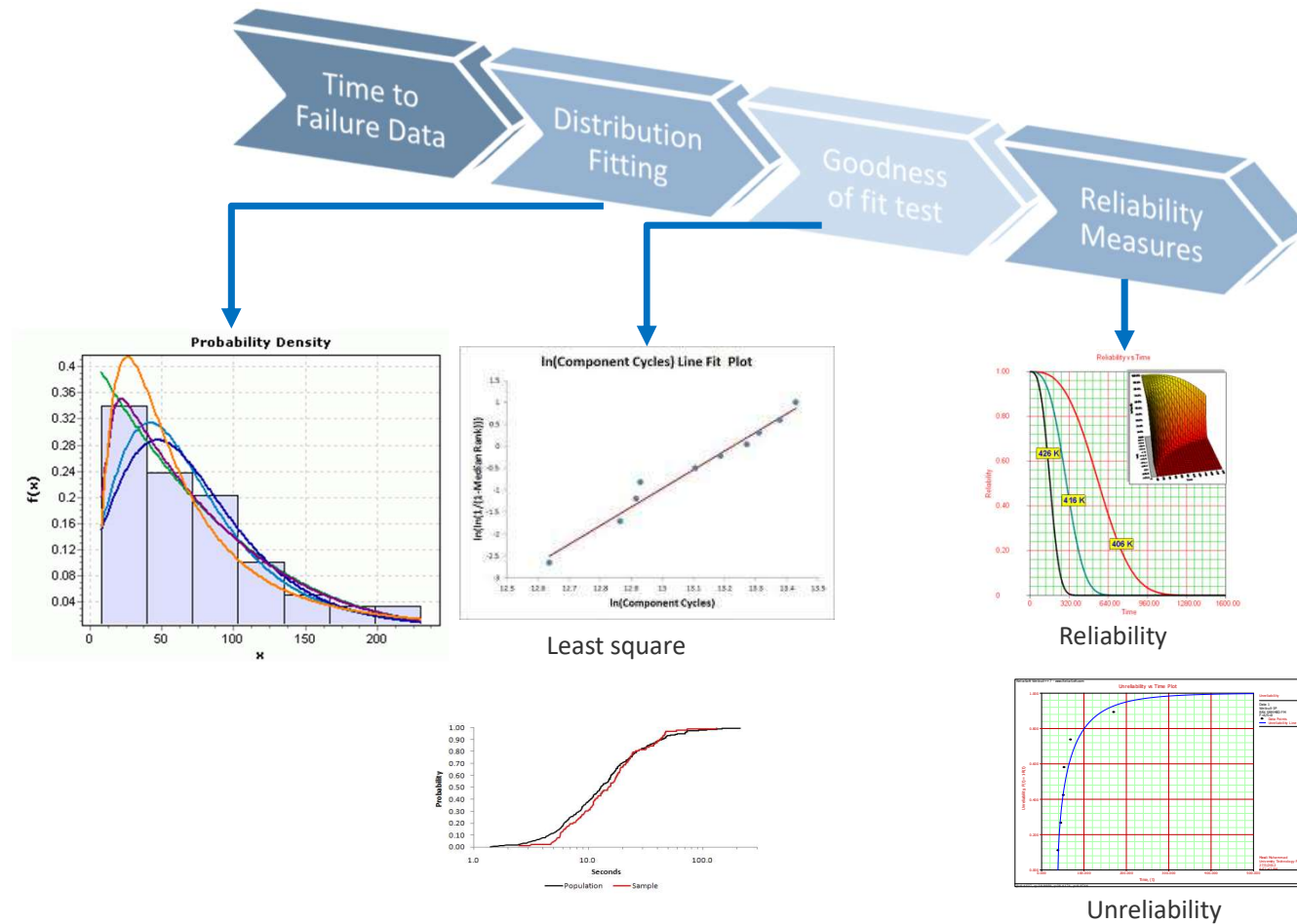
PM types	Downtime data (hrs)	Average	Std	90% confidence interval
4K	44	59.75	12.6	30.02 – 80.48
	55.5			
	67.5			
	72			
8K	68.5	124.9	41.6	56.47 – 193.33
	78			
	94			
	104			
	174			
Engine wash	3	9.4	3.75	3.23 – 15.57
	5			
	5.4			
	9			
	9.5			
	10			
	11			
	11			
	11			
	13			
	16			

Questions:

1. Determine the best distribution for 4K, 8K ppm and Engine Wash
2. Based on those proposed dist., determine the maintainability at 90% for all PM types

Degradation Analysis

RECALL: Steps involved in analyzing non-repairable system/component



Objectives

At the end of the section, the participants should be able to :

- Understand the concept of Degradation Analysis
- Apply degradation analysis for inspection parameters to estimate time to failure (or remaining life)
- Estimate the reliability metrics for the equipment:
 - The mean life
 - The B10 life (i.e. the time by which 10% of the equipment will fail)

What is Degradation?

- Degradation is the reduction in performance, reliability, and life span of assets.
- Most assets degrade or deteriorate as they age as a result of some internal and external factors.
- Many failure mechanisms can be traced to an underlying degradation process.
- Assets fail when their level of degradation reaches a specified failure threshold.

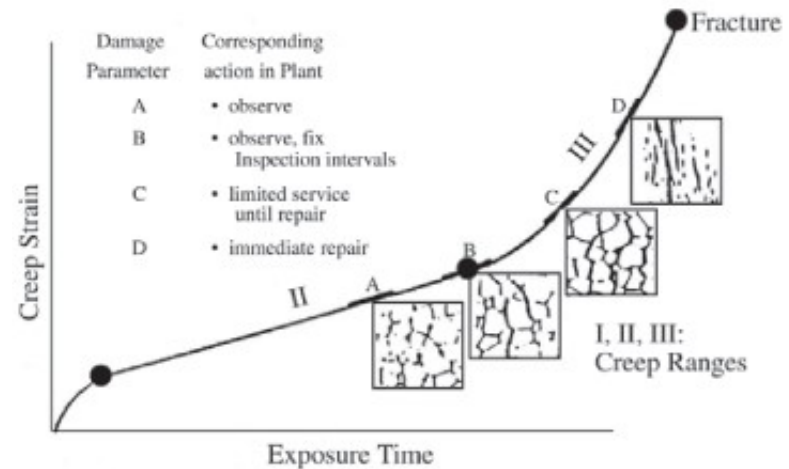


Figure 1. Neubauer's classification of creep damage from observation of replicas and consequent action to be taken.

Degradation Models in Reliability Analysis

The fundamental research about the life of an asset is to predict **how much time is left before a failure** occurs given the current asset condition and past operation profile.

The time left before observing a failure is usually termed as Remaining Useful Life (RUL).

In some industry applications, especially when a failure is catastrophic (e.g. in nuclear power plants, airplanes, bridges, dams), it would be more **imperative to predict the chance that an asset operates without a failure** up to some future time (e.g. next inspection time) or a specified failure threshold given the current asset condition and past operation profile.

Degradation models are one of the suitable approaches to deal with this type of prediction.

Degradation Models in Reliability Analysis

A failure can be produced by different causes that can be classified either as internal or external

- Internal failures
 - occur due to the inner structure of systems
 - e.g. ageing and quality of materials
- External failures
 - occur due to the environmental conditions in which systems operate
 - e.g. vibrations, humidity, and pollution

Degradation Models in Reliability Analysis

Generally, failures can be divided into two groups:

- Gradual failure
 - predicted by one or several condition monitoring indicators
 - also called soft (or degradation) failure

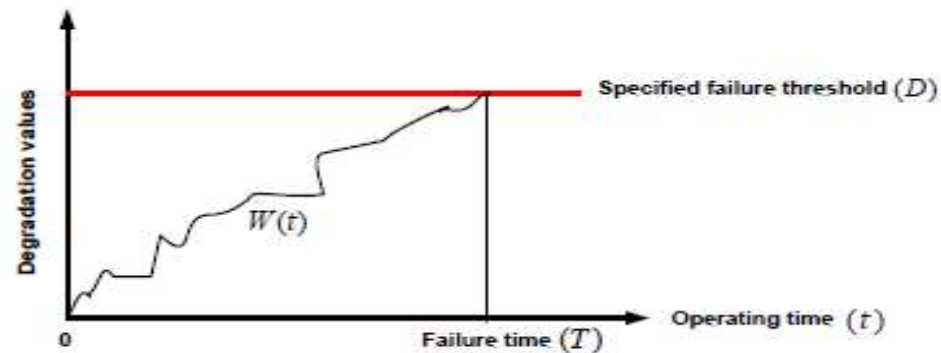
- Sudden failure
 - completely random
 - cannot be predicted by either condition monitoring indicators or by measuring the age of the asset.

Degradation Models in Reliability Analysis

There are two types of degradation:

- natural degradation
 - age or time-dependent
 - the term ageing refers to an internal process in systems where gradual degradation occurs, thus it brings the systems closer to failure
- Forced degradation
 - external to the systems
 - its loading gradually increases in response to increased demand so that a point is reached beyond which the systems can no longer safely carry the load

Figure illustrates the basic notion in degradation models. This notion is that assets fail when their levels of degradation hit a specified failure threshold.



Time Series Degradation Analysis

- Extrapolation can be done using the one of the following models:

- Linear model

$$y = ax + b$$

- Exponential model

$$y = be^{ax}$$

- Power model

$$y = bx^a$$

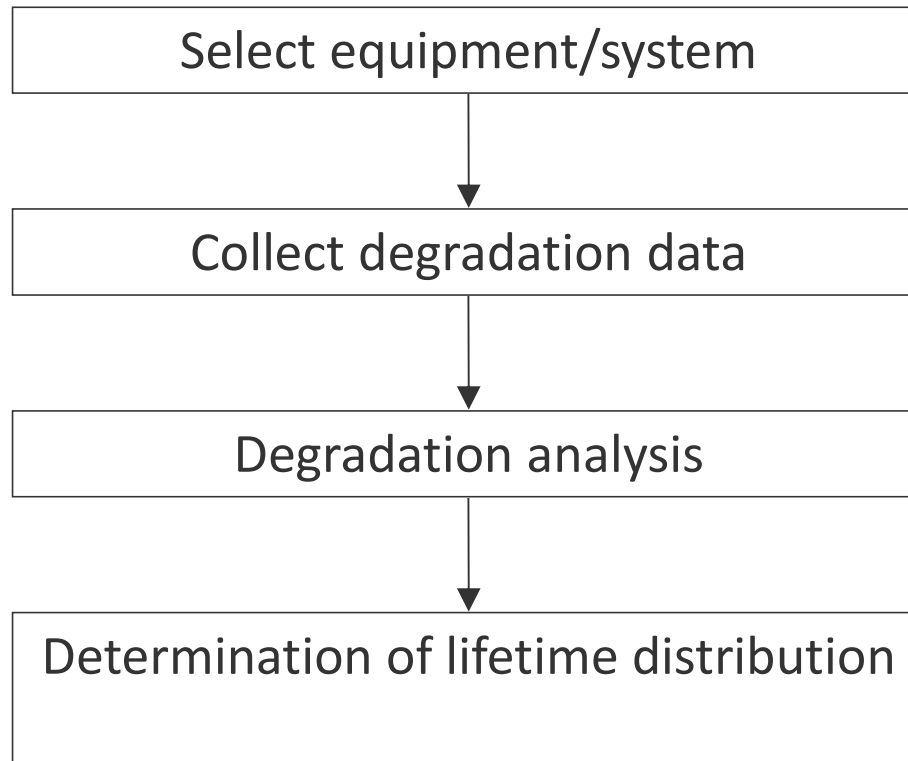
- Logarithmic model

$$y = a \ln(x) + b$$

where y represents the degradation, x represents time and a and b are model parameters to be solved for.

- Once the model parameters have been estimated for each sample i , a time-to-failure can be extrapolated.

Methodology

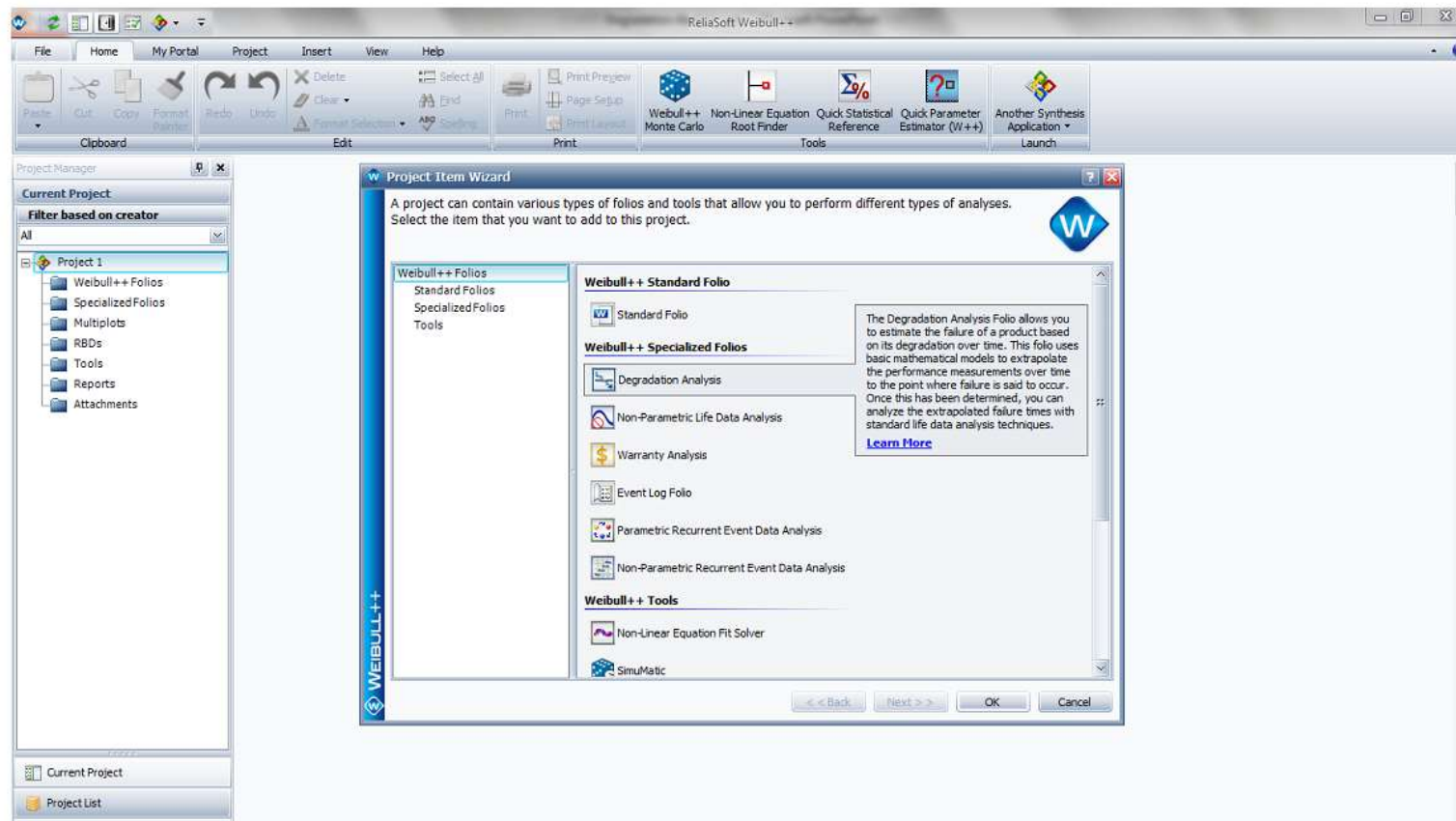


Case Study-Corrosion

Barringer [7] published the pipe wall thickness measurement data from a periodic inspection program. Data was recorded using ultrasonic testing and was taken from a pipe cross section over a period of time reflecting the age/use of the pipe at six locations in measurement plane. The original data which is in inches was converted to millimeter as presented in the Table below. The data shows small variations within each year.

Location	1	2	3	4	5	6
Year 1	7.90	8.03	7.82	7.75	8.08	8.15
Year 2	7.90	8.08	7.62	7.75	7.90	8.08
Year 4	7.90	7.87	7.57	7.67	7.72	7.95
Year 6	7.97	8.00	7.65	7.72	7.75	7.95
Year 8	7.85	7.75	7.49	7.49	7.62	7.82
Year 10	7.82	7.62	7.39	7.47	7.59	7.75
Year 12	7.82	7.67	7.49	7.37	7.49	7.62
Year 14	7.74	7.57	7.42	7.34	7.24	7.49

Degradation Analysis in Weibull++



STEP 1: Data Entry

The screenshot shows the ReliaSoft Weibull++ software interface. The main window displays a data table for 'Degradation (W)1'. The table has three columns: 'Inspection Time (Yr)', 'Measurement', and 'Unit ID'. The data is organized into rows, with the first row being a header and subsequent rows containing numerical data. The left sidebar shows the Project Manager with a tree view of the project structure. The right sidebar shows the Degradation Model settings, including the Linear model and Life Data Model settings.

Inspection Time (Yr)	Measurement	Unit ID
4	7.57	3
6	7.65	3
8	7.49	3
10	7.39	3
12	7.49	3
14	7.42	3
1	7.75	4
2	7.75	4
4	7.67	4
6	7.72	4
8	7.49	4
10	7.47	4
12	7.37	4
14	7.34	4
1	8.08	5
2	7.9	5
4	7.72	5
6	7.75	5
8	7.62	5
10	7.59	5
12	7.49	5
14	7.24	5
1	8.15	6
2	8.08	6
4	7.95	6
6	7.95	6
8	7.82	6
10	7.75	6
12	7.62	6
14	7.49	6

Step 2: Choose the right model

The screenshot displays the ReliaSoft Weibull++ software interface. The main window shows a data table with columns for Inspection Time (Hr), Measurement, and Unit ID. The Degradation Model Wizard is open, showing the 'Main' tab with a list of models to be analyzed. The 'Analysis Details' tab is also visible, showing the results of the analysis.

Degradation Model Wizard - Main Tab

Models and Rankings:

- ☒ Linear
- ☒ Exponential
- ☒ Power
- ☒ Logarithmic
- ☒ Gompertz
- ☒ Lloyd-Lipow

Select All Clear All

Select the models you want to consider, then click the Analyze button.

Degradation Model Wizard - Analysis Details Tab

Analysis is complete. The results and detailed rankings are displayed on the Analysis tab. Click 'Implement' to use the model with the best rank.

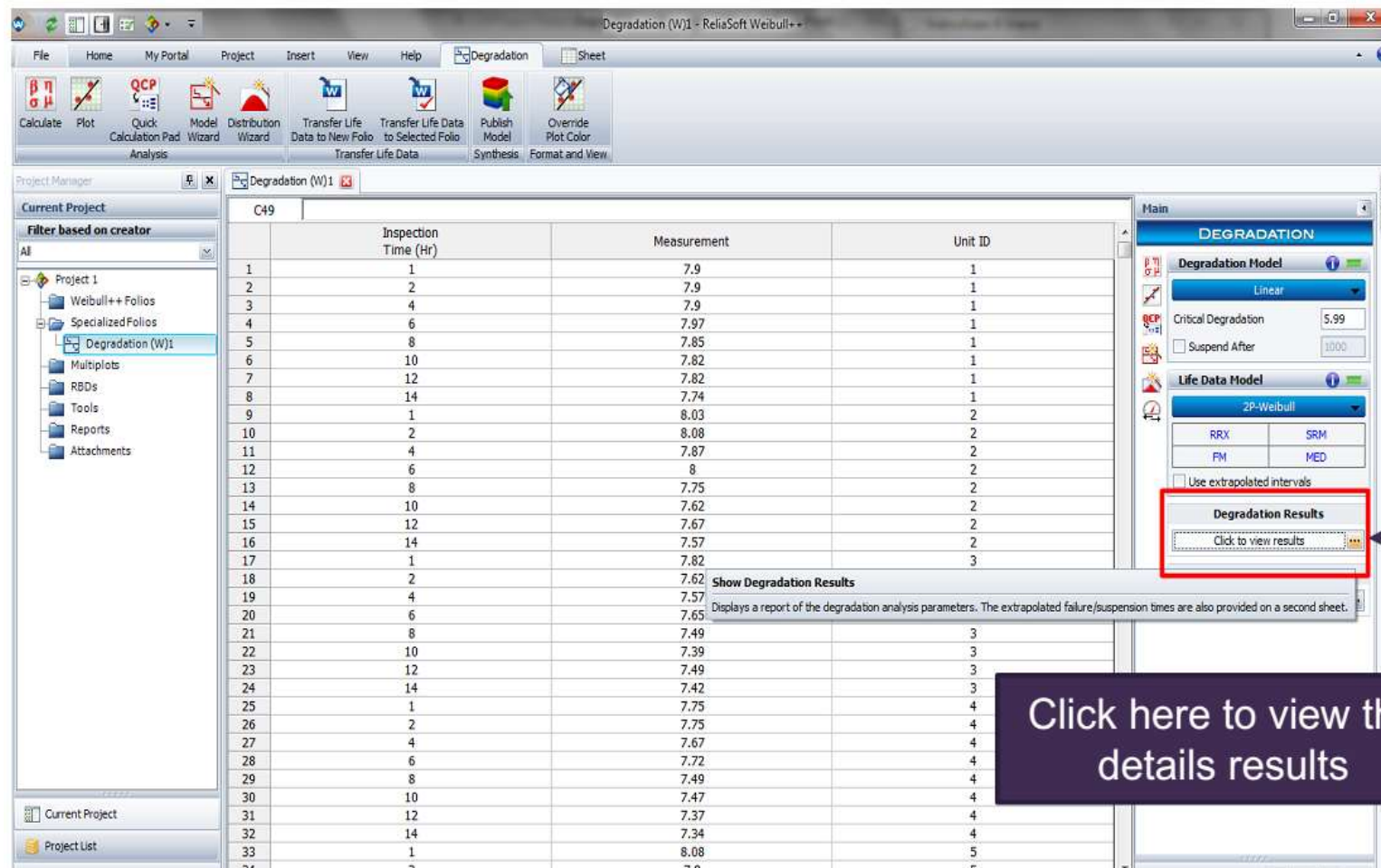
Models and Rankings:

- ☒ Linear 1
- ☒ Exponential 2
- ☒ Power 4
- ☒ Logarithmic 3
- ☒ Gompertz 6
- ☒ Lloyd-Lipow 5

Select All Clear All

Analyze Implement Close

Step 3: Generate Result



DEGRADATION

Degradation Model

Linear

Critical Degradation: 5.99

☐ Suspend After: 1000

Life Data Model

2P-Weibull

RRX SRM

FM MED

☐ Use extrapolated intervals

Degradation Results

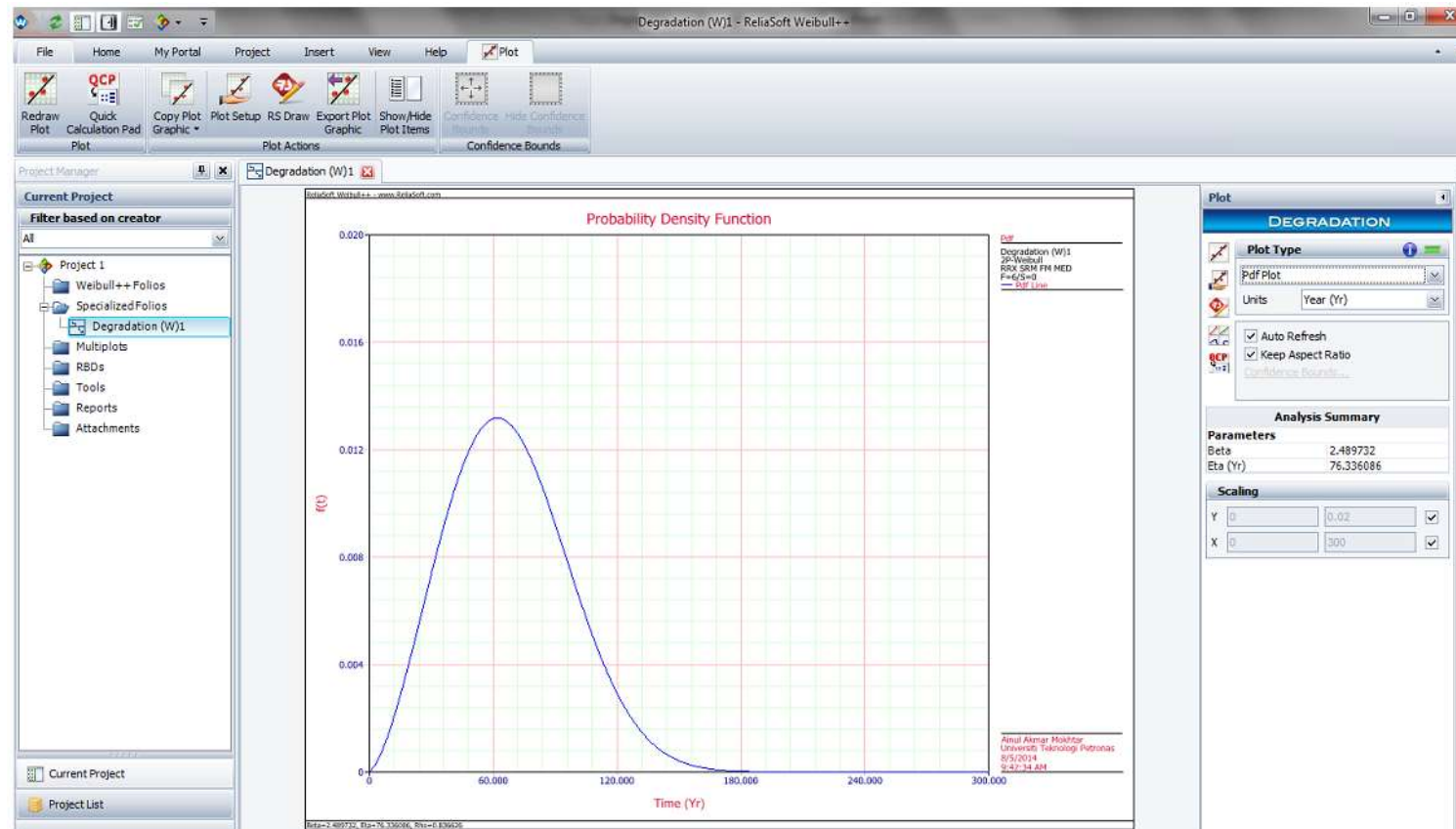
Click to view results

Show Degradation Results

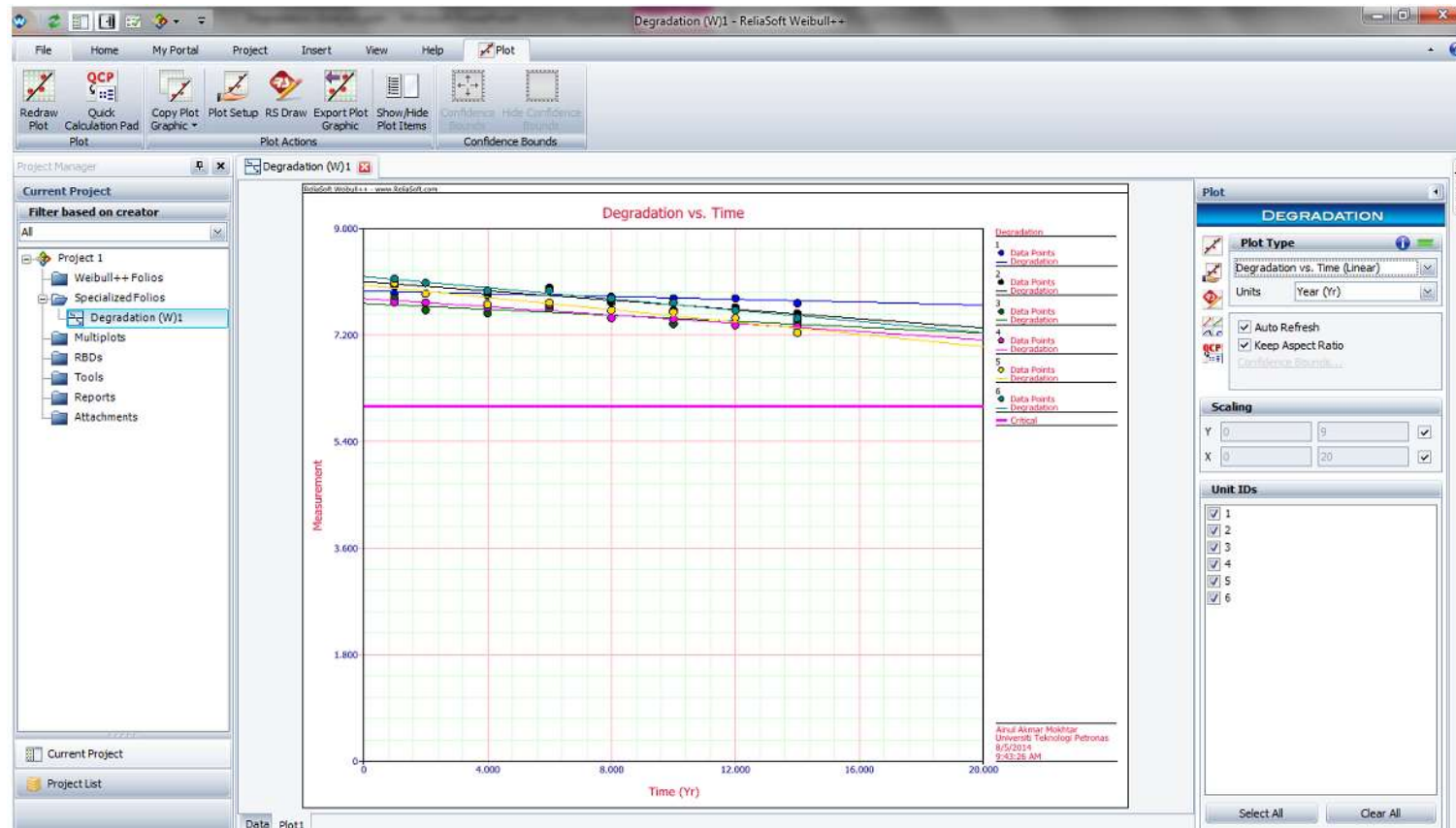
Displays a report of the degradation analysis parameters. The extrapolated failure/suspension times are also provided on a second sheet.

Item ID	Inspection Time (Hr)	Measurement	Unit ID
1	1	7.9	1
2	2	7.9	1
3	4	7.9	1
4	6	7.97	1
5	8	7.85	1
6	10	7.82	1
7	12	7.82	1
8	14	7.74	1
9	1	8.03	2
10	2	8.08	2
11	4	7.87	2
12	6	8	2
13	8	7.75	2
14	10	7.62	2
15	12	7.67	2
16	14	7.57	2
17	1	7.82	3
18	2	7.62	
19	4	7.57	
20	6	7.65	
21	8	7.49	3
22	10	7.39	3
23	12	7.49	3
24	14	7.42	3
25	1	7.75	4
26	2	7.75	4
27	4	7.67	4
28	6	7.72	4
29	8	7.49	4
30	10	7.47	4
31	12	7.37	4
32	14	7.34	4
33	1	8.08	5
34	2	7.6	5

Step 4: Review Result



Step 4: Review Result



Case Study-Crack

- Five turbine blades were collected and tested for crack propagation. The units were cyclically stressed and inspected every 100,000 cycles for crack length in mm. The threshold for the crack length is 30 mm. Following is the table of the test results:

Cycle (x1000)	Blade A	Blade B	Blade C	Blade D	Blade E
100	15	10	17	12	10
200	20	15	25	16	15
300	22	20	26	17	20
400	26	25	27	20	26
500	29	30	33	26	33

- All blades have more than 30mm crack length. Propose the inspection time (ie after how many cycles)

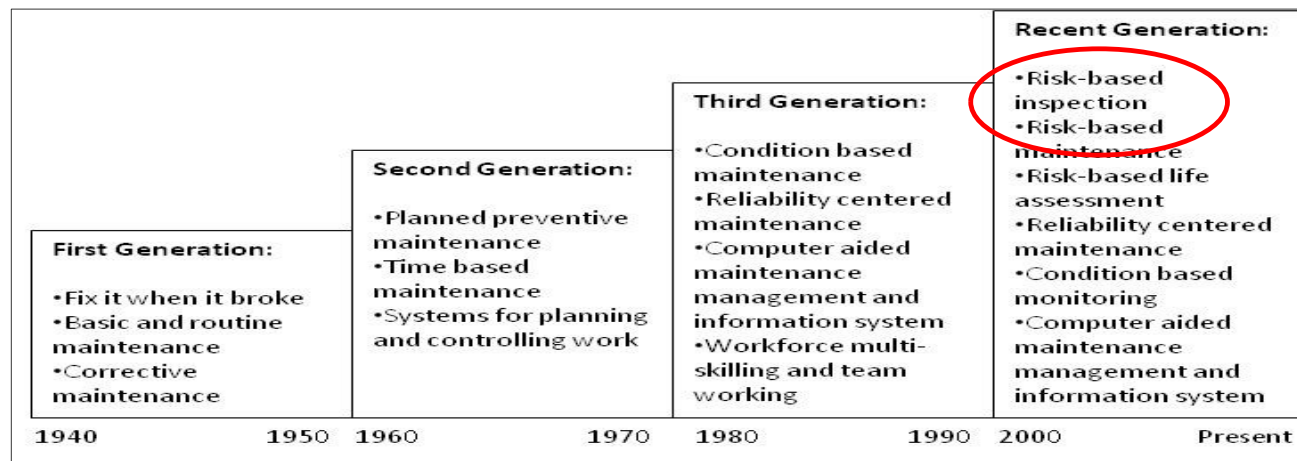
Summary

- Degradation analysis is useful as many failure mechanisms are directly linked to the degradation parameters
 - Wall thickness data for corrosion
 - Crack length for crack propagation
 - Strain for creeps
- Degradation model allows the user to extrapolate the time to failure for the specific failure mechanism (or estimating the remaining life)

Weibull Analysis in API 581 Risk-Based Inspection

Background

- Over the past few decades, the strategies in inspection and maintenance evolved from the breakdown maintenance to the more sophisticated strategies such as condition monitoring and reliability centered maintenance.
- The paradigm shift was motivated by the need to implement new maintenance strategies which would increase the effectiveness and profitability of the business.
- Another link to the evolution is the introduction of risk-based approach to inspection and maintenance.



Background

- The first initiatives for the developments of risk based approaches to the inspection and maintenance planning were directed towards the inspection planning for welded connections subject to fatigue in fixed steel offshore structures (Skjong, 1985; Madsen et al., 1989; Fujita et al., 1989).
- Later, the same methodology was applied to other structures such as ships and tankers (Soares & Garbatov, 1996; Paik et al., 2004, Vanem & Skjong, 2008), floating, production, storage and off-loading facilities (Lotsberg et al., 1999; Goyet et al., 2002), semi-submersibles, pipelines (Willcocks & Bai, 2000; Desjardins, 2002; Dey & Gupta, 2001), process plants (Geary, 2002; Montgomery & Serratella, 2002; Kallen, 2002; Kallen & Noortwijk, 2006; Khan et al., 2006), bridges (Frangopol et al., 2004), and breakwaters (Noortwijk & Phajm, 1996) to name a few.

What is Risk-Based Inspection?

- Risk-based inspection (RBI) has been an industry standard for prioritizing inspection of static equipment, such as pressure vessels, tanks, heat exchanger, piping systems, relief valves and control valves (Dey, 2001; Khan & Haddara 2003; Dey, Ogunlana & Naksuksakul, 2004; Krishnasamy et al., 2005; Khan et al., 2006).
- Although, the application of RBI is more on static equipment, it has also been applied to rotating equipment (Fujiyama et al., 2004).
- RBI is a systematic process for evaluating risk and factoring it into decisions concerning how, where and when to inspect.
- RBI is a method which uses risk as a basis to prioritize and manage an inspection program, where equipment items to be inspected are ranked according to their risk.

What is Risk?

- Risk = Probability of Failure (PoF) $\times$ Consequence of the failure (CoF)
- Definition of Failure in API RBI - a loss of containment from the pressure boundary resulting in leakage to the atmosphere or rupture of a pressurized component.
- As damage accumulates in a pressurized component during in-service operation the risk increases.
- At some point, a risk tolerance or risk target is exceeded and an inspection is recommended of sufficient effectiveness to better quantify the damage state of the component.

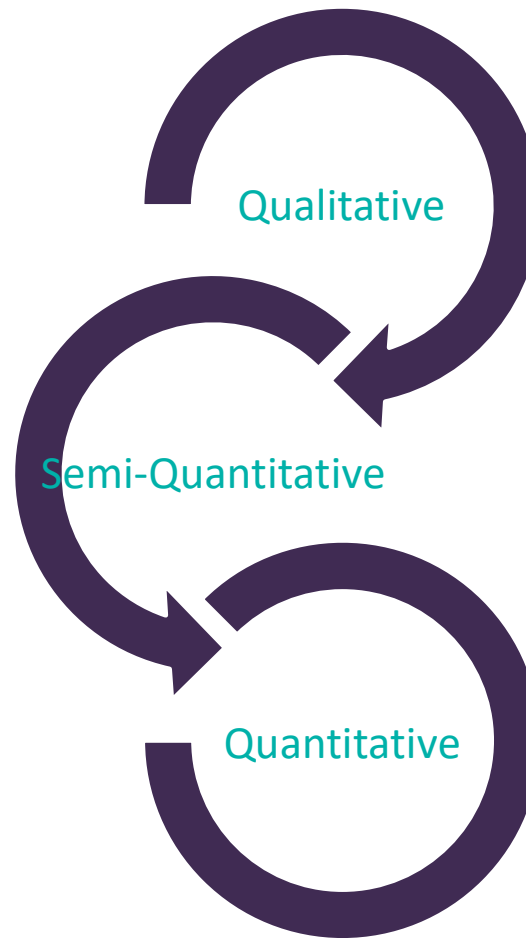
What is Risk?

- According to Giribone and Valette (2004), the main driver for scheduling inspection interval is the probability of failure.
- The probability of failure is defined as the mean frequency or rate with which the specified failure event would be expected to occur in a given period of operation, normally one year.
- Therefore, estimating the probability of failure is an important input to RBI analysis.

Inspection Plans

- Inspection plans are the primary output of an RBI assessment
- Elements of an inspection plan
 - Level of risk of current operation
 - Plans to reduce the risk to tolerable levels
 - Cost effective means of risk management

RBI Approaches



Qualitative Approach

- Based on engineering judgments made by experts.
- The failure probability is described using terms such as very unlikely, unlikely, possible, probable or highly probable where subjective scores are assigned to different factors which are thought to influence the probability of failure.
- To ensure that this term will be used consistently, criteria for the descriptive categories should be defined.
- Several studies have been found using the qualitative assessment to generate the failure probability (Muhlbauer, 1992; Cagno et al., 2000; Dey, 2001; Dey & Gupta, 2001; Dey, 2004; Dey et al., 2004).
- Highly subjective and involves a lot of randomness and uncertainty.

Semi-Quantitative Approach

- POF should generally be more numerically based and detailed than the qualitative approach, but still contain a large element of engineering judgments.
- The common method is based on the guidelines by American Petroleum Institute, API 581 Risk-based Inspection Base Resource Document (API, 2000).
- Several papers have been found in the literature in applying the semi-quantitative approach in failure probability assessment (Khan & Haddara, 2003a; Khan & Haddara, 2003b; Willcocks & Bai, 2000; Khan & Haddara, 2004; Khan et al., 2004; Noori & Price, 2005; Krishnasamy et al., 2005; Noori et al., 2005).
- Since this approach relies on the database which is built using records from all plants within a company or from various plants within an industry, from literature sources, past reports, and commercial data bases in general, the values represent may not reflect the true failure frequencies for a specific plant and equipment.

Quantitative Approach

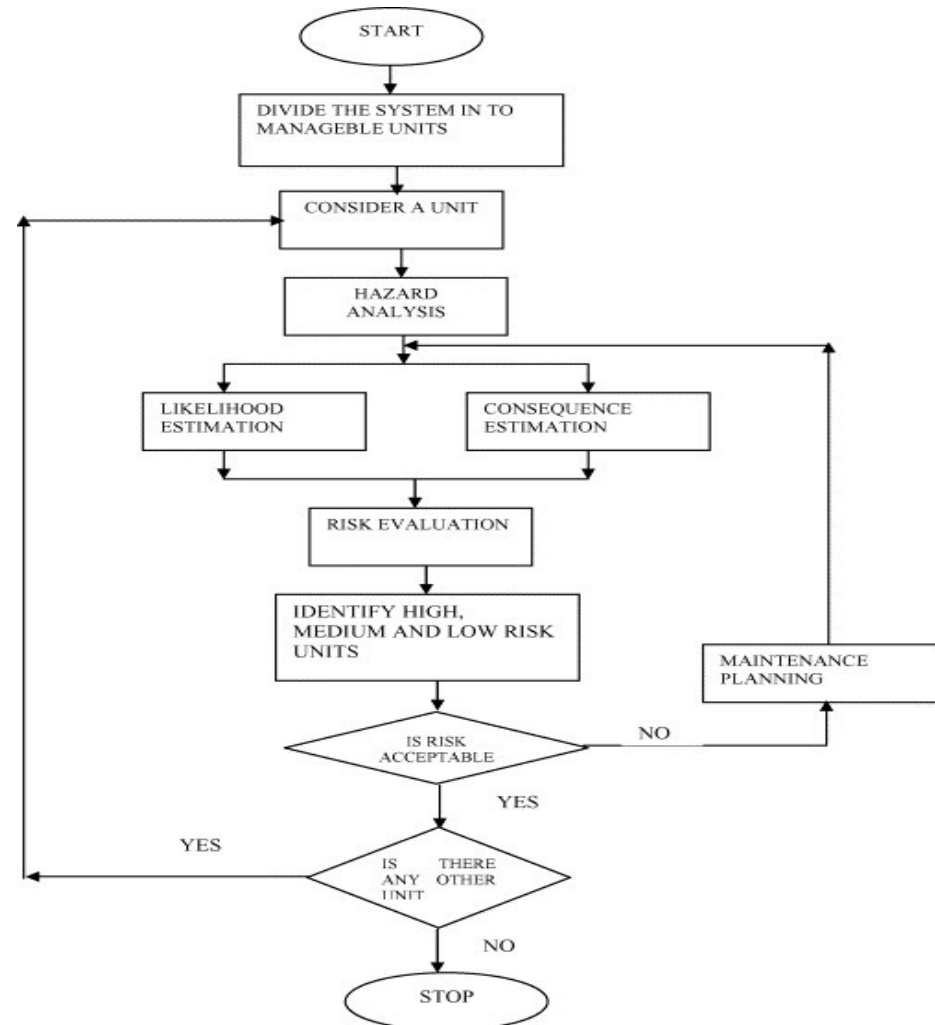
- These approaches determine the level of risk based on direct estimates of the probability and/or consequences of failure.
- Current quantitative risk assessment approaches focus on a single aspect of the consequence associated with failure. Published studies deal with either loss of life risk or economic risk (Hill, 1992; Concord, 1993; Nessim & Stephens, 1995; Pandey, 1998; Nessim et al., 2000). Integration of environmental damage, life safety, and economic risks has not been addressed adequately.
- Another limitation of quantitative risk assessment approaches is that they typically base the failure probability estimates on historical failure rates.

Quantitative Approach

- Publicly available databases do not usually allow subdivision of the failure data according to the attributes of a specific process component and where adequate subdivision is possible, the amount of data associated with a particular attribute set is very limited because of the rarity of the failures.
- Failure probabilities estimated from public data are, therefore, not sufficiently specific to represent a given failure in a specific process component.
- Another approach to calculate failure probability is based on the concept of structural reliability that includes Markovian models and hot spots on a component-by-component basis.

RBI Framework

- Two main phase:
 1. Risk assessment
 - risk assessment is of prime importance and critical phase, as the inspection decisions refer to the risk as a center.
 2. Inspection planning based on risk
- In risk assessment phase, there are 5 main steps, which are:
 1. Hazard analysis
 2. Likelihood analysis
 3. Consequence analysis
 4. Risk estimation
 5. Risk acceptance



RBI Framework

Phase	Steps	Objectives
Risk Assessment	Hazard analysis	To identify failure scenarios.
	Likelihood analysis	To calculate the frequency of failure for defined period of time.
	Consequence analysis	To quantify the potential consequences of the failure scenario e.g. production loss (performance loss and operational loss), asset loss, environmental loss, health and safety loss
	Risk estimation	To compute risk based on the result obtained in likelihood and consequence assessment modules.
	Risk acceptance	To compare the computed risk against the acceptable criteria.
Inspection planning based on risk	Inspection planning	To plan inspection intervals based on the calculated risk.

Risk Matrix

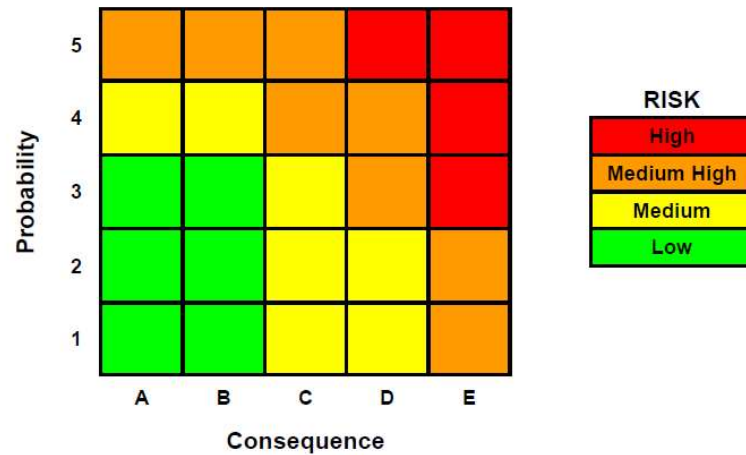


Table 4.1 – Numerical Values Associated with Probability and Area-Based Consequence Categories in API RBI

Probability Category (1)		Consequence Category (2)	
Category	Range	Category	Range (ft ²)
1	$D_{f-total} \leq 2$	A	$CA \leq 100$
2	$2 < D_{f-total} \leq 20$	B	$100 < CA \leq 1000$
3	$20 < D_{f-total} \leq 100$	C	$1000 < CA \leq 3000$
4	$100 < D_{f-total} \leq 1000$	D	$3000 < CA \leq 10000$
5	$D_{f-total} > 1000$	E	$CA > 10000$

Notes:

- In terms of the total damage factor, see [Part 2](#), paragraph 4.4.2.
- In terms of consequence area, see [Part 3](#), paragraphs 5.11.4.

RBI Standards

- API RP 580 RISK-BASED INSPECTION
 - provides guidance on developing a risk-based inspection program for fixed equipment in the refining and petrochemical, and chemical process plants.
- API RP 581 RISK-BASED INSPECTION TECHNOLOGY
 - provides quantitative procedures to establish an inspection program using risk based methods for pressurized fixed equipment including pressure vessel, piping, tankage, pressure relief devices, and heat exchanger tube bundles. . The intent of these publications is for API RP 580 to introduce the principles and present minimum general guidelines for RBI while this recommended practice provides quantitative calculation methods to determine an inspection plan.

API 581 methodology

- The API RBI methodology is presented in a three part volume.
 - a) **Part 1 – Inspection Planning Using API RBI Technology**
 - The methods used to obtain an inspection plan are provided in Part 1 for fixed equipment including pressure vessels, piping, atmospheric storage tanks, pressure relief devices and heat exchanger tube bundles. The pressure boundaries of rotating equipment may also be evaluated using this Part.
 - b) **Part 2 – Determination of Probability of Failure in an API RBI Assessment**
 - The probability of failure for fixed equipment is covered in Part 2. The probability of failure is based on the component type and damage mechanisms present based on the process fluid characteristics, design conditions, materials of construction, and the original construction code.
 - c) **Part 3 – Consequence Modeling in API RBI**
 - provides methods for computing the consequence of failure. Two methods are provided. The first method, or Level 1, is based on closed form solutions generated for a limited set of reference fluids or fluid groups. The second method, Level 2, is a general, more rigorous method that can be used for any fluid stream composition.

PRESSURE RELIEF DEVICE (PRD)

Weibull Analysis in API 581 Risk-Based Inspection

Overview

- Fundamental approach: to use a demand rate for the device combined with a Probability of Failure on Demand (POFOD) determined from plant-specific data if available, or starting with default data.
- These inputs are used to generate a POF as a function of time via a Weibull statistical approach.
- The consequence of device failure is determined based on methods outlined in Part 3, but modified to include overpressure demand cases, the amount of expected overpressure upon PRD failure, and the additional consequences associated with device leakage.

Overview

- The risk of a PRD is directly related to the equipment that is protected by the PRD.
- The consequence analysis is performed for the protected equipment at an elevated pressure representing the estimated overpressure upon PRD failure to open upon demand.

Overview

- The combination of consequence with a time-based POF results in a risk value that increases with time between tests. This allows test intervals to be determined based on risk targets.
- The flow chart shown (i.e. Figure 7.1 in API 581) illustrates the basic methodology required for the determination of a risk-based inspection and test schedule.
- The basic data required for the evaluation is listed in Table 7.1

PRD Failure Modes

For the PRD, the failure modes are grouped into two categories:

1. Fail to Open (FAIL)

- a. Stuck or fails to open (FTO)
- b. Device partially open (DPO)
- c. Opens above set pressure (OASP)

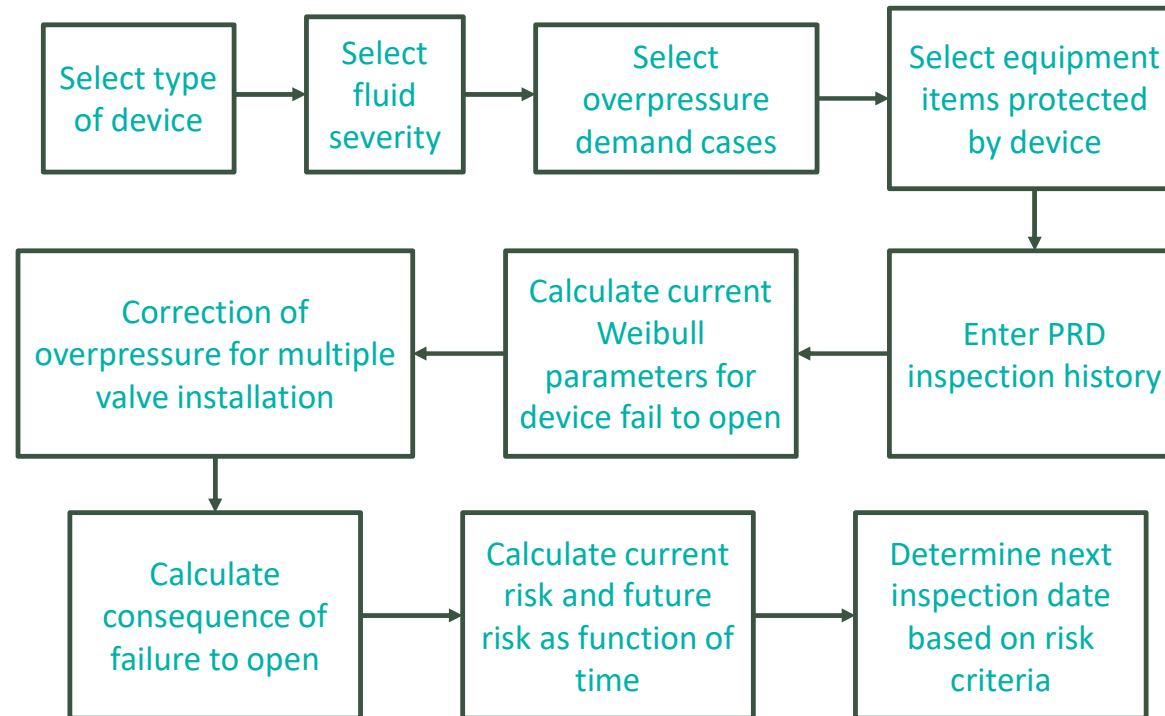
2. Leakage Failure (LEAK):

- a. Leakage past device (LPD)
- b. Spurious/premature opening (SPO)
- c. Device stuck open (DSO)

PRD Failure Modes

- **Primary concern:** Fail to open (FAIL)
 - This failure results in the potential of over-pressuring protected equipment and loss of containment.
 - Included in this case is the instance when a device only opens partially (DPO), where consequences are not as severe, but equipment may be over-pressured nonetheless.
 - API RP 581 conservatively considers any data indicating a partially open device as a failure to open as discussed in Section 7.2.2. Data that indicates a device OASP is included in the FAIL failure mode, since the POF curves are based on bench test data where a failure is defined as any test requiring a pressure greater than 1.3 times the set pressure.
 - A test point that opens above set pressure but does not exceed 1.3 times the set pressure is considered a successful test and is not included in the FAIL case.
 - Consequences associated with the FAIL failure mode include the effects of fires and explosions on personnel and equipment and the exposure of personnel to toxic fluids as a result of loss of containment.

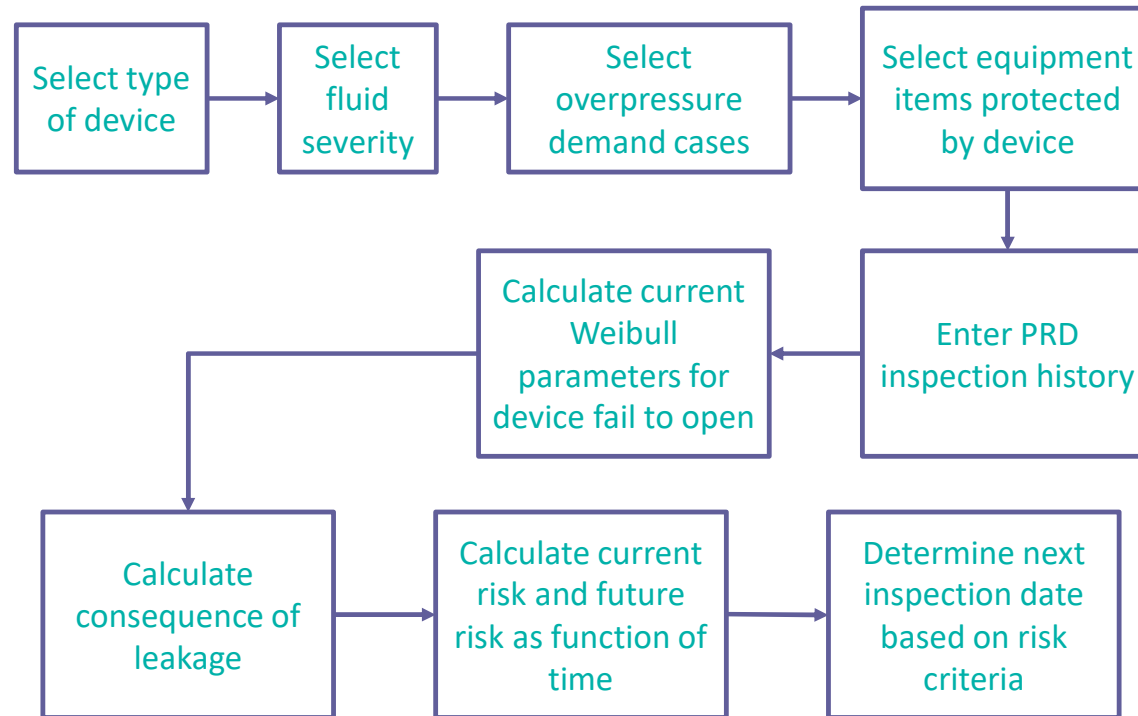
PRD Methodology (Fail to Open)



PRD Failure Modes

- **Secondary concern:** Leakage Failure (LEAK)
 - Failure of the PRD to contain the process while operating at normal conditions.
 - The API RP 581 methodology groups the remaining three failure modes together into the LEAK category
 - Leakage past the device (LPD), spurious or premature opening (SPO), and devices stuck open (DSO) are failures that will not result in overpressure or loss of containment from the protected equipment, but will result in potential unacceptable leakage from the process system.
 - The consequences of leakage through a PRD can range from a minor nuisance, causing some loss of product, to a more severe consequence resulting in a process shutdown to repair or replace the PRD. If the PRD discharges to the atmosphere, additional consequences may be environmental impact and potential for fires, explosions, and toxic exposure.

PRD Methodology (Leakage)



Use of Weibull Curve

- It is necessary to express the probability of failure to open and the probability of leakage as a function of time for risk-based planning of inspections and tests.
- Weibull functions are suitable for this task with the added advantage that they may be used to evaluate large populations of data points to seek trends.
- In the absence of large sets of failure data, the functions are still useful as a starting point since the parameters involved describe both the manner of failure and the time to failure.

How to calculate Probability of Failure (FAIL)

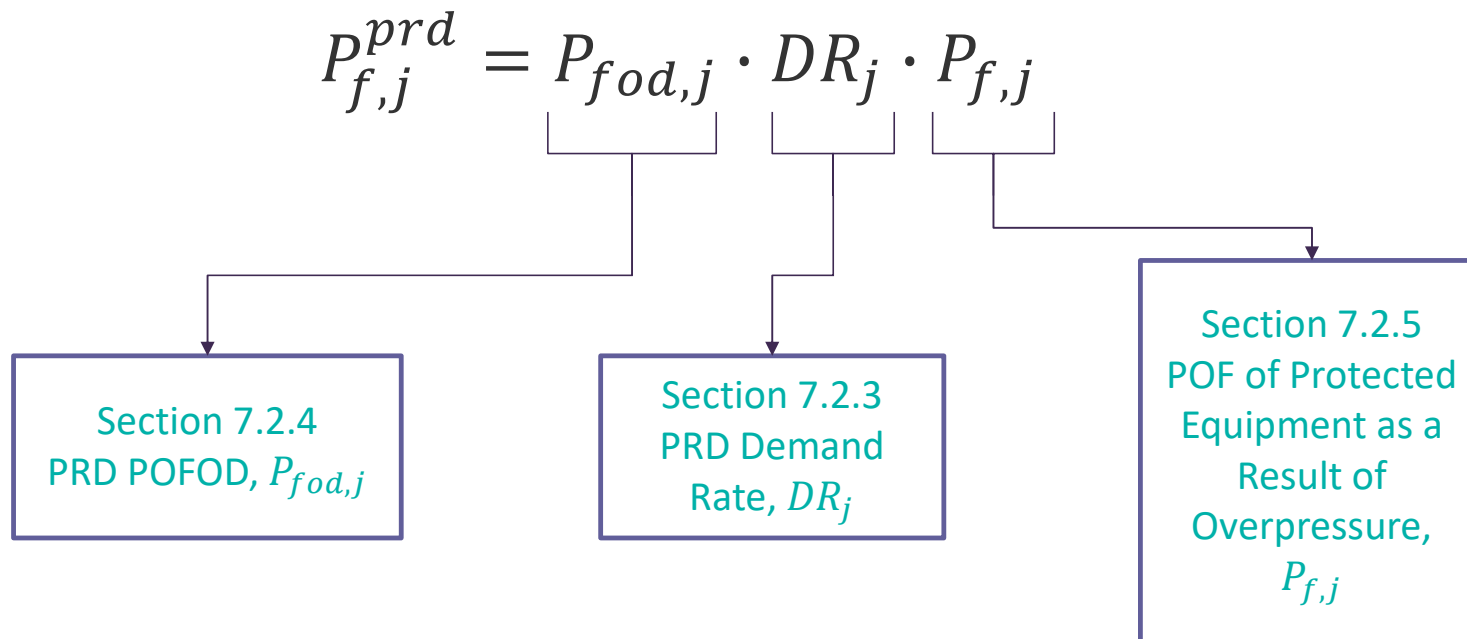


Table 7.5 – Default Weibull Parameters for POFOD

Table 7.5 – Default Weibull Parameters for POFOD

Fluid Severity	Conventional and Balanced Bellows PRVs ¹		Pilot-Operated PRVs ²		Rupture Disks ³	
	β	η_{def}	β	η_{def}	β	η_{def}
Mild	1.8	50.5	1.8	33.7	1.8	50.5
Moderate	1.8	23.9	1.8	8.0	1.8	50.5
Severe	1.8	17.6	1.8	3.5	1.8	50.5
Notes: 1. The η_{def} parameter values for conventional PRVs are reduced by 25% if the discharge is to a closed system or to flare, see paragraph 7.2.4.q. 2. The η_{def} parameter values for pilot-operated valves are currently based on the conventional PRV data; however, reduced by a factor of 1.5, 3, and 5 for Mild, Moderate and Severe services, respectively, see paragraph 7.2.4.e. 3. Without any failure rate data for rupture disks, the conventional PRV values for Mild services were used. This assumes that the RD material has been selected appropriately for the fluid service, see paragraph 7.2.4.f.						

Note: These parameters were determined using industry failure rate data. The majority of the available data indicated successful performance during the interval that the PRD was in service. The successful test points are referred to as suspensions and were included with the failure data in determination of the Weibull parameters.

Table 7.12 – Default Weibull Parameters for POL

Table 7.12 – Default Weibull Parameters for Probability of Leakage (POL)

Fluid Severity	Conventional PRVs ¹		Balanced Bellows PRVs ¹		Pilot-Operated PRVs ²		Rupture Disks ³	
	β	η_{def}	β	η_{def}	β	η_{def}	β	η_{def}
Mild	1.6	17.5	1.6	16.0	1.6	17.5	1.6	17.5
Moderate	1.6	15.5	1.6	14.0	1.6	15.5	1.6	17.5
Severe	1.6	13.1	1.6	11.5	1.6	13.1	1.6	17.5
Notes: 1. The η_{def} parameter values are increased by 25% for Conventional and Balanced PRVs that have soft seats. 2. The η_{def} parameter values for pilot-operated valves are currently based on the conventional PRV data, since there is currently no failure rate data to support otherwise. 3. Without any failure rate data for rupture disks, the conventional PRV values for Mild service were used.								

Note: These parameters were determined using industry failure rate data. The majority of the available data indicated successful performance during the interval that the PRD was in service. The successful test points are referred to as suspensions and were included with the failure data in determination of the Weibull parameters.

Calculation Procedure – POL at Specified Inspection Interval (Section 7.3.3)

STEP 1 – Select an inspection interval, t_{insp} .

STEP 2 – Determine the default values for the Weibull parameters, β and η_{def} using Table 7.12.

Table 7.12 – Default Weibull Parameters for Probability of Leakage (POL)

Fluid Severity	Conventional PRVs ¹		Balanced Bellows PRVs ¹		Pilot-Operated PRVs ²		Rupture Disks ³	
	β	η_{def}	β	η_{def}	β	η_{def}	β	η_{def}
Mild	1.6	17.5	1.6	16.0	1.6	17.5	1.6	17.5
Moderate	1.6	15.5	1.6	14.0	1.6	15.5	1.6	17.5
Severe	1.6	13.1	1.6	11.5	1.6	13.1	1.6	17.5

Notes:

1. The η_{def} parameter values are increased by 25% for Conventional and Balanced PRVs that have soft seats.
2. The η_{def} parameter values for pilot-operated valves are currently based on the conventional PRV data, since there is currently no failure rate data to support otherwise.
3. Without any failure rate data for rupture disks, the conventional PRV values for Mild service were used.

STEP 3 – Determine the adjustment factor, F_s , using Section 7.3.2.g.

g) Adjusted Default POL Curve for PRVs Containing Soft Seats

Soft seats (O-rings) are often added to spring-loaded PRVs to reduce the potential for leakage across the seat. When a conventional or balanced bellows PRV contains a soft seat design, the η parameter for the default POL Weibull curve is increased by a factor of 1.25 in accordance with the following factors:

$$F_s = 1.25 \quad \text{for soft seated designs}$$

$$F_s = 1.0 \quad \text{for all other cases}$$

Calculation Procedure – POL at Specified Inspection Interval (Section 7.3.3)

STEP 4 – Determine the environmental adjustment factor, F_{env} , using Table 7.6.

Table 7.6 – Environmental Adjustment Factors to Weibull η Parameter

Environment Modifier	Adjustment to POFOD η Parameter	Adjustment to POL η Parameter
Operating Temperature 200<T<500°F	1.0	0.8
Operating Temperature > 500°F	1.0	0.6
Operating Ratio >90% for spring-loaded PRVs or >95% for pilot-operated PRVs	1.0	0.5 ¹
Installed Piping Vibration	1.0	0.8
Pulsating or Cyclical service, such as Downstream of Positive Displacement Rotating Equipment	1.0	0.8
History of Excessive Actuation in Service (greater than 5 times per year)	0.5	0.5 ²
History of Chatter	0.5	0.5
Notes: 1. Some Pilot-operated PRVs operate extremely well with operating ratios approaching 98%. In these cases, the environmental factor should not be applied (reference API 520 Part 1). 2. This factor should not be applied if the environmental factor for operating ratio is already applied.		

STEP 5 – Calculate the modified characteristic life, η_{mod} , using Equation (1.34) and the factors obtained in STEPs 3 and 4.

$$\eta_{mod} = F_s \cdot F_{env} \cdot \eta_{def} \quad (1.34)$$

Calculation Procedure – POL at Specified Inspection Interval (Section 7.3.3)

STEP 6 – Assemble the PRD's inspection history. Grade each record using the inspection effectiveness table, Part 2, Annex 2.C, Table 2.C.3.1. Record the results of each inspection record; NO LEAK/LEAK and determine the confidence factors, CF_i , as applicable, for each inspection history based on the results of the test. Determine the time duration, $t_{dur,i}$ of each inspection cycle.

2.C.3.1 Tables

Table 2.C.3.1 – Inspection and Testing Effectiveness for Pressure Relief Devices

Inspection Effectiveness	Component Type	Description of Inspection
Highly Effective A	Pressure Relief Device	A bench test has been performed on the PRD in the as-received condition from the unit and the initial leak pressure, opening pressure and the reseal pressure has been documented on the test form. The inlet and outlet piping has been examined for signs of excessive plugging or fouling.
	Rupture Disk	No inspection methods are available to meet the requirements for an A level inspection.
Usually Effective B	Pressure Relief Device	A bench test has been performed, however, the PRD was cleaned or steamed out prior to the bench test. Additionally, a visual inspection has been performed where detailed documentation of the condition of the PRD internal components was made. An in-situ test has been performed using the actual process fluid to pressurize the system.
	Rupture Disk	The rupture disk is removed and visually inspected for damage or deformations.
Fairly Effective C	Pressure Relief Device	A visual inspection has been performed without a pop test, where detailed documentation of the condition of the PRD internal components was made. A trevtest or in-situ test has been performed where the actual process fluid was not used to pressurize the system.
	Rupture Disk	The space between the disk and the PRD is monitored for leakage in accordance with the ASME Code and API RP 520 Part 2.
Ineffective D	Pressure Relief Device	Valve overhaul performed; No pop test conducted/documentated
	Rupture Disk	No details of the internal component were documented.

NOTE: This table does not prescribe specifically to the five categories as discussed in this annex. However, given the methodology presented, it is in agreement with the division of those categories.

Calculation Procedure – POL at Specified Inspection Interval (Section 7.3.3)

STEP 7 – Starting at the earliest inspection record, update the modified characteristic life, η_{mod} determined in STEP 5 as follows:

STEP 7.1 – Calculate the prior probability of leakage using Equation (1.17).

$$P_{f,prior}^{prd} = 1 - \exp \left[- \left(\frac{t}{\eta_{mod}} \right)^\beta \right] \quad (1.17)$$

Note that for the first inspection record, the modified characteristic life, η_{mod} , is used. Subsequent inspection records will use the updated characteristic from STEP 7.5 below.

STEP 7.2 – Calculate the prior probability of passing using Equation (1.18).

$$P_{p,prior}^{prd} = 1 - P_{f,prior}^{prd} \quad (1.18)$$

Calculation Procedure – POL at Specified Inspection Interval (Section 7.3.3)

STEP 7.3 – Determine the conditional probability of leakage and conditional probability of pass using Equations (1.19) and (1.20), respectively.

$$P_{f,cond}^{prd} = (1 - CF_{pass}) \cdot P_{p,prior}^{prd} \quad (1.19)$$

$$P_{f,cond}^{prd} = CF_{fail} \cdot P_{f,prior}^{prd} + (1 - CF_{pass}) \cdot P_{p,prior}^{prd} \quad (1.20)$$

STEP 7.4 – Calculate the weighted probability of leakage, $P_{l,wgt}^{prd}$, using the appropriate equation from Table 7. 9.

Table 7. 9 - Inspection Updating Equations

Inspection Effectiveness and Result	Equation for Weighted POFOD
Highly Effective Pass	$P_{f,wgt}^{prd} = P_{f,prior}^{prd} - 0.2 \cdot P_{f,prior}^{prd} \left(\frac{t}{\eta} \right) + 0.2 \cdot P_{f,cond}^{prd} \left(\frac{t}{\eta} \right)$
Usually Effective Pass	
Fairly Effective Pass	
Highly Effective Fail	$P_{f,wgt}^{prd} = P_{f,cond}^{prd}$
Usually Effective Fail	
Fairly Effective Fail	$P_{f,wgt}^{prd} = 0.5 \cdot P_{f,prior}^{prd} + 0.5 \cdot P_{f,cond}^{prd}$

Calculation Procedure – POL at Specified Inspection Interval (Section 7.3.3)

STEP 7.5 – Determine the updated characteristic life, η_{upd} , using Equation (1.21) and the weighted probability of leakage, $P_{l,wgt}^{prd}$, established in STEP 7.4

$$\eta_{upd} = \frac{t}{\left(-\ln[1 - P_{f,wgt}^{prd}]\right)^{\frac{1}{\beta}}} \quad (1.21)$$

STEP 7.6 – Repeat these steps for each of the inspection records available for the PRD until a final updated value for the characteristic life, η_{upd} , is determined.

STEP 8 - Calculate the probability of leakage for the specific PRD in accordance with Equation (1.35).

$$P_i^{prd} = 1 - \exp\left[-\left(\frac{t}{\eta_{upd}}\right)^{\beta}\right] \quad (1.35)$$

Heat exchanger

Weibull Analysis in API 581 Risk-Based Inspection

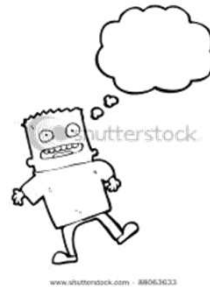
Case Study: Heat Exchanger Tube Bundles

Table 8.2 in API RBI shows the reliability data for HEX tube bundle

Table 8.2: Example - Matching Bundles from Reliability Database

Bundle Tag #	In-Service Duration (years)	Failure Reported
191-X-25A-T1	18	Yes
191-X-25A-T2	22	Yes
191-X-25A-T3	16	No
E101-A-T1	10	Yes
E322-A-T1	12	No
E322-A-T2	13	No
HE-115-T1	14	Yes
HE-115-T2	25	No
PR6419-T1	8	Yes

How to analyze this data?



LIFE DATA ANALYSIS (LDA)

Definition of Bundle Failure

- A failure is defined as a tube leak for the purposes of RBI.
- When the controlling damage mechanism for the bundle is general corrosion, the current condition or remaining life of a bundle can usually be quantified or expressed as a percent of the original wall thickness. Where inspection records document average remaining wall thickness, this data may be used to determine when failure occurred or to predict the point in the future where a failure is probable.
- Typically, bundles are replaced or repaired prior to failure due to a deteriorated condition. In these situations, one approach for estimating bundle failure is to assume that the bundle was replaced at 75% of the bundle life. A 75% bundle life assumption is recommended and may be modified, if desired.

Calculation of POF

1. **Method 1** – Using matching criteria to filter on the reliability library, (see Section 8.3.2) – If a value of the MTTF has not been specified, then the Weibull distribution is assumed and a Weibayes' analysis is performed on the matching bundles in the reliability library.
2. **Method 2** – Owner-User Specified Weibull Parameters (see Section 8.3.4) – This involves specification of the Weibull β and η parameters for the exchanger bundle being evaluated.
3. **Method 3** – Owner-User Specified MTTF (see Section 8.3.4.2) – This involves the owner-user providing the MTTF for the bundle being evaluated. The method will convert the MTTF to a Weibull curve using a β value of 3.0. As an option, the Weibull β parameter in addition to the MTTF is specified.
4. **Method 4** – Specific Bundle Inspection History (see Section 8.3.4.3) – Once enough inspection history is accumulated, the statistics could be determined for the bundle.

Method 1: Using matching criteria to filter on the reliability library

A reliability database or Weibull library is extremely important to the evaluation of the risks associated with bundle failure. Certain exchanger details are required to be gathered for each bundle entry in the reliability database. Minimum basic data required for the database are indicated in Table 8.1. Optional data is also shown in Table 8.1 that could be collected to provide additional matching/filtering capability for the reliability library.

Method 2: Owner-User Specified Weibull Parameters

- As an option, the owner-user can provide the β and η parameters for the exchanger bundle and use Equation (1.60) to determine the POF for the bundle as a function of time.

$$P_f^{tube} = 1 - R(t) = 1 - \exp\left[-\left(\frac{t}{\eta}\right)^\beta\right]$$

Method 3: Owner-User Specified MTTF

- When sufficient inspection information exists for a bundle such that a *MTTF* may be determined, the analyst can specify the bundle *MTTF*. A Weibull distribution can still be used.
- If the β parameter (slope) is known (default is 3.0), the η parameter (characteristic life) can be calculated using the gamma function as shown in Equation (1.61).

$$MTTF = \eta \cdot \Gamma \left[1 + \frac{1}{\beta} \right]$$

Method 4: POF Calculated using Specific Bundle History

- Once the bundle has accumulated at least two life cycles with inspection data, a Weibayes' analysis can be performed keeping the β parameter the same as determined from the matching bundle criteria. The η parameter (characteristic life) can be recalculated using Eq. (1.62).

$$\eta = \left(\sum_{i=1}^N \frac{t_{dur,i}^{\beta}}{r} \right)^{\frac{1}{\beta}}$$

Where N is the number of past bundles, $t_{dur,i}^{\beta}$ is the time in service for each bundle in years, r is the number of failed bundles, and β is the Weibull slope parameter.

Method 4: POF Calculated using Specific Bundle History

- For the 191-X-25A exchanger in our example (see Table 8.2), there were failures recorded after 18 and 22 years. The current bundle (T3) has been in-service for 16 years without failure (suspension). The modified characteristic life may be recalculated using Equation (1.63) as demonstrated below:

$$\eta = \left[\frac{(22)^{2.568} + (18)^{2.568} + (16)^{2.568}}{2} \right]^{\frac{1}{2.568}} = 22.16 \text{ years}$$

- Note that this is slightly higher than the 20.45 year characteristic life calculated using the matching bundles from the reliability database, see Equation (1.65). The user should be cautioned that this method assumes that the bundle has not been redesigned over its life time. Changes in metallurgy, process conditions, or bundle design need to be considered before assuming that all of the past failure history is representative of the current bundle being evaluated.

Method 4: POF Calculated using Specific Bundle History

Table 8.2 – Example - Matching Bundles from Reliability Database

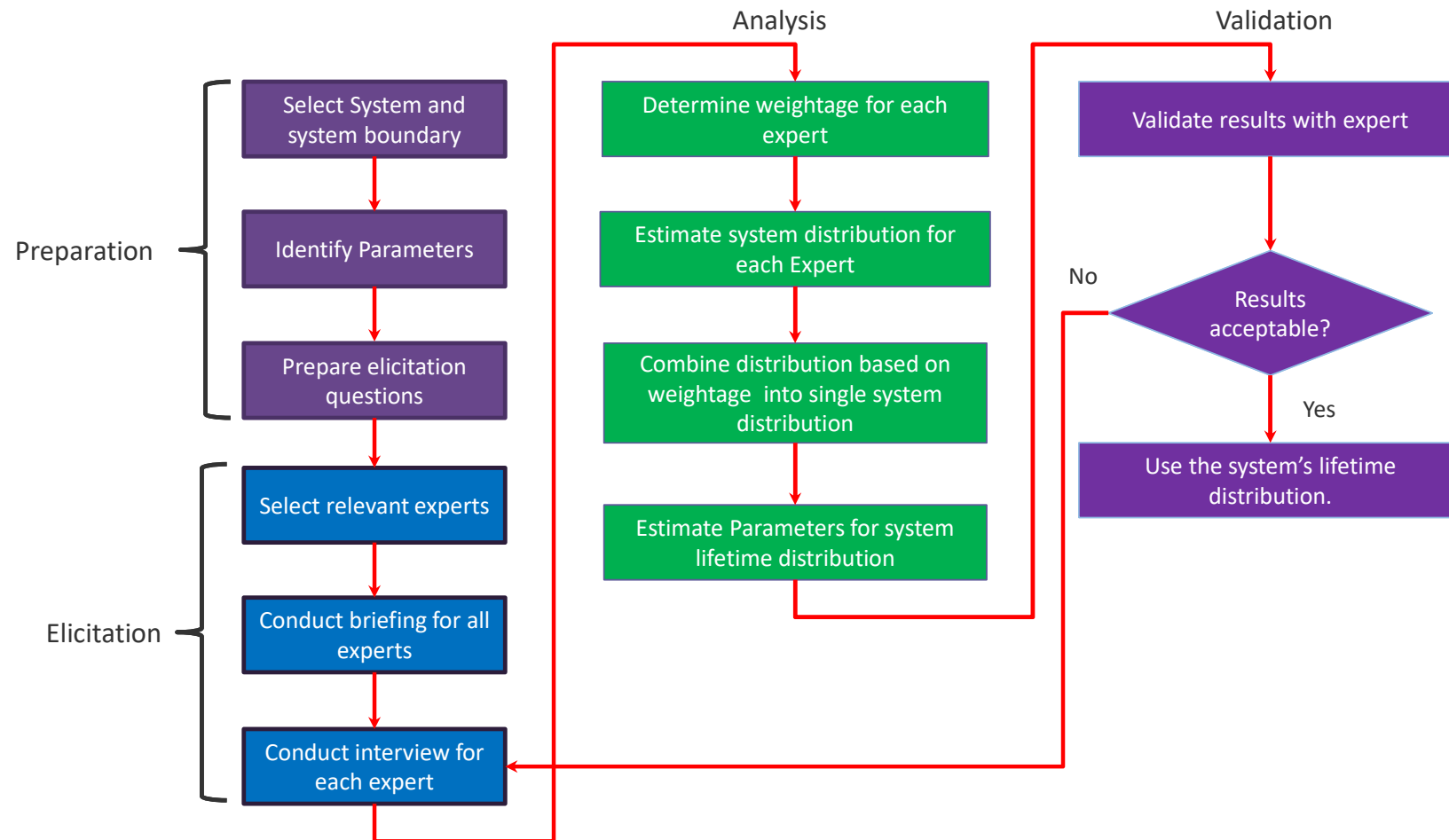
Bundle Tag #	In-Service Duration (years)	Failure Reported
191-X-25A-T1	18	Yes
191-X-25A-T2	22	Yes
191-X-25A-T3	16	No
E101-A-T1	10	Yes
E322-A-T1	12	No
E322-A-T2	13	No
HE-115-T1	14	Yes
HE-115-T2	25	No
PR6419-T1	8	Yes

Expert Elicitation

Recall: Sources of Data

- Historical Data – test and field data on the same components /equipment
- Vendor data – Data from manufacturer / vendor / consultant
- Test data – experimental data of the parts
- Operational data – Field data collected under actual operating conditions
- Handbook data – theoretical data from standard engineering handbook, Reliability database i.e. OREDA, MIL-HDBK 217F
- **Judgmental data** – information based on expert opinion inputs when there is no other data available

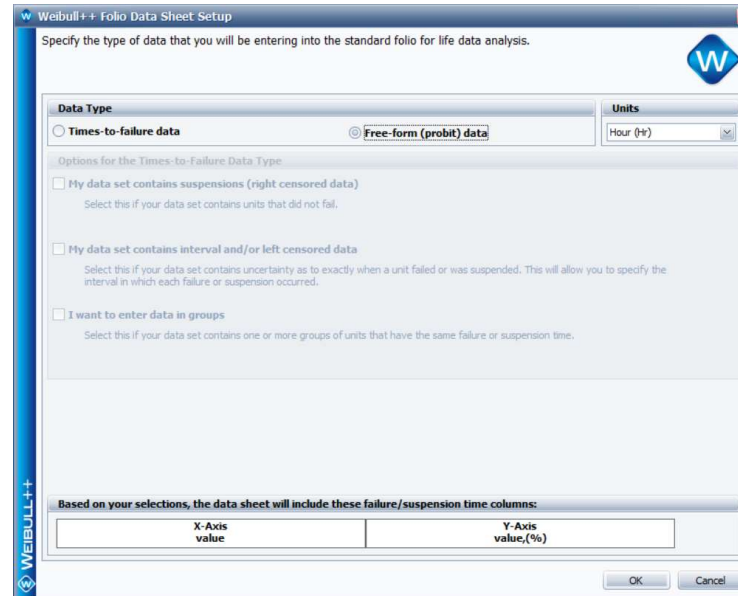
Expert Elicitations Methodology



Expert Elicitations in Weibull ++

Estimated time to failure (Day)	Expected probability of failure (%)
100	30
200	35
300	50
400	70
500	80
600	90

Based on average from a number of experts



The image shows the 'Weibull++ Folio Data Sheet Setup' dialog box. It is titled 'Weibull++ Folio Data Sheet Setup' and has a 'W' logo in the top right corner. The main text says 'Specify the type of data that you will be entering into the standard folio for life data analysis.' Below this, there are two main sections: 'Data Type' and 'Units'. In the 'Data Type' section, there are two radio buttons: 'Times-to-failure data' and 'Free-form (probit) data'. The 'Free-form (probit) data' option is selected. Below these radio buttons, there are three checkboxes under the heading 'Options for the Times-to-Failure Data Type': 'My data set contains suspensions (right censored data)', 'My data set contains interval and/or left censored data', and 'I want to enter data in groups'. All three checkboxes are currently unchecked. The 'Units' section has a dropdown menu showing 'Hour (hr)'. At the bottom, there is a section titled 'Based on your selections, the data sheet will include these failure/suspension time columns:' which contains two columns: 'X-Axis value' and 'Y-Axis value, (%)'. At the very bottom, there are 'OK' and 'Cancel' buttons.

Expert Elicitations in Weibull ++

C7			
	X-Axis value (Day)	Y-Axis value, (%)	Subset ID 1
1	100	30	
2	200	35	
3	300	50	
4	400	70	
5	500	80	
6	600	90	
7			
8			
9			
10			
11			
12			
13			
14			
15			
16			
17			
18			
19			
20			
21			
22			
23			
24			
25			
26			

Data1 Plot of Data1

STANDARD FOLIO

Distribution

2P-Weibull

Analysis Settings

RRY

SRM

MED

Analysis Summary

Parameters

Beta

1.069067

Eta (Day)

336.200874

Other

Rho

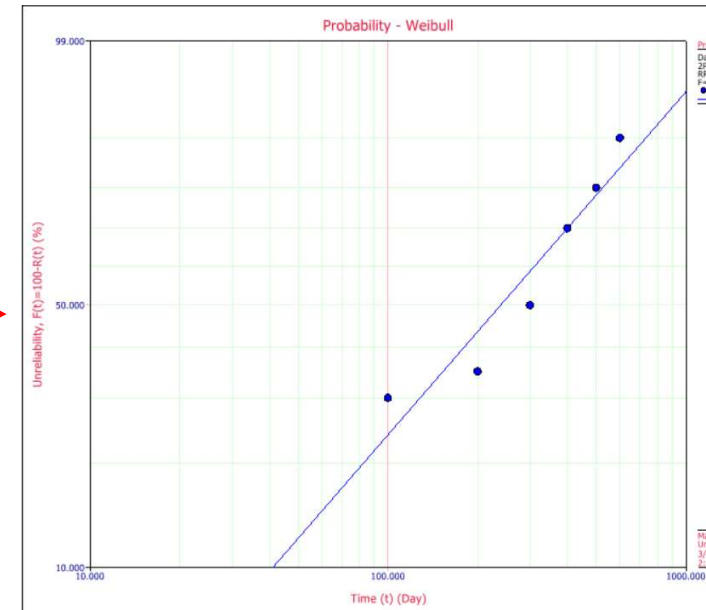
0.951657

Failures/Suspensions

F/S

6/0

Comments



Once the failure distribution is identified, other reliability measures can be estimated.

*Thank
you*



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